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NORTH AMERICAN ELECTRIC  
RELIABILITY CORPORATION

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# Power System Oscillation Monitoring and Mitigation

Technical Reference Document

**MM DD** 2025

RELIABILITY | RESILIENCE | SECURITY



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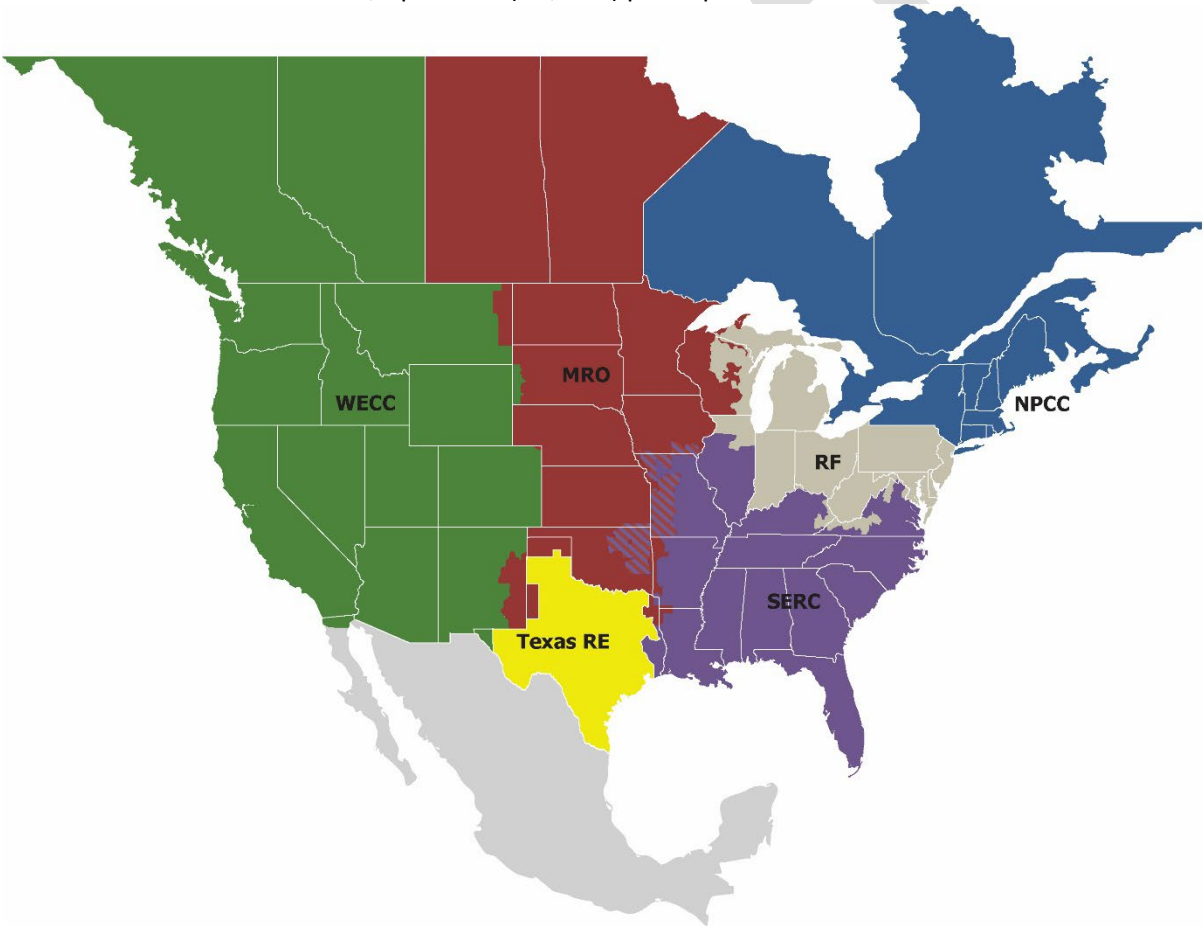
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# Preface

Electricity is a key component of the fabric of modern society and the Electric Reliability Organization (ERO) Enterprise serves to strengthen that fabric. The vision for the ERO Enterprise, which is comprised of NERC and the six Regional Entities, is a highly reliable, resilient, and secure North American bulk power system (BPS). Our mission is to assure the effective and efficient reduction of risks to the reliability and security of the grid.

Reliability | Resilience | Security  
*Because nearly 400 million citizens in North America are counting on us*

The North American BPS is made up of six Regional Entities as shown on the map and in the corresponding table below. The multicolored area denotes overlap as some load-serving entities participate in one Regional Entity while associated Transmission Owners/Operators (TO/TOP) participate in another.



|                 |                                      |
|-----------------|--------------------------------------|
| <b>MRO</b>      | Midwest Reliability Organization     |
| <b>NPCC</b>     | Northeast Power Coordinating Council |
| <b>RF</b>       | ReliabilityFirst                     |
| <b>SERC</b>     | SERC Reliability Corporation         |
| <b>Texas RE</b> | Texas Reliability Entity             |
| <b>WECC</b>     | WECC                                 |

# Executive Summary

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Recent oscillation events, such as the January 11, 2019, forced oscillation event in Florida that interacted with a natural system mode of the Eastern Interconnection (EI) and led to propagation of the oscillation across the Interconnection, have highlighted the need for increased monitoring and consistency in the monitoring of oscillation disturbances. The key recommendations from the report<sup>1</sup> on the event included the need for Reliability Coordinators (RC) and Transmission Operators (TOPs) to utilize real-time oscillation detection tools. RCs and TOPs should have real-time oscillation detection tools in place to identify when oscillations are occurring and determine if the oscillations are limited locally within their footprint or are more widespread. Equally important is the ability to distinguish between forced oscillations, poorly damped natural oscillations, or scenarios where forced oscillations may be propagating across a wider area due to resonance conditions. There was a recognition that RCs should improve their communication with other RCs in the event of widespread oscillation disturbances in the BPS and when operating procedures could be an effective means of ensuring this coordination upon the identification of an oscillation.

The NERC Synchronized Measurement Working Group (SMWG) was also requested to develop guidance on oscillation analysis methods to encourage consistency in the system quantities that are monitored for oscillation events and the respective thresholds for alarms. The detection and alarming of oscillations and their classification in a consistent manner is critical in ensuring coordinated mitigation of both local and widespread oscillation disturbances in the BPS.

It is also important to identify the types of operator actions necessary for different types of oscillations. These actions can range from locating the source of forced oscillations to reducing power transfers across major transmission paths. Actions can also include making topological changes to improve damping of widespread natural system oscillations or combinations of those actions if there are interactions of forced oscillations with natural system modes that result in propagation across the Interconnection.

The following are key findings and recommendations of this technical reference document:

- For monitoring of natural oscillations, various methods exist that can utilize ambient or post-disturbance data to determine the system modes that are significant. These assessments are recommended to be done annually or following significant changes in the system. After identifying the significant modes, additional analyses can be performed to determine the locations where the modes are more observable and the respective thresholds for alarming, which may call for operator actions, based on damping and the energy of the modes. In addition, analyses using power flow cases and the associated dynamic models can be utilized to validate the modeling of the observed significant modes and to determine what mitigation actions might be effective in improving the damping of those modes.
- Forced oscillations can be detected using various methods and use thresholds established by prior analyses to differentiate between sustained forced oscillations and normal ambient changes in system conditions. Once the forced oscillations are detected, various methods exist to determine the locations from where the forced oscillations originate.
- Under certain conditions, forced oscillations can propagate across an Interconnection due to resonance with a natural system mode. Mitigation of such widespread oscillations can require a combination of mitigation actions ranging from locating and eliminating the source of the oscillation to taking actions to reduce the impacts across the system by improving the damping of the impacted system mode.
- Mitigation of local and widespread oscillation disturbances and their impact requires effective tools and coordination between RCs and TOPs to determine the type of oscillation and the appropriate mitigation actions.

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<sup>1</sup> Lesson Learned: Interconnection Oscillation Disturbances: [https://www.nerc.com/pa/rrm/ea/Lessons Learned Document Library/LL20210501\\_Interconnection\\_Oscillation\\_Disturbances.pdf](https://www.nerc.com/pa/rrm/ea/Lessons%20Learned%20Document%20Library/LL20210501_Interconnection_Oscillation_Disturbances.pdf)

## Introduction

The SMWG, formerly the Synchronized Measurement Subcommittee, has provided well-accepted guidance on what oscillations are, why they occur, and what are the various tools to monitor them. Various previously published reports on Interconnection oscillation analyses<sup>2</sup> and forced oscillation events<sup>3</sup> provide recommendations on what RCs and TOPs should do to monitor oscillations, emphasizing the need to coordinate actions and develop tools. Monitoring of oscillations requires the setup and configuration of real-time tools backed by a certain level of analyses and preparation to determine the quantities for monitoring and their respective alarming thresholds followed by the corresponding operator actions.

The analysis required can vary depending on whether a tool is being set up to monitor a natural oscillation or a forced oscillation. The levels at which operator actions are necessary to intervene and mitigate for any potential reliability impacts of oscillations can also vary based on the type and location of the oscillation. Several RCs and TOPs have developed operating procedures to supplement the monitoring of oscillations, while others are in the process of developing them. These operating procedures contain specific actions for operators to follow during critical oscillatory conditions. Model-based simulations provide an opportunity to determine required mitigation actions for consistency in developing operating procedures and mitigation plans.

This technical reference document addresses two distinct types of power system oscillations, natural and forced, and is divided into six chapters:

- **Chapter 1** provides an overview of oscillations.
- **Chapter 2** discusses natural oscillations, a.k.a. electromechanical oscillations as they arise from the electro-mechanical dynamics inherent to synchronous generator-dominated power systems.
- **Chapter 3** discusses forced oscillations which are the response of a system to a periodic input that continuously excites a system, such as a generator with a steam valve cycling on and off continuously.
- **Chapter 4** discusses the case where forced oscillations resonate with a system's natural mode and results in inter-area forced oscillations observed across a wide area.
- **Chapter 5** provides recommended guidelines for operators to address wide-area oscillations.
- **Chapter 6** discusses examples of existing practices in different RCs and grid operators.

In summary, the chapters of this technical reference document aim at providing a framework of methods to do the following:

- Conduct natural and forced oscillation analysis by using examples
- Determine quantities to be monitored and quantify their boundaries and how to implement these boundaries as monitoring thresholds in tools
- Establish distinction between local and system issues
- Determine and validate mitigation actions

<sup>2</sup> Interconnection Oscillation Analysis: [https://www.nerc.com/comm/RSTC/SMWG/Interconnection\\_Oscillation\\_Analysis.pdf](https://www.nerc.com/comm/RSTC/SMWG/Interconnection_Oscillation_Analysis.pdf)

<sup>3</sup> Eastern Interconnection Oscillation Disturbance: [https://www.nerc.com/pa/rrm/ea/Documents/January\\_11\\_Oscillation\\_Event\\_Report.pdf](https://www.nerc.com/pa/rrm/ea/Documents/January_11_Oscillation_Event_Report.pdf)

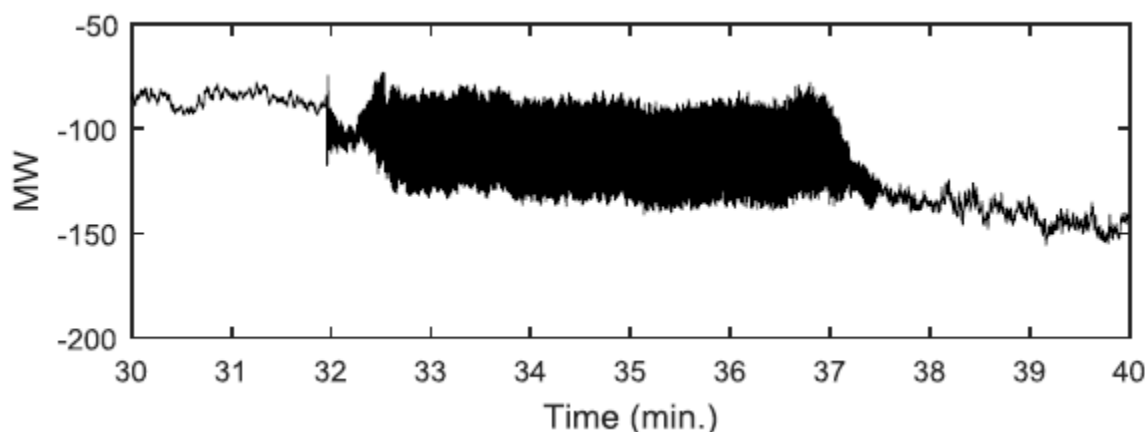
## Chapter 1: Introduction to Oscillations

Oscillations are inevitable in power systems and have been a threat to reliability for years. The monitoring and mitigation measures that are appropriate for an oscillation depend on various factors, most importantly the type of oscillation. Oscillation refers to the periodic variation of a system variable, such as voltage, frequency, or rotor angle, around an equilibrium point. The variations can be sustained by periodic external inputs of disturbances (forced oscillations) or spontaneously arise from the system's intrinsic dynamic features (natural oscillations). Hence, the two broad categories of oscillations considered in this document are *natural* and *forced*. This reference document discusses the analysis of natural and forced oscillations with both model- and measurement-based<sup>4</sup> approaches.

Natural oscillations are the natural response of a power system to system disturbances. Some of these oscillations are continuously occurring due to random load changes. This kind of response results in low-level oscillations referred to as the system's *ambient* response since they are usually buried inside the noise and can only be extracted by using statistical methods. Visible natural oscillations usually occur following a sudden large disturbance, such as a fault, line trip, generator trip, or load rejection.

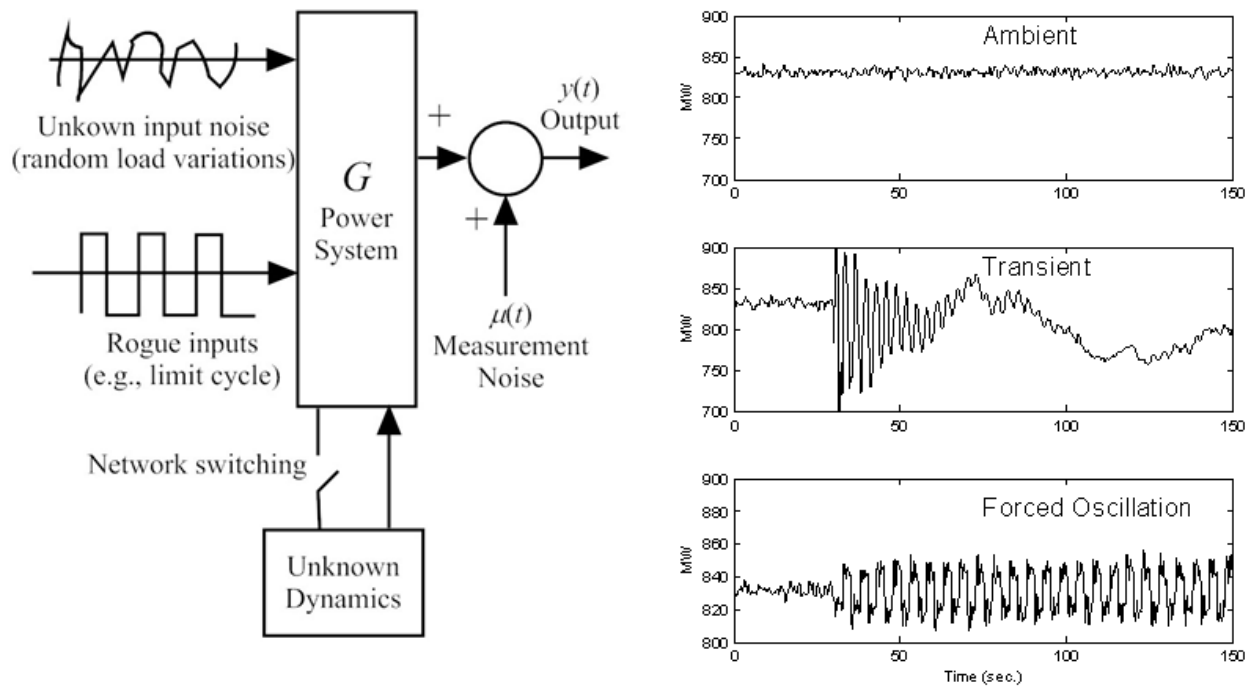
In contrast, forced oscillations are the response of a system to a "rogue" input subjecting the system to a periodic disturbance. Examples of forced inputs include cyclic loads, a malfunctioning steam valve cycling on and off, or an arc furnace inducing its dynamics on the system. The frequency of the forced oscillation is governed by the frequency of the forcing input rather than the system's natural dynamical property. As the forced oscillations are a system response to forced input, these oscillations do not decay like natural oscillations and remain in the system unless the forcing input is removed from the system, as depicted in [Figure 1.1](#).

A summary of the ambient, transient, and forced responses of a power system is provided in [Figure 1.2](#). In the following subsections, additional discussion of the significance of these oscillations from a BPS reliability perspective is provided.



**Figure 1.1: MW Flow on Transmission Line during Forced Oscillation Event Caused by Control Failure**  
(Source: Montana Tech)

<sup>4</sup> In the past, many oscillation events went by unnoticed due to an inability to detect oscillations using supervisory control and data acquisition (SCADA) measurements. Synchrophasors provide the resolution and accuracy necessary to capture oscillations (up to around 5 Hz) in support of off-line studies and real-time monitoring and mitigation. Further details on measurement technologies are provided in [Appendix B](#).



**Figure 1.2: Types of System Response – Ambient, Transient, Forced Oscillation**  
(Source: Montana Tech)

## Natural Oscillations

Natural oscillations are the inherent characteristics of power systems, that are dominated by synchronous generators and governed by the system's electromechanical modes. Mathematically, the modes are given by the complex eigenvalues of the system state matrix and their associated eigenvectors. They dictate how generators oscillate against each other through the transmission network. The modes are characterized by the following properties:

- a. **Mode Frequency:** The number of cycles per second. Common causes of oscillations at different frequencies are described below:
  - i. 0.01–0.15 Hz: governor, plant controller, or automatic generation control (AGC)
  - ii. 0.15–1.00 Hz: electromechanical inter-area and some local plant oscillations
  - iii. 1.00–5.00 Hz: local generator control oscillations, excitation controls, dc circuit controls
  - iv. 5.00–50.00 Hz: high (subsynchronous) frequency torsional, dc controls, inverter-based generation controls, subsynchronous interactions
- b. **Mode Damping/Damping Ratio:** The damping describes how quickly an oscillation decays (or grows when negative) once initiated. In this way, the damping serves as an indicator of the system's small signal stability margin. In practice, because oscillation risks also depend on frequency, the damping effect of different frequencies can be made comparable by normalizing the damping by the magnitude of the mode to derive the damping ratio. Under normal operating conditions, the damping ratio of dominant inter-area modes tend to remain between 5% and 25%.
- c. **Mode Shape:** Relative perception of an oscillation mode in different parts of a power grid. The shape is defined by the amplitude and phase of the mode at specific measurement locations. The amplitude describes the visibility of the mode at that location, and the angles describe the relationships between different locations whether they are oscillating together or against each other.

The dominance of a system mode is largely determined by its damping ratio and shape. As described above, the damping ratio is a direct indicator of system stability margin. Modes having damping ratios near 0% are a significant threat to system reliability because contingencies can push the damping ratio to become negative, leading to growing oscillations in frequency and power that will increase until protective devices engage. This type of instability can lead to system breakup. See [Appendix D](#) for an example.

Inter-area natural modes (two or more large groups against each other) of oscillation are particularly significant because they dictate how generators in different parts of the system oscillate against each other. Oscillations with a more localized shape (one or a small group against the rest of the system) are typically less consequential from a BPS reliability standpoint but should still be treated accordingly.

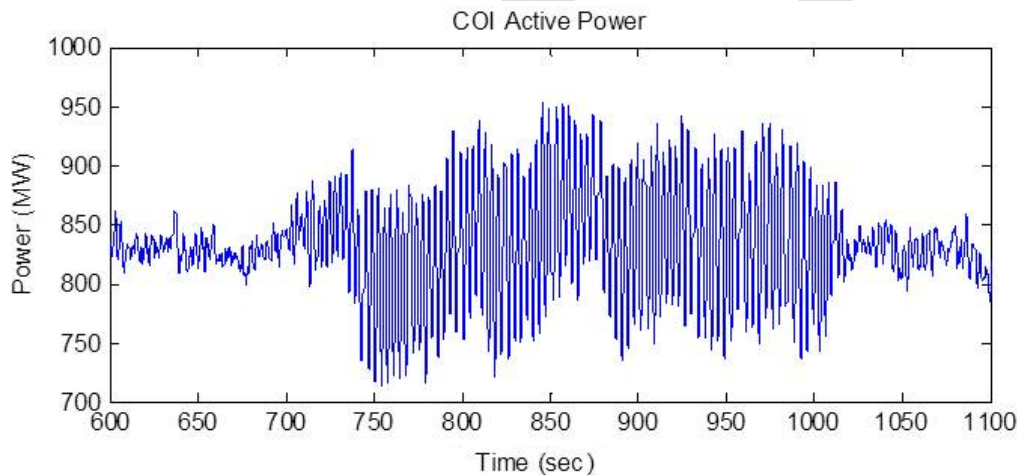
## Forced Oscillations

Forced oscillations can occur in the power grid under a wide variety of circumstances, such as equipment failure, inadequate control designs, and abnormal generator operating conditions. Depending on the frequency of the forced oscillations, the impact of these oscillations may be experienced locally near their source or across wide geographical regions of the system.

Forced oscillations with frequencies over 1 Hz tend to be local in nature and may be seen in a few plants near the source of the oscillations. Forced oscillations with frequencies under 1 Hz may interact with natural oscillatory modes of the power grid and can lead to wide-area oscillations across an interconnected power system. For instance, a 0.25 Hz forced oscillation at 20 MW in Alberta, Canada in 2005 led to 200 MW resonant oscillations on California-Oregon

Intertie (COI) lines 1,100 miles away from the source<sup>5</sup> (Figure 1.3). More recently, oscillations at 0.27 Hz at a nuclear plant in Mississippi in 2016 interacted with an EI natural oscillatory mode and led to widespread oscillations that were seen from New York to Florida for over 30 minutes.

The occurrence of large resonant oscillations far away from the source is especially a concern for system operation since they may lead to unnecessary or inadvertent tripping of lines and generators that may be simply reacting to the forced oscillations originating from a remote source. Simulation studies show that if a forced oscillation interacts with a system mode that has weak damping, it can lead to wide-area resonant oscillations of large amplitude that can contribute to potential disturbances, such as the August 10, 1996, outage in the Western Interconnection.<sup>6</sup> The sustained presence of significant forced oscillations on the BPS could lead to long-term effects as well, such as equipment fatigue and potential damage to rotor shafts exposed to such sustained, high-magnitude oscillations. Power quality may also be a concern depending on the amplitude and frequency of the forced oscillations. These types of issues signify the importance of monitoring for these oscillations, understanding how they interact with the BPS, and finding potential mitigation measures for these events. Chapter 3 provides an in-depth analysis of forced oscillations.



**Figure 1.3: MW Flow on California-Oregon Intertie during Forced Oscillation Event (Source: Bonneville Power Administration (BPA))**

<sup>5</sup> S. A. Nezam Sarmadi, V. Venkatasubramanian and A. Salazar, *Analysis of November 29, 2005, Western American Oscillation Event*, in IEEE Transactions on Power Systems, vol. 31, no. 6, pp. 5210-5211, Nov. 2016, doi: 10.1109/TPWRS.2016.2521319.

<sup>6</sup> D. Kosterev, C. Taylor, W. Mittelstadt, *Model Validation for the August 10, 1996, WSCC System Outage*, IEEE Trans. Power Systems, Vol. 14, No. 3, pp. 967-979, Aug 2002.

# Chapter 2: Natural Oscillations

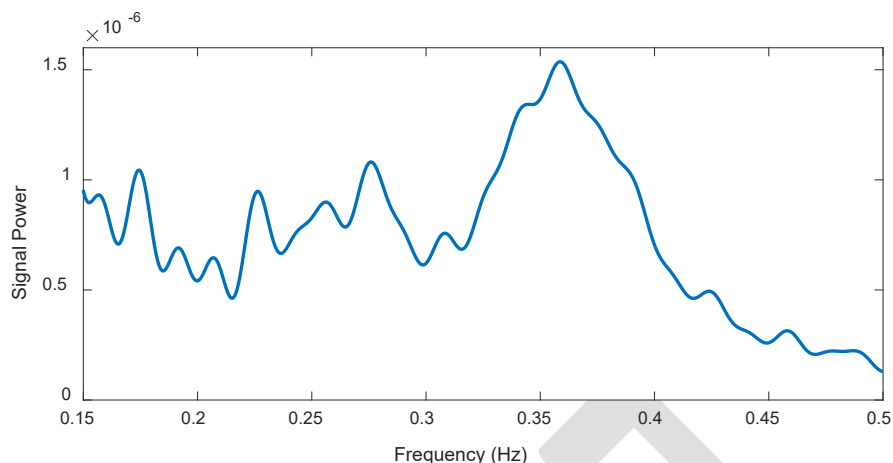
Natural oscillations are the power system’s natural responses to disturbances; their characteristics are governed by the system’s electromechanical or natural modes. Natural oscillations can take on different forms depending on how the system’s dynamics are excited: Ambient oscillations result from continuous perturbation by random load changes, while ringdown oscillations occur after an impulsive disturbance, such as a generator tripping off-line. As demonstrated by the *2019 Interconnection Oscillation Analysis Reliability Assessment*,<sup>7</sup> different Interconnections have different known modes. In fact, there are many modes, but only a handful are dominant and have a noticeable effect on electrical signal dynamics of the system. This reduction usually leads to roughly a half dozen or so modes in each Interconnection (depending on grid size and complexity) that are worth understanding and tracking. Some of the commonly known modes are shown in [Table 2.1](#), which demonstrates the differences between the Interconnections and the expected frequency range for these oscillations.

| Table 2.1: Major Modes in Various Interconnections |                       |                           |
|----------------------------------------------------|-----------------------|---------------------------|
| Interconnection                                    | Mode Name             | Mode Frequency Range (Hz) |
| Eastern                                            | N–S                   | 0.16–0.22                 |
|                                                    | NW–S                  | 0.29–0.32                 |
|                                                    | NE–NW–S               | 0.23–0.24                 |
| Texas                                              | N–SE                  | 0.62–0.73                 |
| Western                                            | North-South A (NSA)   | 0.20–0.30                 |
|                                                    | North-South B (NSB)   | 0.35–0.45                 |
|                                                    | East-West A (EWA)     | 0.35–0.45                 |
|                                                    | British Columbia (BC) | 0.50–0.72                 |
|                                                    | Montana               | 0.70–0.90                 |

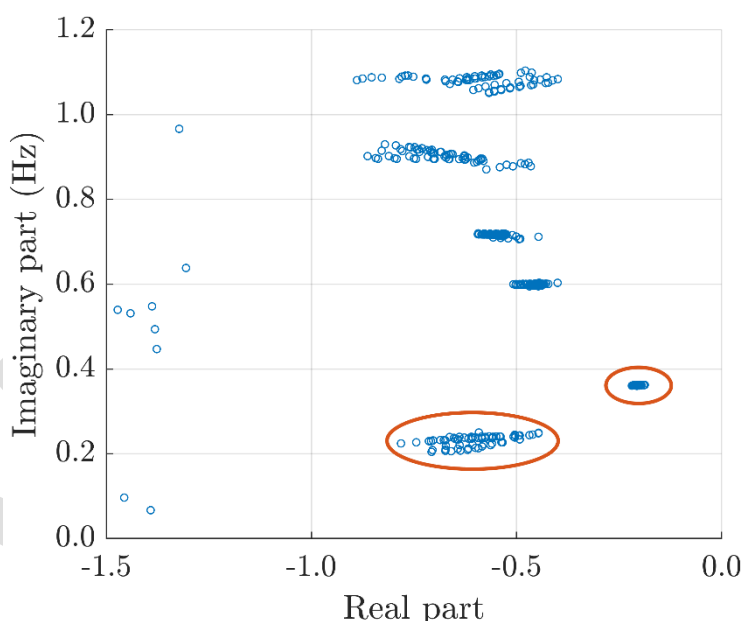
The next section provides a summary of known methods that can help determine the following:

- The modes of significance in an Interconnection that are recommended to be monitored for a given system topology and operating condition, typically appear as consistent peaks in the signal spectrum (see [Figure 2.1](#)). When mode monitoring algorithms (both ringdown and ambient measurement-based) are used, the estimated modes tend to form clusters (as shown in [Figure 2.2](#)) over time or across multiple signals on a complex plane. In addition, significant modes tend to have much higher pseudo modal energy<sup>8</sup> compared with other estimated modes from the same measurement window. This can be verified by looking at curve-fitting errors of the reconstructed curve using only the selected dominant modes.
- Methods for identifying the measurement channels having high observability for dominant modes should be used to effectively monitor the dominant modes.
- Methods for determining the effective mitigation actions should be used to increase damping of the dominant modes.

<sup>7</sup> Interconnection Oscillation Analysis: [https://www.nerc.com/comm/RSTC/SMWG/Interconnection\\_Oscillation\\_Analysis.pdf](https://www.nerc.com/comm/RSTC/SMWG/Interconnection_Oscillation_Analysis.pdf)  
<sup>8</sup> Trudnowski, Daniel J., John W. Pierre, Ning Zhou, John F. Hauer, and Manu Parashar. *Performance of three mode-meter block-processing algorithms for automated dynamic stability assessment*. IEEE Transactions on Power Systems 23, no. 2 (2008): 680–690.



**Figure 2.1: Estimate of the Spectral Content of the Frequency Signal**



**Figure 2.2: Example of Clusters Formed by Eigenvalue Estimates**

## Interconnection-Wide Analysis to Determine Significant Modes

This section categorizes the known small-signal stability analysis methods used to determine the inter-area modes of significance, based on the algorithms they use and the input data they require. Guidance is also provided on how often these studies should be performed. The 2019 *Interconnection Oscillation Analysis*<sup>9</sup> report also presented various types of oscillation analysis techniques for modal identification. More comprehensive details on these methods are available in the IEEE Power and Energy Society (PES)-developed technical report, *Identification of Electromechanical Modes in Power Systems*.<sup>10</sup> Several of the methods described in this section have been used in commercial tools and/or deployed by system operators. Examples are provided in [Chapter 6](#).

Major modal identification methods fall into three categories:

<sup>9</sup> Interconnection Oscillation Analysis: [https://www.nerc.com/comm/RSTC/SMWG/Interconnection\\_Oscillation\\_Analysis.pdf](https://www.nerc.com/comm/RSTC/SMWG/Interconnection_Oscillation_Analysis.pdf)

<sup>10</sup> IEEE Task Force on Identification of Electromechanical Modes, *Identification of Electromechanical Modes in Power Systems*, IEEE Technical Report PES-TR15, June 2012 (<https://resourcecenter.ieee-pes.org/publications/technical-reports/pestr15>).

**Ringdown Methods:** These are data-based methods used to analyze visible natural oscillations that result from large disturbances on the BPS. These methods can use phasor measurement data from on-line recordings or simulated data from off-line dynamic simulations. The post-disturbance trajectories of relevant power system states, or sometimes combinations of states, are commonly referred to as “ringdowns.” Analysis of ringdowns provides valuable insight into the frequency, damping ratio, and shape of the oscillation. It can also serve as a useful tool in model validation (i.e., ensuring that the modal characteristics of the planning cases match those of the actual system).

**Ambient Methods:** These are also data-based methods but used to analyze noise-looking signals during normal, steady-state conditions to identify the underlying system modes, where the primary excitation to the system is random load changes. The data source is typically recorded phasor measurements with a longer time window. They can be used to track the frequency, damping ratio, shape of specific modes, or to identify periods of low damping in any mode. They can also be used to analyze ambient periods surrounding disturbance data to establish a baseline for the ringdown analysis.

**Eigenanalysis Methods:** These are model-based methods that are used to calculate the modes of a dynamic system model under a certain dispatch. The mode frequency, damping and associated mode shape, controllability, and participation factor of participating generators can be determined. These methods can be used on either off-line transmission planning or operational planning models or on near real-time dynamic models.

**Table 2.2** shows a summary of the methods along with the types of data or model to which they are applicable.

| Table 2.2: Methods for Modal Analysis      |                  |                 |                       |
|--------------------------------------------|------------------|-----------------|-----------------------|
| Data Type                                  | Ringdown Methods | Ambient Methods | Eigenanalysis Methods |
| Synchrophasor Data (Ambient Data)          |                  | X               |                       |
| Synchrophasor Data (Post-Disturbance Data) | X                |                 |                       |
| System Dynamic Model                       | X                |                 | X                     |

The results of the analyses help to determine the important modes to monitor. The modes are typically defined by a frequency and a mode shape that lists the participating generators and the respective areas in the mode. The analysis also provides the damping<sup>11</sup> of the respective modes that can vary significantly based on the system operating conditions and the generator controller configurations. Most well-known data-based methods are summarized in **Table 2.3** for a quick comparison of the algorithms used and the data required.

| Table 2.3: Ringdown and Ambient Methods |                                                                                                                      |                       |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------|
| Name of Method                          | Algorithm                                                                                                            | Data required         |
| Prony Methods <sup>12</sup>             | <ul style="list-style-type: none"> <li>Estimates damped sinusoidal components in a linear system response</li> </ul> | Post-Disturbance Data |

<sup>11</sup> Modes are often represented as complex numbers in rectangular form:  $\lambda = \sigma + j\omega$ . A mode's frequency is given by  $f = \frac{\omega}{2\pi}$  Hz and its damping ratio is often expressed in percentage as  $\zeta = \frac{-\sigma}{\sqrt{\sigma^2 + \omega^2}} \times 100\%$ . A system maintains a positive small-signal stability margin if all of its modes have a positive damping ratio.

<sup>12</sup> D. J. Trudnowski, J. M. Johnson and J. F. Hauer, *Making Prony analysis more accurate using multiple signals*, in IEEE Transactions on Power Systems, vol. 14, no. 1, pp. 226–231, Feb. 1999.

**Table 2.3: Ringdown and Ambient Methods**

| Name of Method                                        | Algorithm                                                                                                                                                                                                                                                                                                                                        | Data required         |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|                                                       | <ul style="list-style-type: none"> <li>Expressing the system outputs as linear combinations of fundamental sinusoidal modal components</li> <li>Singular value decompositions (SVD) for handling the noise measurement and for reducing the computational burden</li> </ul>                                                                      |                       |
| Eigensystem Realization Algorithm (ERA) <sup>13</sup> | <ul style="list-style-type: none"> <li>SVD of a matrix whose entries are samples of the system impulse response (Hankel matrix)</li> <li>Using the decomposition, a reduced linear system realization is computed (i.e., a reduced system state matrices)</li> </ul>                                                                             | Post-Disturbance Data |
| Matrix Pencil <sup>14</sup>                           | <ul style="list-style-type: none"> <li>Compute a pseudo-inverse of a matrix using an SVD technique (This also includes a built-in filter for leaving out noise related phenomena in the SVD formulation.)</li> </ul>                                                                                                                             | Post-Disturbance Data |
| Variable Projection (VARPRO) <sup>15</sup>            | <ul style="list-style-type: none"> <li>General nonlinear least-squares optimization technique</li> </ul>                                                                                                                                                                                                                                         | Post-Disturbance Data |
| Hankel Total Least Squares (HTLS) <sup>16</sup>       | <ul style="list-style-type: none"> <li>Formulate a Hankel matrix from the observed phasor measurement unit (PMU) measurements of the event</li> <li>Use a total least squares approach to evaluate the eigenvalues again using a SVD computation</li> </ul>                                                                                      | Post-Disturbance Data |
| Yule Walker <sup>17</sup>                             | <ul style="list-style-type: none"> <li>Method operates by first estimating the autocovariance sequence of the measured data</li> <li>It then fits a model that describes the relationship between the autocovariance sequence at different lag values</li> <li>The parameters of this model are associated with a rational polynomial</li> </ul> | Ambient Data          |

<sup>13</sup> J. J. Sanchez-Gasca, *Identification of power system low order linear models using the ERA/OBS method*, in IEEE PES Power and Systems Conference and Exposition, 2004.

<sup>14</sup> Guoping Liu, J. Quintero and V. M. Venkatasubramanian, *Oscillation monitoring system based on wide-area synchrophasors in power systems*, 2007 iREP Symposium - Bulk Power System Dynamics and Control - VII. Revitalizing Operational Reliability, Charleston, SC, 2007, pp. 1–13.

<sup>15</sup> A. R. Borden and B. C. Lesieutre, *Variable Projection Method for Power System Modal Identification*, in IEEE Transactions on Power Systems, vol. 29, no. 6, pp. 2613–2620, Nov. 2014.

<sup>16</sup> J. J. Sanchez-Gasca and J. H. Chow, *Computation of power system low-order models from time domain simulations using a Hankel matrix*, in IEEE Transactions on Power Systems, vol. 12, no. 4, pp. 1461–1467, Nov. 1997.

<sup>17</sup> R. W. Wies, J. W. Pierre and D. J. Trudnowski, *Use of ARMA block processing for estimating stationary low-frequency electromechanical modes of power systems*, in IEEE Transactions on Power Systems, vol. 18, no. 1, pp. 167–173, Feb. 2003, doi: 10.1109/TPWRS.2002.807116.

**Table 2.3: Ringdown and Ambient Methods**

| Name of Method                                         | Algorithm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Data required |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                                                        | <p>whose poles correspond to the power system's electromechanical modes</p> <ul style="list-style-type: none"> <li>An extension allows the parameters of electromechanical modes and forced oscillations to be identified simultaneously<sup>18</sup></li> </ul>                                                                                                                                                                                                                                                        |               |
| Least Squares <sup>19</sup>                            | <ul style="list-style-type: none"> <li>Fits a model that describes the current measurement in terms of past measurements and the current random input</li> <li>This model parameterized as a rational polynomial whose poles correspond to the power system's electromechanical modes</li> <li>An extension allows the parameters of electromechanical modes and forced oscillations to be identified simultaneously<sup>20</sup></li> </ul>                                                                            | Ambient Data  |
| Frequency Domain Decomposition <sup>21</sup>           | <ul style="list-style-type: none"> <li>Power spectral density (PSD) functions of the ambient measurements are first estimated in the frequency domain</li> <li>SVD then used to combine and extract the principal singular values of the multiple PSD estimates</li> <li>Local peaks among the singular values to correspond to frequencies of system modes and oscillations observed in the data</li> <li>Modal properties estimated by analyzing these principal singular values near the peak frequencies</li> </ul> | Ambient Data  |
| Stochastic subspace identification (SSI) <sup>22</sup> | <ul style="list-style-type: none"> <li>Formulate the PMU measurements as outputs of a linear system being excited by unknown random load fluctuations, modeled as independent white noise inputs.</li> </ul>                                                                                                                                                                                                                                                                                                            | Ambient Data  |

<sup>18</sup> U. Agrawal, J. Follum, J. W. Pierre and D. Duan, *Electromechanical Mode Estimation in the Presence of Periodic Forced Oscillations*, in IEEE Transactions on Power Systems, vol. 34, no. 2, pp. 1579–1588, March 2019.

<sup>19</sup> N. Zhou, J. W. Pierre, D. J. Trudnowski and R. T. Guttromson, *Robust RLS Methods for Online Estimation of Power System Electromechanical Modes*, in IEEE Transactions on Power Systems, vol. 22, no. 3, pp. 1240–1249, Aug. 2007.

<sup>20</sup> J. Follum, J. W. Pierre and R. Martin, *Simultaneous Estimation of Electromechanical Modes and Forced Oscillations*, in IEEE Transactions on Power Systems, vol. 32, no. 5, pp. 3958–3967, Sept. 2017.

<sup>21</sup> Guoping Liu and V. Venkatasubramanian, *Oscillation monitoring from ambient PMU measurements by Frequency Domain Decomposition*, 2008 IEEE International Symposium on Circuits and Systems, Seattle, WA, 2008, pp. 2821–2824.

<sup>22</sup> S. A. Nezam Sarmadi and V. Venkatasubramanian, *Electromechanical Mode Estimation Using Recursive Adaptive Stochastic Subspace Identification*, in IEEE Transactions on Power Systems, vol. 29, no. 1, pp. 349–358, Jan. 2014

**Table 2.3: Ringdown and Ambient Methods**

| Name of Method | Algorithm                                                                                                                                         | Data required |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                | <ul style="list-style-type: none"> <li>Essential features of the linear system model describing the power system can then be estimated</li> </ul> |               |

The recommendation is to perform these analyses on a regular basis and specifically after disturbance events using the recorded synchrophasor data. Model-based eigenanalysis can be performed on a seasonal or yearly basis due to model availability. Significant system changes can also be used as triggers for performing these analyses, such as the following:

- Topology changes in system
- New or retired generation facilities
- Other significant changes in generation dispatch patterns

The modes can also be validated using recorded data during actual disturbance tests. The annual Chief Joseph Brake insertion test in the Western Interconnection (WI) is an example of such efforts.

### **Commonalities on Interconnection Oscillation Analysis Using Modal Analysis of Ringdown Data**

This section uses an example to discuss concepts that are generally applicable to all ringdown methods. The notion of curve fitting is central to modal analysis of ringdowns. In broad terms, curve fitting is the process of specifying a model whose output provides the best fit to the measured data under certain metrics. Specifically, the method seeks to identify a minimal set of sinusoids that represent the dominant modes of the system.

A crucial step in any curve-fitting procedure is to specify the structure of the model being fit to the data. For power system applications, common choices include autoregressive models and discrete-time state-space models. Often, choosing a specific model structure helps to determine which algorithm is best suited to the problem. For instance, Prony's method identifies an autoregressive model to determine the coefficients of the characteristic polynomial. In contrast, the Eigensystem Realization Algorithm identifies a discrete-time state-space model.

### **Data Collection and Pre-Processing**

In general, it may not be possible to analyze all the modes of interest by using data collected for a single disturbance. The reason for this limitation is that a given disturbance (and the system dispatch at the time) may not excite all the possible modes, or it may not excite them enough to permit estimation with the desired accuracy. For example, following a Chief Joseph Brake insertion in the WI, it may be possible to estimate the frequency and damping of the North-South B mode but not the East-West A mode. Best practice is to use disturbances from different parts of the system during different operating conditions.

Modal analysis may be performed by using data collected during a wide variety of disturbances; however, special considerations arise in the case of transient disturbances during which the operating point of the system may move from one equilibrium to another. For example, in the case of resource or load trips, the trajectories of rotor speeds and bus frequency measurements contain considerable low-frequency content (slow trend). It is generally desirable to remove this slow trend (detrending) of the system response to better highlight oscillatory phenomena.

To get the best possible results from any modal analysis technique, care must be taken in collecting and preparing the input data. When analyzing data collected from actual disturbances, it is often beneficial to use a well-designed lowpass filter to mitigate the impact of high-frequency measurement noise and/or process noise. When circumstances allow, as in a posteriori analysis, high-order finite impulse response filters are preferred because they can be designed with a linear phase response, preserving a constant group delay across the entire frequency band.

The corner of this lowpass filter should be placed such that the oscillatory phenomena of interest fall firmly within the passband. This step may be omitted when analyzing simulated data when noise is not a concern.

To maximize the accuracy of mode frequency estimates, the input data may be resampled via a combination of anti-aliasing and decimation. A useful rule of thumb is to set the sampling rate of the input data to approximately 10 times the highest frequency of interest. For example, if the highest mode frequency of interest is 1 Hz, this would imply a sampling rate in the range of 10–12 sps (samples per second). During this process, the reporting rate of the original data should also be considered. Resampling generally yields the best results when the final sampling rate is an integer factor of the original rate. For instance, if the original data is sampled at 60 sps, it would be advisable to resample the data at 10 or 12 sps, as opposed to 11 sps. Choosing the final sampling rate in this way obviates the need to upsample (i.e., to interpolate the original data, which is not advisable in modal analysis applications). Before downsampling the original data, it must be passed through an anti-aliasing (lowpass) filter that limits its bandwidth to satisfy the Nyquist-Shannon sampling theorem. For example, if 60 sps data is going to be resampled at 12 sps, the stopband of the anti-aliasing filter should begin at 6 Hz. As with noise reduction filters, high-order linear phase finite impulse response filters are preferred for this application where possible.

### Curve-Fitting Sensitivities

Most curve-fitting algorithms for modal analysis are designed to work well on the so-called “free and linear response” of the system (i.e., the period in which the input or forcing function has gone to zero, and the system response is largely linear). Thus, the position of the curve-fitting window must be aligned with the free and linear response period for best accuracy. For example, if the system stimulus is a dynamic brake insertion, the starting point of the curve-fitting window should be no sooner than the instant when the brake is removed. In general, it is good practice to allow some additional time to elapse to ensure that the dynamics within the curve-fitting window correspond to the free response. In special circumstances where the forcing function is known (e.g. probe testing), this rule for positioning the curve-fitting window may be relaxed; however, the analysis algorithm may require modification. The positioning of the ending point of the curve-fitting window is equally important. A useful rule of thumb is that the duration of the window should be approximately 3–4 cycles of the lowest frequency oscillation of interest if possible. For example, if the lowest mode frequency of interest is 0.25 Hz, this would imply a curve-fitting window length of approximately 12–16 s, providing that the system has not settled down yet.

Most curve-fitting algorithms also rely on knowing the correct order (number of dominating modes) of the system. While this information is not known beforehand, engineering judgment is needed to set the correct order.

There are multiple ways to quantify a curve-fitting error, some of which depend on whether the analytical formulation is single-channel or multi-channel in nature. Two commonly employed approaches are the  $\ell_2$ -norm and the closely related mean squared error (MSE). Both methods correspond to so-called least-squares error minimization. In the single-channel case, the  $\ell_2$ -norm is given by

$$f(z, \tilde{z}) = \sqrt{\sum_{k \in \mathcal{K}} (z_k - \tilde{z}_k)^2},$$

Where  $z_k$  is the input data and  $\tilde{z}_k$  the output of the model. Here  $k$  denotes the sample index and  $\mathcal{K}$  the set of points in the analysis window. Similarly, the MSE is defined as

$$f_{\text{mse}}(z, \tilde{z}) = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} (z_k - \tilde{z}_k)^2,$$

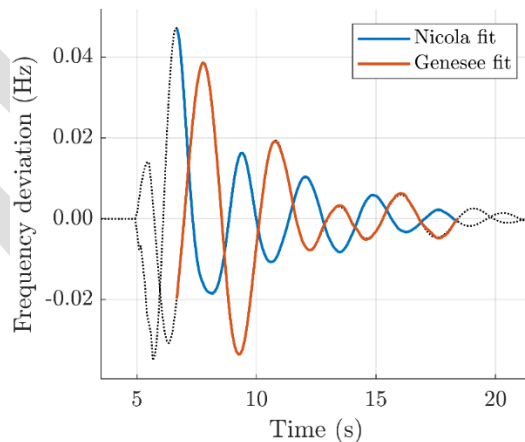
Where  $|\mathcal{K}|$  is the total number of samples in the window. These methods may be extended to the multi-channel case by summing over not only time but also the various signal channels. This type of least-squares minimization is implicit in many algorithms, such as Prony’s method and dynamic mode decomposition (DMD).

When performing modal analysis of ringdowns, it is generally advisable to sample the space of possible input parameter combinations. For example, an input parameter combination may comprise a given model order, analysis window, and trade-off parameter value. At the start of the procedure, a set of possible combinations is defined that spans some portion of the space of interest. Then, the results of analysis are recorded for each combination that creates a collection of mode estimates. Eigenvalue estimates may be categorized first according to their position in the complex plane and then according to the corresponding eigenvectors. This process produces clusters of eigenvalue estimates in the complex plane, each that roughly takes the shape of an ellipse. Categorizing estimates based on the eigenvectors helps to distinguish modes that are close to one another in the complex plane but have different shapes. For example, the North-South B mode and the East-West A mode reside at very similar frequencies in the WI; however, they have different mode shapes. Final mode estimates may be derived by averaging individual estimates that have been confirmed to have the same shape.

### Examples

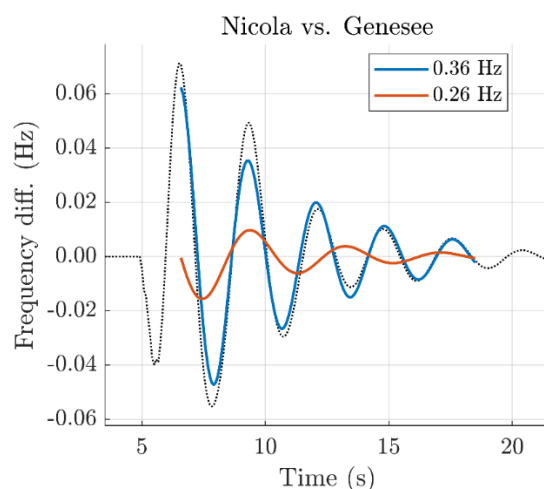
This subsection presents practical examples stemming from modal analysis of a simulated Chief Joseph Brake insertion in the WI. To generate this simulated data, the dynamic brake was inserted at the 5s mark for a duration of 0.5s. In these examples, the inputs to the curve-fitting routine were bus frequency measurements recorded at 26 points distributed geographically throughout the system. Each bus frequency was calculated by using a backward difference derivative approximation applied to the voltage angle. To model the effect of a bandlimited sensor, the output of the derivative approximation was then passed through a first-order lowpass filter. Modal analysis was performed by using an optimization-based, multi-channel curve-fitting algorithm. As with ERA, this method identifies a reduced-order state-space model that can then be fed into Eigenanalysis routines.

Figure 2.3 shows the frequency deviation (from nominal) measured at Nicola, British Columbia, and Genesee, Alberta. The dashed traces show the measured trajectories, and the colored traces show the output of the model constructed from the algorithm. In this case, the curve-fitting window begins at the 6.5s mark and is 12s in length.



**Figure 2.3: Example Curve Fit**

Figure 2.4 shows the frequency difference measured between Nicola and Genesee. In this case, computing the frequency difference between two points can make modal analysis easier by canceling out common mode behavior. The dashed trace shows the relative frequency itself, and the colored traces show the two dominant modal components of the ringdown, the North-South A and B (NSA and NSB) modes. This decomposition can be performed by solving for the minimum-norm solution to an overdetermined system of linear equations as in Prony's method.



**Figure 2.4: Decomposition of the Relative Frequency**

The curve fitting procedure was repeated for 64 unique combinations of input parameters that spanned eight different analysis windows, and eight values of a trade-off parameter used in the optimization. [Figure 2.2](#) shows the eigenvalue estimates generated using this approach. The orange ellipses highlight the estimates of the North-South A and B Modes generated from this disturbance. As discussed above, after the estimates were clustered according to their position in the complex plane, the eigenvectors were checked to distinguish the estimates within a given ellipse. In general, the variance associated with mode frequency estimates is lower than the variance associated with damping estimates; this is because the frequency is much easier to estimate than the damping from the signals. Furthermore, it is generally true that the variance associated with damping estimates increases as the true damping of the underlying mode increases. For this disturbance, the North-South B mode may be estimated with a higher degree of confidence than the North-South A mode because the variance is lower in both dimensions.

[Figure 2.5](#) shows the shape of the North-South A mode overlaid on a map of the WI. Likewise, [Figure 2.6](#) shows the shape of the North-South B mode. Recall that mode shapes correspond to the right eigenvectors of the linearized state matrix. The shape of a mode provides information about how observable it is in a particular state or at a particular location. The shape also provides information about the phase of the oscillation, which can be used to determine which complexes of generators (or other components) are oscillating against one another. In the maps, the area of each marker is proportional to the magnitude of the entry of the right eigenvector corresponding to state measured at that location. Likewise, the arrow emanating from the center of each marker shows the precise phase angle of the oscillation. The color gradient indicates which states are oscillating against one another, as delineated in the key. This analysis is useful in verifying that the modal properties of dynamic models used in operations and planning match those of the actual system.

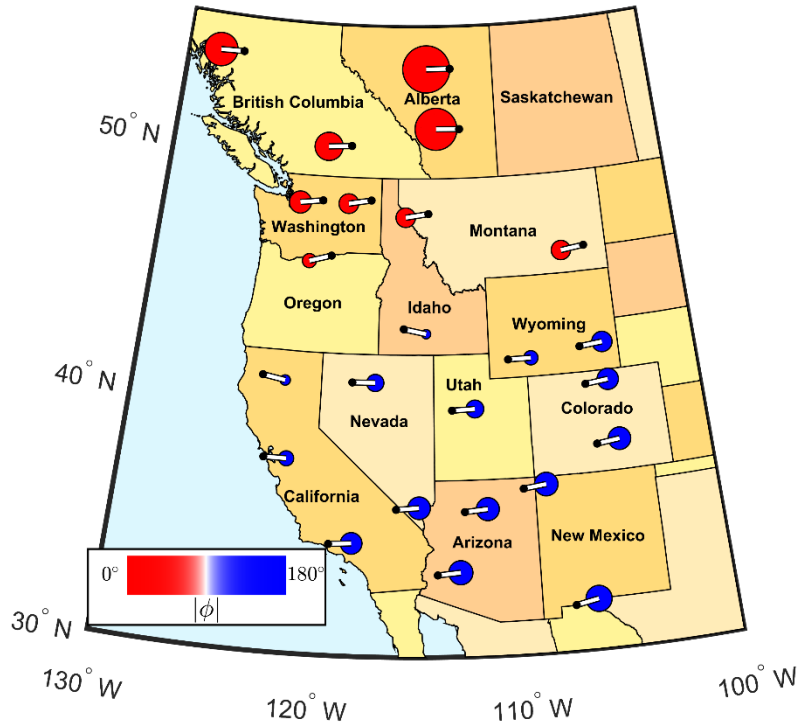


Figure 2.5: Map of the Shape for the 0.26 Hz North-South A Mode

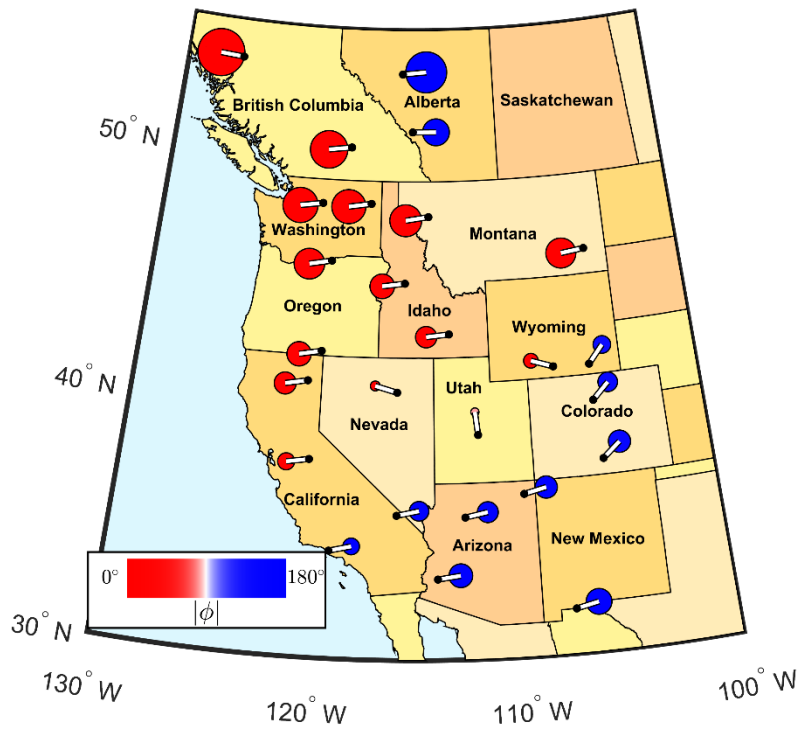
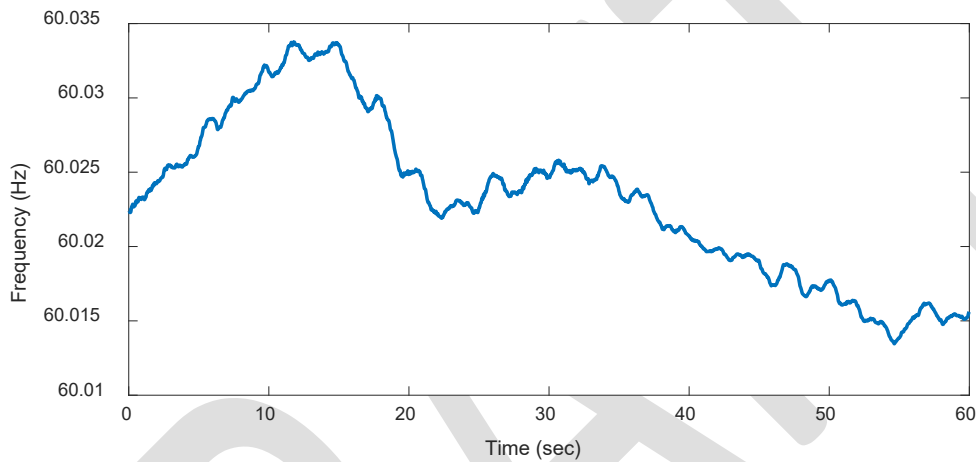


Figure 2.6: Map of the Shape for the 0.36 Hz North-South B Mode

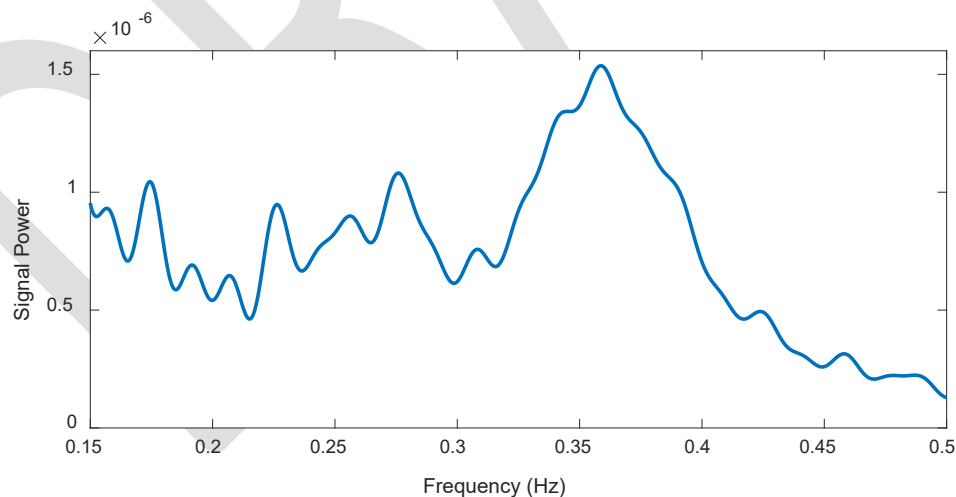
## Commonalities on Interconnection Oscillation Analysis Using Modal Analysis of Ambient Data

This section describes ambient analysis techniques and the commonalities among them. The fundamental assumption behind modal analysis using ambient data is that the power systems' dynamics are continuously excited by random and small disturbances (e.g., load changes) and the systems' small-signal oscillation characteristics can be extracted from analyzing the ambient data.

A sample frequency measurement from a PMU under ambient condition is plotted in [Figure 2.7](#). Modal oscillations are not apparent from visual inspection; however, the impact of the system's modes is present in the signal's random variation. To better see this, consider the estimate of the signal's frequency-domain spectrum in [Figure 2.8](#). This estimate was generated by analyzing 30 minutes of data, including the 60 seconds shown in [Figure 2.7](#), by applying a method based on the Discrete Fourier Transform (DFT). Note the spectrum's peak near 0.35 Hz. This peak corresponds to the well-known North-South B mode in the WI.



**Figure 2.7: Plot of a Sample Frequency Measurement from a PMU under Ambient Condition**



**Figure 2.8: Estimate of the Spectral Content of the Frequency Signal**

The observability of modes in the frequency-domain spectra of PMU measurements forms the basis for several ambient modal analysis algorithms. The Least Squares and Yule-Walker algorithms used in mode meters are spectral estimation methods. The methods determine model parameters to estimate the spectrum. These models can then be evaluated to extract estimates of the system's modes. Similarly, the fast frequency domain decomposition (FFDD) algorithm used in oscillation monitoring tools is based on estimation of power spectrum densities from all available

PMU measurements. Further details are provided in [Detection Method 2: Detection of Sustained Oscillations of Any Type](#).

Ambient algorithms were primarily developed to provide improved situational awareness to system operators in real-time environments, identifying and tracking modes that need to be monitored. Ambient algorithms can also be applied to historical data to obtain mode estimates that are updated regularly, even at one-minute intervals. This approach provides a much more available and granular view of each mode's behavior over a time period than those obtained using transient data. Ambient analysis can also be used to understand how modes change during different loading conditions, system topologies, and seasons. This information can be critical in understanding which modes need to be monitored and what conditions may lead to poor damping.

## Determination of Mitigation Actions

In addition to monitoring, operators would need guidance on how to mitigate inter-area oscillations. It is important to determine and validate mitigation actions in a consistent manner. These actions are developed ahead of time in operational planning studies and summarized in operating guides. There are two main model-based methods to determine and validate mitigation actions that can be included in the operating guidelines provided to operators:

- Eigenvalue analysis
- Ringdown analysis with simulated disturbances

## Using Eigenvalue Analysis to Determine Mitigation Actions

Eigenvalues and eigenvectors can be calculated for a system model under specific operating conditions or after contingencies. Specifically, left eigenvectors quantify the degree to which individual system components can affect the behavior of each mode, offering insight into their controllability within the dynamic response. This information can help identify generators that have more control over a specific mode and therefore helps identify strategies to increase the damping ratio of that mode, such as tuning the parameters of the power system stabilizers (PSS). This approach can be applied to both off-line and real-time conditions. This model-based approach of determining mitigation actions has some advantages:

- It allows for real-time validation of the predefined mitigation actions in operating guidelines.
- It facilitates the assessment of system response under various conditions, such as altered transfer levels across major transmission corridors, modified topology conditions (e.g., transmission lines and series capacitors status), and generation redispatch scenarios.
- It allows testing the impact of contingencies on the damping of the significant modes.

The basic approach in this method is to calculate the eigenvalues within a certain frequency and damping range. It is recommended to pick a frequency range of 0.1 to 0.95 Hz to capture all possible inter-area modes and make the damping range large enough to capture the known significant modes. It is also recommended to filter out all modes that have a damping ratio of more than 25% to focus on the modes of interest that show lower damping. Once a known significant mode is identified, mode tracing can be utilized to trace the impact of transfer across a path and variation of other topological conditions or equipment status, such as PSS on the damping of the known significant mode. This allows the development of mitigation actions for the operators.

[Figure 2.9](#) and [Figure 2.10](#) are examples of eigenvalue analysis results from the Small Signal Analysis Tool (SSAT) using the above approach. The power flow model was a real-time state estimator snapshot, paired with the associated dynamic models at RC West. Mapping the GPS coordinates to each of the resources allows the development of the mode shape plots. Mode frequency, damping, and shape can be used to identify a new mode or verify against a known mode.

SSAT

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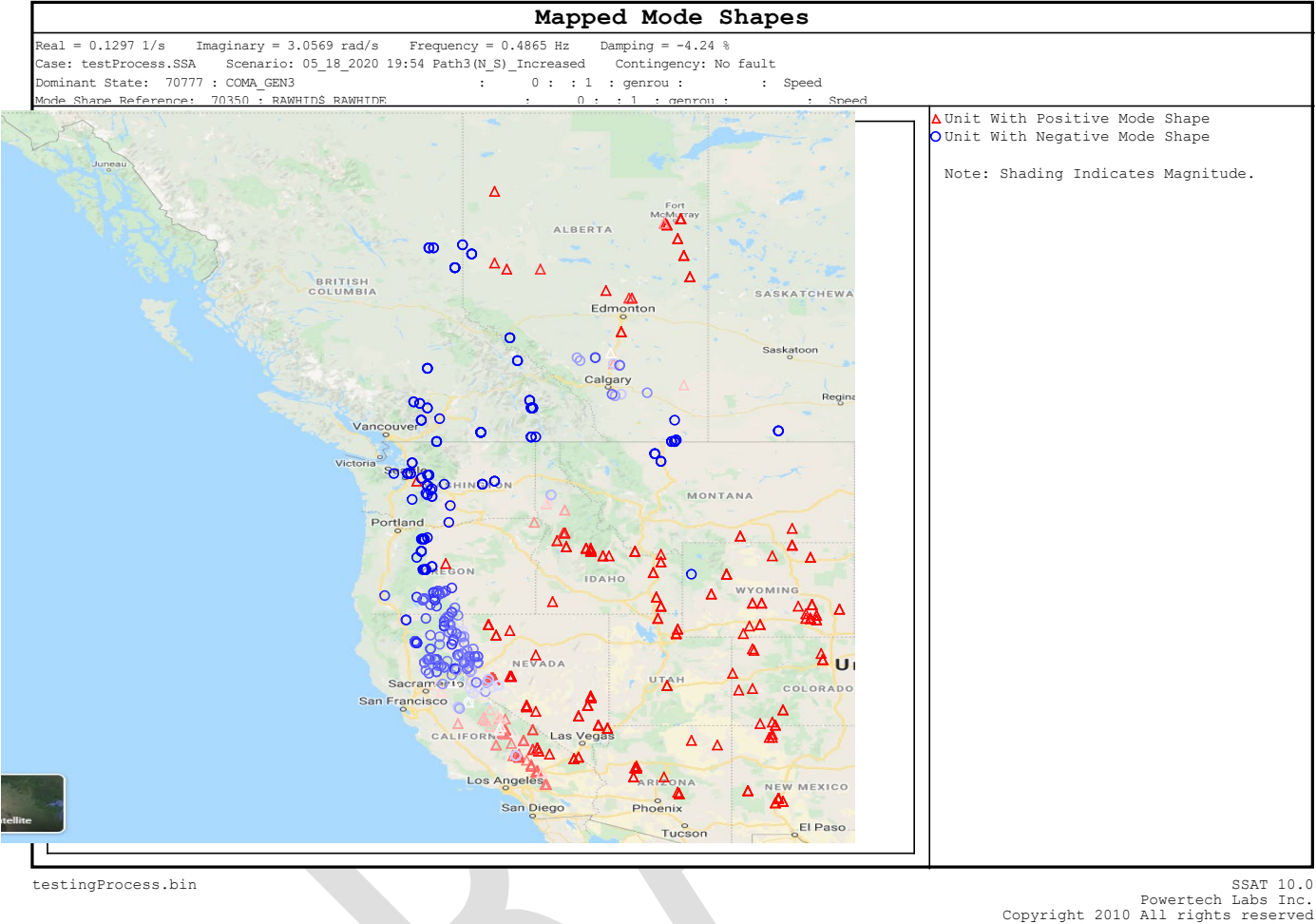
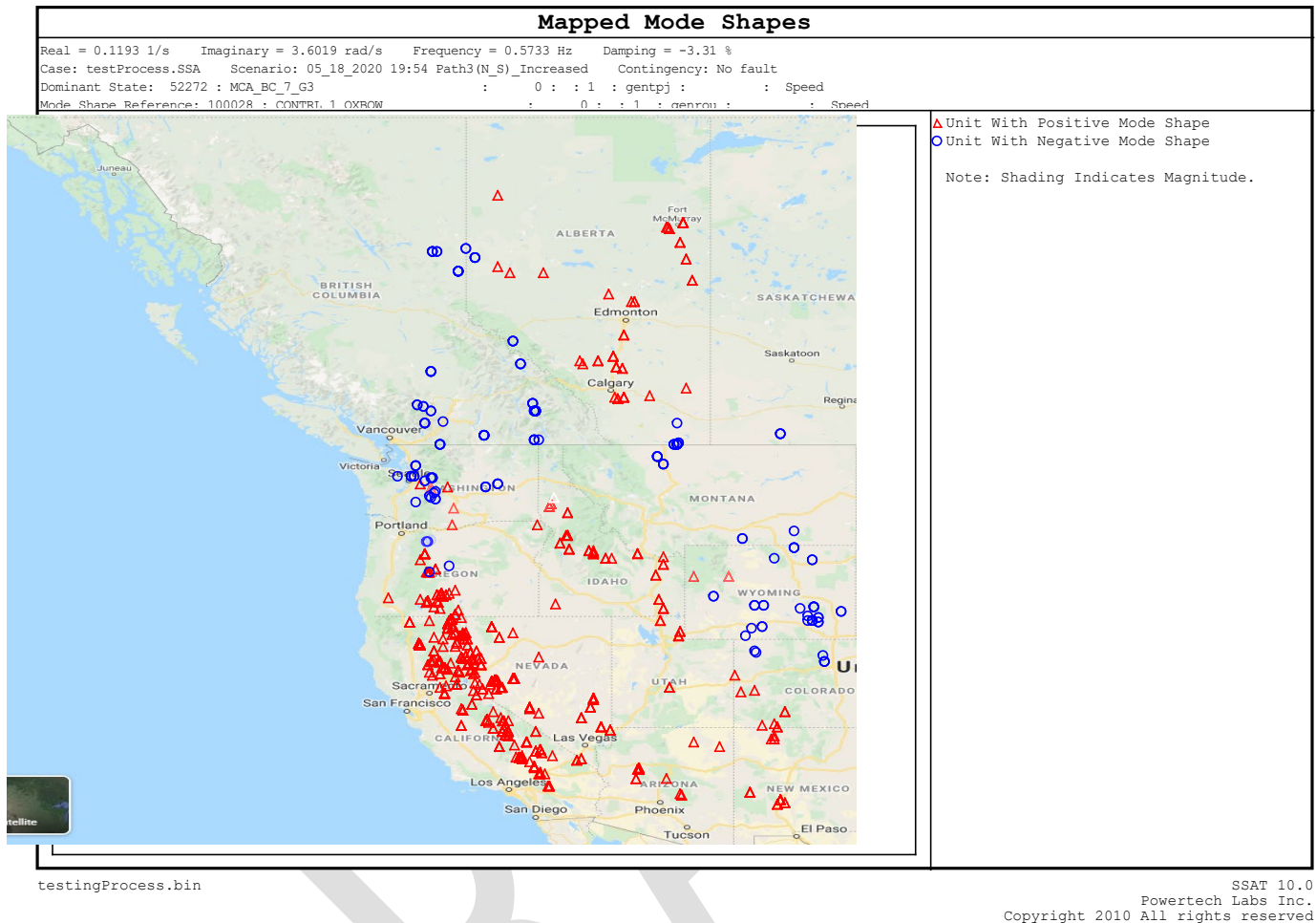


Figure 2.9: Mode Frequency, Damping, and Shape Estimated Using Eigenvalue Analysis

# SSAT

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**Figure 2.10: Mode Frequency, Damping and Shape Estimated Using Eigenvalue Analysis**

## Using Ringdown Analysis to Determine Mitigation Actions

Another way to validate the designed mitigation actions from the small-signal analysis is to simulate a disturbance and extract the modes from the model response using ringdown analysis. The proposed mitigation actions, such as changes in interface MW transfer levels or in topological conditions, would be simulated to help determine whether there are any changes in the damping of the modes being tracked while also ensuring that the mode shape is more or less constant. This iterative process helps to determine and validate the mitigation actions.

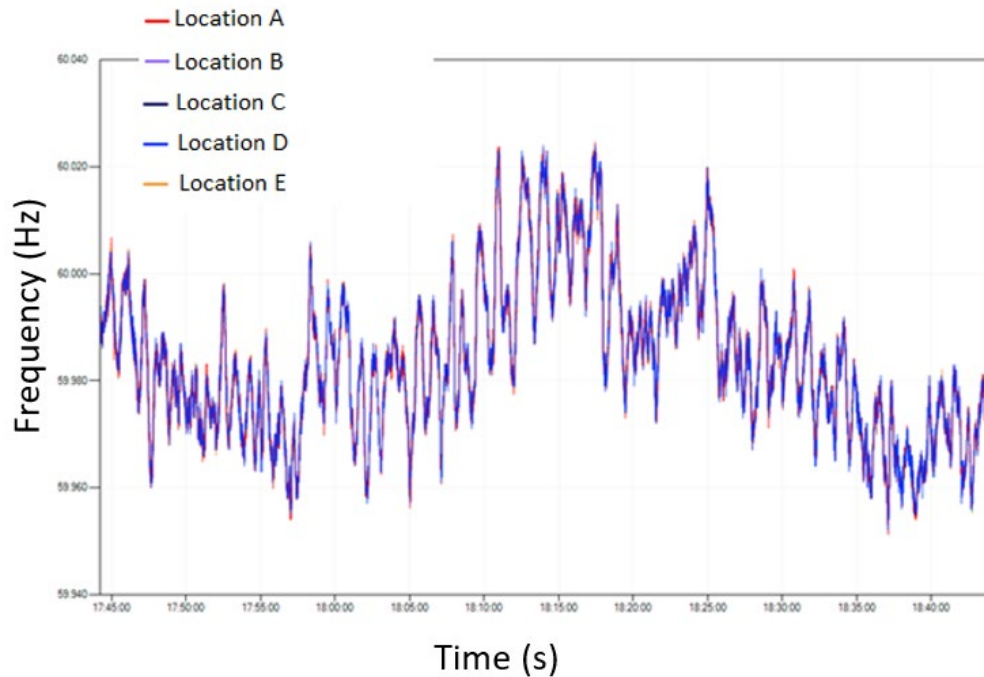
## Monitored Quantities Informed by Analysis

Monitoring the modes of significance requires determining which synchrophasor data should be used in tools to monitor the modes. Various quantities can be used, such as voltage angle pair, voltage magnitude, and active or reactive power. It is important for system operators to monitor modes in a consistent manner. This section provides guidance on selecting the monitored quantities.

While the number of PMU measurements are limited, all the system variables' dynamics, to some extent, are affected by each mode. This means that measurements of electrical quantities, to some extent, will contain information on the modes that can be extracted/estimated for analysis and tracking purposes. This does not mean that all measurements are the same when it comes to mode detection; some measurement points are affected by certain

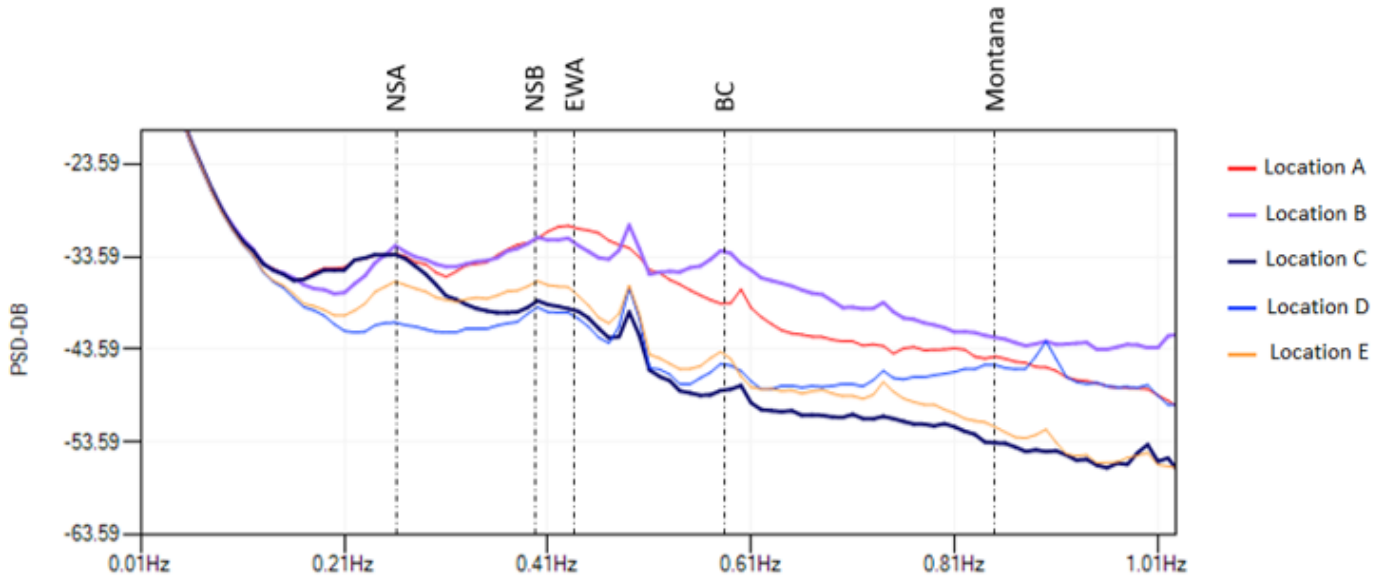
modes more than others or have more unique properties. For example, some measurements have higher observability, determined by the right eigenvector. For inter-area mode estimation/tracking, angle difference between a pair is a good choice because it represents the flow on the equivalent network between two locations. There are many methods for modal analysis and signal selection. Below is just one example of the process for choosing signals for a mode meter setup.

Consider the following frequency trend lines at different substations spread out through the WI over a one-hour duration, shown in [Figure 2.11](#).



**Figure 2.11: Sample Frequency Trends in the WI**

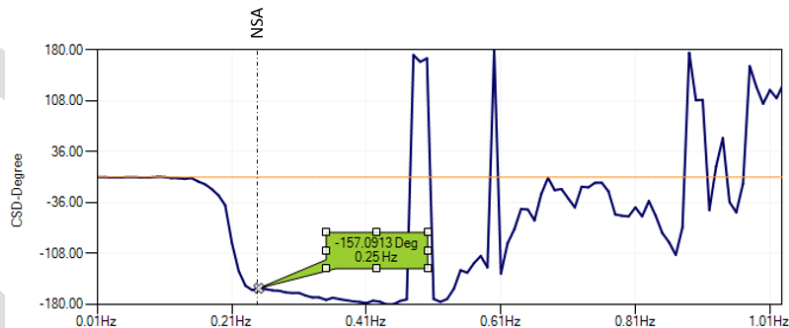
At first glance, this looks like one signal because all of them are overlapping one another. However, after spectra analysis, their energy as a function of frequency shows the difference. [Figure 2.12](#) shows the dominant modes (broad peaks) as well as the how their energies compare in different signals. For example, the Montana mode is more apparent in Location D's frequency signal spectrum than in that of Location C's. In addition, at Location A, the East-West A Mode (EWA) mode clearly overpowers all other modes.



**Figure 2.12: Mode Observability at Different PMU Locations**

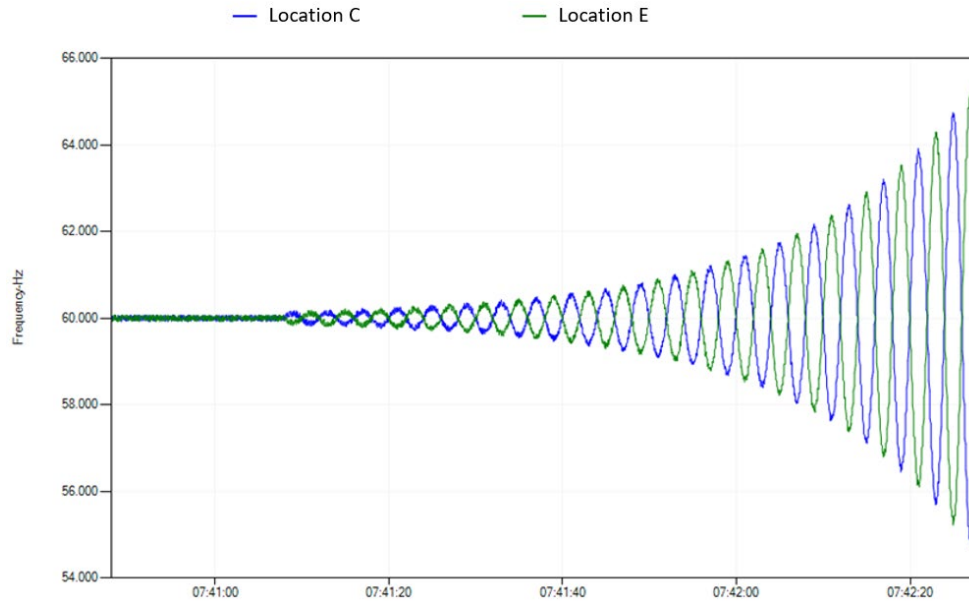
Analyzing the mode shape or phase angle across the spectrum for each mode also helps explain the observability of the mode in each generation unit. Units oscillating in phase tend to form coherent groups, while those exhibiting an approximate 180-degree phase difference oscillate in opposition to the group. Groups around 90 degrees are uncorrelated. To put this concept into practice, consider the example scenario below. According to the power spectrum in [Figure 2.12](#), Location C and E are grid locations where the NSA mode clearly presents itself (~0.25 Hz). According to the mode phase data shown in [Figure 2.13](#), Location E is nearly -180 degrees from Location C (approx. -157 degrees).

Mode Phase Between Location C and E



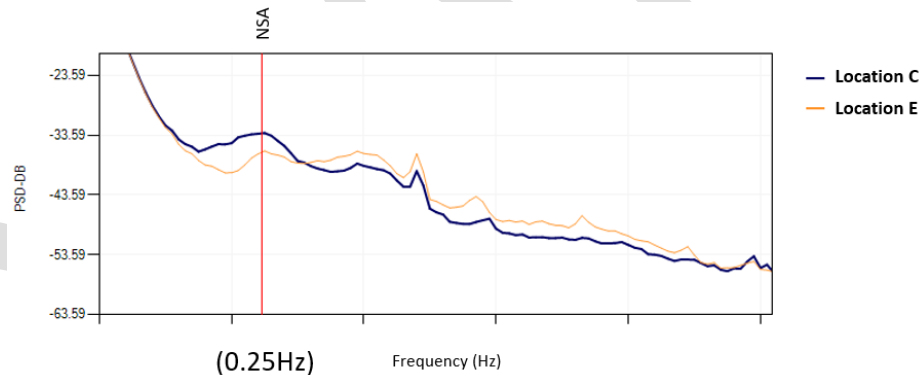
**Figure 2.13: Example of Mode Phase Between Two Locations**

By taking the phase and mode energy data from the estimated power spectrums, a “mode shape” plot describes how all the signals that are being influenced by the mode are related to one another. To visually show this, imagine a PMU at Locations E and C that shows the ideal frequency signals of 60 Hz when suddenly the NSA mode becomes unstable and negatively damped. Consequently, a near-0.25 Hz oscillation starts to build and, due to the locations’ relative mode phase as the signal at Location C swings up, the signal at Location E does exactly the opposite as shown in [Figure 2.14](#). This is an example of an “unstable inter-area natural oscillation” caused by a negatively damped mode.



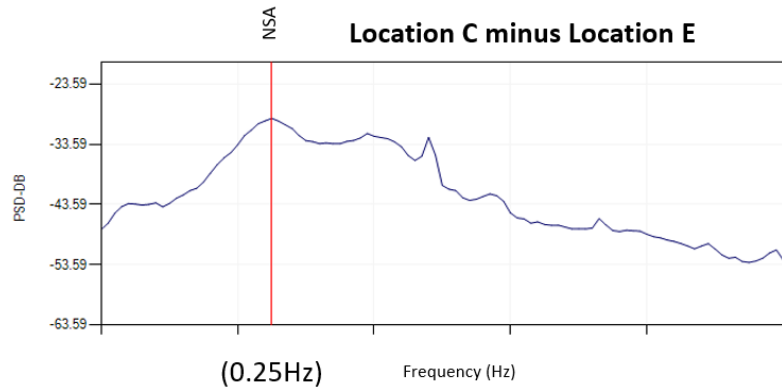
**Figure 2.14: Example of Frequency at Different Locations during an Inter-Area Oscillation**

Since both signals exhibit the mode and have a relative mode phase near 180 degrees, these two signals can be used in a special way to gain an advantage for tracking a mode with a mode meter. Shown again in Figure 2.15 are the sample estimated power spectrums of the frequency signals at Locations C and E.



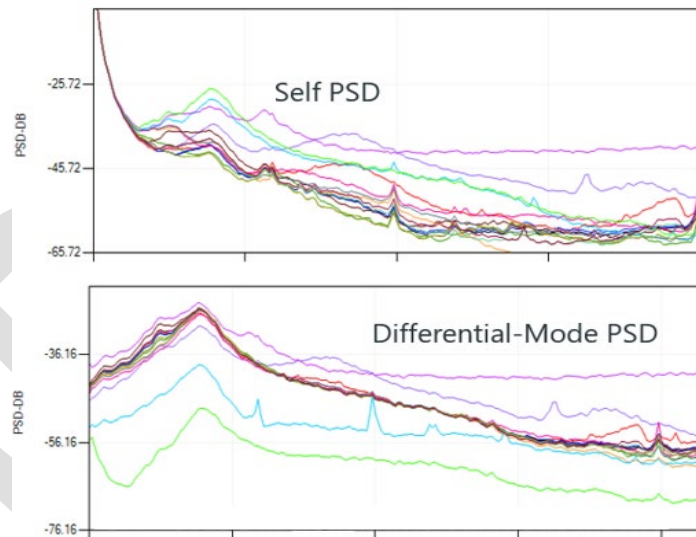
**Figure 2.15: Sample Estimated Power Spectrums of Frequency Signals**

By subtracting these two frequency (or numerical derivative of PMU voltage angle signal) signals at Locations E and C creating a differential “angle pair” signal, we can see that the new signal makes the NSA mode “pop-out” as shown in Figure 2.16 by attenuating common properties and amplifying differing properties in the Locations C and E frequency signals.



**Figure 2.16: Sample Angle Pair Differential Pair Signal for North-South A Mode**

As seen above, using the angle-pair differential signals can provide an effective way to monitor a given mode in mode-meters. Therefore, it can more easily calculate and estimate frequency and damping ratio of a given mode if the system is in ambient conditions. If we do the same procedure over a large set of signals, we can better determine an acceptable input signal for a given mode meter. See an example in [Figure 2.17](#).



**Figure 2.17: Sample Self and Differential Spectrums for Multiple Signals**

So far, this paper has discussed a maximum of two signals. If two dominant modes are close in terms of natural frequency, it might be necessary to either add more input signals to create a more unique output signal that distinguishes the two close modes or change the signal type since all signal types exhibit different network mode shapes. By utilizing the gain properties of differential and common signals, a unique signal for a given mode can be created that provides the best input to a given mode meter for a given set of available PMUs. For example, the sum of active power flows in two lines interconnecting Mexico to the rest of Central America is used to monitor and control unstable modal activities for opening the interconnection between Guatemala and El Salvador when an unstable operating condition is detected.<sup>23</sup>

<sup>23</sup> Espinoza, José Vicente, Armando Guzmán, Fernando Calero, Mangapathirao V. Mynam, and Eduardo Palma. *Wide-area protection and control scheme maintains Central America's power system stability*. In 39th Annual Western Protective Relay Conference. 2012.

## Thresholds for Damping and Energy

Simulation studies and review of historical mode estimates can help determine how much a contingency or credible set of contingencies can reduce a particular mode's damping ratio. This information can then be used to select alert and alarm thresholds to ensure that a sufficient stability margin is maintained in pre-contingency conditions to preserve stability in post-contingency. Off-line studies are helpful to determine thresholds, but additional margins should be included to account for unforeseen contingencies and system conditions. Though alert and alarm thresholds should be based on studies, a damping ratio below 5% typically warrants investigation, while corrective action is likely necessary if the damping ratio falls below 3%. The mode estimate should be validated before taking action to ensure that the low damping estimate is not the result of poor data quality, the presence of forced oscillations, or normal variation in the estimate. Off-line modal baseline analysis can help determine the threshold for modal energy in different operating conditions.<sup>24</sup> Peak energy levels in the signal spectrum for a mode in a normal operating condition can be used to set the thresholds for energy.<sup>25</sup> As discussed next, mode shape and system conditions can also be evaluated to validate mode estimates.

## Monitoring Mode Shape

In addition to monitoring the mode frequency and damping, it is also recommended that tools be set up to monitor the mode shapes of the various monitored modes. This will allow operators to validate that the mode where damping and energy are reaching critical levels is the same for which operators have been provided operating guidance.

## Monitoring System Conditions

The system conditions identified for increasing damping and reducing the energy of the monitored modes should be monitored along with the damping, energy, and the mode shape. This is to ensure that system conditions are contributing to the damping and the energy of the monitored modes that reach critical levels where the next contingency can cause instability. Therefore, path flows and topology conditions (e.g. line status, series capacitor status, and generator status) should be monitored along with damping and energy of the modes.

<sup>24</sup> Trudnowski, Daniel J., and Ferryman, Tom, Modal baseline Analysis of the WECC system for the 2008/9 Operating Season, Technical Report, September 2010

<sup>25</sup> Donnelly, Matt, Dan Trudnowski, James Colwell, John Pierre, and Luke Dosiek. *RMS-energy filter design for real-time oscillation detection*. In 2015 IEEE Power & Energy Society General Meeting, pp. 1–5. IEEE, 2015.

## Chapter 3: Forced Oscillations

One of the many challenges faced by the electric power industry is determining the source location of these forced oscillations in real time. The existence of these oscillations is increasing as the grid characteristics change with the addition of wind, solar, and distributed technologies both on grid and behind the meter. Many oscillations have been observed in the power system over the past couple of years across the North American system. While the source of a forced oscillation may be any component connected to the BPS, recent research has shown that sources of oscillations typically include the following:

- **Traditional Generating Resources:** Power plants (e.g., fossil, hydro, and nuclear power plants) have many controls with primary electric generator functions that maintain mechanical power to the generator and generator field current at the desired level. Forced oscillations often result from malfunctioning power plant controls and/or devices and mechanical failure.<sup>26</sup> Generators can also oscillate during short periods of time at specific operating conditions (e.g., during startup or shutdown, hydro generator “rough zone”).
- **Variable Energy Resources:** Forced oscillations can also originate from resources that employ newer technologies, such as inverter-based generation. Oscillations can be induced by a number of factors, including wind turbine controls, inverter controls, plant-level controls, and other types of controls or interactions. The significantly faster and advanced controls of these types of resources, coupled with the fact that they are often placed in “weaker” areas of the BPS, make these resources common sources of forced oscillations.
- **Loads:** Industrial loads are highly cyclic in nature<sup>27</sup> and can induce forced oscillations into the BPS. These can be caused by a range of processes, controls, or malfunctions within these large customers connected at either the transmission or distribution system.
- **Malfunctioning Grid Controls:** Elements on the BPS may fail, malfunction, or misoperate. Periodic failure or malfunction of these types of devices or their controls may induce oscillations into the BPS. Examples of these types of elements include switched capacitors, thyristor-controlled devices, such as Static Var Compensators (SVC) and HVDC circuits, voltage-source converter technology such as Static Synchronous Compensator (STATCOM) and some HVDC, and series compensation devices.

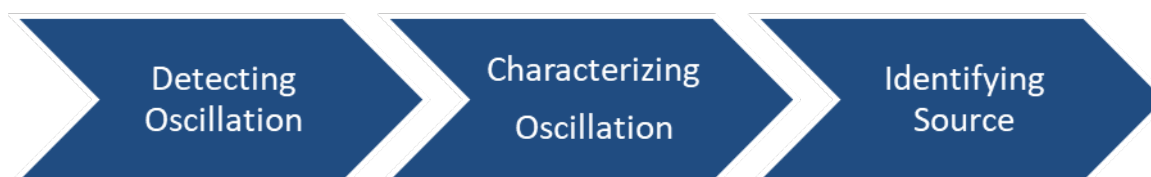
The occurrence of these oscillations can be persistent or intermittent and with low energy or high energy. The presence of these oscillations may lead to undesired operation of the power system, including equipment tripping that causes further stress on the system. Therefore, the early detection and mitigation of oscillations in real time is crucial to ensuring reliable operation of the grid. A general framework for analyses is provided in the next section, followed by a more detailed discussion of detection and source localization methods in [Forced Oscillation Detection Methods](#) and [Determination of Oscillation Source](#), respectively.

### Framework for Forced Oscillation Mitigation

Understanding forced oscillations and mitigating their impact on reliable operation of the BPS can be separated into three distinct phases, as shown in [Figure 3.1](#): detecting the forced oscillation, characterizing the mechanism of the oscillation, and identifying the source(s) of the forced oscillation. Once these phases are accomplished, appropriate and effective mitigating action can be taken. Forced oscillations on the power system should be mitigated, to the extent possible, since these forced perturbations on the system are unplanned, unstudied, and may have short- and/or long-term reliability impacts.

<sup>26</sup> Examples of malfunctioning controls include misoperation or abnormal valve controls swinging in a limit cycle and hydroelectric generation plants operating in a “rough zone” resulting in resonance effects linked to the turbine penstock.

<sup>27</sup> Such as electrolytic process in aluminum smelting.



**Figure 3.1: Steps Toward Mitigating Forced Oscillations**

This section provides a framework based on this approach and also includes some insights and considerations for each step in the process. Step 1 focuses on detection, Steps 2 and 3 relate to characterizing the oscillation, and Steps 4 and 5 seek to identify the source and root cause of the oscillation.

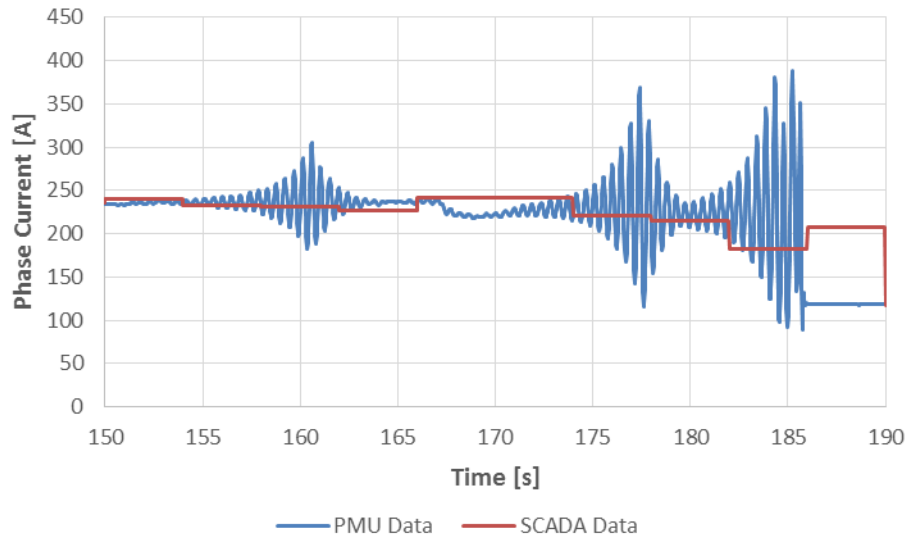
### Step 1: Detect the Forced Oscillation

The first step in the process is to identify an oscillation that is occurring. This can be done through advanced tools or simply by observing oscillatory behavior in electrical signals, such as frequency, voltage magnitude, or power flows. With high-resolution data, advanced algorithms use various techniques to detect the oscillation (e.g., energy, oscillation, coherency, matched filter detectors). This allows automated and fast detection of any abnormal oscillations occurring on the BPS. While high-resolution data provides the clearest picture of the oscillation since aliasing is avoided, in some cases SCADA data can also be used to identify these oscillations.<sup>28</sup> Figure 3.2 and Figure 3.3 show examples of SCADA and PMU data overlaid<sup>29</sup> on each another to show how the oscillation looks in each of the data sources.<sup>30</sup> Figure 3.4 shows another example of SCADA data where the occurrence of the oscillations can clearly be seen from the change in the nature of the plots from “steady-state” to “noisy” in the shaded portion of the plots.

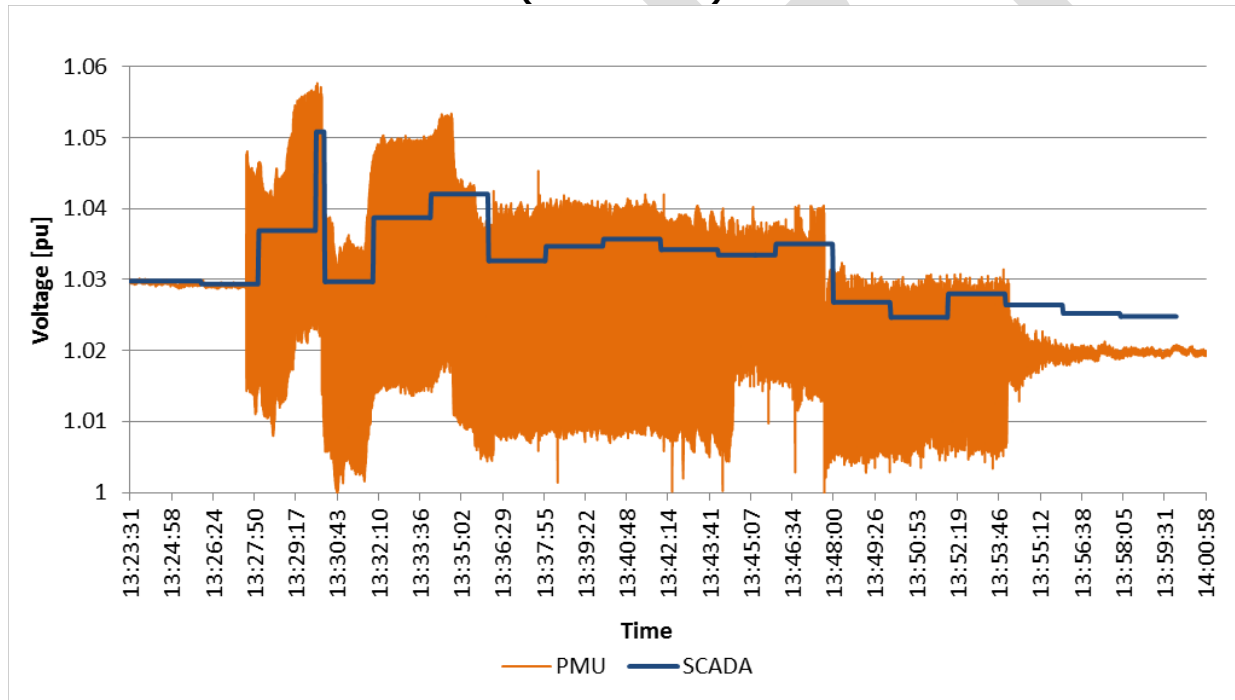
<sup>28</sup> SCADA data may or may not identify a forced oscillation depending on the SCADA scan rate and the forced oscillation frequency. The SCADA data may be aliasing the actual waveforms and may not detect the oscillation. In other instances, SCADA may be able to identify some form of oscillatory behavior. Abnormal movement in SCADA values may be indicative that an oscillation is occurring (assuming good quality of SCADA measurements),

<sup>29</sup> The data may not overlay exactly due to compression and exception processing of the SCADA data vs. the PMU data.

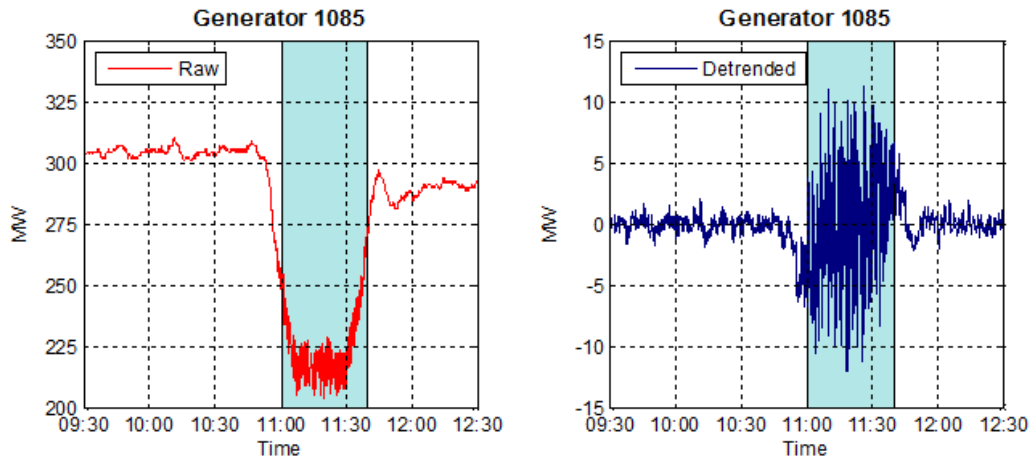
<sup>30</sup> NASPI Control Room Solutions Task Team (CRSTT) has a number of video examples that show SCADA versus PMU data for oscillation events. Available: <https://www.naspi.org/crstt>



**Figure 3.2: Forced Oscillation Observed with SCADA and PMU Data  
(Source: ATC)**



**Figure 3.3: Forced Oscillation Observed with SCADA and PMU Data  
(Source: OG&E)**



**Figure 3.4: Raw (Left) and Detrended (Right) SCADA Data from Generator Oscillating (Shaded) (Source: WSU)**

It is useful to record, if possible, the times when the oscillation started and stopped. This information can be useful for correlating with events on the system or at specific generation sites, for example, to aid in detecting where and how the oscillation manifested. Also worth noting is whether this is a recurring oscillation and whether the oscillation has any cyclic nature to it (e.g., pulses every second, shows up during the day, shows up when it's windy).

### Step 2: Determine Oscillation Frequency and Magnitude

Advanced tools will likely be able to detect the oscillation and provide the operator or operations engineer useful information about the oscillation, including the oscillation frequency. However, if this information is unavailable, a TOP or RC can simply get a rough estimate of the frequency by observing time series data and determining the period,  $T$ , of oscillation. The dominant<sup>31</sup> frequency,  $f$ , of the oscillation is then determined as the reciprocal of the oscillation period,  $f = \frac{1}{T}$ .

The oscillation frequency is an important characteristic for identifying and mitigating its source. As described in previous sections, different oscillation frequencies can be tracked to different electromechanical aspects of the BPS. If the forced oscillation is occurring at the frequency near a system mode frequency, this can cause additional reliability risks since that forced oscillation can propagate across the BPS.

In addition, note the magnitude of the oscillation and whether it appears to pose any reliability risk to the system. This can be as simple as a qualitative point of reference to quickly understand its impact. For example, is the oscillation in real power a couple MWs or hundreds of MWs? Is the voltage oscillation one or two kV, or is it tens of kV? Key locations in the system to observe these oscillations are at major generating resources, tie lines, and critical load buses throughout the system. If the magnitude is noticeably large, this is indicative that a more severe problem has arisen and should be addressed more rapidly.

As described in previous sections, forced oscillations generally have a very low or zero damping ratio. So, it is normal to have tools to alarm of an undamped oscillation when these forced oscillations occur. This is not system instability and should be treated differently than a poorly damped inter-area system oscillation. As described in the previous section, the interaction between forced oscillations and system modes can be a challenge in determining potential risks to the BPS.

<sup>31</sup> This should be handled carefully in the presence of harmonic content and nonlinear characteristics in the signal.

**Step 3: Determine Extent of Oscillation**

TOPs and RCs who have detected a forced oscillation should identify the extent to which the oscillation is seen across their footprint. Is the oscillation more localized (observed in electrical quantities within a contained area of the system) or more widespread (observed across a significantly wider area)? If the oscillation is localized yet occurring near the border of a TOP footprint, the TOP should coordinate with its RC to identify the extent of the oscillation, the impact it may be having on reliability, and possible sources of the oscillation. If the oscillation is occurring near borders of an RC footprint, the RC should coordinate with neighboring RCs to identify its extent. If the oscillation is occurring across the TOP or RC footprint, then the TOP and RC should coordinate with neighboring RCs to understand the extent of the oscillation.

When coordinating with the RC or neighboring RCs, the following considerations may help in determining the source and extent of the oscillation:

1. Does the RC or neighboring RC have oscillation detection tools or the capability to quickly extract or monitor high-resolution PMU data on a wide-area basis?
2. Is the neighboring RC seeing the oscillations in its own data?
3. Is the neighboring RC able to identify the oscillation near the affected border using any of its tools and capabilities?
4. Is it possible to mutually determine whether and where the oscillation is largest? This may be helpful<sup>32</sup> in determining the location of the oscillation.
5. Are there parts of the system(s) that are not experiencing the oscillation? This is useful for larger TOPs and RCs who have a significantly large footprint.

**Step 4: Determine Location or General Proximity of Oscillation**

Once the general vicinity and affected entities have been identified, the source of the oscillation can be pursued. In general, if the oscillation frequency is not at a system mode frequency, the source is close to the measurement location with the largest amplitude oscillation. However, if the forced oscillation is at or near a system mode frequency, the forced oscillation shape changes to the relevant system mode shape and the amplitude may be misleading. Again, this is dependent on whether the forced oscillation is observed in a localized manner or observed across larger parts of the system. If high-resolution monitoring from PMUs is not available directly at the source of the oscillation, it can be a challenge to quickly identify the source location. For example, a PMU is monitoring the 500 kV network, but multiple generators are connected at the 230 kV system in close proximity. Regardless, identifying the source of the oscillation to the extent possible is useful.

For localized forced oscillations, TOPs and RCs can use available PMU data and possibly SCADA data (if useful) to determine where the oscillation magnitude is largest and home in on an exact location. The use of SCADA measurements can be limited for localized forced oscillations having frequencies greater than 1 Hz as the sampling frequency of SCADA is usually 1–2 samples per second. In such cases, SCADA measurements may not be useful in identifying the location of the source of the forced oscillations as these oscillations may not be fully captured in the SCADA measurements. Using high-resolution PMU data for this task is critical to have a clear understanding of the oscillation magnitude. If PMU data is available at the source of the oscillation (e.g., the point of interconnection of a power plant), this helps to identify the source. However, once the PMU data has been used to its fullest extent, lower-resolution SCADA data can sometime be useful to identify the true source depending on the frequency of the forced oscillations and sampling frequency of SCADA measurements. For example, as described above, if the PMU data points to a specific PMU on the EHV system (e.g., 500 kV) with multiple generating units in the vicinity or connected at lower voltage, the RC or TOP can extract SCADA data for the suspect generating units. Useful electrical quantities

<sup>32</sup> Again, this should be handled carefully when the frequency of the forced oscillation is near one of the known system mode frequencies.

include voltage, current, frequency, active power, and reactive power. In some cases, these units may be at the same power plant, and the RC or TOP may need to coordinate with the Generator Operator (GOP). In other cases, these may be separate power plants and coordinating with both entities may be required.

### Step 5: Determine Specific System Component Causing Oscillation

If a BPS element can be determined as the source of the oscillation, the RC and TOP should notify the owner of the element to further investigate the cause of the oscillation. For example, if a single power plant is identified, this plant may have multiple units connected to the BPS. Similarly, a large end-use load customer may have multiple processes that may be causing or contributing to the oscillation. This analysis, in many cases, will require investigation of SCADA data or data collected by the equipment owner for specific components within the plant or at the site. In addition, the cause of the oscillation can sometimes be quickly identified based on the electrical quantities experiencing the oscillation. Oscillations within one equipment owner's facility could lead to damage of their own equipment or potentially damage of others' equipment—hence the importance of mitigating forced oscillations and correctly identifying the source of these oscillations. Considerations that may be made based on the specific component on the system experiencing the oscillatory behavior are provided below.

## Forced Oscillation Detection Methods

There are several methods used to detect sustained oscillations that include forced oscillations or undamped natural oscillations. Most of the time, sustained oscillations are forced in nature as the system modes are mostly well-damped. Examples of methods available to detect sustained oscillations are described below.

### Detection Method 1: Energy Bands Monitoring Approach

Detection of oscillations in the power system requires not only identifying the existence of oscillatory signatures in the real-time measurements but also defining the thresholds to be used to differentiate oscillatory signatures from ambient changes in the measurements. Key metrics of power system oscillations can be monitored and analyzed to define the severity of oscillations and therefore establish the detection criteria. The main parameters for power system sustained oscillations are as follows:

- **Oscillation Frequency:** Rate of change of fluctuations seen in the signal over time
- **Oscillation Energy:** indicator of severity of oscillation (high energy oscillations require operator attention)

### *Oscillation Frequency*

It has been observed from oscillations on the system that similar devices lead to oscillations relatively close in frequency. Therefore, monitoring the oscillation frequency can be an indicator of type of oscillation (e.g., inter-area, local, control system problems, Flexible Alternating Current Transmission System (FACTS) device settings). In this approach, the frequency spectrum is divided into multiple bands where each band of frequency is used to indicate a common root cause of the oscillation. Through observation of previous oscillation events and correlating the frequency to the identified root cause, four bands of oscillations are defined corresponding to various types of oscillations. The oscillation frequency bands are useful to indicate the most likely cause of the oscillation. Frequency band details along with common causes are shown in the [Table 3.1](#).

**Table 3.1: Frequency Bands of Oscillations and Likely Causes**

| Frequency Band | Likely Cause of Oscillations                                                                                         |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| 0.01–0.15 Hz   | Governor, plant controller, automatic generation control                                                             |
| 0.15–1 Hz      | Electromechanical inter-area                                                                                         |
| 1–5 Hz         | Local plant modes, intra-plant modes, local generator control oscillations, excitation controls, dc circuit controls |

**Table 3.1: Frequency Bands of Oscillations and Likely Causes**

| Frequency Band | Likely Cause of Oscillations                                                  |
|----------------|-------------------------------------------------------------------------------|
| >5 Hz          | Torsional oscillations, sub-synchronous oscillations, fast acting controllers |

### *Oscillation Energy*

Amplitude is the most direct indicator of the severity of an oscillation at a particular frequency. When frequency bands are considered, the energy in each band can be monitored as a proxy for amplitude. Oscillations with high peak-to-peak amplitudes will result in high energy within their frequency band. Oscillations can be monitored by alerting or alarming when the energy in a band exceeds predefined thresholds. Monitoring energy in a band<sup>33</sup> can be more practical than estimating the amplitude of individual oscillations because oscillations may have time-varying frequencies and be accompanied by harmonics in the same frequency band.

This sets the stage for monitoring the oscillations in real time. For effective real time oscillation monitoring, it is important to do the following:

- Establish frequency bands for oscillation monitoring:
  - What frequency bands to configure for monitoring?
- Establish thresholds for minimum energy:
  - At what level is an oscillation identified and detected?

In order to establish the parameters for real-time oscillation monitoring, periodic analysis and baselining of key power system metrics is required to establish the proper oscillation thresholds for monitoring. The thresholds are usually system-specific and depend on the normal behavior of the power system. To identify the monitoring parameters, it is realistic to baseline the power system oscillations from a rich source of archived high-resolution phasor data. This requires mining historical phasor data to search for oscillations and gather the prerequisites required to monitor in real time. Phasor data mining studies would reveal both known and unknown oscillations. The unknown oscillations can be added to the bucket of known oscillations for additional monitoring until mitigation actions are executed to terminate the oscillations. The key steps in baselining study are as follows:

1. Access to archived phasor data
2. Data conditioning (filtering and conditioning, eliminating bad measurements)
3. Establishing mining criteria (frequency bands, energy, oscillation period)
4. Scanning data and identifying events that meet mining criteria
5. Reviewing and interpreting mining results to identify types of oscillations
6. Preparing recommended parameters for real time oscillation monitoring—frequency bands, energy thresholds, damping, and key monitoring locations.

The basic requirement for data mining is high-resolution phasor data across the system from multiple locations for a minimum of three months to tabulate different oscillations based on the availability of events. The study can be extended to additional months to extend the research of new additional oscillations. The process could also be made iterative with a relatively short baseline study period to establish low thresholds and utilize prolonged periods of adjusting/increasing thresholds while learning of the system oscillatory properties.

The data mining process includes the following:

<sup>33</sup> Real Time Dynamics Monitoring System – Electric Power Group: <http://www.electricpowergroup.com/rtdms.html>

1. Scan through phasor data, detect low damped oscillations, and record the associated damping and energy value
2. Calculate statistics for each oscillation identified, including pattern of occurrence, highest energy value with timestamp, and PMU measurement
3. Record baseline oscillation characteristics: location, minimum energy level, and frequency band for additional monitoring in real time

### Data Analysis Procedure

The analysis procedure for establishing oscillations alarm thresholds steps are as follows:

1. Assess the quality of the phasor data using PMU quality flags. This step identifies the bad or unusable data (as indicated by the PMUs), which is then removed from the baselining analysis.
2. Identify additional bad data (not indicated by PMUs) using range check and stale check filters. The data dropouts in the communication links are eradicated in this step. Additionally, engineering judgment can be used to identify voltage signals that show significant deviations from the respective base voltages.
3. Perform oscillation analysis on the remaining good data. This step identifies oscillation in the dataset and categorizes the events by location, severity, duration, and count to provide an event library for use in the study. Visual inspection of detected events and validation of oscillation energy detected through peak-peak oscillation variation is helpful in this step to eradicate false events.
4. Establish the oscillation alarm thresholds by calculating the oscillation energy in each frequency band for the dataset during ambient conditions. The average energy for each signal can be used to establish the suggested alarm thresholds for the bands.

### Detection Procedure and Methodology

All available frequency, voltage magnitude, current magnitude, and voltage phase angle signals are to be initially used to mine for oscillation events. Bad data is to be removed prior to mining by using PMU quality flags, range check filters, and stale check filters. Oscillation events are detected based upon the event filters listed in [Table 3.2](#) below.

**Table 3.2: Example Oscillations Mining Criteria**

| Event Filter Type          | Filter Value                   | Description                                                                                                 |
|----------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------|
| Oscillation Frequency (Hz) | $\geq 0.1$ Hz                  | An oscillation frequency filter is used to retain oscillation events above a certain oscillation frequency. |
| Duration (minutes)         | $\geq 1$ Minute                | An oscillation event filter is used to retain sustained oscillations above certain duration.                |
| Energy                     | $> 3$ times Standard Deviation | An oscillation energy filter is used to detect the significant energy in the oscillation.                   |

### Establishing Alarm Thresholds

The oscillation energy is calculated in each frequency band during ambient conditions for each of the four oscillation bands. When oscillations occur in a signal, the oscillation energy for that signal increases, indicating the existence of oscillatory signature in the signal. The mean energy and standard deviation for each signal in each of the oscillation bands is calculated to establish the suggested alarm thresholds. It is suggested that two levels of alarm threshold be established per the formulae below in [Table 3.3](#). This method ensures that only significant oscillation events trigger alarms in real-time and provides multiple levels of alarms to differentiate oscillations requiring monitoring from the ones requiring mitigation. These thresholds should be evaluated using the data analysis approach described earlier

on a regular basis to ensure that all significant oscillations are detected and alarms for insignificant oscillations are kept to a minimum.

**Table 3.3: Example Criteria to Establish Alarm Threshold**

| Alarm Level (Each Band) | Alarm Threshold (Each Band)                                                |
|-------------------------|----------------------------------------------------------------------------|
| Level-1 (Less Severe)   | Mean Energy of Ambient Data Set+(3×Standard Deviation of Ambient Data Set) |
| Level-2 (More Severe)   | Mean Energy of Ambient Data Set+(4×Standard Deviation of Ambient Data Set) |

### ***Detecting and Identifying Local-Area Oscillations***

Forced oscillations can be either localized to a small footprint in the Interconnection, or wide area, resulting in multiple regions swinging against each other. Depending on the oscillation spread, the methods that are available to detect and identify the source may vary. Therefore, it is important to identify whether the oscillations detected are local oscillations or inter-area in nature.

Determining whether an oscillation is local or inter-area can be achieved by answering the following questions:

- What frequency band is generating the oscillation alarm? Is the oscillation frequency detected by the oscillation detector close to one of the system modal frequencies?
- Is my PMU coverage spread out across the system? If so, is the pattern of alarms widespread across most PMUs or is it localized to a specific geographical area? Are the other entities in the Interconnection observing similar alarms?

The first indicator of local oscillations is the range of oscillation frequency. Based on the ranges defined in [Table 3.1](#), if an oscillation is detected and is sustained for a period,<sup>34</sup> an alarm will be generated from a specific band for each PMU that detects the high-energy oscillations. Referring to [Table 3.1](#), local oscillations mostly reside in Band-3 or Band-4, with oscillation frequencies in the range of 1–5 Hz and 5–15 Hz, respectively. Oscillation alarms can be validated by observing the oscillation energy of the band at the PMU that generated the alarm. The energy trend should be increasing to exceed the alarming threshold, indicating the existence of an oscillation in that frequency band. Further validation can be done by observing the trend of the metric, which is used by the Oscillation Detection Module (ODM) to perform calculations at the PMU locations.

Secondly, local area oscillations are contained to a specific geographical footprint and are detected by PMUs nearby the oscillation source. Therefore, only the PMUs relatively close to the source should detect and alarm for the event. That said, detection at the individual PMU level depends not only on the oscillation energy seen by the PMU but also on the alarm levels associated with the oscillation detection for the PMU. Hence, depending on the severity of the oscillation observed from different PMU locations, some PMUs relatively close to the oscillation might not alarm for the oscillation if the energy does not exceed the defined thresholds.

### **Detection Method 2: Detection of Sustained Oscillations of Any Type**

Sustained oscillations can occur in power systems either from the presence of undamped natural oscillations related to power grid dynamics or from the introduction of periodic disturbances in the system, such as from rough zone-related vortex oscillations or from control failures. Thus, algorithms initially intended to detect modal oscillations can also be used to detect sustained forced oscillations. Subsequent analysis would be needed to distinguish whether the detected oscillations are related to poor damping of natural modes or from the presence of forced oscillations. This is important to ensure proper application of the appropriate mitigation measures, either improving a mode's damping

<sup>34</sup> Waiting period is configurable and can be set to dynamically change depending on event severity. Waiting period ensures filtering of transients and post-events ringdown behavior of the system.

(**Determination of Mitigation Actions**) or locating and disabling the forced oscillation's source (**Forced Oscillation Detection Methods**).

Multi-dimensional methods are effective for implementing this approach since they can point to oscillations present in any of the PMU signals included in the analysis. This includes algorithms, such as the FFDD and fast stochastic subspace identification (FSSI). Specifically, an FFDD that is based on estimation of power spectrum densities from all available PMU measurements is very effective in detecting sustained oscillations.

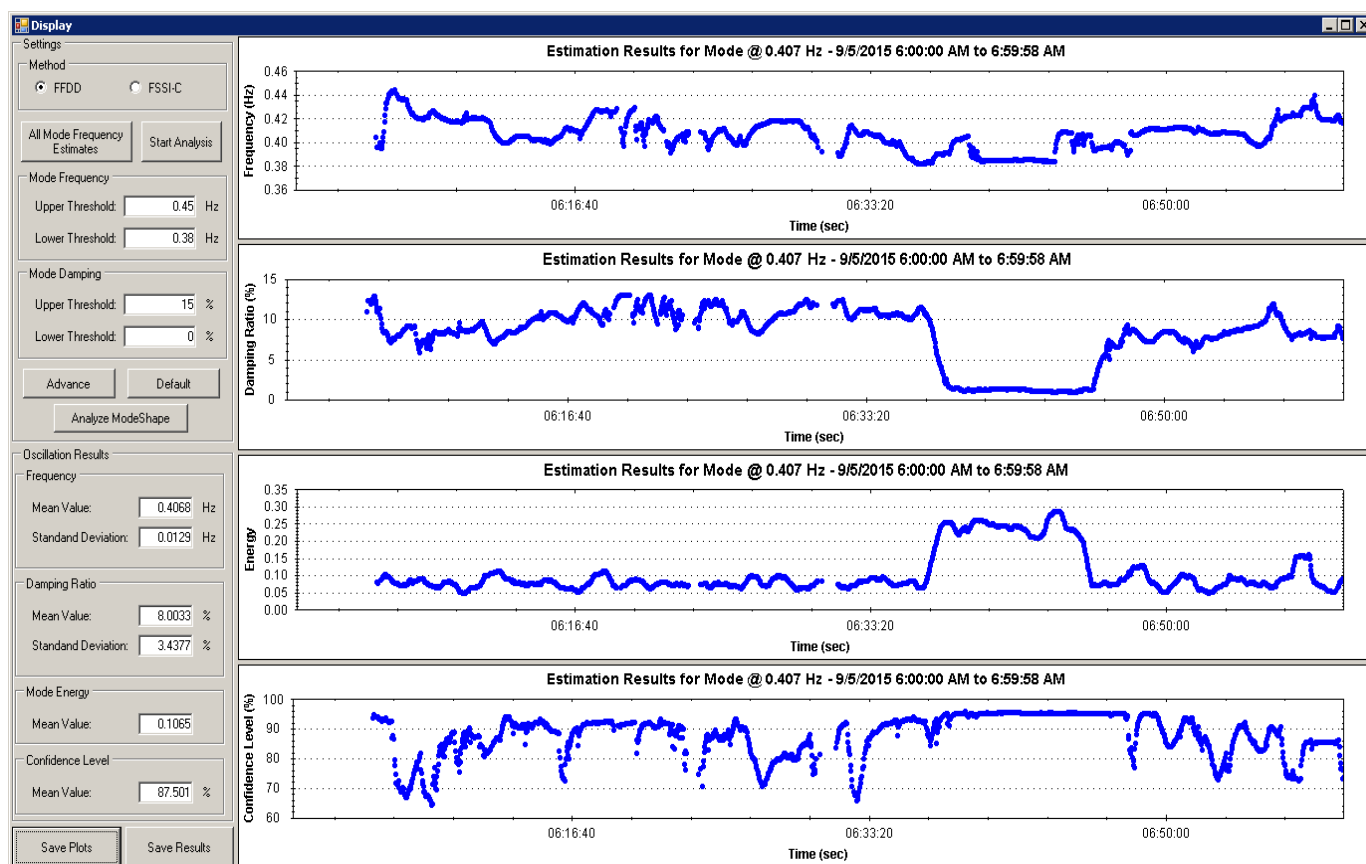
The FFDD first estimates a net energy estimate for the system in the frequency domain, called the Complex Mode Identification Function (CMIF),<sup>35,36</sup> from all the available signals for each frequency value of interest. Then the local peaks in CMIF can be shown to be the dominant system modes and oscillations that are observable in the PMU data at that time in the system. Then using the shape of the CMIF near the local peaks, the FFDD associates a damping estimate for each of the peaks.

In this sense, sustained oscillations can be easily distinguished by the presence of sharp peaks in the CMIF estimate, and correspondingly, the damping estimates associated with sustained oscillation frequencies can be shown to be near zero. Therefore, the FFDD-based approach can directly detect the presence of any sustained or forced oscillation in the frequency range of interest and only the oscillations that have significant energy compared to the other natural modes in the CMIF estimate are selected for detection. Therefore, the method is self-calibrating, in this sense, and does not require any baselining studies. Moreover, the FFDD directly provides the mode shape or oscillation shape directly from the power spectrum matrix as part of the FFDD estimation procedure, and the shape can be useful in understanding the nature and potential source of the sustained oscillation.

The FFDD was tested extensively at Peak Reliability as part of the Forced Oscillation Detection and Source Locator (FODSL) that used PMU based FFDD for oscillation detection and SCADA based engines for source location using generator MW and MVAR SCADA outputs. The FFDD was monitoring over a thousand current magnitude and MW power signals and was very effective at detecting hundreds of sustained oscillation events at Peak Reliability. The FFDD has been implemented as an action adapter into openPDC and has been tested at several other utilities in North America and in Europe. **Figure 3.5** shows an illustration of FFDD estimates showing a forced oscillation event that occurred on September 5, 2015.

<sup>35</sup> H. Khalilinia, L. Zhang and V. Venkatasubramanian, *Fast Frequency-Domain Decomposition for Ambient Oscillation Monitoring*, in *IEEE Transactions on Power Delivery*, vol. 30, no. 3, pp. 1631–1633, June 2015

<sup>36</sup> G. Liu and V. Venkatasubramanian, *Oscillation monitoring from ambient PMU measurements by frequency domain decomposition*, *Proc. IEEE ISCAS*, pp. 2821–2824, May 2008



**Figure 3.5: Illustration of FFDD Estimates Showing a Forced Oscillation Event that Occurred on September 5, 2015**

### Detection Method 3: Mode Frequency Band Monitoring Method

Interactions of different plants and controllers in power systems can result in small or large oscillations in the system. For a stable system, oscillations should decay by the passage of time. In an unstable system, oscillations can grow to dangerous levels where remedial actions are needed to mitigate the adverse effects of the oscillations.

Oscillations appear in power systems in different situations. One of the types of oscillations is forced oscillations (which are due to continuous cyclical excitation) as opposed to ambient oscillations (which are caused by system response to random load changes). Forced oscillations can jeopardize the stability of the system when the amplitude of these oscillations is large or when these oscillations interact with system poorly damped natural modes resulting in high magnitude of the oscillations across a wide area.

### Power Dynamic Extraction

The Power Dynamics Extraction (PDX)<sup>37</sup> analysis method processes power system PMU signals to determine the presence of oscillatory components or system modes and their key parameters. For each oscillatory component, PDX determines these key parameters: frequency, damping ratio, mode shape, and amplitude of the mode. PDX down-samples the input signals with any sampling frequency to 10 Hz. The algorithm uses autoregressive models to obtain characteristics of the modes. PDX processing is applied to a window of the most recent data, and the dynamic characteristics of the system are derived for that time window. The analysis is updated at regular intervals. There are two variations of the algorithm:

<sup>37</sup> PhaorPoint, GE Digital: <https://www.ge.com/digital/applications/transmission/phasorpoint>

- PDX-1 uses a short window (3 minutes) and updates every 5 seconds. This window is used for alarming as it can respond quickly to changes in amplitude and damping ratio.
- PDX-2 uses a longer window (more than 20 minutes) and updates every 20 seconds. This window gives more accurate and stable results. This could be used for model validation or for comparing different cases of well damped modes.

### Mode Bands Management

To track modes in the frequency range of interest, mode bands should be defined. For example, a range may be constructed to contain a single persistently occurring low-frequency mode. Defining effective mode boundaries is facilitated by collecting prior PDX-1 and PDX-2 processed data. Histograms containing large timespans of data (up to three days) can then be used to decide initial mode boundaries.

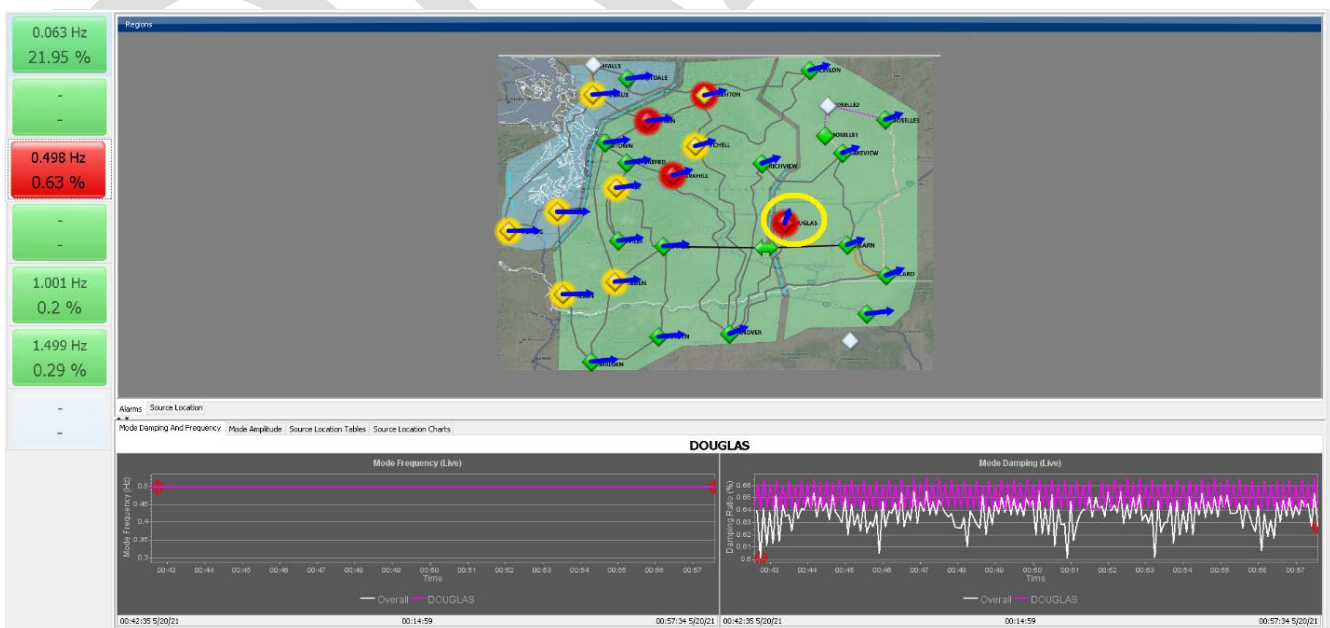
### Alarms and Alerts

For each frequency band and each signal, some thresholds for estimated mode characteristics can be set for alarms and alerts. Damping ratio and amplitude thresholds are the most important limits. Alerts or alarms are issued when the damping ratio of a mode is below the threshold, and the amplitude is larger than the threshold. There is another threshold by means of which the modes with small amplitudes are removed from alarming, regardless of their damping ratio. Similar threshold for damping ratio exists, very well-damped modes with high damping ratios are excluded from alarming regardless of the amplitude. Hysteresis limits can be set for alert and alarm events by requiring the threshold breach to persist for a defined period (in seconds) before an alert or alarm event is triggered.

### Sustained Oscillation Detection

Any type of sustained oscillation, such as natural near-zero damping mode or forced oscillation, can be detected by PDX method. These types of sustained oscillations are detected and presented by a near-zero damping ratio mode by the PDX engine. However, for an appropriate mitigation plan, further analysis and investigation are required to understand whether the sustained oscillation is the result of a zero-damping natural mode or a forced oscillation.

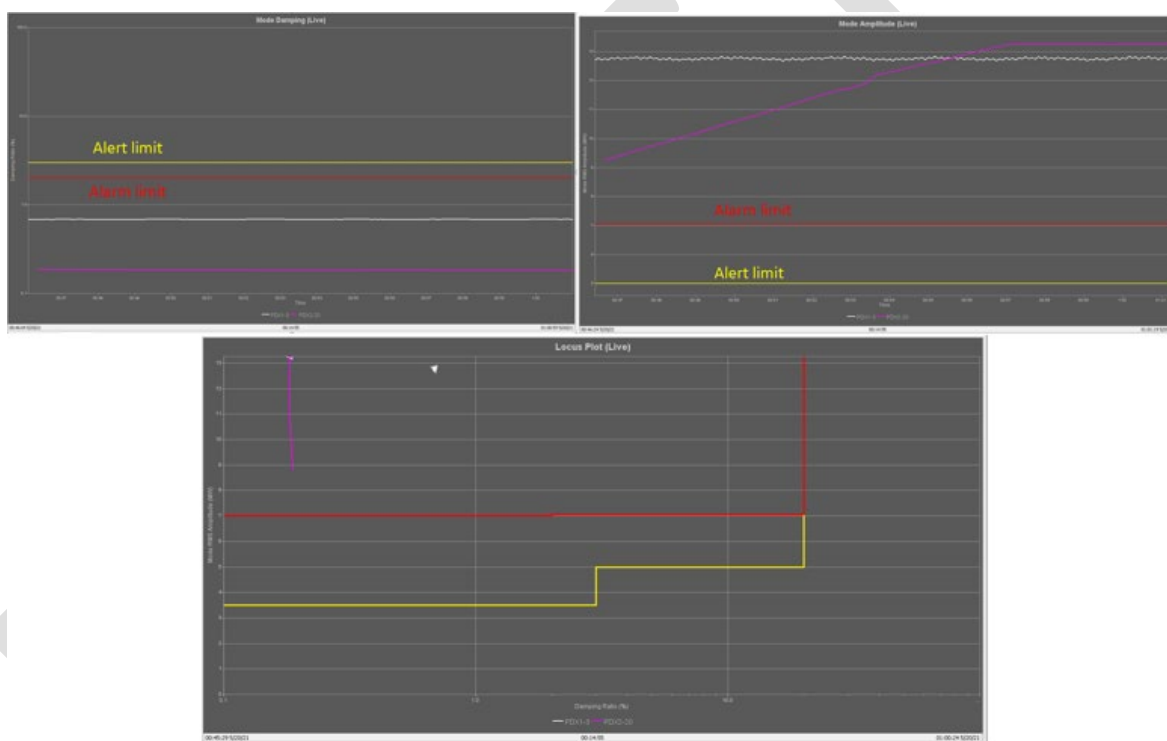
In the following simulation example, the understudy system is ESCA60 and a forced oscillation with the frequency of 0.5 Hz is injected to the system from a generator at Douglas substation. In [Figure 3.6](#), a poorly damped oscillation with a frequency of about 0.5 Hz is identified and an alarm is issued for this oscillation.



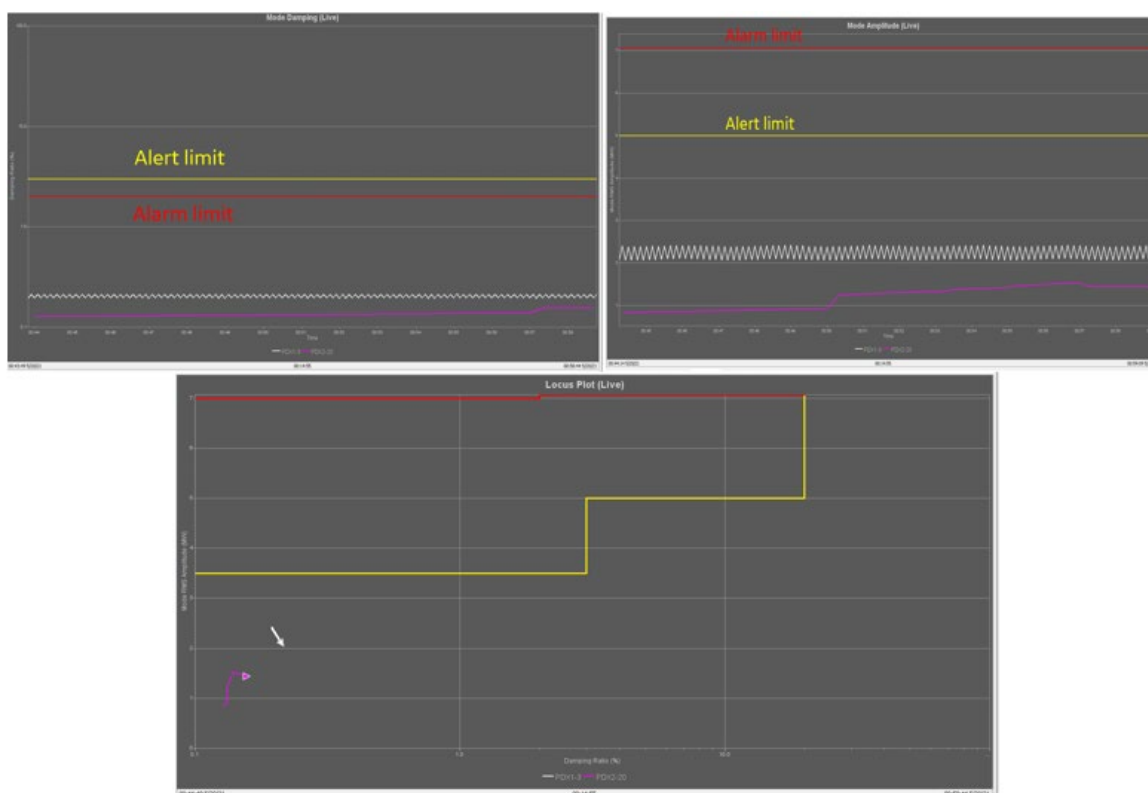
**Figure 3.6: Identification of Forced Oscillation and Triggering an Alarm by PDX Method**

From estimated modes, there are poorly damped oscillations at 1 Hz and 1.5 Hz. These oscillations, which are harmonics of the forced oscillation at 0.5 Hz, are other indications of the presence of forced oscillation in the system. PDX could identify the forced oscillation and its harmonics that help operators to validate the presence of forced oscillation and distinguish forced oscillation from system inter-area electromechanical oscillation.

As can be seen in Figure 3.6, the alarm is issued only for the estimated poorly damped oscillation at 0.5 Hz (the estimate that corresponds to the main forced oscillation) and not for the 1 Hz and 1.5 Hz (harmonics). Figure 3.7 illustrates the estimated damping ratio and amplitude (at the location of source) for a 0.5 Hz oscillation along with the alarms and alerts thresholds. Both time plots and locus plot show that the estimated damping ratio and amplitude are in the alarm zone. In contrast, alarms or alerts are not triggered for harmonics at 1 Hz and 1.5 Hz despite their near-zero damping ratio. Figure 3.8 shows the time plots and the locus plot of estimated damping ratio and amplitude (at the location of source) for the oscillation at 1 Hz. As is evident in Figure 3.8, for the estimated oscillation at 1 Hz, the amplitude is below the alert and alarm thresholds.



**Figure 3.7: Example of Estimated Damping Ratio and Amplitude of Forced Oscillations**



**Figure 3.8: Time Plot and Locus Plot of Estimated Damping Ratio and Amplitude of the Harmonic of Forced Oscillation at 1 Hz Along with Alarm and Alert Limits**

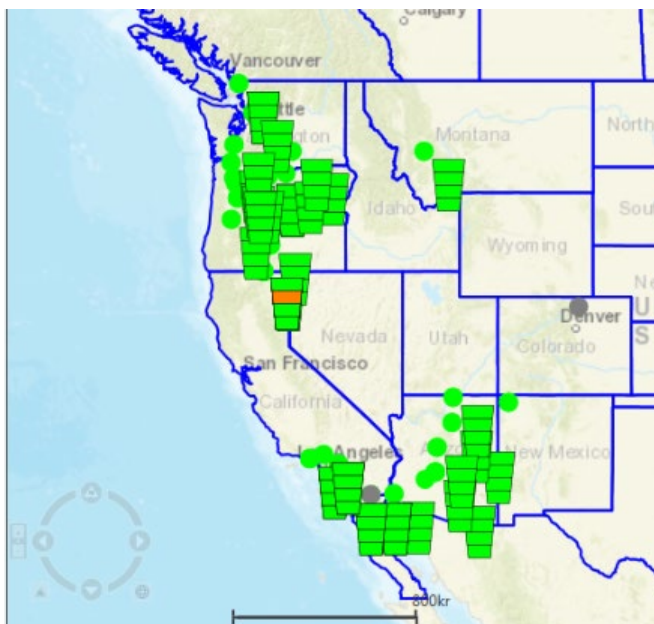
## Determination of Oscillation Source

Determining the location of the source of forced oscillations can be challenging especially when the oscillation is widespread or there may not be sufficient PMU coverage to locate the exact source. For example, an RC may only be monitoring BPS level synchrophasor data, which would be utilized for alarming the operators. However, the RCs can always coordinate with the impacted TOPs to determine the source. The impacted TOPs would have a more granular view as they would be monitoring more synchrophasor data specific to the respective TOP area. In scenarios where the synchrophasor data itself is scarce, the available synchrophasor data can help to locate the local area from where the oscillations originate; however, determining the exact source location would require further review of SCADA data and coordination and communication with the impacted TOPs, Balancing Authorities, and Generator Owners (GO).

When synchrophasor or telemetered data are available, the following methods help to determine the exact source location of forced oscillations or the local regions from where the oscillations originate.

### Local Forced Oscillations

**Figure 3.9** shows an example of a local forced oscillation where the source of the oscillation can be easily traced to a single location. The unit causing the oscillation may exist either at the location shown on the operator dashboard or may be in the underlying system connected to the location with the alarm if the operator dashboard does not have every location shown since synchrophasor data may not be available from every generator location. This gives the impacted RC and TOP a starting point to coordinate to determine the source.



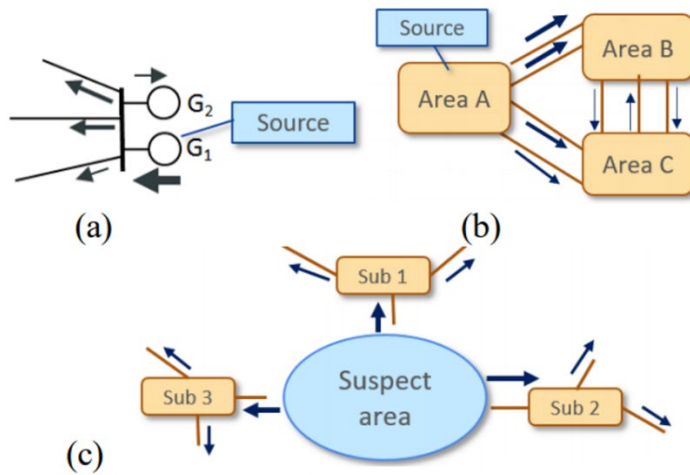
**Figure 3.9: Example of Local Forced Oscillation**

### ***Forced Oscillations Impacting a Wide Area***

When forced oscillations are observable over a wider area, the location of the oscillation source can be more challenging. In certain cases, an oscillating unit can cause another unit to oscillate, which makes it difficult to pinpoint the suspect unit. The following methods help to locate oscillation sources when the forced oscillation is observable over a wider area.

### Dissipating Energy Method

The method is based on tracing the flow of transient dissipating energy through the power system network (Maslennikov & Litvinov, 2020). By tracing the flow of energy back to its source, the equipment generating the oscillation is identified. When synchrophasor data is available at a sufficient number of units, the method can point to the exact source of oscillation; the source will have the unique characteristic of dissipating energy flowing out of the generator. Similarly, when sufficient synchrophasor data does not exist at the terminals of the generator, the suspect area can be localized by utilizing synchrophasor data from line flows. Examples of the interpretation of dissipating energy flow (DEF) results are presented in Figure 3.10. This method was developed and is in use at ISO New England.



The use cases of the DE pattern interpretation (a) PMU monitors POI of a power plant, (b) PMU monitor tie-lines between areas, (c) localization of suspect area non-observable by PMUs

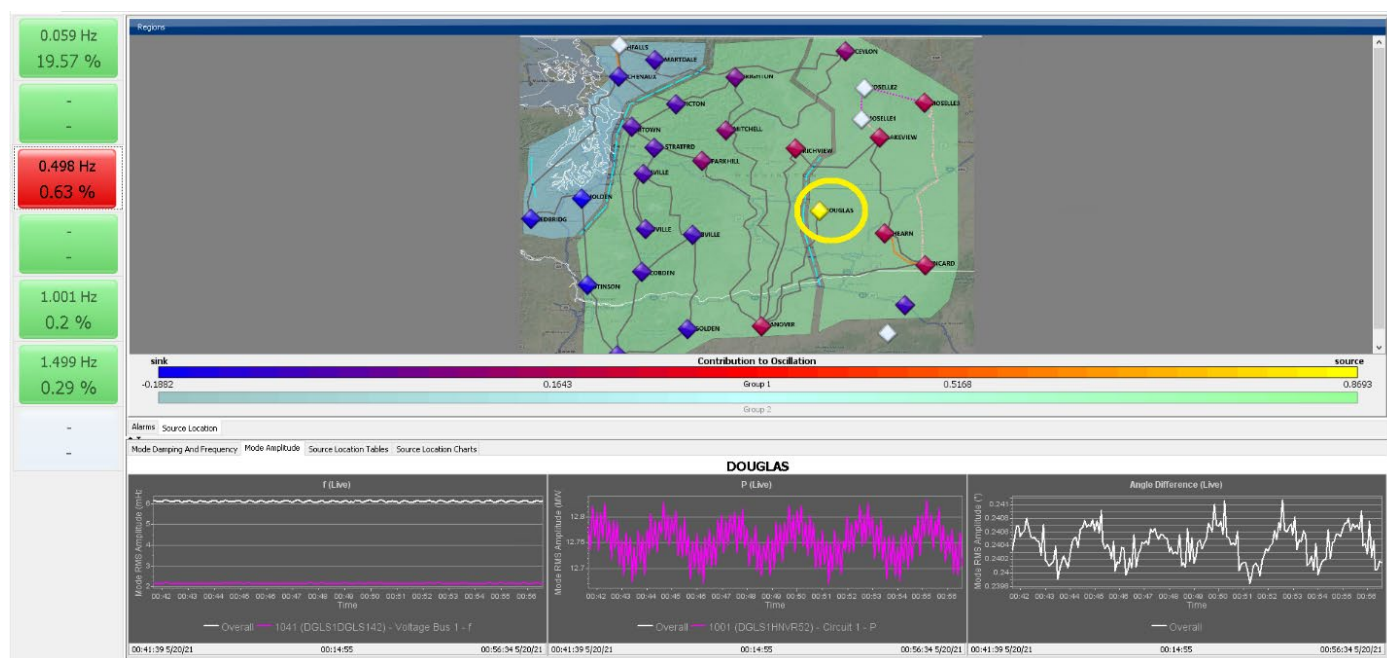
**Figure 3.10: Use Cases of the Dissipation Energy Method**

### Source Location Using the Phase of Oscillation

For the oscillation source location (OSL), voltage angle measurements are used to identify the PMU in the substation or region that is the closest to the sources of oscillations. This is achieved by analyzing the phase of oscillations in the voltage angles from around the system. A leading phase indicates less damping contribution, and the source is the location with the lowest damping contribution.<sup>38</sup>

Figure 3.11 shows an example of an oscillation source location map. As is evident, the source is identified and shown by the yellow circled area in the map.

<sup>38</sup> Al-Ashwal, N., D. Wilson, and M. Parashar. *Identifying sources of oscillations using wide-area measurements. Proceedings of the CIGRE US National Committee 2014 grid of the future symposium, Houston*. Vol. 19. 2014.



**Figure 3.11: Identification of the Source of Forced Oscillation**

### *SCADA Data-Based Source Location*

SCADA data can also be used to detect potential oscillation locations using generator MW and MVAR telemetered data. SCADA data is easily available, and, despite the lower resolution, SCADA data can sometime help to point to the potential sources of oscillation based on the magnitude and the frequency of the oscillation. This was one of the methods deployed at Peak Reliability for identifying sources of oscillation and validated with GOPs for real events.<sup>39</sup>

### *Other Oscillation Source Location Methods*

Several other oscillation location methods have been developed to identify source locations under various imperfect signal conditions and practical event scenarios. These include a method<sup>40</sup> that combines dissipated energy flow (DEF) with a dynamic state estimator (DSE), a machine-learning pattern recognition (ML-PR) method<sup>41</sup> that uses DEF results as inputs, and a cross power spectral density (CPSD)-based method.<sup>42</sup>

<sup>39</sup> Source Location of Forced Oscillations Using Synchrophasor and SCADA

Data, <https://scholarspace.manoa.hawaii.edu/server/api/core/bitstreams/2fde8564-15a3-4885-98ba-f9462a71005c/content>

<sup>40</sup> Combined Dissipated Energy Flow Method, [https://www.naspi.org/sites/default/files/2021-10/D3S8\\_02\\_estevez\\_fiuba.pdf](https://www.naspi.org/sites/default/files/2021-10/D3S8_02_estevez_fiuba.pdf)

<sup>41</sup> Machine-Learning Pattern Recognition method, [https://www.naspi.org/sites/default/files/2021-10/D3S8\\_03\\_zheng\\_ge\\_20211007\\_0.pdf](https://www.naspi.org/sites/default/files/2021-10/D3S8_03_zheng_ge_20211007_0.pdf)

<sup>42</sup> Cross Power Spectral Density based Method, [https://www.naspi.org/sites/default/files/2021-10/D3S8\\_04\\_osipov\\_rpi\\_20211007\\_0.pdf](https://www.naspi.org/sites/default/files/2021-10/D3S8_04_osipov_rpi_20211007_0.pdf)

**Key Takeaways:**

- Forced oscillations are relatively common yet sporadic in nature. They can occur at any time and in any place in the BPS where the system is subjected to a periodic disturbance that continuously excites the system.
- Forced oscillations are not described by modes or eigenvalues.
- Forced oscillations can occur at any frequency; the fundamental frequency of the oscillation depends on the nature of the driving cyclic force.
- Forced oscillations typically have harmonics of the forced oscillation frequency since the cyclic input waveform is generally not sinusoidal.
- Forced oscillations often have a near-zero damping ratio since their source of oscillation perturbing the system is persistent (no decay). This does not mean the system is unstable.
- Forced oscillations are not generally described by a system mode shape; however, forced oscillations have an associated oscillatory characteristic for that forcing function which can be represented like a perceived oscillation mode shape.
- The driving cyclic force is usually not accounted for in dynamic models; therefore, traditional modal analysis and transient stability analysis are not applicable to studying this phenomenon. Rather, forced oscillations are usually detected through dynamic measurement devices, such as PMUs or other DDRs.

## Chapter 4: Forced Oscillations Causing Inter-Area Oscillations

This section addresses the phenomenon that leads to widespread impact of forced oscillations due to resonance conditions involving system modes. Also described is an example of how model-based simulations can be utilized to observe the impact of widespread oscillations during resonance conditions.

### Conditions that Lead to This Phenomenon

Forced oscillations that occur on the BPS are generally locally contained and can be identified accordingly. However, if a forced oscillation occurs at an oscillation frequency near the frequency of an inter-area system mode, the forced oscillation can “excite” the mode so that the oscillations propagate through wide geographical domains where the system mode is observable. If the mode’s damping is poor and the mode is highly controllable at the forced oscillation’s source, the interactions can also lead to a “resonance” effect so that the resulting system-wide oscillations are much larger than the oscillations at the source. In this case, even small oscillations can induce larger, system-wide oscillations from resonance amplification. To summarize, the conditions that lead to this effect are the following:<sup>43 44</sup>

- The forced oscillation frequency should be at or near a system mode frequency.
- The system mode should be poorly damped.
- The source is near a strong participation location of that system mode, such as the distant ends of the system mode. The distant end locations are the strongest participants in the system mode.

Examples of wide area forced oscillations can be found in [Appendix D](#), particularly Events. An important property of such resonance conditions is that the larger magnitude forced oscillation can be observed in multiple locations across the grid, particularly far away from the actual source where the magnitude of the oscillation can be relatively small. That significantly complicates the identification of the source of oscillations in practice when there is no prior knowledge of the location and nature of the forced signal. Using the magnitude/energy of the oscillations as an indicator of the source location could be misleading in these situations. In these situations, it is important for RCs to coordinate with one another to simultaneously identify the oscillation and attempt to determine a source and mitigating procedure if necessary.

During resonance conditions between forced oscillations and a system mode, the interaction between the forced oscillation and natural modes can sometime be misunderstood. For clarification purposes, the oscillations observed across a wide area during the resonance conditions are forced oscillations and not natural oscillations. The interaction simply means that the forced oscillations’ observability will be determined by the observability of the mode having frequency close to that of forced oscillations and the generating units participating in the mode are likely to participate in these forced oscillations. Also, during this interaction, modemeter algorithms not designed to operate in the presence of forced oscillations, such as YW-ARMA and LS-ARMA, can introduce bias in the estimate of the damping ratio of that mode toward 0%.<sup>45,46</sup> However, system mode damping is not affected during a forced oscillation event, at least not unless the forced oscillations cause some cascading trips and that eventually affects the damping ratio of the mode. Therefore, a close to 0% damping ratio estimate of the mode seen during the resonance condition does not reflect the true damping ratio of the mode and is merely a limitation of the algorithms.

<sup>43</sup> S. Arash Nezam Sarmadi, Vaithianathan Venkatasubramanian, Armando Salazar, *Analysis of November 29, 2005, Western American Oscillation Event*, *Power Systems IEEE Transactions on*, vol. 31, no. 6, pp. 5210–5211, 2016.

<sup>44</sup> S. A. N. Sarmadi and V. Venkatasubramanian, *Inter-area resonance in power systems from forced oscillations*, *IEEE Trans. Power Syst.*, vol. 31, no. 1, pp. 378–386, Jan. 2016.

<sup>45</sup> U. Agrawal, J. Follum, J. W. Pierre and D. Duan, *Electromechanical Mode Estimation in the Presence of Periodic Forced Oscillations*, in *IEEE Transactions on Power Systems*, vol. 34, no. 2, pp. 1579–1588, March 2019.

<sup>46</sup> J. Follum, J. W. Pierre and R. Martin, *Simultaneous Estimation of Electromechanical Modes and Forced Oscillations*, in *IEEE Transactions on Power Systems*, vol. 32, no. 5, pp. 3958–3967, Sept. 2017.

The interplay between forced oscillations and a power system's natural oscillatory properties drives the need to understand the electromechanical modes in each of the North American interconnections to be prepared for these types of situations where the oscillations could interact with one another. Understanding the potential interactions is critical when trying to identify the source of oscillations, particularly when they are observed throughout the BPS.

## Testing Possible Mitigations

[Appendix A](#) provides an example of how the impact of the January 11, 2019, forced oscillations can be determined by using model-based simulations and how possible mitigation plans can be determined for utilization in guidance provided to operators.

Since wide-area resonant forced oscillations can be simulated in dynamic analysis, the effect of various plant controls and operator actions can also be tested; see the following examples:

- Switching Audio/Video Receivers to manual mode
- Generation redispatch
- Curtailing power transfers
- Commissioning more PSS

The recommended steps are as follows:

1. Modify the load flow and dynamic cases to match the system conditions prior to the event
2. Perform a baseline eigen-analysis to determine the characteristics of the subject oscillatory mode
3. Model and initiate the forced oscillation
4. Perform a second eigen-analysis to determine changes in subject oscillatory mode
5. Simulate various mitigating actions and determine the most effective course of action

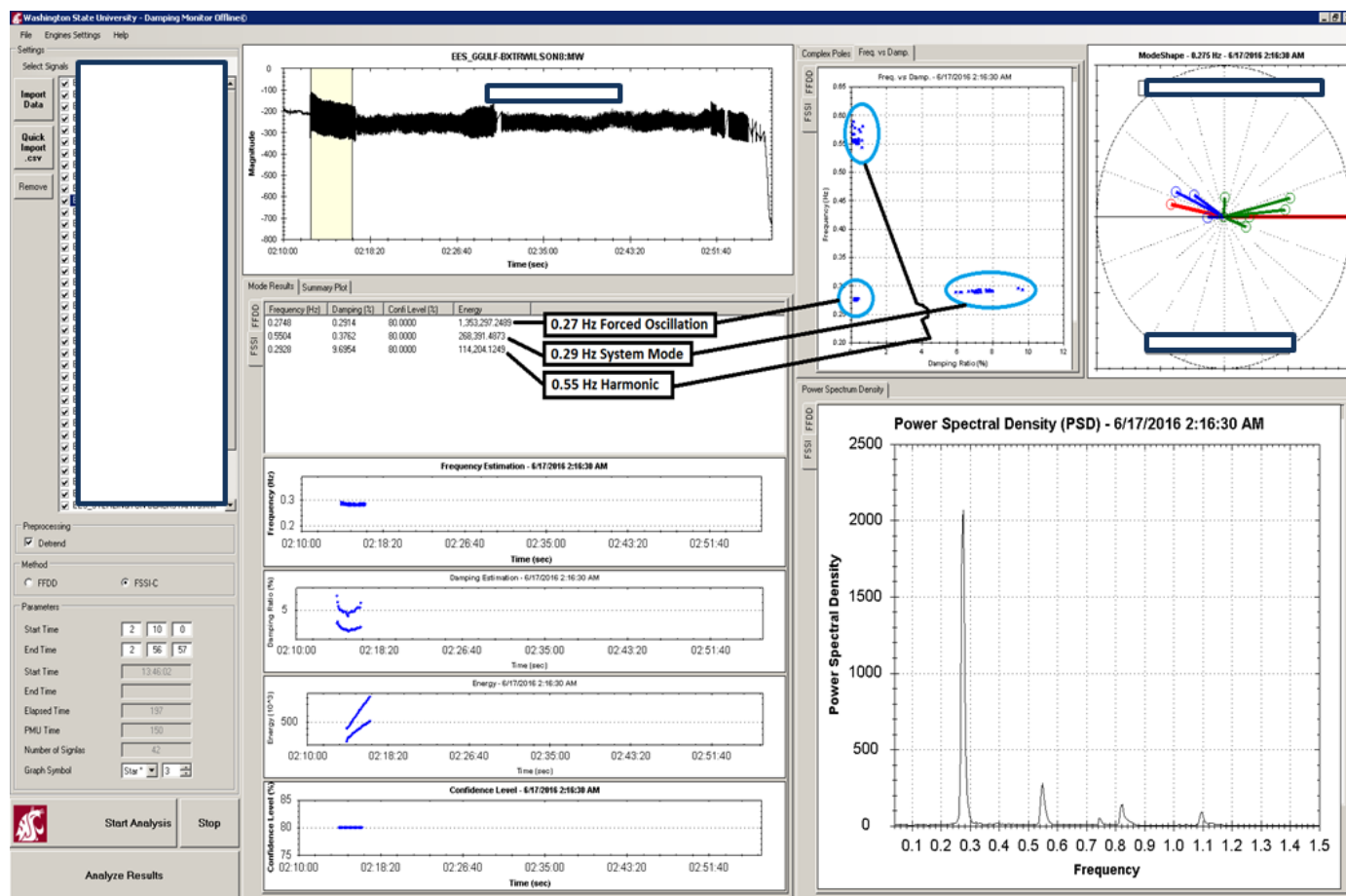
## Chapter 5: Guidelines for Addressing Wide-Area Oscillations

This section addresses the analyses, tools, and procedures that RCs/TOPs need during a wide-area oscillation event. To begin, approaches for distinguishing between a natural and forced oscillation are described in [Framework for Forced Oscillation Mitigation](#). In the relatively rare case where the oscillation is due to a poorly damped inter-area electromechanical mode, the mitigation actions developed with methods described in [Determination of Mitigation Actions](#) should be implemented. If the oscillation is identified as forced, the second step is to determine if action should be taken; guidelines for making this determination are described ([Framework for Forced Oscillation Mitigation](#)) then followed by a discussion of potential mitigation actions and their validation. The section concludes with a summary of the tools and procedures needed within and across RC/TOP footprints to enable the previously described analyses and actions.

### Determining if an Oscillation is Natural or Forced

As described previously, forced oscillations become widespread when they excite one or more of a power system's inter-area electromechanical modes. For this to occur, the frequency of the forced oscillation must be close to the frequency of an inter-area system mode. Thus, when a sustained oscillation occurs in the frequency range of electromechanical modes, it may not be immediately apparent whether the oscillation is forced or natural. The appropriate responses to natural and forced oscillations differ, so it is important to either classify the oscillation as forced or natural before taking action or ensure control and operator responses are appropriate for either type of oscillation. Currently, many of the entities operating grids are doing neither because methods for classifying sustained oscillations are not yet widely available in commercial tools. Additionally, sustained natural oscillations are a rare occurrence as the system modes are mostly well-damped with sufficient small-signal stability margin under normal operating conditions and for credible contingencies. Also, the operating conditions that could potentially lead to these oscillations are typically identified and addressed in planning studies. Though relatively uncommon, undamped inter-area natural oscillations do occur and pose a significant threat to reliable system operation.

One readily observable difference between natural and forced oscillations that is widely agreed upon is the presence of harmonics. Forced oscillations are often accompanied by harmonics due to the periodic nature of the driving input. For example, a 0.25 Hz oscillation may be accompanied by harmonics at 0.75 Hz, 1.25 Hz, etc. The appearance of harmonics is a clear indication of a forced oscillation because inter-area electromechanical modes do not possess harmonics. Some commercial tools may provide frequency estimates for each of an oscillation's harmonics. Others may enable spectral analysis so that the harmonics can be picked out by examining a plot. However, not all forced oscillations have harmonics, and they may be too small to detect if they do. Thus, harmonics are an important characteristic to consider but are not sufficient to make the distinction in all cases. [Figure 5.1](#) shows SSI estimates during a forced oscillation event that shows the presence of harmonics as zero damping oscillation estimates and as sharp peaks in the power spectrum plot. The forced oscillation at 0.28 Hz leads to harmonic peaks at 0.56 Hz, 0.84 Hz, and 1.12 Hz in the power spectrum plot (in the lower right side) and as 0.56 Hz, zero damping oscillation estimates in the mode estimation table (in the center).



**Figure 5.1: Example of Harmonics Present as Zero Damping Oscillation Estimates**

The methods used to continuously monitor inter-area electromechanical modes described in [Data Collection and Pre-Processing](#) can also help determine if an oscillation is natural or forced. These tools offer situational awareness that can help a well-trained engineer or operator understand what type of event is occurring. Forced oscillations are often (but not always) blue-sky events because they can be caused by a single piece of equipment. In contrast, poorly damped inter-area natural oscillations are often (but not always) associated with high-stress conditions and/or multiple contingencies. Monitoring tools can provide an early warning of modes headed toward instability. If sustained oscillations occur following a progressive decline in a mode's damping, the oscillation is likely natural. If a sustained oscillation occurs under low-stress conditions apart from contingencies, there is a greater chance that it is forced. These are not hard-and-fast rules, but they can help operations staff interpret their tools appropriately.

In addition, some oscillation monitoring tools are capable of tracking forced oscillations and relatively well-damped electromechanical modes of oscillation simultaneously. If only the sustained oscillation is found at the frequency of the known electromechanical mode, there is increased likelihood that poor mode damping is leading to the oscillation. However, if the sustained oscillation and the known mode with sufficient damping are both identified, the sustained oscillation is most likely forced and has been observed before in RC operations.<sup>47</sup> Again, this approach requires training to help operations staff interpret results. Such approaches will be necessary while commercial tools that explicitly classify sustained oscillations as natural or forced are developed.

<sup>47</sup> H. Zhang, J. Ning, H. Yuan and V. Venkatasubramanian, *Implementing Online Oscillation Monitoring and Forced Oscillation Source Locating at Peak Reliability*, 2019 North American Power Symposium (NAPS), 2019, pp. 1–6

## Mitigating Actions

Once an oscillation is detected, a decision about whether to take mitigating actions must be made. Defining thresholds to determine if an oscillation is a reliability threat requiring action is a nontrivial task, but the key parameter is the oscillation's amplitude. Offline dynamic studies and reviews of relay settings can be used to set well-informed thresholds. In addition, regular investigation of oscillatory events can build an institution's understanding of how oscillations of various amplitudes, durations, and frequencies may impact reliability. As much as possible, oscillation scenarios should be studied in advance using measurements and models to determine appropriate action.

When a wide-area oscillation occurs, effective communication among RCs/TOPs can help maintain system reliability by ensuring that only well-coordinated and effective mitigation actions are performed. Local oscillations can typically be addressed by an RC/TOP acting alone, but if the oscillation is observed throughout an RC/TOP footprint, coordination should begin quickly. TOPs should contact their RC, and RCs should reach out to neighboring RCs. In this way, the affected portion of the Interconnection can be quickly identified.

Any oscillation observed across an RC/TOP footprint warrants coordination. This coordination is important even when the source is readily apparent, and the oscillation is not large enough to threaten reliability within the RC/TOP footprint from which it is originating. Due to the system's dynamics, a forced oscillation may be larger in far off areas of the system. RCs in those areas need to know that the source has been identified, and the RC/TOP from which it is originating needs to adjust their response if reliability of other portions of the grid are threatened. Even oscillations too low in energy to impact system reliability may be indicative of malfunctioning or mis-operating equipment. When oscillations are found to originate from a power plant, RCs/TOPs can help the power plant operator maintain reliable and safe operation by communicating with them.

With coordination efforts in place, mitigation measures can be enacted effectively. Inter-area natural oscillations should be mitigated based on the results of studies as discussed in [Determination of Mitigation Actions](#). If the mode of oscillation is unknown, reducing flows along tie-lines may reduce system stress and improve stability. In the case of wide area forced oscillations, the most effective mitigation measure is to identify and disable the forcing input. Disabling the input does not necessarily mean that equipment needs to be taken offline. Certain areas of operation, such as the rough zone on hydro units, may lead to oscillations. In some cases, unintended control interactions leading to oscillations can also be mitigated by adjusting an operating point. Whether the mitigating action is simply adjusting an operating point or tripping a power plant offline, the source must first be identified. Methods for identifying the source of an oscillation were discussed in Forced [Oscillation Detection Methods](#).

When a severe oscillatory event occurs and reliability is threatened, a potential action is to increase the damping of the excited electromechanical mode. This action can be performed while searching for the oscillation's source, which should be the primary action of all impacted RCs/TOPs. As listed previously, the three conditions leading to widespread forced oscillations are as follows:

- Proximity of the frequencies of the forced oscillation and an electromechanical mode of oscillation
- Low damping of an electromechanical mode with similar frequency
- Controllability of an electromechanical mode with similar frequency at the source of the forced oscillation

Of these three conditions, only the damping of the electromechanical mode can be effectively adjusted by RCs/TOPs through changes in the system's operating point. Increasing the damping ratio of a mode excited by a forced oscillation will not fully address the forced oscillation, which will continue until its source is disabled. However, increasing the damping ratio of the excited mode will reduce the forced oscillation's amplitude in other parts of the system that participate in the mode.

To implement this approach, studies must be conducted ahead of time to identify the operating conditions, such as flow along transmission corridors that can be adjusted to improve the damping of each mode. When a forced oscillation occurs near the frequency of a system mode, the measures designed for that mode can be implemented to improve damping. In many cases, these actions will be similar or identical to those designed to mitigate natural oscillations due to poor system damping, as described in [Determination of Mitigation Actions](#).

DRAFT

## Chapter 6: Examples of Oscillation Monitoring Implemented by RCs and TOPs

In this chapter, overviews of existing tools and procedures that RCs and TOPs have put into place are discussed. PMUs and the oscillation monitoring tools provided to system operators for situational awareness of concerns that could make a 15-minute impact on BPS equipment or operational decisions should be evaluated for NERC CIP applicability.

### Bonneville Power Administration

The following overview of Bonneville Power Administration's oscillation detection capabilities draws upon two key references: Implementation and Operating Experience with Oscillation Detection - Application at Bonneville Power Administration,<sup>48</sup> and Synchrophasor Technology at BPA: From Wide-Area Monitoring to Wide-Area Control,<sup>49</sup> an overview of the oscillation applications in use at Bonneville Power Administration (BPA). Together these works provide critical insight into the deployment and evolution of synchrophasor-based applications for wide-area monitoring and control.

BPA successfully deployed an oscillation detection application<sup>50</sup> on its control center video wall in October 2012 that uses an "RMS Energy Filter"<sup>51</sup> as shown in Figure 6.1. The application scans multiple signals (power, frequency, voltages) across the grid for indication of growing or sustained high-energy oscillations. The application monitors the point of interconnection (POI) of power plants, the Pacific HVDC Intertie, SVCs, and the BPA 500 kV grid internal to the Pacific Northwest. The application's primary purpose is to detect unanticipated oscillations that result from control system failures, local power plant instability, forced oscillations, excitation of inter-area modes, or a generating unit in an unstable operating region.



**Figure 6.1: RMS Energy Filter in Oscillation Detection Application**  
(Source: BPA)

A PMU-derived input signal (e.g., active power or voltage magnitude) is formed and passed through a bandpass (BP) filter that focuses on the desired bandwidth for oscillation detection. The application extends classic energy detector methods<sup>52</sup> to multiple frequency bands, including the following:

- **0.01 to 0.15 Hz band:** for governor, plant controller, or AGC control oscillations
- **0.15 to 1 Hz band:** for electromechanical inter-area and some local plant oscillations, and also oscillations caused by sustained operation of a hydro-power plant in rough zone
- **1 to 5 Hz band:** for local generator oscillations, generator excitation controls, PDCI controls, etc.
- **5 to 15 Hz high frequency band:** for steam-generator torsional, PDCI, wind generator control oscillations, sub-synchronous interactions

<sup>48</sup> Implementation and Operating Experience with Oscillation Detection - Application at Bonneville Power Administration: <http://cigre-usnc.org/wp-content/uploads/2016/10/Kosterev.pdf>

<sup>49</sup> Synchrophasor Technology at BPA: From Wide-Area Monitoring to Wide-Area Control: <https://www.energy.gov/oe/articles/hamody-hindi-bonneville-power-administration>

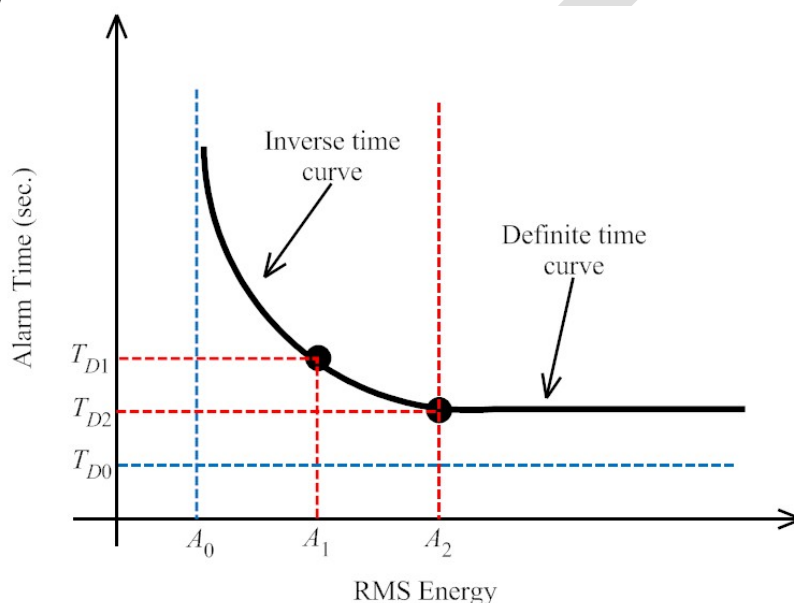
<sup>50</sup> BPA Oscillation Detection Application developed by Montana Tech University and University of Wyoming based on original work of Dr. John Hauer.

<sup>51</sup> M. Donnelly, D. Trudnowski, J. Colwell, J. Pierre, and L. Dosiek, *RMS-Energy Filter Design for Real-Time Oscillation Detection*, Proceedings of the 2015 IEEE PES General Meeting, July 2015.

<sup>52</sup> S.M. Kay, *Fundamentals of Statistical Signal Processing: Volume 2, Detection Theory*, Prentice Hall, 1998.

After filtering, the signal is then squared, passed through an averaging filter, and then square-rooted. The goal of the averaging filter is to estimate the mean of the squared signal matched to the BP filter. The resulting output will be an estimate of the RMS of the input signal in the bandwidth of the BP filter. An oscillation must be persistent for several minutes for the application to issue an alarm. An inverse-time characteristic, often used in protective relaying, could be considered.

The current alarm system used in the BPA control center is a definite-time alarm strategy. If the RMS energy for a given band exceeds a trigger level for a fixed amount of time, an alarm is set. Future versions being considered use a combination of inverse-time and definite-time alarms, as shown in Figure 6.2. The black curve is the alarm curve. The  $A_i$  are alarm levels set by the user.



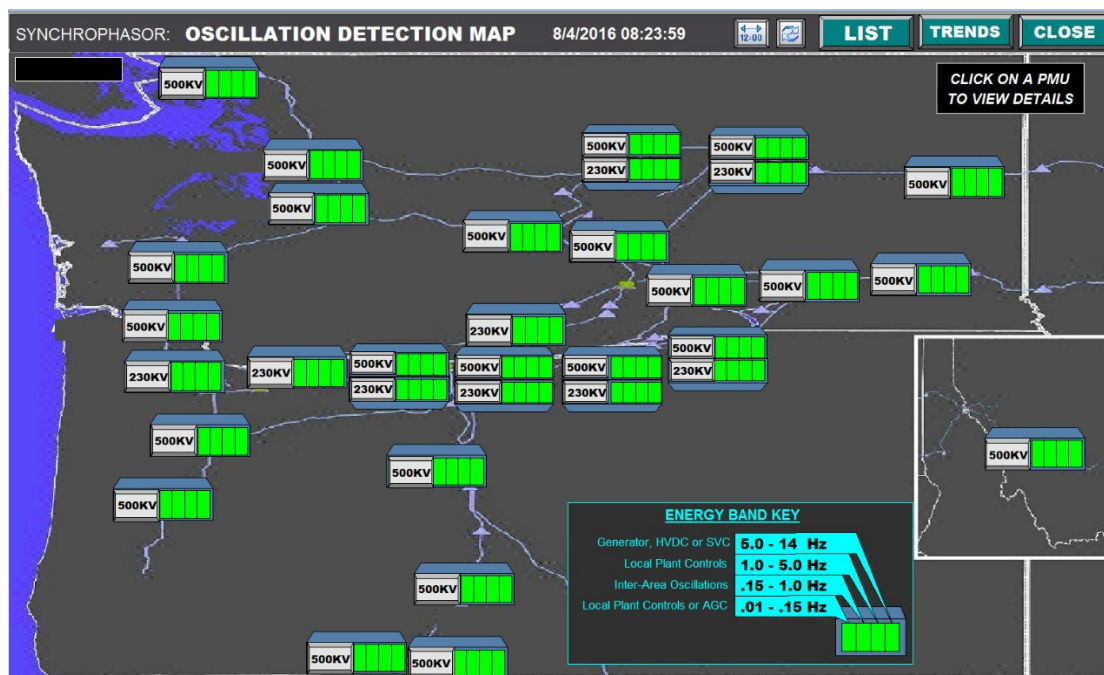
**Figure 6.2: Combined Inverse, Time and Definite, and Time Alarm Curve (Source: BPA)**

Figure 6.3 shows an overview of BPA display for the oscillation detection application. PMUs at locations where oscillations are monitored are placed on a geographic map. Each PMU has four frequency bands as described above. Should an oscillation alarm occur, a corresponding frequency band at that PMU will turn red. The display provides very effective visual indication on whether the oscillation is local or wide area. For local oscillation, only one or a few PMUs in the vicinity of the oscillation source will go into alarm state. For wide-area oscillations, multiple PMUs will go into alarm state over a large geographic area. The display also provides initial indication of the type of oscillation based on the frequency band alarmed.

BPA dispatchers and operations technical staff can drill down further into the oscillation by selecting the PMU, which brings up a trend display with monitored signals. The signals that triggered the alarm will be identified on the display.

The display includes signal trend, alarm status trends, oscillation energy trend, and alarm settings for all four frequency bands. BPA developed dispatcher standing orders with instructions for operators on actions to be taken when an oscillation alarm is issued. BPA technical staff conducted several training sessions for dispatchers on oscillation basics; the oscillation detection application and the associated operating procedures went into effect in June 2016.

The BPA oscillation detection application has detected several oscillation events since it was implemented in 2012. Most of the events are local forced oscillations due to equipment issues. However, there were several wide-area events due to oscillatory instability of generators.



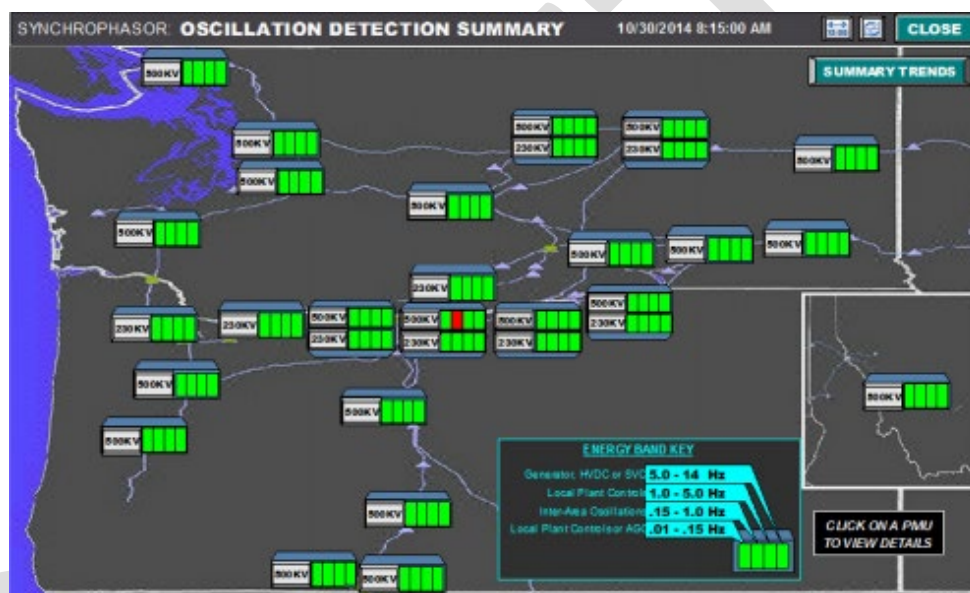
**Figure 6.3: Overview Display for Oscillation Detection Application**  
(Source: BPA)



**Figure 6.4: Third Layer for Oscillation Detection Application with Energy Bands**  
(Source: BPA)

BPA's oscillation detection application has been deployed in the BPA control room since 2013, and it operates by monitoring PMU measurements for increased energy. This monitoring is performed for four frequency bands as described in [Framework for Forced Oscillation Mitigation](#). When a signal's energy exceeds the threshold, it is indicated on a geographical map in red as in [Figure 6.4](#). An example where the oscillation is detected across a wide area is displayed in [Figure 6.5](#). These displays provide operators with an initial indication of how widespread the oscillation is and what its type may be based on the frequency band. Once a detection is displayed, the operator can click on specific locations to obtain more detailed information as shown in [Figure 6.5](#) and [Figure 6.6](#).

The BPA established operating procedures to accompany their oscillation detection application in 2016. If an alarm is issued for a single location, the system operator contacts the local operator or field staff to investigate further. If the alarm is issued for multiple locations, the oscillation is considered widespread. Depending on the situation, the RC may be contacted to identify the cause and develop a course of action. Potential operator actions, governed by BPA's operating procedures, include inserting series capacitors, inserting transmission lines that are out of service for voltage control, moving generation to increase system inertia, and curtailing schedules.



**Figure 6.5: Control Room Display Indicating a Localized Oscillation in Band 2**

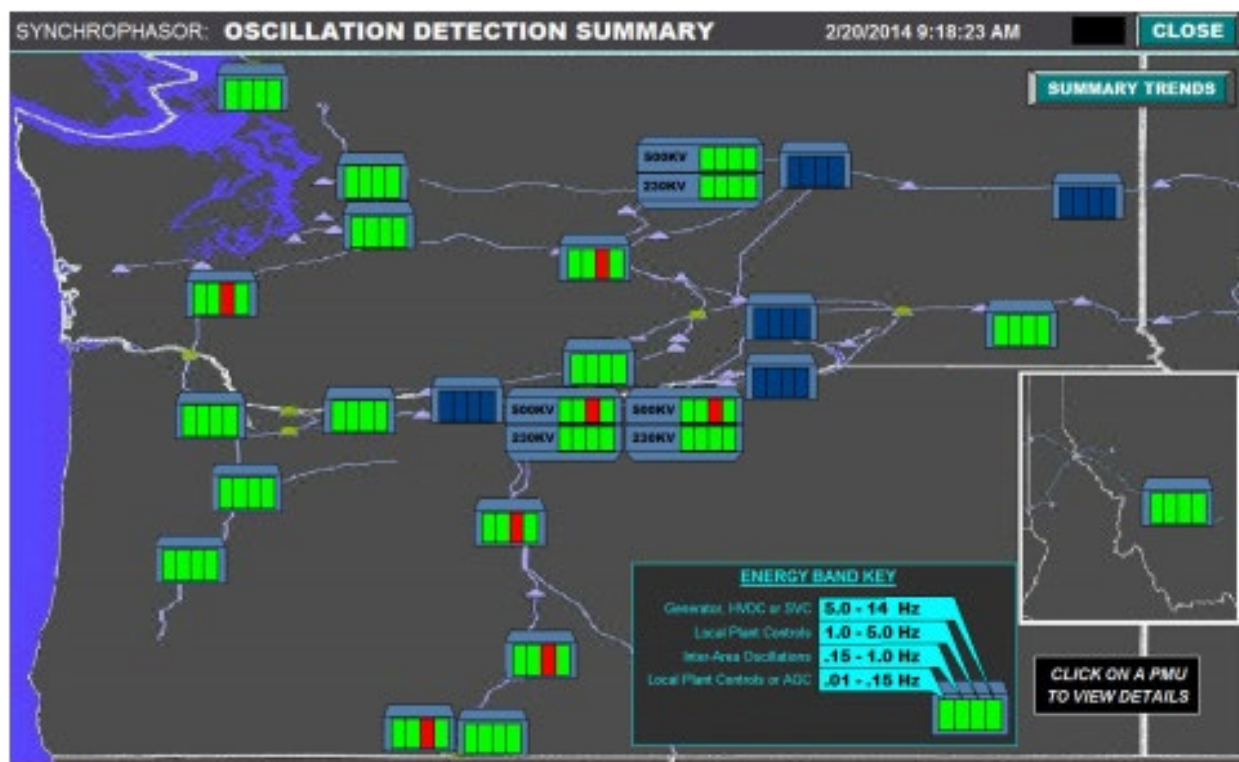


Figure 6.6: Control Room Display Indicating a Wide-Area Oscillation in Band 3

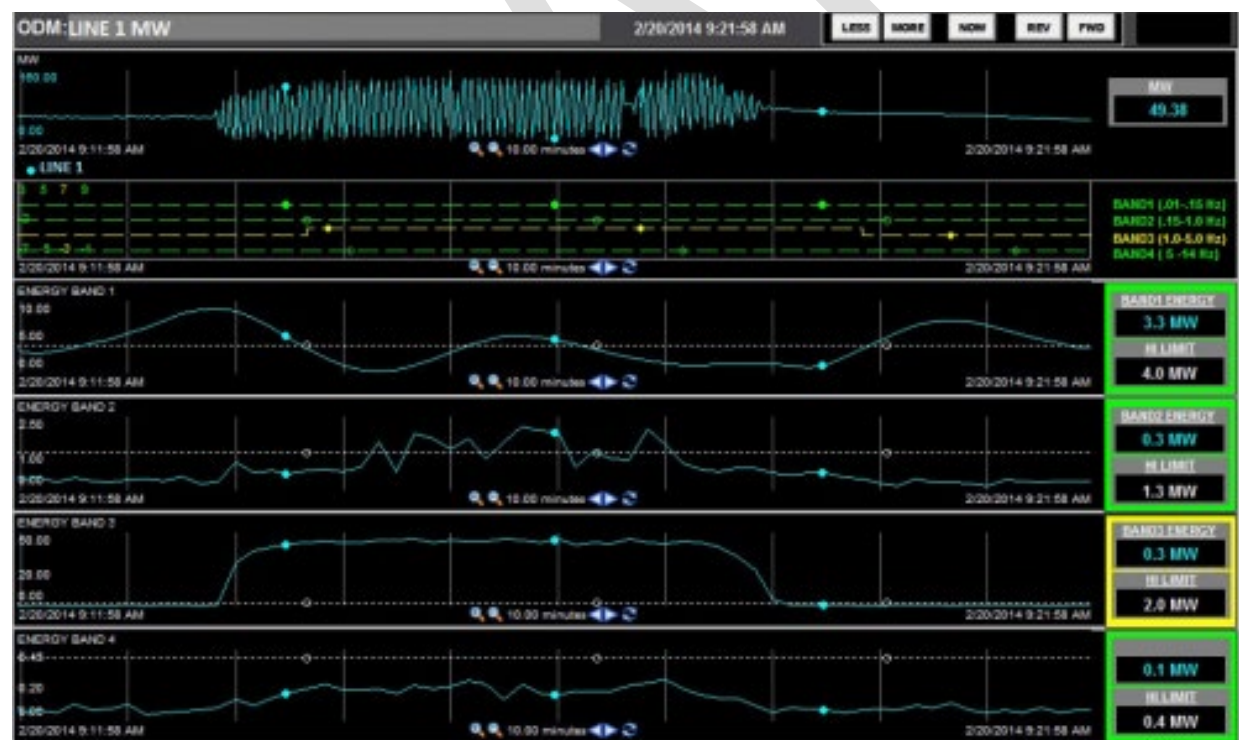
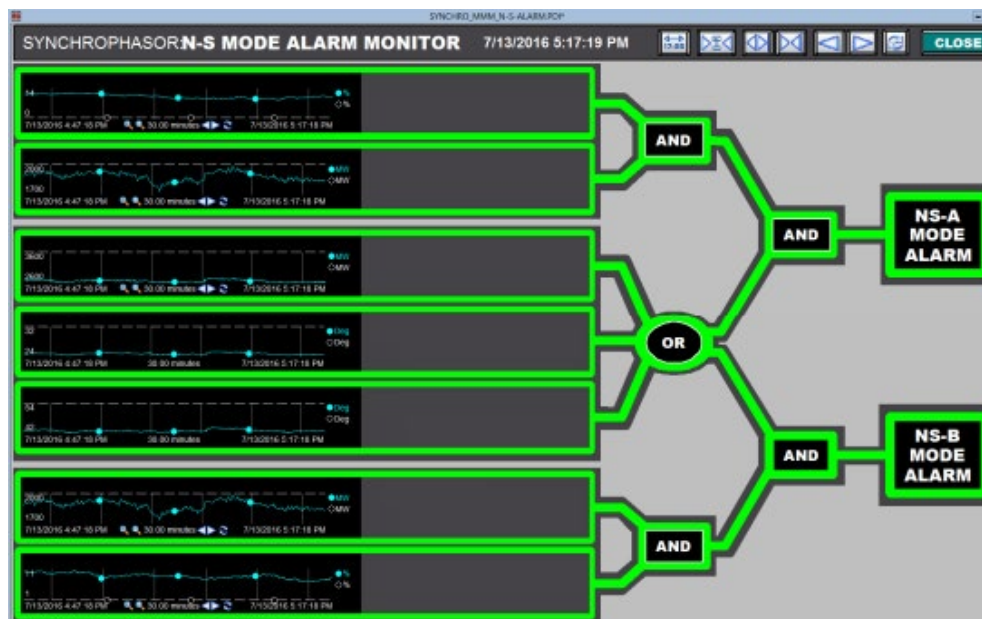


Figure 6.7: Control Room Display Providing Additional Information About a Band 3 Oscillation Alarm

BPA has also deployed a mode meter application to provide continuous tracking of known system modes. By regularly updating estimates of mode damping, the application could potentially be used to provide an early warning of inter-area oscillation problems due to high system stress; alarming and operating procedures are under development. The concept for a composite alarm that accounts for damping ratio estimates, power flows on major interfaces, and phase angle differences is displayed in Figure 6.7. Power flow and angle difference values have been removed from the figure.



**Figure 6.8: Example of a composite Alarm for Monitoring of Modes**

## ISO New England (ISO-NE)

*ISO New England Experience in Locating the Source of Oscillations Online*<sup>53</sup> provides an overview of oscillation applications in use at ISO-NE.

ISO-NE has used the PhasorPoint application by GE for detecting and characterizing oscillations since 2012. PhasorPoint estimates magnitude, frequency, and damping of several dominant modes in the range from 0.05 Hz to 4 Hz. The entire frequency range is split into several adjustable sub-bands, and each sub-band has predefined alarm/alert thresholds based on magnitude and damping of oscillations. PhasorPoint generates alarms and alerts when the detected oscillations exceed the corresponding thresholds for all types of oscillations regardless of their forced or natural classifications. Detection and characterization of oscillations at ISO-NE is considered a well-established and reliable process.

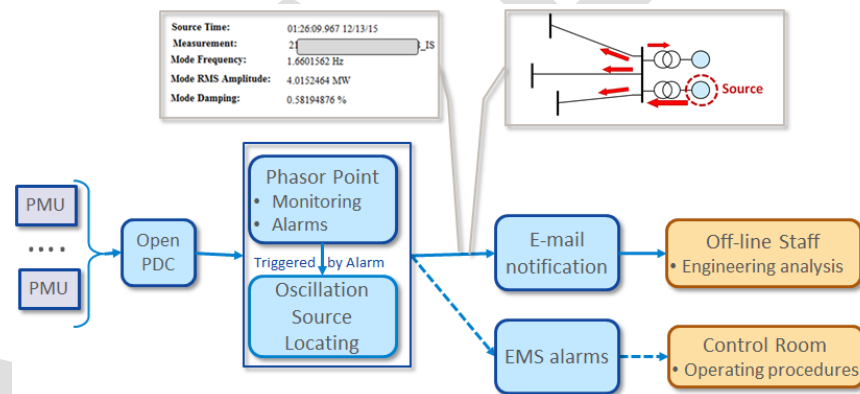
Multiple instances of sustained oscillations with significant magnitude in the frequency range from 0.05 Hz to 2 Hz, and some instances of oscillations up to 8 Hz, have been observed since installing PMUs. Regardless of the forced or natural classification, sustained oscillations with significant magnitudes should be mitigated. The key step for successful mitigation is to identify the source of sustained oscillations. To apply the proper remedial measures, the identification process should answer the question of whether the source is located inside or outside of the control area. If the source is located inside, then the identification should provide the resolution up to the specific power plant or even to a specific unit within a power plant.

<sup>53</sup> S. Maslennikov and E. Litvinov, *ISO New England Experience in Locating the Source of Oscillations Online*, in IEEE Transactions on Power Systems, vol. 36, no. 1, pp. 495–503, Jan. 2021

Several methods were evaluated by ISO-NE, and the energy-based method<sup>54</sup> was selected as the candidate for practical use. The method is based on the primary attribute of oscillations (i.e., energy) and is generally more robust in multiple possible situations while other methods that are based on other attributes of oscillations (e.g., magnitude, phase angles, propagation speed) and statistical signature experience deficiencies. The method has solid theoretical justification; however, the proposed original implementation does not account for actual PMU characteristics and thus is not sufficiently robust for practical use.

ISO-NE developed the DEF method,<sup>55</sup> combining the original energy-based method with PMU data processing. The DEF method has demonstrated efficiency in identifying the source of forced oscillations for a variety of possible situations. It is equally efficient for natural or forced oscillations, resonance conditions of forced oscillations with local or inter-area modes, and multiple sources of oscillation. Using PMU measurements in tie-lines between control areas, the method can identify whether the source is located inside or outside the control area. In testing, the method has correctly identified the source in all situations of the specifically created library of simulated cases to test the source locating methods,<sup>56</sup> and in more than 30 actual events from ISO-NE and in two actual events from WECC.

The use of the DEF method is expected to be a significant milestone in the application of the PMU data for creation of actionable items for the control room information. ISO-NE plans to develop the oscillation source locating (OSL) tool for online use based on the DEF method. OSL tool will be an important part of the oscillation management process (see Figure 6.9).



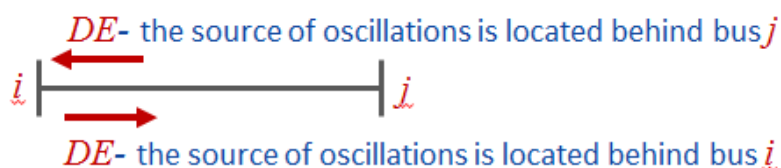
**Figure 6.9: Oscillations Management Process at ISO-NE**

The DEF method uses standard PMU data and calculates the flow of dissipating energy (DE) in any network element monitored by PMU. DE flow can be viewed similarly to regular MW power flow. The sign of DE indicates the direction of where the DE is coming from, and the DE value indicates the amount of energy (Figure 6.10). DE calculated for multiple transmission elements allows tracing the source of an oscillation in network. If the source of an oscillation is a generator monitored by PMU, the DEF methods will identify this generator as the source of the oscillation.

<sup>54</sup> L. Chen, Y. Min, W. Hu, *An energy-based method for location of power system oscillation source*, *IEEE Transaction on Power Systems*, 28(2):828-836, 2013.

<sup>55</sup> S. Maslennikov, B. Wang, E. Litvinov, *Dissipating Energy Flow Method for Locating the Source of Sustained Oscillations*, *International Journal of Electrical Power and Energy Systems*, issue 88, 2017, pp.55–62.

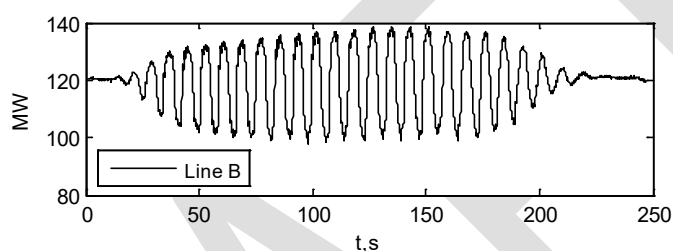
<sup>56</sup> See: <http://current.ute.edu/research/test-cases>.



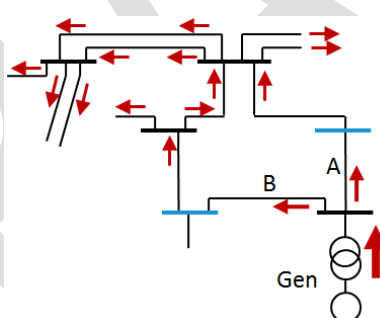
**Figure 6.10: Interpretation of DE<sup>57</sup> for Locating Source of Oscillations**

Below are some examples of using the DEF method for actual events in ISO-NE.

**April 5, 2013, Event:** Rapidly growing in magnitude oscillations of 0.12 Hz were observed in a significant part of the system. The magnitude of oscillations reached up to 100 MW peak to peak in some transmission lines. Oscillations gradually disappeared in 3 minutes. **Figure 6.11 (a)** shows MW flow in one of the lines. The DEF method has clearly identified one power plant as a source of the oscillations, and **Figure 6.11 (b)** illustrates the DE flow in the 345 kV network around that power plant.



**(a) MW flow in line B**

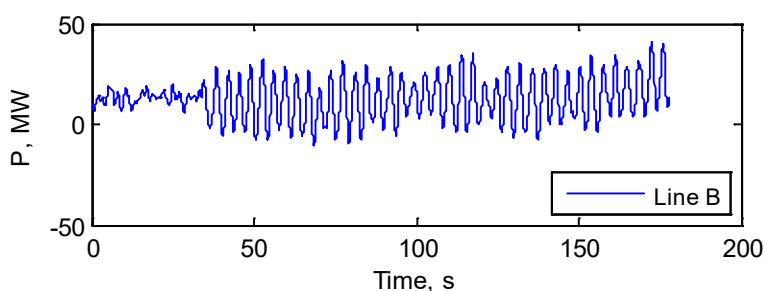


**(b) DEF in 345 kV network. Arrows indicate calculated DE values. Blue color substations do not have PMU measurements.**

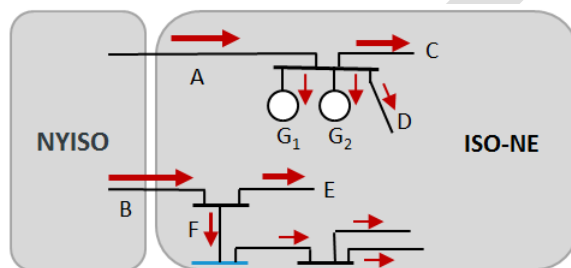
**Figure 6.11: Oscillation Source Location for April 5, 2013, Event**

**June 17, 2016, Event:** Oscillations with magnitude up to RMS = 11 MW and frequency fluctuating from 0.22 Hz to 0.28 Hz were detected in multiple locations of ISO-NE lasting 45 minutes. **Figure 6.12 (a)** shows the active power flow in one of the two 345 kV lines connecting ISO-NE and New York ISO (NYISO) during the initial stage of the event. The mode shape showed that the majority of ISO-NE generators were oscillating in phase, suggesting that ISO-NE PMUs allow the observation of only a part of the system wide event. The DEF method was applied for this event by using available PMU data from ISO-NE. **Figure 6.12 (b)** shows the DE values for the 345 kV part of network connecting ISO-NE and NYISO. DE flow clearly indicates that the source of oscillations was located outside ISO-NE.

<sup>57</sup> DE calculated by PMU measurement at bus i.



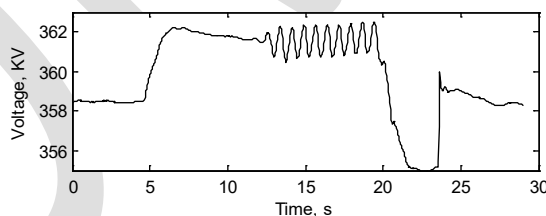
(a) MW flow in 345 kV line B between NYISO and ISO-NE



(b) Red arrows indicate DE in the 345 kV network

**Figure 6.12: Locating Source of Oscillations for June 17, 2016 Event**

**June 30, 2016, Event:** Three generators (G1, G2, and G3) at one of the power plants were operating in the same conditions, producing 237 MW each. A failure suddenly occurred at the excitation system of G3 that resulted in 1.3 Hz oscillations with RMS magnitude of 18 MW and consequential tripping of this unit. Figure 6.13 shows power plant terminal voltage. Excitation failure at G3 started at  $t = 5$  s and the unit was disconnected at  $t = 24$  s. All three units were monitored by PMU, and Table 6.1 shows the results of the DE results. Unit G3 is reliably identified as the source of oscillations producing energy while other two units were contributing into damping by dissipating energy. Note that the G3 has the highest DE flow in the network. This example illustrates the ability of the DEF method to identify a specific unit as the source from multiple units within a power plant.



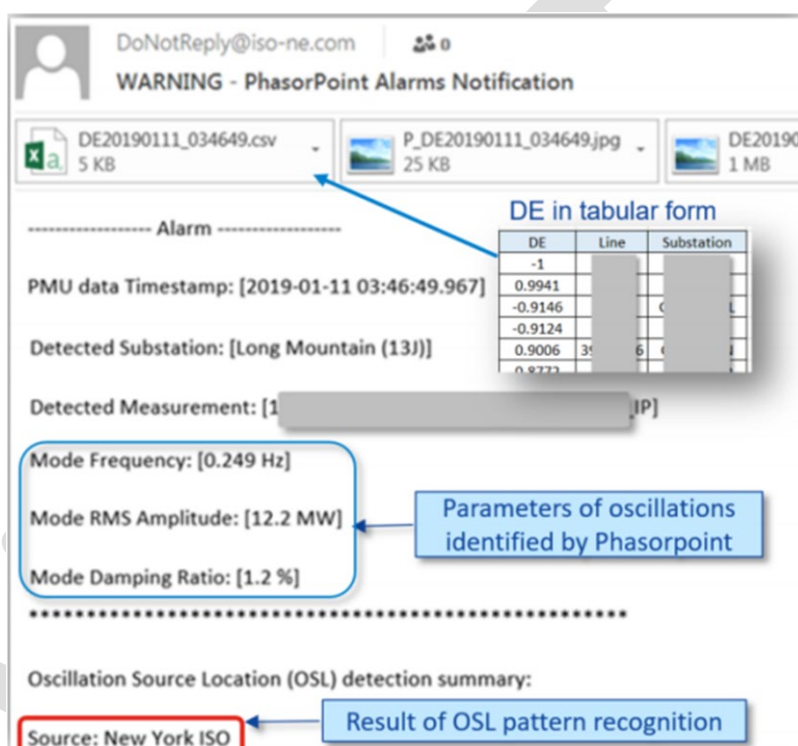
**Figure 6.13: Power Plant Terminal Voltage During June 30, 2016, Events**

**Table 6.1: DE of Generators for June 30, 2016 Event**

| Generator | G1    | G2    | G3  |
|-----------|-------|-------|-----|
| DE        | -0.01 | -0.01 | 1.0 |

ISO-NE utilizes the online OSL application to process events. The OSL application is based on the DEF method and has processed over 1,000 oscillatory events. As mentioned earlier, the equipment generating the oscillation is identified by tracing the flow of energy back to its source. Tie-lines between ISO-NE and neighboring system operators are also monitored so that ISO-NE can determine if an oscillation is coming from outside of their territory. The OSL's pattern recognition module converts the DEF in the network into a text message identifying a specific generator, power plant, or area as the containing the source of oscillation. The pattern recognition compares the energy flow with a set of pre-defined topology-based energy flow templates for all potential oscillation sources that could be uniquely identified based on system observability with PMU measurements.

ISO-NE developed an alarm notification service to deliver the PhasorPoint and OSL results by email and text messaging. An example email is displayed in Figure 6.14.



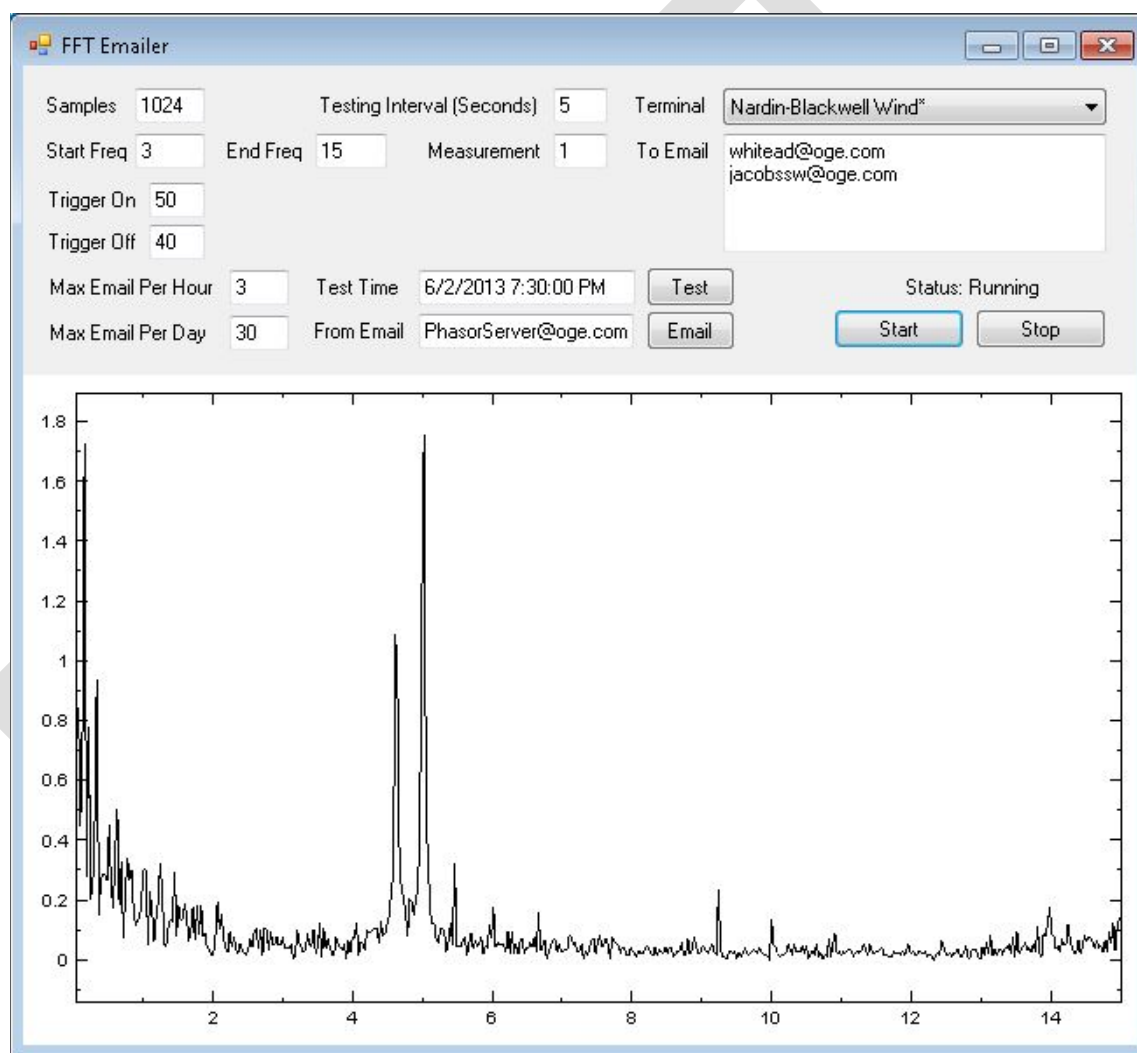
**Figure 6.14: Example Email from the Alarm Notification Service Detailing the Characteristics and Source of an Oscillation**

Emails from the alarm notification service are automatically generated and sent to ISO-NE control room staff and operation support engineers. If a specific power plant or generator is identified as the source, an email is also sent to that generator's personnel and the local control center overseeing the generator. If the oscillation has a large magnitude, ISO-NE operational staff communicates with the operator of the source generator to apply remedial actions. Potential mitigating measures include disconnecting the source generator from the network, adjusting the generator's MW output, or changing its control mode. If the oscillation is small, the mitigation process is shifted offline. Mitigation is again based on communication between ISO-NE personnel and the source generator's operator/owner. There is no formal distinction between a "large" and "small" oscillation at ISO-NE, but any persistent oscillation with a magnitude above the power system's ambient noise is investigated.

## Oklahoma Gas & Electric (OG&E)

OG&E, along with many other utilities in the U.S. Great Plains region has a large wind generation resource potential. Many large-scale wind farm facilities, varying in size from 100 MW to 300 MW, have been brought on-line with total wind capacity now exceeding 5,300 MW on the OG&E system. Each new wind farm facility brought on-line in OG&E's service territory is accompanied by PMU measurements at the point of interconnection. In December 2010, the utility began observing sub-synchronous oscillations on the transmission system in a concentrated portion of the grid in northwestern Oklahoma as shown in [Event 19: Woodward Wind 13 Hz Oscillation](#).

This phenomenon could not be observed with traditional SCADA monitoring and would have taken much longer to identify and resolve without synchrophasor technology. After observing these instances of voltage oscillations, the company implemented a fast Fourier transform (FFT) detection program to detect the oscillations and send email notifications when it requires corrective action. (See [Figure 6.15](#))

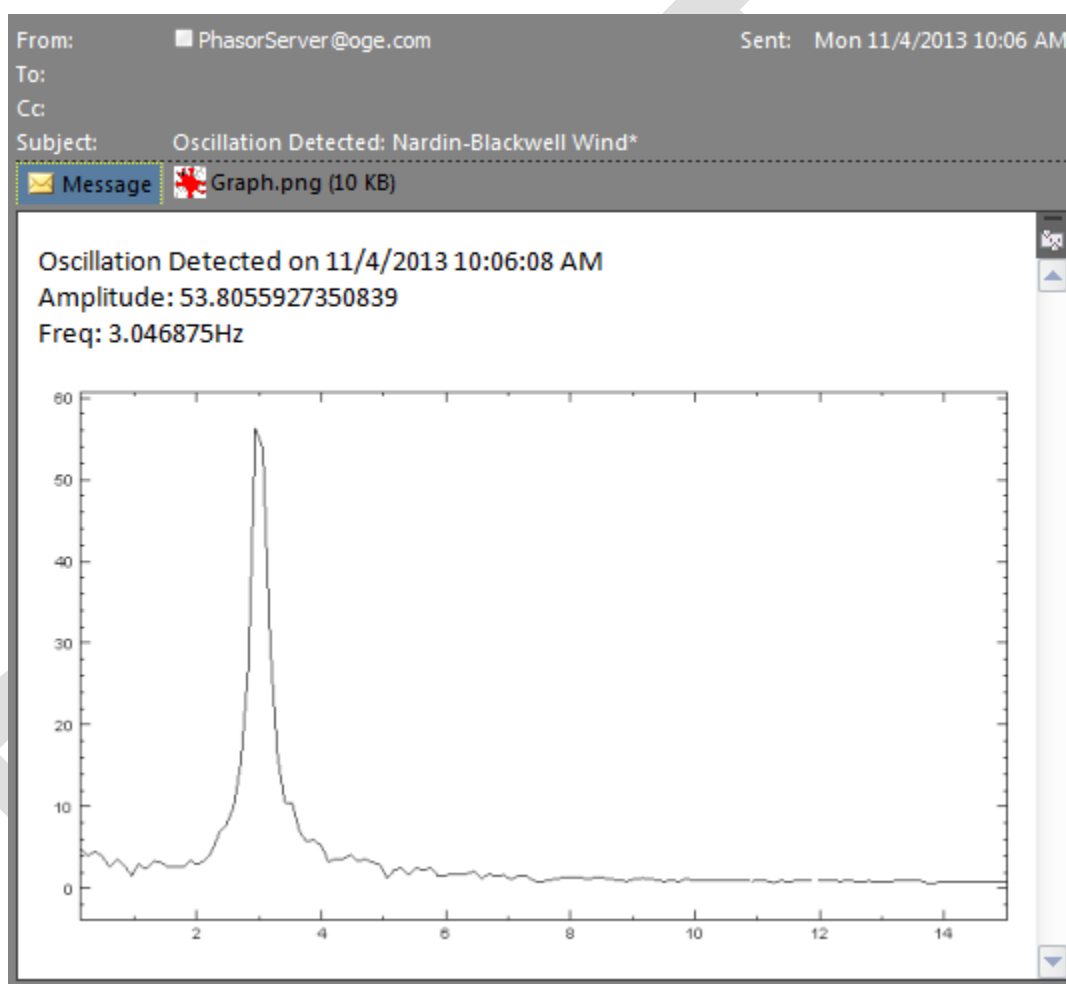


**Figure 6.15: FFT Emitter Application  
(Source: OG&E)**

The application was developed in VB.net and utilizes the OSS exocortex library for FFT oscillation detection (<http://www.exocortex.org/dsp>). To validate that the program was correctly calculating the FFT, the output was

compared with MathCad's FFT algorithm,<sup>58</sup> which proved to be a match. To provide some additional details on how the algorithm is used, the input parameters to the program will be discussed. The user can set the number of samples to be queried (in this case, 1,024) and how often the query is repeated. The user then specifies the PMU to take measurements from and the frequency range of interest. The trigger on/off settings allow the user to specify how sensitive the detection should be. A set of email recipients are specified with some reasonable limits to be sure that the users are not flooded with nuisance emails. The user can also test the parameters against a historical data sample to tune the parameters and make sure they are detecting only events of interest.

The output of the program comes via email (see Figure 6.16) to operations support engineers, who will investigate and determine if action should be taken. If it is determined that mitigation is necessary, they will contact the transmission control center and recommend a mitigation strategy to the operators.



**Figure 6.16: Email Notification Example  
(Source: OG&E)**

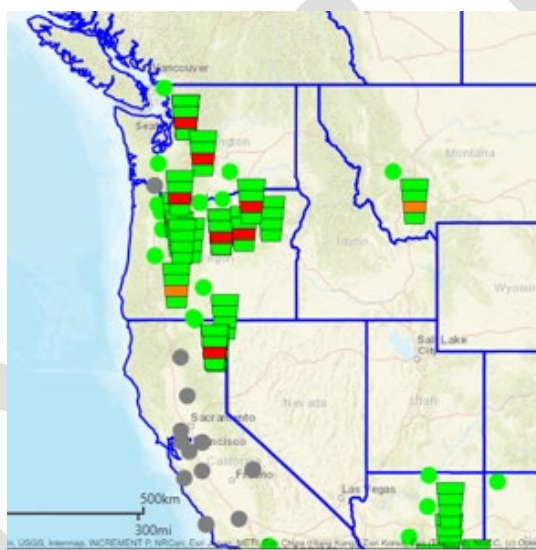
<sup>58</sup> [www.ptc.com/product/mathcad](http://www.ptc.com/product/mathcad)

## RC West

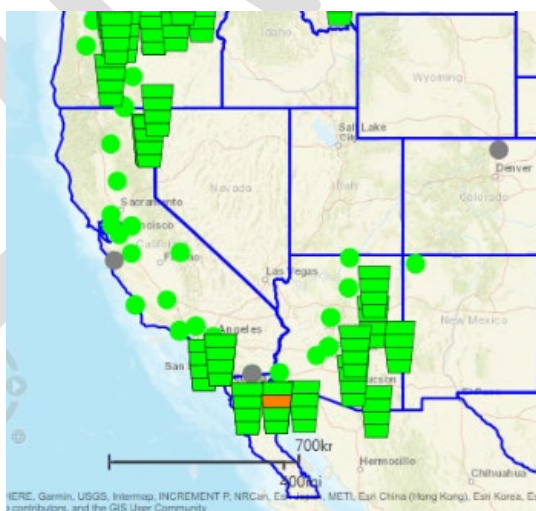
RC West utilizes the Electric Power Group Real Time Dynamic Monitoring System (RTDMS) for monitoring of forced and natural oscillations. The monitoring of oscillations is supplemented by an RC operating guideline that provides guidelines to the operators for the following three scenarios:

- Forced oscillations
- Natural oscillations
- Forced oscillations causing inter-area oscillations

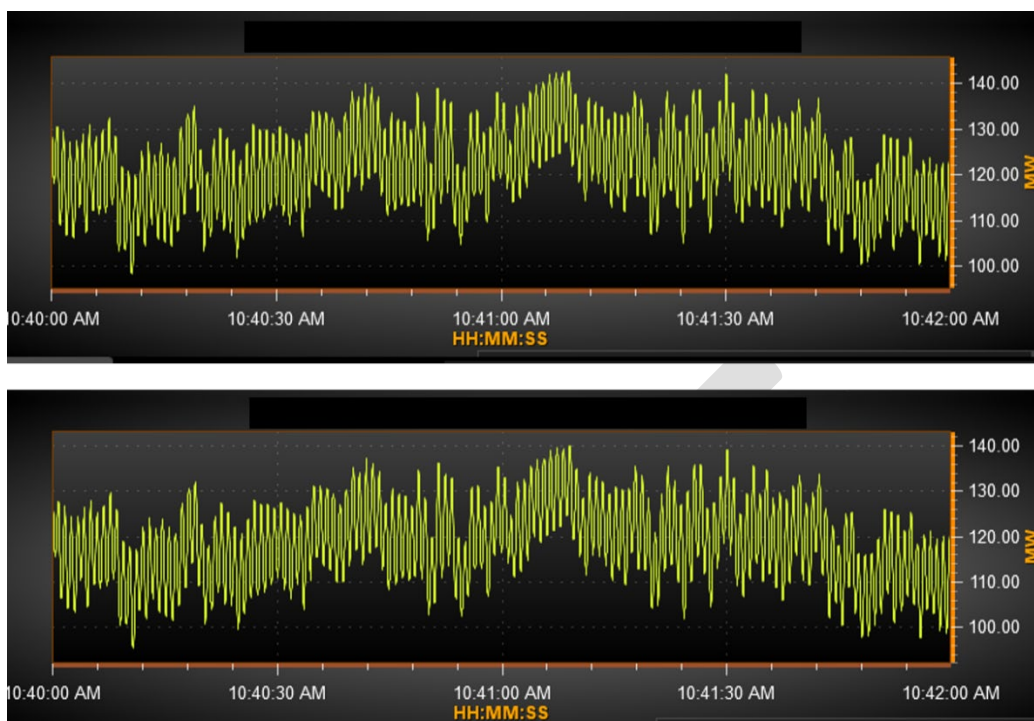
The forced oscillation monitoring is accomplished using the ODM that monitors oscillations in the four energy bands at various locations across the RC West footprint. [Figure 6.17](#) and [Figure 6.18](#) show examples of forced oscillations observed over a wide area and a local area. [Figure 6.19](#) and [Figure 6.20](#) show examples of forced oscillations observed flows and bus voltages.



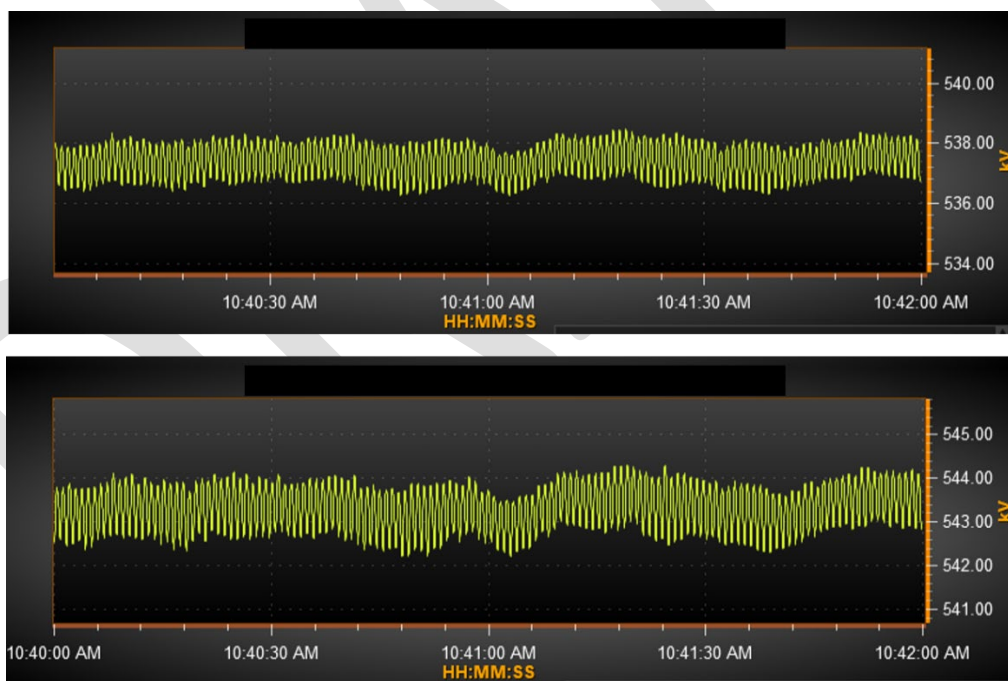
**Figure 6.17: Wide-Area Forced Oscillations**



**Figure 6.18: Local Forced Oscillations**

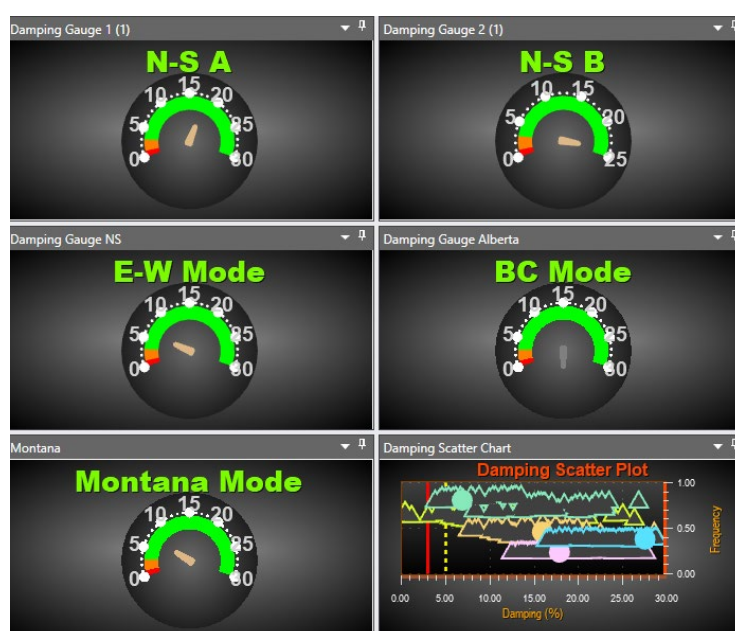


**Figure 6.19: Forced Oscillations Observed on Line Flows**



**Figure 6.20: Forced Oscillations Observed on Voltages in RC West**

In addition to monitoring forced oscillations, the RTDMS setup at RC West monitors the five significant modes in the WI as shown in [Figure 6.21](#): Operators have been provided guidelines on path flows, or generators, relevant to each of these modes that can be utilized to increase damping in the event there is low sustained damping on any of these monitored modes.



**Figure 6.21: Mode Meter Monitoring**

The mitigation actions for increasing damping of the modes are validated by performing SSAT analysis with real-time state estimation cases that are used along with the dynamic data applicable for the season. RC West has a running real-time transient stability analysis setup that provides the framework to perform small-signal stability analysis with the same input data from state estimation and dynamic data that is used for the transient stability analysis. The SSAT analysis, as described in [Using Eigenvalue Analysis to Determine Mitigation Actions](#), allows operators to determine and validate relevant path flows or status of equipment and generators that can be adjusted to mitigate observed sustained low damping on any of the monitored modes.

## Southwest Power Pool (SPP)

SPP uses PMU technology and Electric Power Group RTDMS for real-time non-decision-making oscillation monitoring to ensure wide-area situational awareness of Interconnection electrical signal dynamics. SPP utilizes the ODM and mode meter engines within RTDMS to extract grid dynamics information for the purposes of oscillation monitoring.

ODM is used to agnostically monitor for manifested oscillations via the use of a sliding root mean square (RMS) window. An increase in signal dynamics directly translates to an increase in the RMS value. Therefore, if an oscillation occurs on a signal measurement, ODM detects this by comparing the real-time RMS value to a pre-set RMS threshold value. If the current RMS value stays above the pre-set threshold for a minimum amount of time (e.g., 60 or 120 seconds), operators are alerted of the possible oscillation event. The pre-set thresholds are set so that they represent an increase in signal dynamics above typical ambient conditions. This information can be extracted by performing statistical analysis on raw and post-processed signals and making a decision that suffices the needs of the end user. Several methods can be used (e.g., 3sigma above mean or more slightly advanced items like Dual-Dirac fitting). If there is an oscillation present and ODM detects it, internal SPP software will alert operators who then use RTDMS displays to assess the situation, such as oscillation location, signal type, oscillation frequency, and oscillation magnitude. From there, RCs will contact entities or other RCs to communicate the detection of the event. [Figure 6.22](#) and [Figure 6.23](#) show an example of a real-time WI forced oscillation from 2019 that showed up in SPP's RTDMS system.

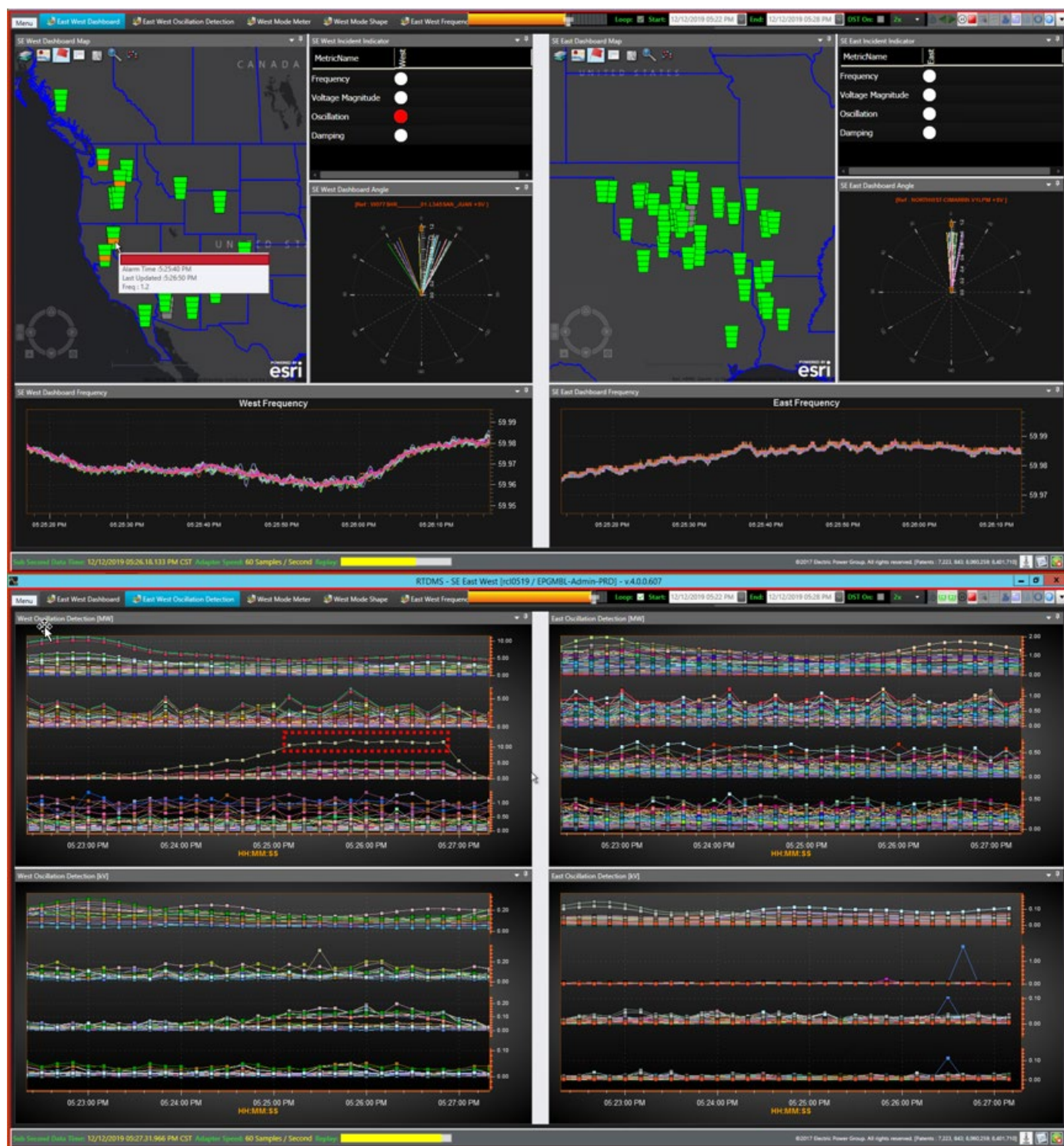
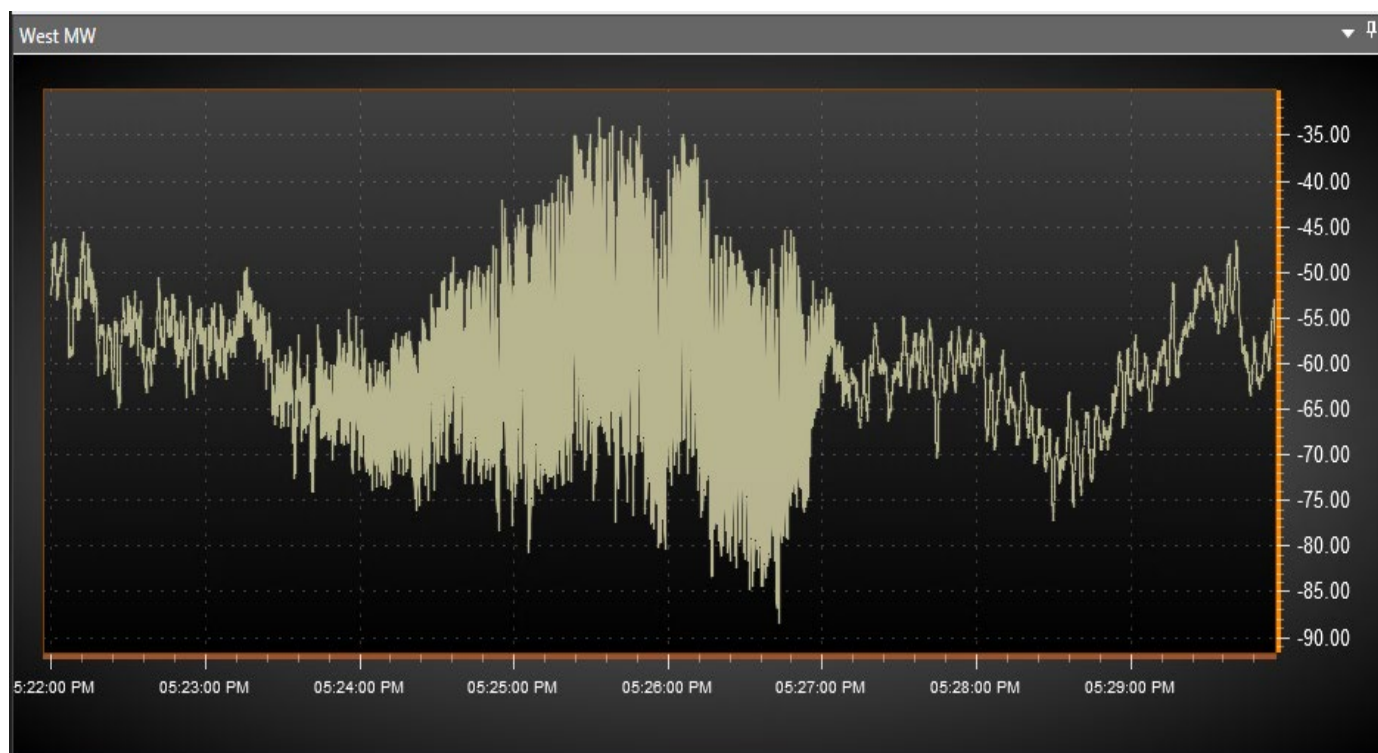


Figure 6.22: Forced Oscillation Observation in SPP



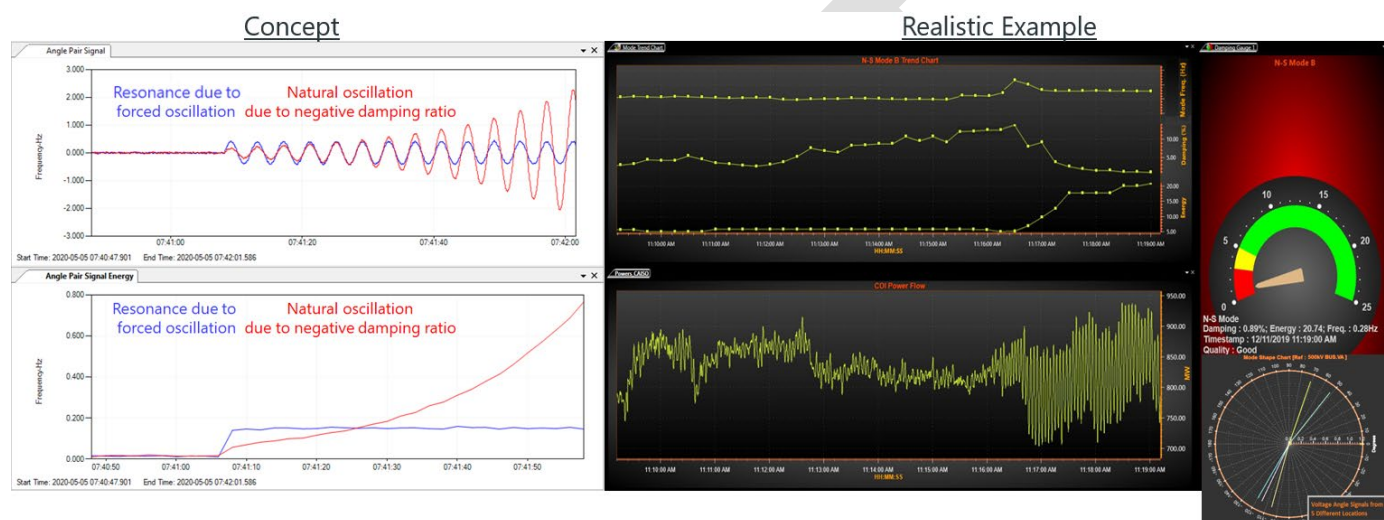
**Figure 6.23: Forced Oscillation Observed on Line-Flows in SPP**

On the other side of the spectrum, a mode meter is used to assess the current stability of known inter-area modes. The main outputs are the estimated damping ratio of the mode and the energy level of the composite signal used as an input to mode meter (SPP currently uses single angle pair composite signals). To assess the stability of the mode, only damping ratios are estimated and monitored in real time. If the damping ratio of a mode drops below 3%, internal SPP software notifies the operator(s) of a potentially weakened mode. Figure 6.24 shows an example of SPP's west mode meter display, showing the real-time states of the five main WI modes.



**Figure 6.24: Mode Monitoring in SPP**

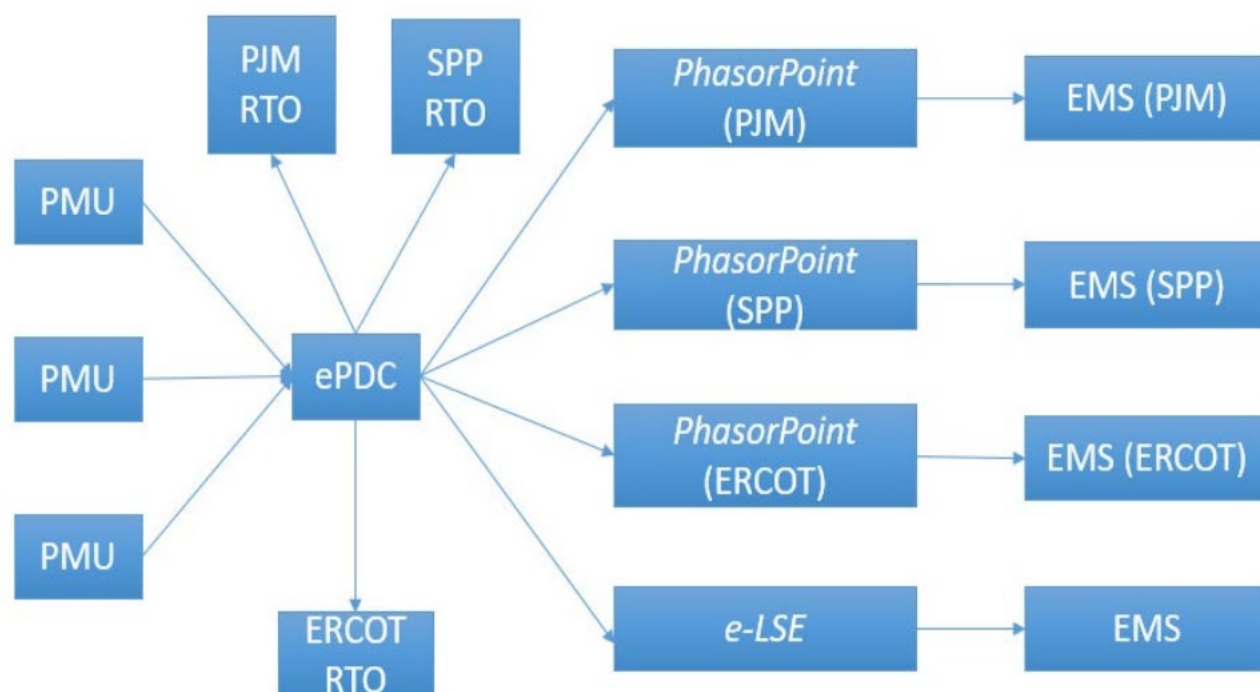
SPP also uses the energy values in addition to damping ratio as a quasi-ODM for detecting system oscillations pertaining to a particular system mode. This can only be done with very careful signal choices for mode meter inputs. The basic premise is that current mode meters (not just in RTDMS) typically do not work well when a forced oscillation is happening and, in that event, will generally have damping ratio estimates drop dramatically and artificially to low levels (e.g., 0% to 3%). While there are currently fixes in place, this artifact can be exploited advantageously. By coupling a low damping ratio percent with an energy threshold (like ODM, above typical ambient conditions), a mode meter can help operators and shift engineers assess whether there is an oscillation happening that pertains to a particular system mode whether that be an unstable mode causing growing oscillations or a case where a natural mode resonance happens. In both cases, estimated damping ratios drop and energy levels rise. **Figure 6.25** is an example conveying the concept as well as a real-use case. In addition, mode shapes are also plotted as another indicator to operators that a power system oscillation pertaining to a particular system mode is in effect.



**Figure 6.25: Mode Monitoring**

## American Electric Power (AEP)

Since early 2018, AEP has used PMU-based applications for online oscillation analysis and offline event studies. Installed and maintained by AEP's protection and control team, field PMU units will stream high sampling data to an enhanced Phasor Data Concentrator (ePDC), which is a data repository. An ePDC will then dispatch PMU data streams to PhasorPoint in which real-time monitoring is enabled, and alarms are generated and transmitted to the energy management system (EMS) side. Figure 6.26 illustrates the configuration of AEP's PMU system.



**Figure 6.26: AEP's PMU Configuration**

AEP has deployed close to 400 PMUs across its three footprints governed by PJM, SPP, and ERCOT. With the large amount of data accumulated, real-time oscillation event detection on frequency and active power was developed and deployed in the control room to enhance situational awareness.

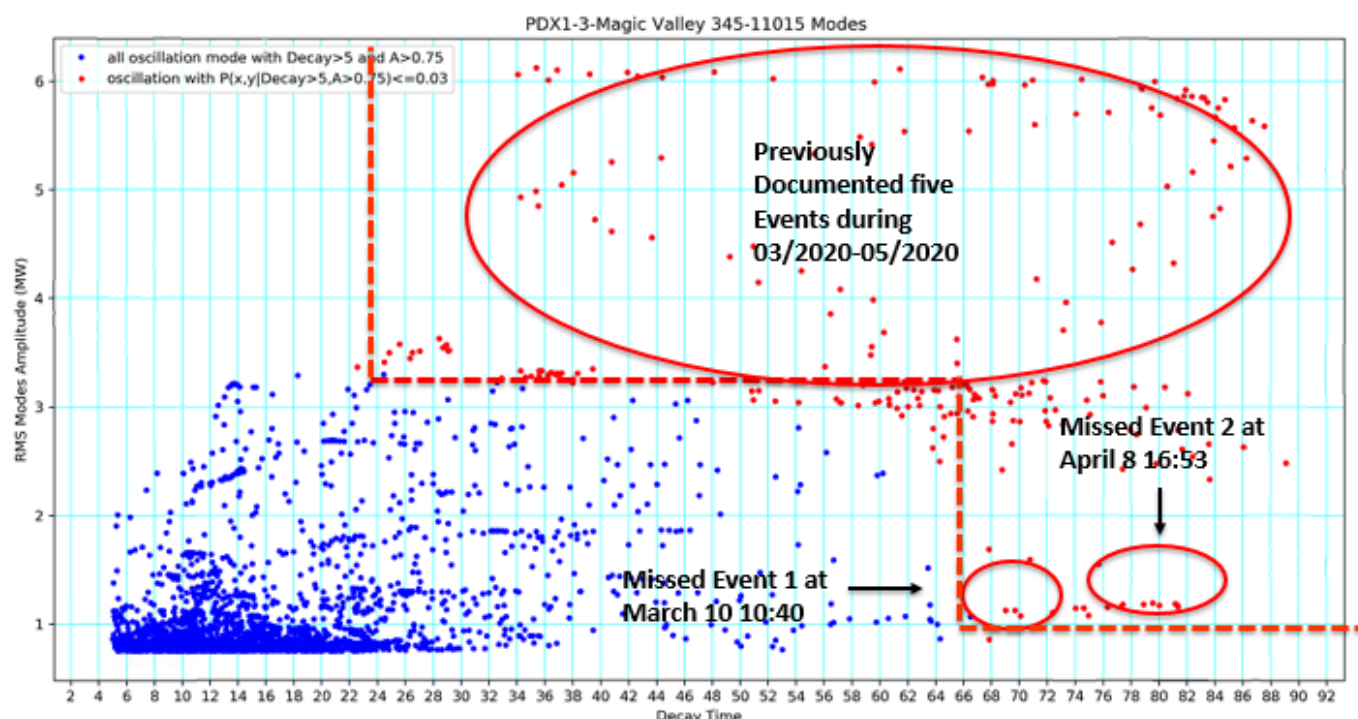
### *AEP Real-Time Oscillation Monitoring and Event Detection*

Taking advantage of the high sampling rate of phasor measurement units, system dynamics are visualized and documented in PhasorPoint. The purpose of developing this event detection mechanism is to help control room personnel quickly identify harmful oscillations from common system variations. With online modal decomposition in PhasorPoint, system dynamics were decomposed in real time with oscillation magnitude and decay time as two critical metrics to measure the severity of the oscillation. The bigger the magnitude and longer the decay time, the more severe the oscillation will be. While it is preferred to avoid alarms on the small and quickly damped oscillations, the lingering ones with big swings are supposed to be caught by operators as quickly as possible.

With this purpose in mind, a kernel density estimation (KDE)-based methodology<sup>59</sup> was developed to detect oscillation events. In this methodology, a cross-validated KDE was adopted to regress historical oscillation data. As a result, a bi-variable probability density function was derived to summarize the distribution of documented oscillation data. Knowing that severe oscillatory disturbances are statistically rare, a cut-off probability initialized at three

<sup>59</sup> Y. Kong, Y. Lu and F. Tu, *Implementation of Synchrophasor Monitoring and Event Analysis at American Electric Power*, 2022 IEEE/PES Transmission and Distribution Conference and Exposition (T&D), New Orleans, LA, 2022, pp. 1–5, doi: 10.1109/TD43745.2022.9816977

standard deviations above the mean was used to identify historical observations of oscillation events. With heuristic<sup>60</sup> training based on a historical event list, this cut-off probability would be finalized, and observations of past oscillation events were picked and located on Locus plot like shown in Figure 6.27.



**Figure 6.27: Locus Demonstration of Observations Representing an Oscillation Event**

With all event observation labeled in red in Figure 6.27, the alarm settings concerning amplitude and decay time were configured. For the particular PMU signal demonstrated in Figure 6.27, its finalized alarm configuration is listed in Table 6.2.

**Table 6.2: Alarm Setting for PMU @Magic Valley**

|               | Decay Time Exclusion | Decay Time Threshold | Amplitude Exclusion | Amplitude Threshold |
|---------------|----------------------|----------------------|---------------------|---------------------|
| Alarm Setting | 24 seconds           | 66 seconds           | 1 MW                | 3.25 MW             |

The alarm configurations obtained from KDE-based methodology were more reliable than the previous intuitive configurations. Historical event studies proved the method's enhanced sensitivity as the method could detect previously missed oscillation events. In addition, over one year of control room deployment has verified long-lasting reliability of the methodology as the rate of false alarms is significantly reduced. Table 6.3 is a performance overview on the KDE-based online event detection.

**Table 6.3: Performance Overview of KDE-Based event detection**

| Footprint | Production Deployment | False Alarm Count | False Alarm Rate (after) | False Alarm Rate (before) | Footprint |
|-----------|-----------------------|-------------------|--------------------------|---------------------------|-----------|
|-----------|-----------------------|-------------------|--------------------------|---------------------------|-----------|

<sup>60</sup> In [mathematical optimization](#) and [computer science](#), **heuristic** (from Greek εὕρισκω "I find, discover") is a technique designed for [solving a problem](#) more quickly when classic methods are too slow, or for finding an approximate solution when classic methods fail to find any exact solution. This is achieved by trading optimality, completeness, [accuracy](#), or [precision](#) for speed. In a way, it can be considered a shortcut.

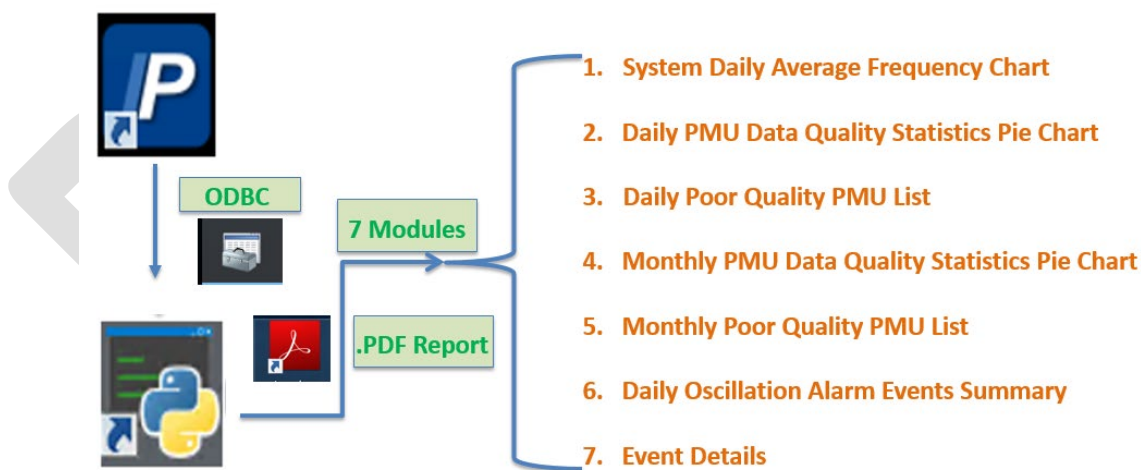
|       |         |     |              |         |       |
|-------|---------|-----|--------------|---------|-------|
| ERCOT | 07/2020 | <30 | Around 6%    | 50+%    | ERCOT |
| SPP   | 09/2020 | <10 | Less than 5% | 45%–50% | SPP   |
| PJM   | 10/2020 | <20 | Less than 5% | 50+%    | PJM   |

With the KDE-based event detection in place, all the oscillation events captured in PhasorPoint will be streamed in real time to control room and documented in daily PMU reports for offline studies.

### ***AEP Auto Daily PMU Report Implementation***

To enhance the situational awareness of AEP's system for control rooms, the daily PMU report is automatically generated as a pdf file and is archived in an internal shared folder. So far, this report includes information of the system average frequency and PMU data quality statistics and lists the poor-quality PMUs that need maintenance. In addition, the reports summarize the oscillation events and have the charts attached of event details, such as related PMU measurement waveform charts and oscillation mode information charts. More information and charts are planned to be included in future daily reports.

As shown in [Figure 6.28](#), the daily reports are created by retrieving data through Phasorpoint SQL database via the Open Database Connectivity (ODBC) connector. Then Python scripts have been written to access the SQL data to make necessary plots and tables, and a pdf file report containing all the information needed will be automatically generated by the script. The purpose of daily reports is to help operators and engineers have better situational awareness of AEP's system operation. Therefore, three daily reports for each of AEP's footprints (ERCOT, SPP, and PJM) are generated daily. So far, there are seven modules in the daily report that each cover the system frequency chart, the daily/monthly PMU data quality statistics pie chart, and the daily/monthly poor PMU quality list as well as the oscillation event summary list and detailed charts. Some examples of the charts in the AEP Daily PMU report are provided in [Figure 6.29](#), [Figure 6.30](#), [Figure 6.31](#), and [Figure 6.32](#).



**Figure 6.28: AEP Auto Daily PMU Report Deployment**

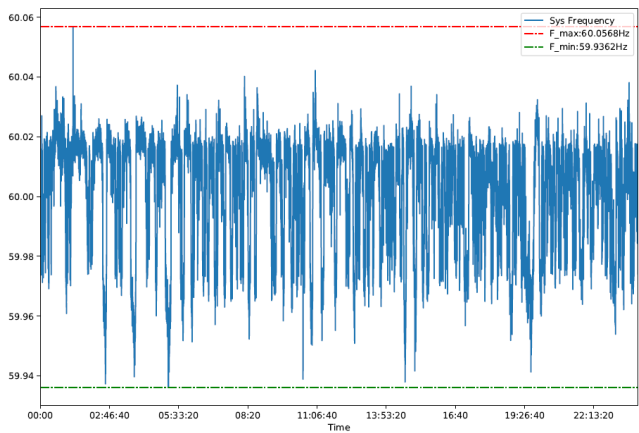


Figure 6.29: System Daily Average Frequency Chart

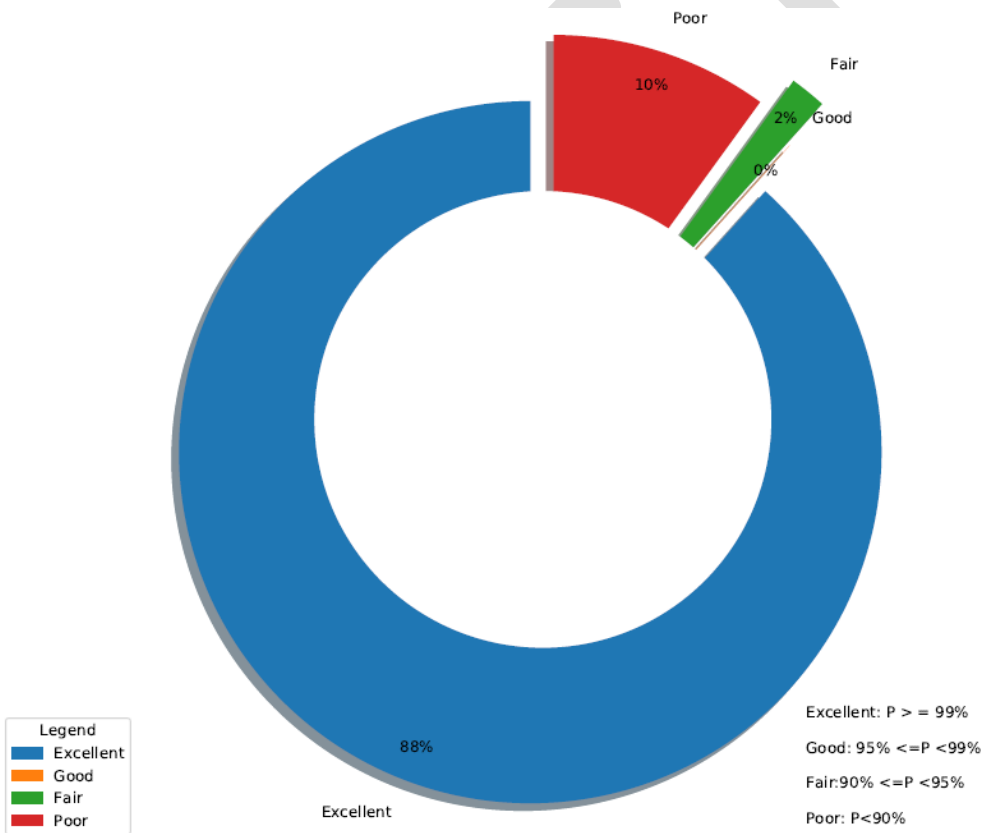


Figure 6.30: Daily PMU Data Quality Statistics

| Index | Date       | Time     | measurement_group | measurement    | parameter | message                   |
|-------|------------|----------|-------------------|----------------|-----------|---------------------------|
| 1     | 2021-04-29 | 10:56:01 | RH                | 12345 (ABC@RH) | P         | PDX1-3 event status alarm |

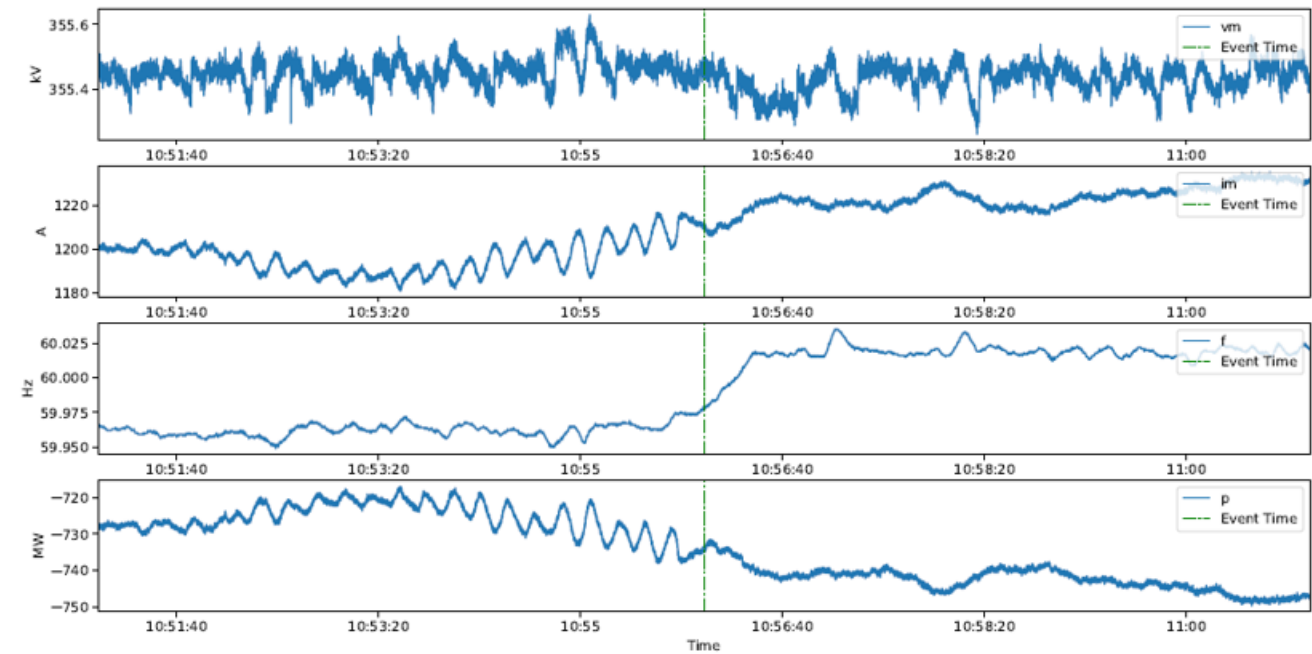


Figure 6.31: Event 1 PMU Measurements

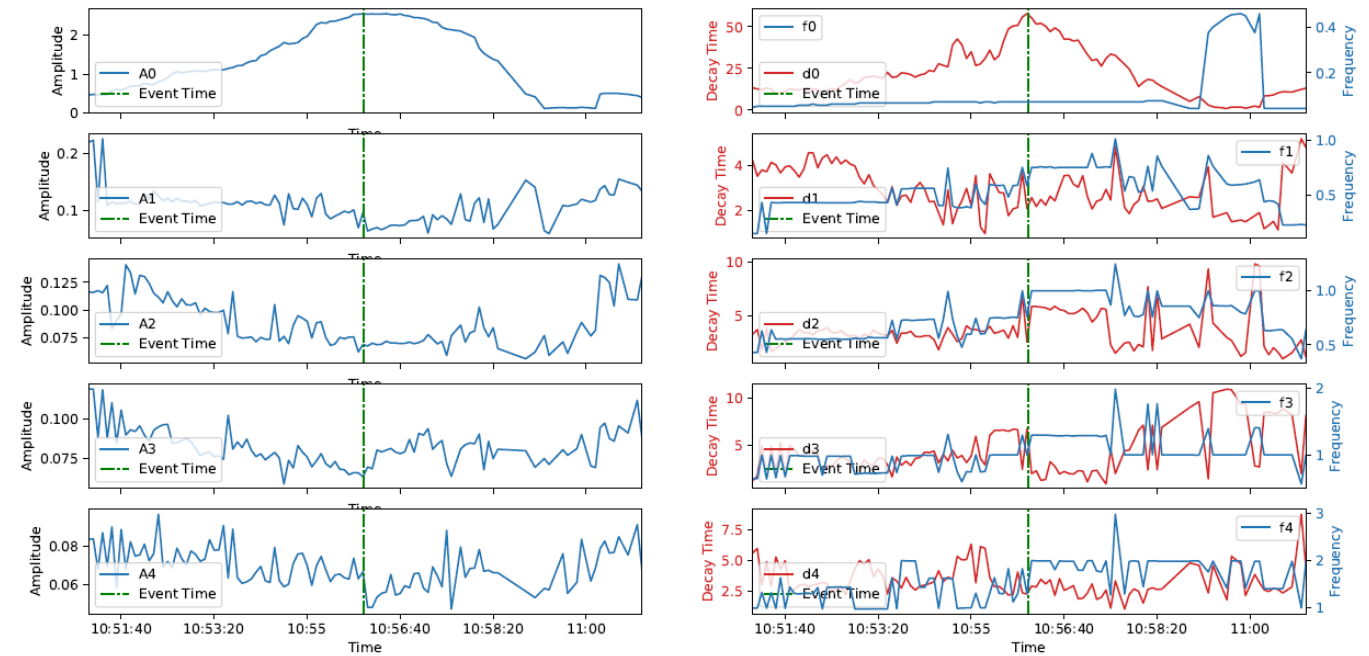


Figure 6.32: Event 1 Oscillation Mode

## Appendix A: Determining the Impact of Forced Oscillations

This appendix uses an example to detail how one can determine the impact of forced oscillations at frequencies closer to the inter-area modes that need to be monitored.

During the January 11, 2019, forced oscillation event, several EI GOs and system operators set their plant AVR to manual control and ramped down online pumped storage plants upon identifying the undamped oscillation. Following the event, there were questions on whether those were the appropriate actions and whether there were any other actions the operators could have taken to mitigate the event. This example describes how a wide-area resonant forced oscillation can be re-created in dynamic simulation and how possible mitigations can be tested. For simplicity, example simulations shown below were performed on a subset of the EI model that does not include a large enough area to capture the modes involved in the January 11, 2019, event.

### Modeling of Forced Oscillations

The source of the oscillation was a steam turbine in Florida that experienced 200 MW peak-to-peak oscillations due to controller failure causing the intercept valves to open and close every 4 seconds. As the intercept valves cyclically open and close, they increase and decrease the flow of steam through the turbine, effectively causing the mechanical power input to the generator to oscillate between full output and zero. To represent this controller failure in dynamic simulation, a user model is needed to oscillate a generator's mechanical power at a defined amplitude and frequency. The FORTRAN code for a Power System Simulation for Engineering (PSSE) user model, named "GOV\_OSCILLATE," accomplishes this as shown in [Figure A.1](#).

Without delving into the details of user model writing in PSSE, the inputs to this model are as follows:

- The machine number  $I$
- The amplitude and frequency of the mechanical power oscillation  $CON(J)$  and  $CON(J+1)$
- The initial mechanical power  $STATE(K)$
- No used variables (VARS) or integer constants (ICONS)

Each PSSE dynamic model performs various computations at different stages of the dynamic simulation:

- In Mode 1, dynamic models are initialized. The initial mechanical power is saved in the state variable,  $STATE(K)$ .
- In Mode 3, governor type models must compute the current value of mechanical power and populate the mechanical power (PMECH) arrays. As shown in line 42 of [Figure A.1](#),  $PMECH(I)$  is only modified if the simulation time is greater than 1 second. After 1 second,  $PMECH(I)$  is by the equation below:

$$P_{Mech} = P_{Mech\_Initial} + \frac{Amplitude}{MVA_{base}} * \sin [2\pi f(t - 1)]$$

```

1 SUBROUTINE GOV_OSCILLATE(I,ISLOT)
2 C Logic to add a 10 MW oscillation on Pmech at time >1s
3 C
4 INCLUDE 'COMON4.INS'
5 C
6 INTEGER I, ISLOT
7 C = MACHINE ARRAY INDEX
8 C ISLOT = ARRAY ALLOCATION TABLE INDEX
9 C J = STRIIN(1, ISLOT) [ USES CON(J) AND CON(J+1) ]
10 C K = STRIIN(2, ISLOT) [ USES STATE(K) ]
11 C
12 INTRINSIC SIN
13 INTEGER J, K
14 REAL PMECH_INITIAL
15 C
16 IF (MODE.EQ.8)
17 . CON_DSCRPT(1) = 'Amplitude'
18 . CON_DSCRPT(2) = 'Frequency'
19 . RETURN
20 ...FIN
21 C
22 C GET STARTING 'CON' INDICES
23 C
24 J=STRIIN(1,ISLOT)
25 K=STRIIN(2,ISLOT)
26 C
27 IF (MODE .GT. 4) GO TO 1000
28 C
29 GO TO (100,200,300,400), MODE
30 C
31 C MODE = 1: INITIALIZE
32 C
33 100 STATE(K) = PMECH(I)
34 RETURN
35 C
36 C MODE = 2: CALCULATE DERIVATIVES
37 C
38 200 RETURN
39 C
40 C MODE = 3: SET PMECH
41 C
42 300 IF (TIME .GT. 1) THEN
43 PMECH(I) = STATE(K) + CON(J)*SIN(6.2832*CON(J+1)*(TIME-1))/MBASE(I)
44 END IF
45 RETURN
46 C
47 C MODE = 4: SET NINTEG
48 C
49 400 NINTEG=MAX(NINTEG,K+1)
50 RETURN
51 C
52 1000 RETURN
53 END

```

**Figure A.1: PSSE USER Model to Create Forced Oscillation**

After compiling the FORTRAN code in [Figure A.1](#) and creating a \*.dll file the user model can be utilized using the statement in [Figure A.2](#). The highlighted variables indicate that the model should be applied to machine 1 at bus 4, the amplitude of oscillation is 100MW (200 MW peak-to-peak), and the frequency of oscillation is 0.67hz. The other parameters are required for PSSE to classify it as a turbine-governor model and reserve the necessary space in the ICONS, CONS, STATES, and VARS arrays.

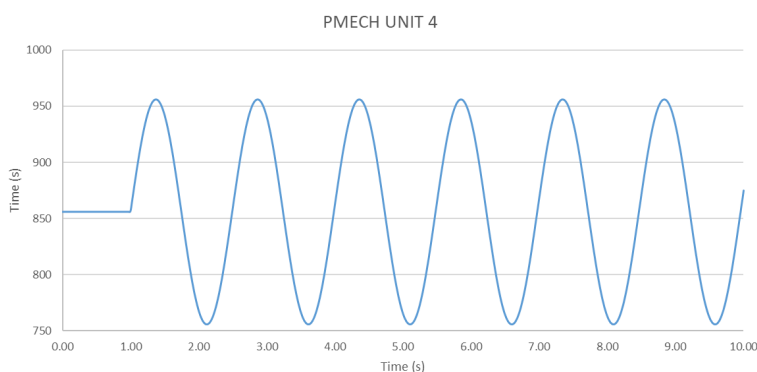
```

TP4_gov_oscillate_rev.dyr  GOV_OSCILLATE_rev2.flx
4 'USRMIL' 1 'GOV_OSCILLATE' 5 0 0 2 1 100 0.67 /

```

**Figure A.2: User Model Calling Statement**

[Figure A.3](#) shows the mechanical power of a unit to which this model is applied.



**Figure A.3: Mechanical Power for Source of Forced Oscillation**

## Modeling Wide-Area Resonant Forced Oscillations

The previous section described how a forced oscillation in mechanical power with a defined amplitude and frequency can be applied to any machine in a PSSE dynamic case. However, there are three conditions required for wide-area resonant forced oscillations:

- A source oscillating at a frequency close to a system mode
- The system mode is poorly damped
- The source is near a strong participation location of that system mode

An essential step to simulating wide-area resonant forced oscillations is to identify natural system modes that are poorly damped and determine which units strongly participate in those modes. Software packages, such as PSSE's SINCAL or Power Tech's SSAT, can be used to perform eigenvalue analysis. The software packages take the load flow and dynamic models as inputs and then provide a list of natural modes within a specified frequency range (e.g., 0.1–2 Hz). The poorly damped modes should be further analyzed to determine their mode shape and participation factors. This will illustrate which source location will most excite the poorly damped modes and which areas of the grid will experience higher oscillations.

**Figure A.4** and **Figure A.5** demonstrate how wide-area resonant forced oscillations can be simulated. Eigenvalue analysis was used to identify a lightly damped natural system mode with a frequency of 0.67 Hz and it is observed that Unit 4 strongly participates in this mode. **Figure A.4** shows system variables throughout a wide area when Unit 4 experiences a cyclical failure at 0.25 Hz. The first condition for resonant forced oscillations is not well satisfied and the wide-area impact is minimal. **Figure A.5** shows the same variables when Unit 4 experiences a cyclical failure at 0.67 Hz, such that all three conditions for resonant forced oscillations are well satisfied. The electrical power oscillations on Unit 4 are higher than the mechanical power oscillations indicating amplification or resonance. The difference between a local forced oscillation and a wide-area resonant forced oscillation is clearly demonstrated by observing oscillations in frequency and power throughout the system. Even the mode shape can be verified by observing which units oscillate against each other.

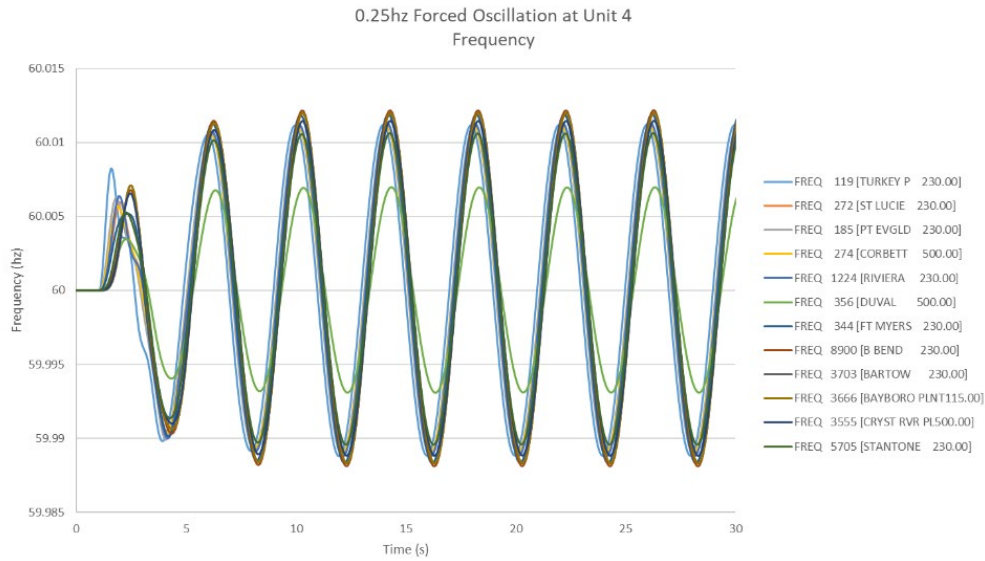
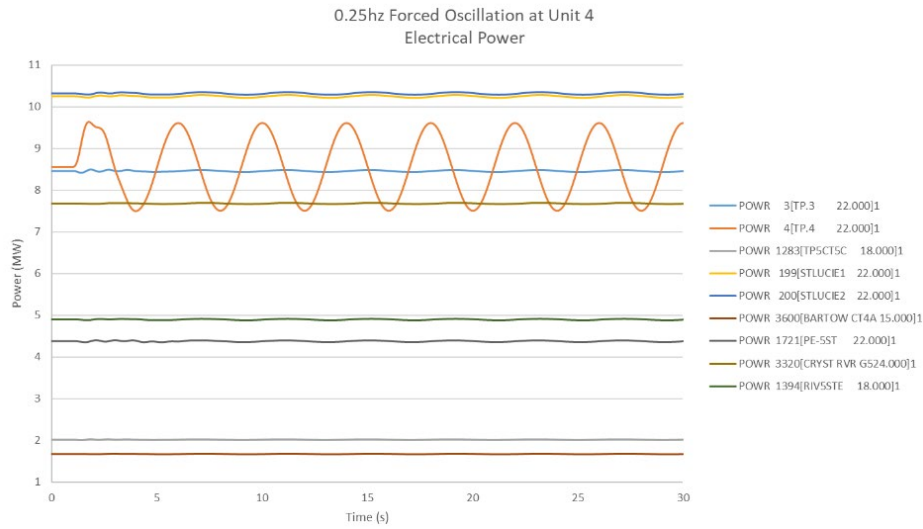
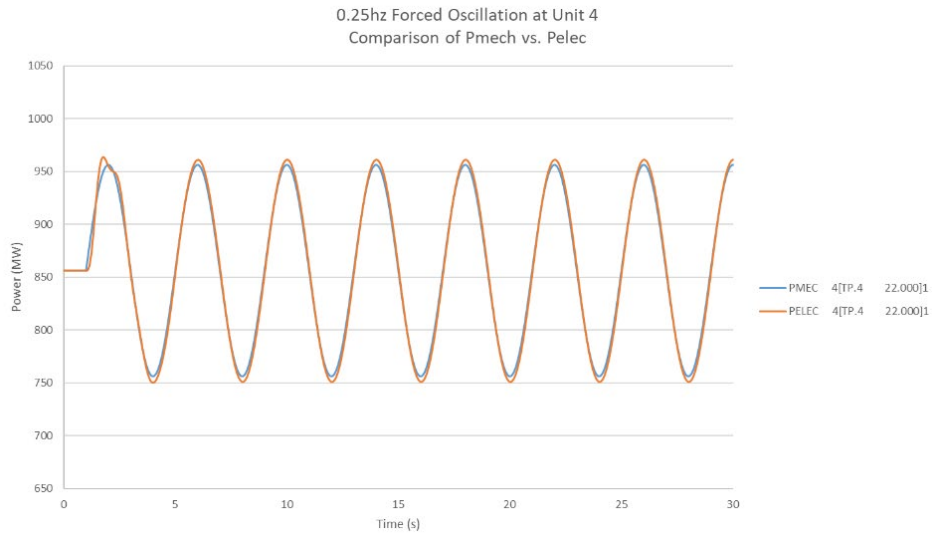


Figure A.4: Local Forced Oscillation

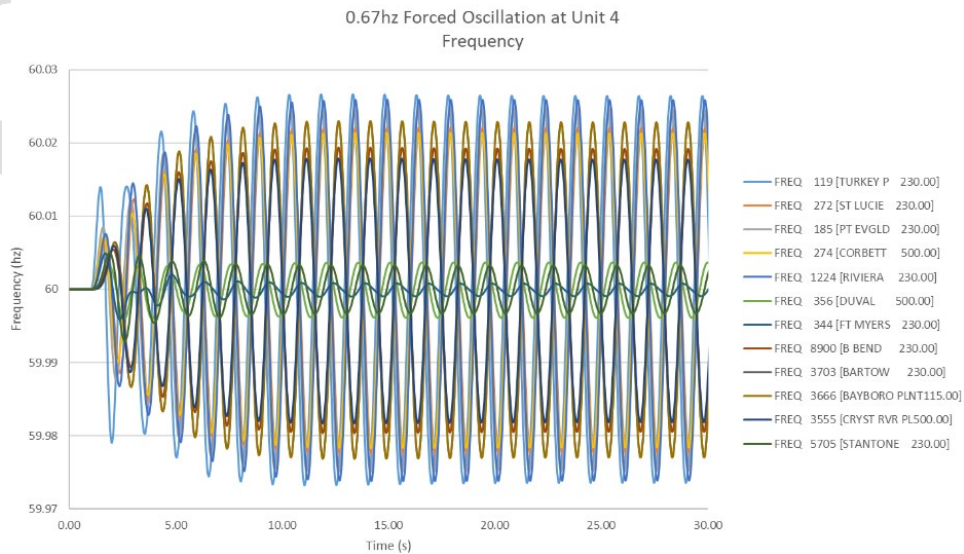
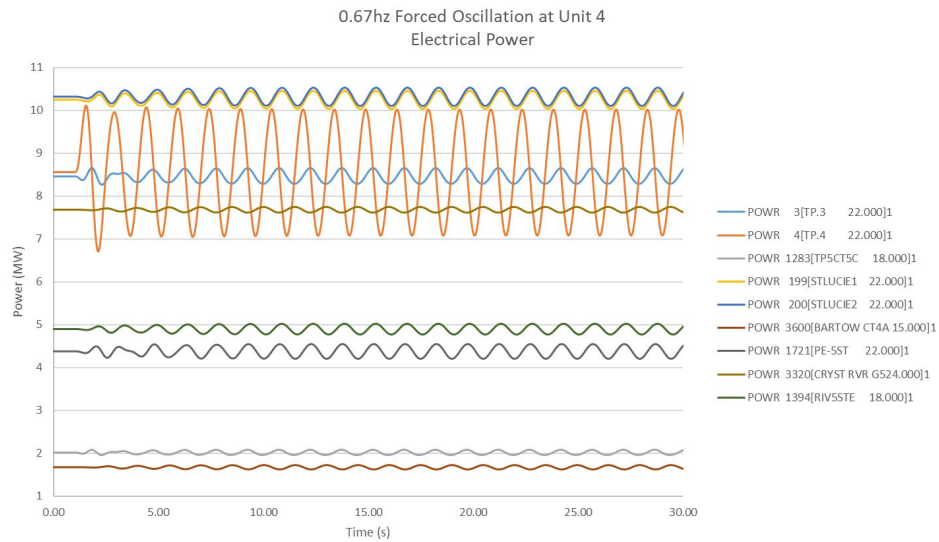
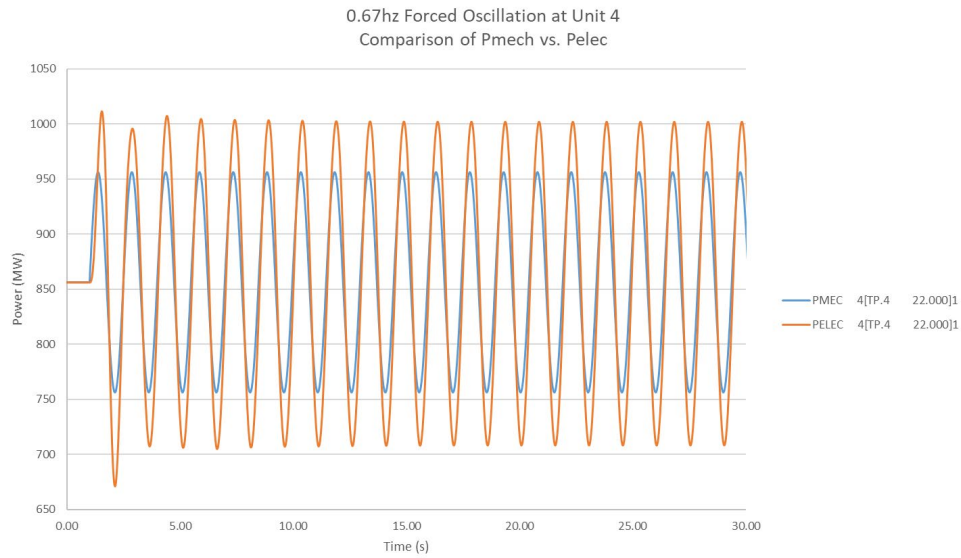


Figure A.5: Wide-Area Resonant Forced Oscillation

## Appendix B: Monitoring Devices for Oscillations

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While the effects of oscillations may be observed using lower-resolution SCADA data, higher-resolution Double Data Rate (DDR) data is useful in fully characterizing oscillations on the BPS. Conversely, very high-resolution data captured from devices, such as fault recorders can generally capture short durations of an oscillation event; however, they typically lack the capability or configuration to capture the event in its entirety. In general, reporting rates of 30 to 60 samples per second, typical of modern DDRs, is sufficient to capture most oscillations observed on the BPS.

DDRs (including PMUs) should be placed at or near generating resources (including wind and solar), large load buses, and grid control devices to capture the location of forced oscillations. Truly locating forced oscillation sources requires considerably wide DDR coverage across a particular interconnection. Having DDR monitoring capability at the borders between TOP and/or RC footprints can help in coordinating the observation of oscillations propagating across multiple TOP and/or RC areas.

Most existing PMUs may not meet the requirement of M class<sup>61</sup> configuration, but experience has proved that these PMUs can be effectively used for oscillation monitoring and analysis. However, if an entity is specifically installing new PMUs for oscillation monitoring, it is recommended to configure as an M class device to accurately capture oscillations in the typical frequency range. With P class filtering, phasor magnitude rolls off as oscillation frequency increases, leading to less of the oscillation captured up to the Nyquist frequency. The P class characteristic is also more susceptible to aliasing, as higher-frequency (beyond the Nyquist) oscillations may not be adequately filtered. The M class characteristic displays less roll off at lower frequency, allowing for more of the oscillation to be captured. Higher frequencies are filtered out, reducing the potential effect of aliasing.

In most cases, oscillations are captured by DDRs in the following electrical quantities: real and reactive power flows, frequency signals, and voltage magnitude signals. These quantities provide physical perspective to grid operators. For example, if the output of a monitoring device identifies that a particular generator is oscillating with an RMS magnitude of 50 MW, this provides physical perspective and alarm limits to the operator. One natural perspective is to alert/alarm operators on forced oscillations that are X MW above the normal ambient level and that persist for more than Y seconds.

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<sup>61</sup> IEEE Standard for Synchrophasor Measurements for Power Systems, IEEE Standard C37.118.1-2011, 2011.

## Appendix C: Forced Oscillation Event Examples

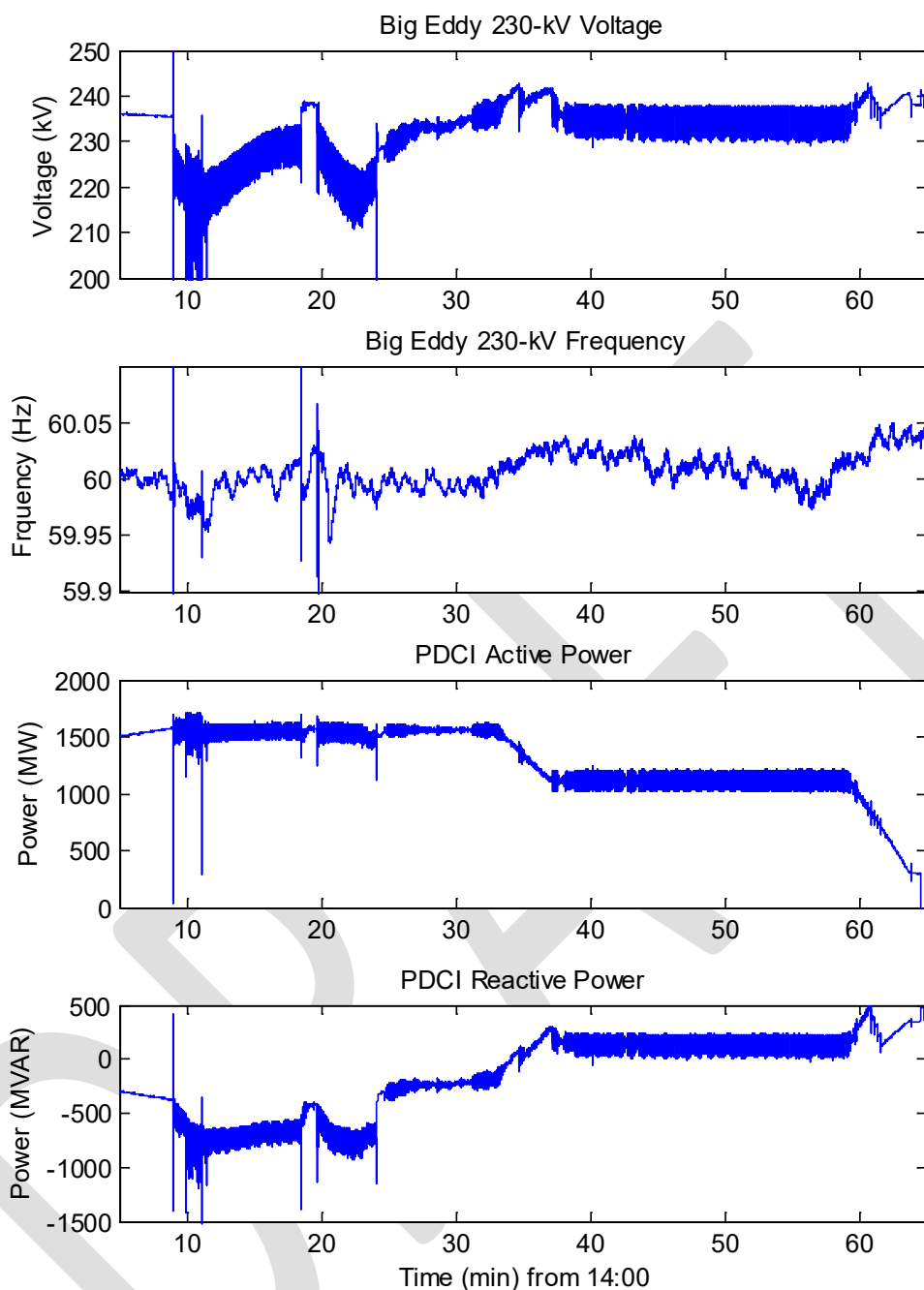
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This section provides illustrative examples of forced oscillation events, how they were identified, and any mitigation measures taken.

### Event 1: Controller Oscillation at Pacific HVDC Intertie

An oscillation event occurred on the Pacific HVDC Intertie (PDCI) on January 26, 2008 ([Figure C.1](#)). At the time of event, the PDCI was operating in the south to north direction with the Pacific Northwest (Celilo) importing about 1,700 MW from Southern California (Sylmar). An outage of two 500/230 kV transformers near the Celilo converter station resulted in low short-circuit ratio conditions, causing the PDCI controls to develop a high-frequency power oscillation. The active power oscillation reached 150 MW, reactive power oscillation was at 200 MVAR peak to peak, and the 230 kV bus voltage oscillation was at 7 kV peak to peak. The oscillation frequency was 4 Hz, indicating a control system oscillation.

BPA did not have an oscillation detection application at the time, so the event lasted for almost an hour. Reports were made about the oscillation causing erratic readings at power plants in Southern California. The oscillation persisted until the PDCI power transfer was lowered below a certain threshold. A combined-cycle plant was tripped in the Pacific Northwest close to the Celilo terminal; however, a link between the trip and the oscillation was not definitively established.



**Figure C.1: January 26, 2008, PDCI Oscillation Event  
(Source: BPA)**

## Event 2: Alberta Forced Oscillation Event

A sustained oscillation developed in the WI at 23:37 PST on November 29, 2005. The oscillation was initiated by a malfunctioning steam extractor control valve at a cogenerating plant in Alberta, Canada. Figure C.2 shows the plant diagram and plant recordings. The steam turbine generator developed a 20 MW peak-to-peak 0.28 Hz oscillation. The plant oscillation resonated with one of the north-south modes of inter-area oscillations, causing a 200 MW peak-to-peak oscillation on the California–Oregon Intertie (Figure C.3). Figure C.4 shows the active and reactive power oscillations on the California–Oregon Intertie. The oscillation went away after six minutes when the steam supply to the industrial process was reduced. BPA did not have the oscillation detection application at that time.

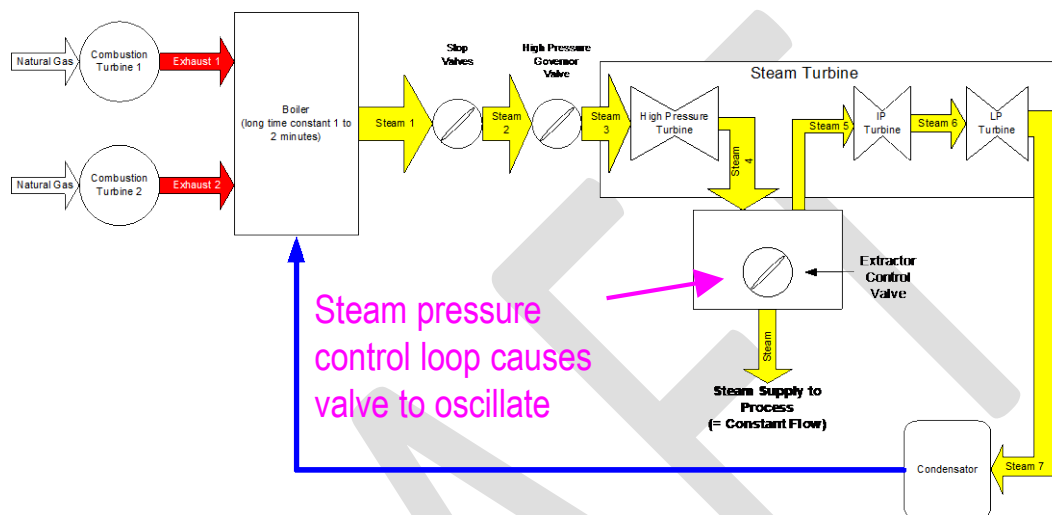


Figure C.2: Combined Cycle Simplified Oneline  
(Source: BPA)

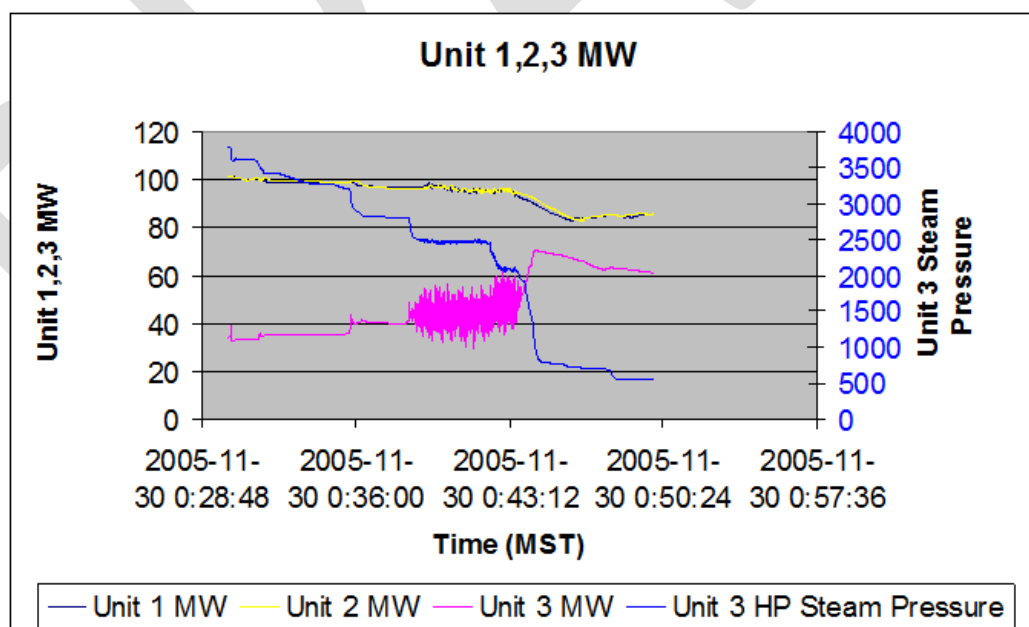
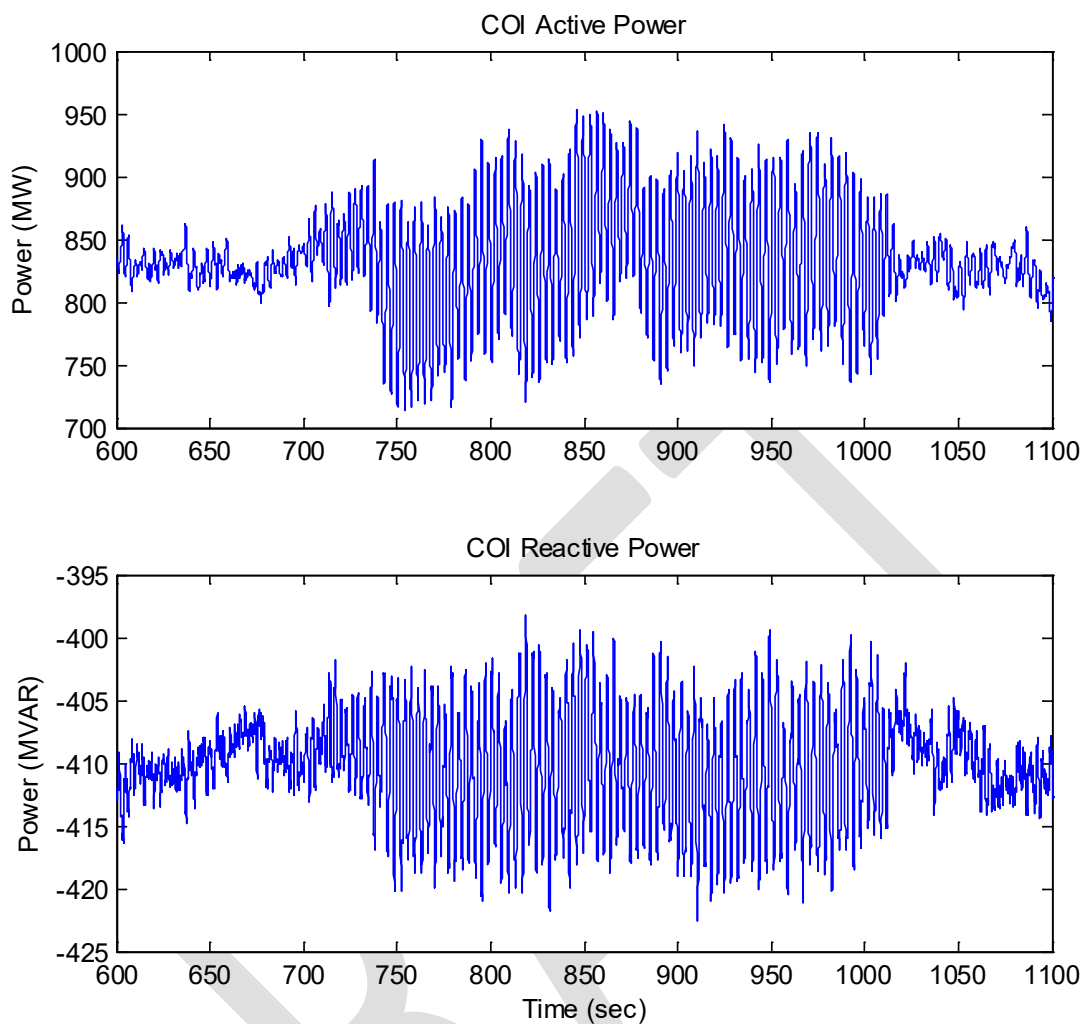


Figure C.3: Alberta Co-Generation Plant Diagram and Plant Recordings—November 29, 2005, Event  
(Source: BPA)



**Figure C.4: Active and Reactive Power Oscillations on California-Oregon Intertie (COI) during Nov 29, 2005, Event (Source: BPA)**

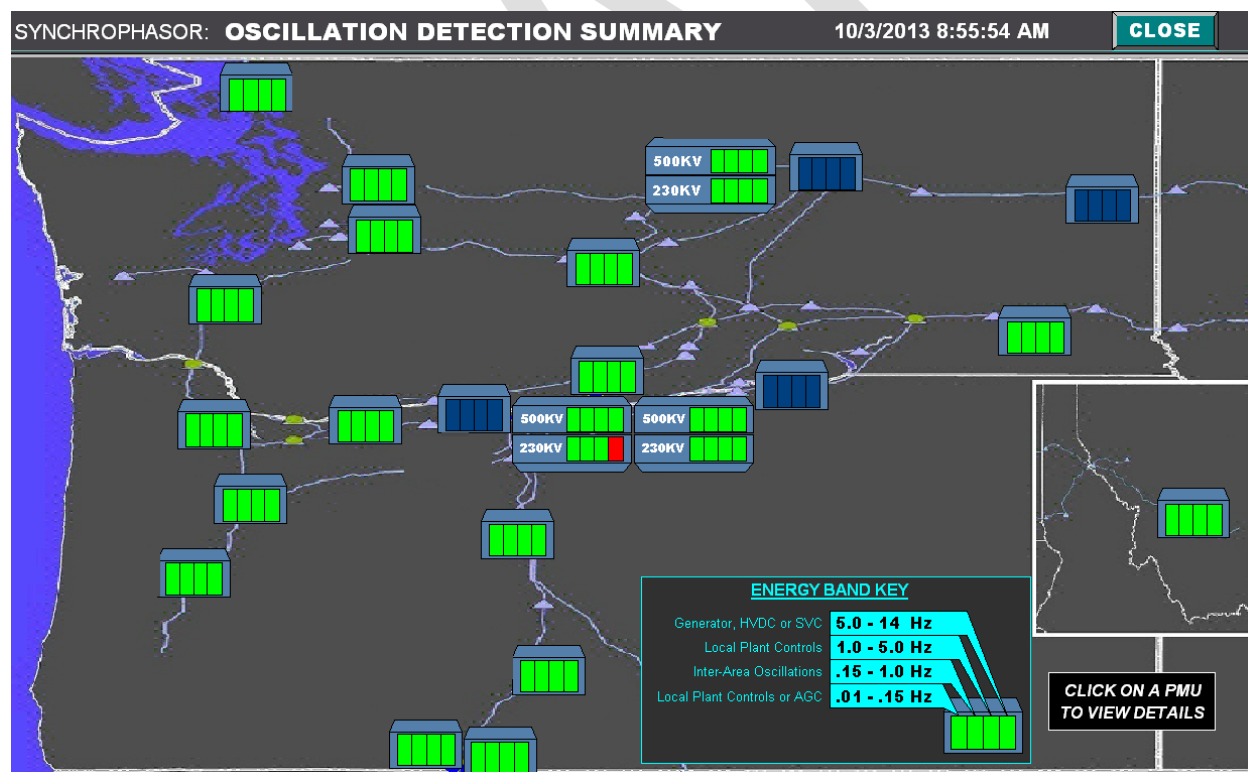
### Event 3: Wind Power Plant High Frequency Oscillations

As early as May 2011, BPA observed high-frequency oscillations during high wind generation operating conditions. However, the older research-grade PMUs installed at that time were not adequate to locate the source of oscillation nor was the PMU bandwidth sufficient to do oscillation analysis. Deployment of newer, high-bandwidth PMUs at the wind power plant enabled BPA engineers to identify a 450 MW wind plant in Northeast Oregon as the source of the oscillation. After the event, BPA was able to link the onset of oscillation events with the energization of the second and third phases of the wind plant consisting of Type 4 full converter wind turbines.

The oscillations were detected on numerous occasions using BPA's oscillation detection application when it became available in Summer 2013. [Figure C.5](#) shows the application display for one of the events. The oscillation is local, confined to the wind generation site. The oscillation frequency is above 5 Hz, suggesting voltage control problems.

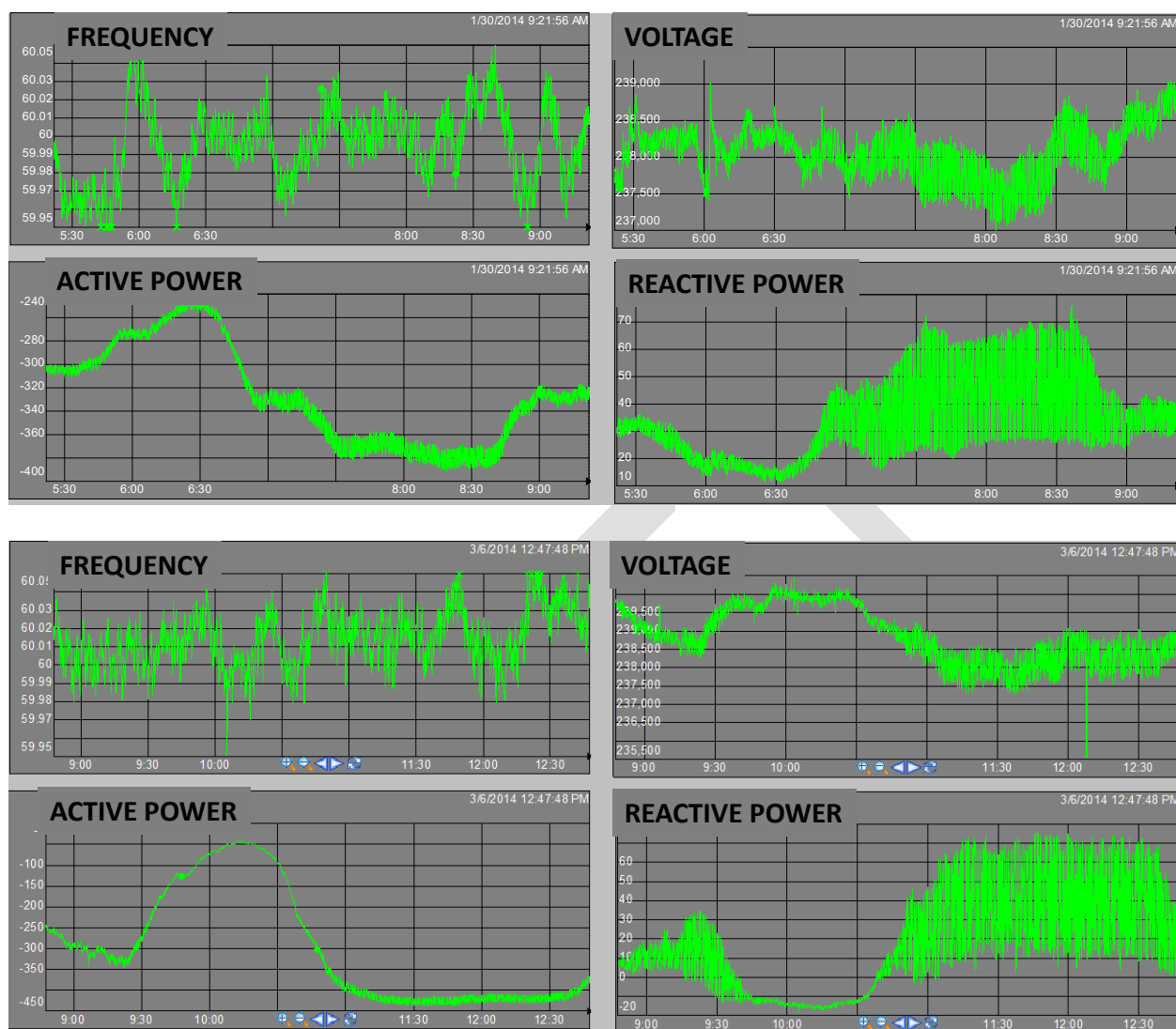
Oscillations developed every time the wind plant was generating above about 85% of its rated output, and reactive power oscillations reached 80 MVAR peak to peak ([Figure C.6](#)). The oscillation frequency was 14 Hz, similar to the oscillation frequency observed during sub-synchronous control interaction events in Texas in 2009. BPA also has series-compensated transmission lines in the area, and there was a concern about the potential of similar interactions should a wind power plant become isolated on a series-compensated line.

BPA engineering staff notified the plant owner about the oscillation and associated risks and the plant owner requested that the wind generator manufacturer upgrade their voltage controls. No oscillations<sup>62</sup> have been detected since April 2014 ([Figure C.7](#)).



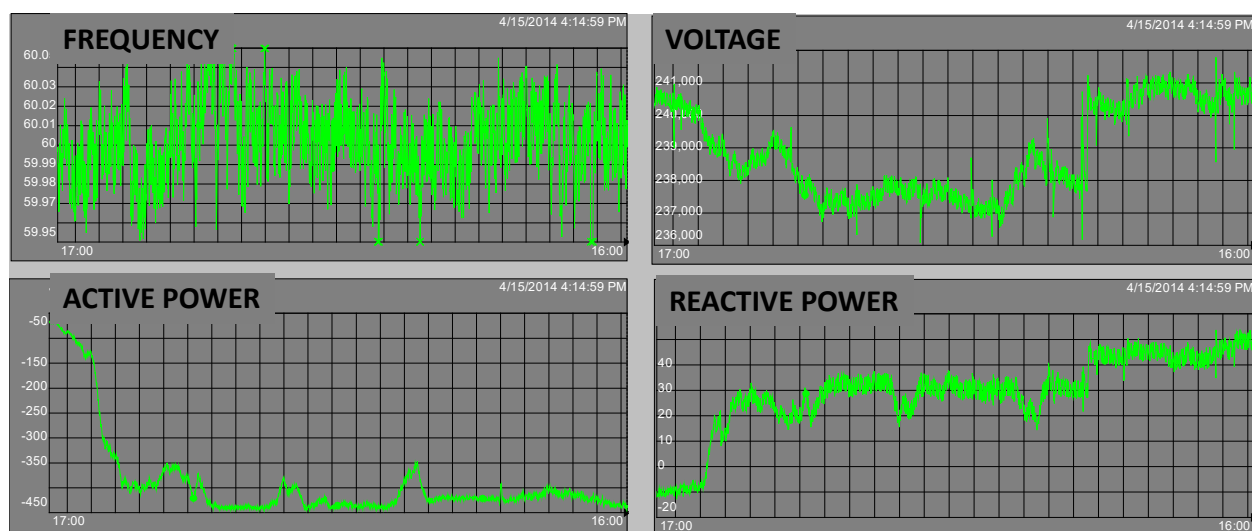
**Figure C.5: Oscillation Detection Application Display Alarming on Wind Plant Oscillation (Source: BPA)**

<sup>62</sup> The magnitude of any oscillations has been drastically reduced and is no longer considered a reliability issue.



**Figure C.6: Two Examples of Controller Oscillation Events (1/20/2014 and 3/6/2014) at a Wind Power Plant<sup>63</sup>**  
(Source: BPA)

<sup>63</sup> Measurements are taken at the BPA substation looking into the plant; negative values refer to power flow from the plant to the substation.



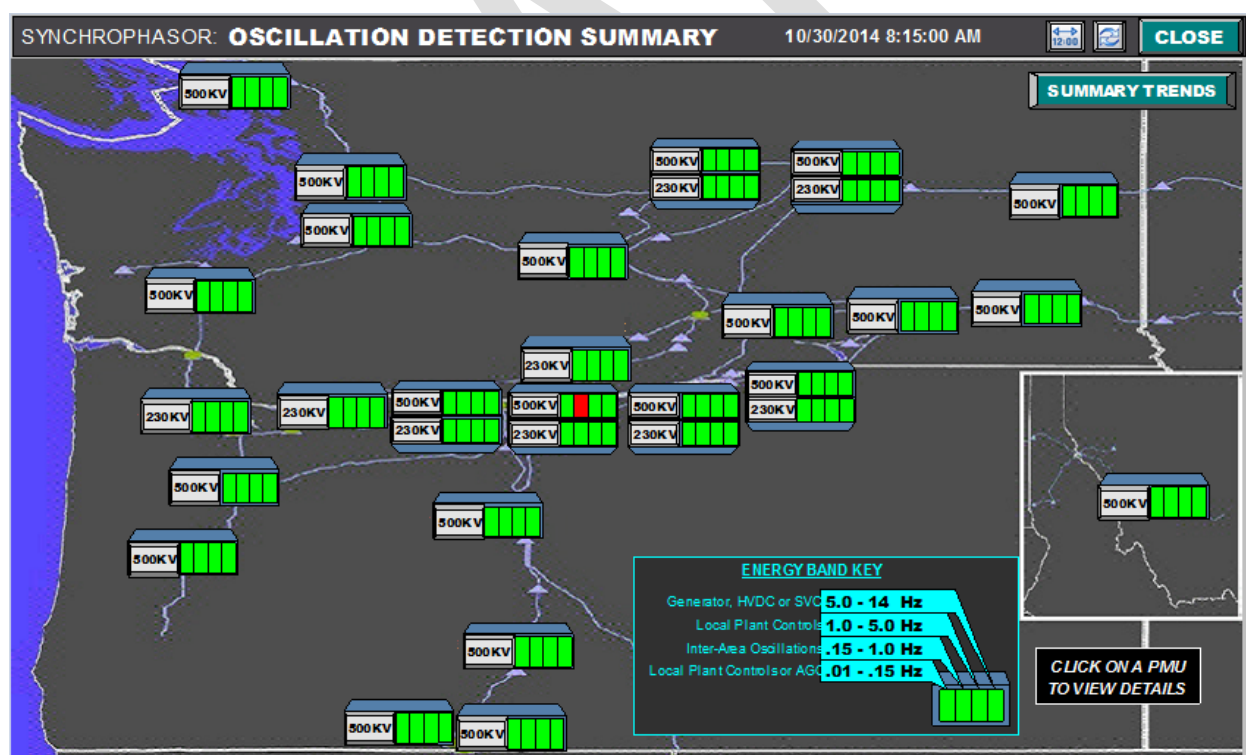
**Figure C.7: Wind Power Plant Ramp to Full Output after Controls Fixed on 4/15/2014  
(Source: BPA)**

## Event 4: Hydro Power Plant Surging Vortex Oscillations

Hydro turbines are designed for the most efficient operation at nominal head and flow. Hydro turbines operating at flows and heads significantly different from their design values can experience rough operation and power swings from water passing through the draft tube in a whirling and spiraling vortex. Hydrodynamic instability occurs at partial load, typically 25 to 60 percent of their rated generator output. The result of the instability is high vibration of the machine components with the possibility of premature wear or even catastrophic damage. The generators go through the “rough zone” during startup loading and shutdown unloading; however, the transition only lasts about 30 seconds in these situations as the unit is ramping through the zone. Continuous operation in the “rough zone” is very undesirable for generator turbines.

BPA’s oscillation detection application issued an alarm of a sustained oscillation at one of its hydro-power plants in October 2014. The oscillation alarm was local and in the second frequency band as shown in Figure C.8. Analysis of the sustained oscillation using the application pointed to powerhouse line #2 of the hydro-power plant being monitored by PMUs (Figure C.9). The oscillation frequency was about 0.38 Hz (Figure C.10) and observable in both active and reactive power outputs.

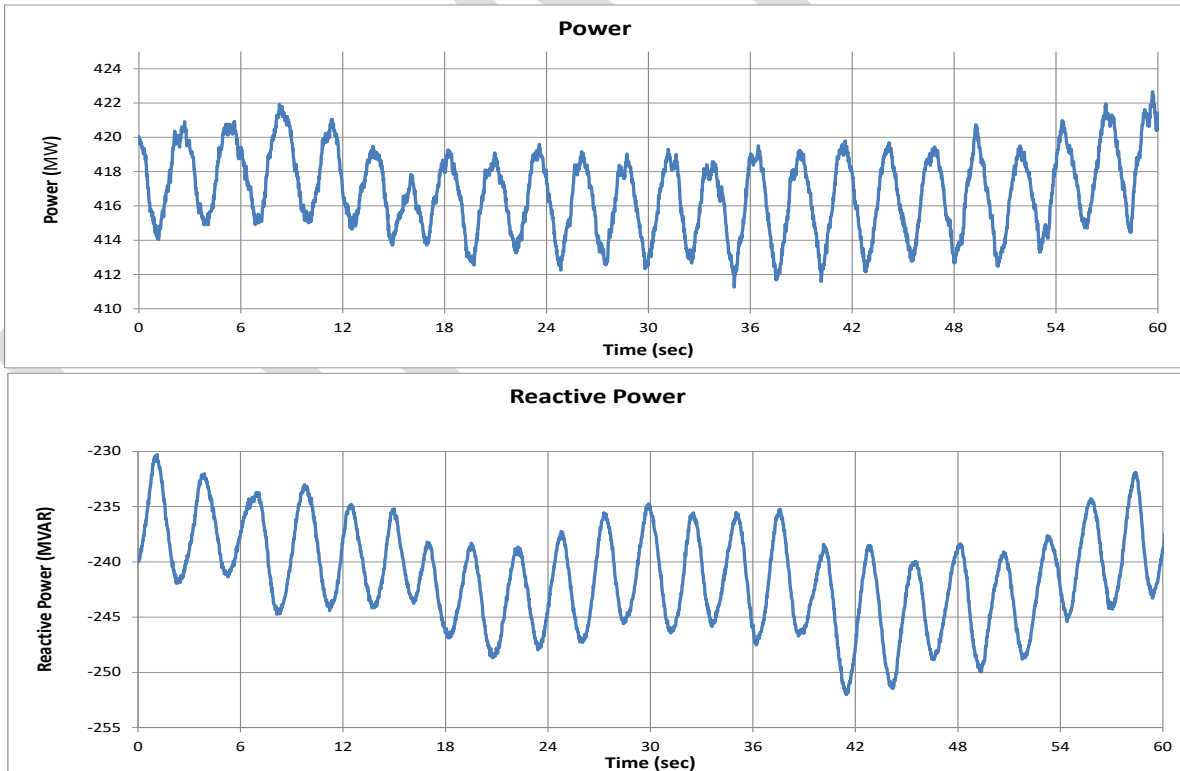
Advised by BPA’s operations engineering staff, BPA dispatchers contacted the plant operators and notified them about the oscillation. Plant operators were not aware of the problem and increased generator active power output above the “rough zone.” Once the unit was maneuvered out of the rough zone, the oscillation went away. There have been several occasions when sustained oscillations due to surging water vortices have been detected at hydro-power plants using the tool.



**Figure C.8: Oscillation Detection Overview Display with Hydro Plant in Rough Zone (Source: BPA)**



**Figure C.9: Oscillation Trend Display**  
(Source: BPA)

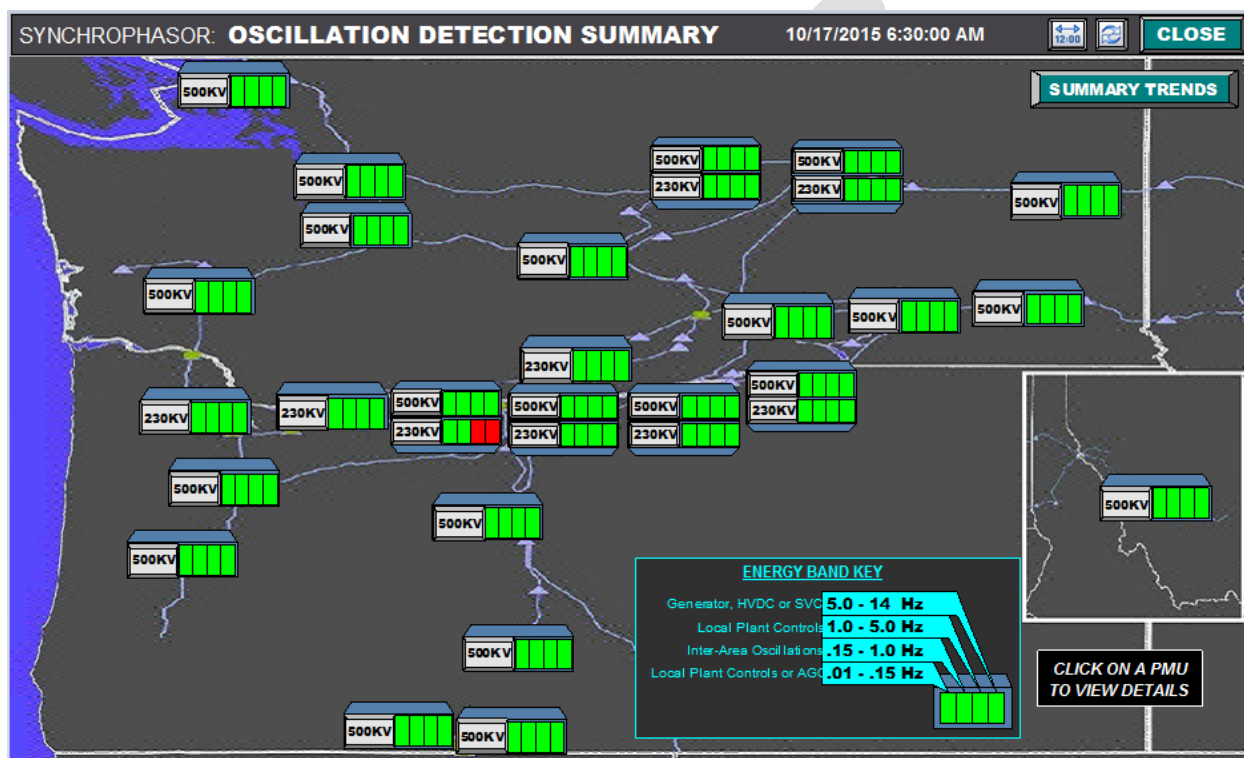


**Figure C.10: Oscillation Time Series**  
(Source: BPA)

## Event 5: Control Interactions Between UEL and PSS

The BPA oscillation detection application detected several occurrences of sustained oscillations at one of its hydro-power plants in fall 2015. The oscillation triggered local alarms in frequency bands 3 and 4 (see [Figure C.11](#), [Figure C.12](#), [Figure C.13](#), and [Figure C.14](#)). The oscillation waveform had a unique beat characteristic (see [Figure C.14](#)). Further analysis indicated that the oscillation was due to interactions between the plant's PSSs and under-excitation limiters (UEL).

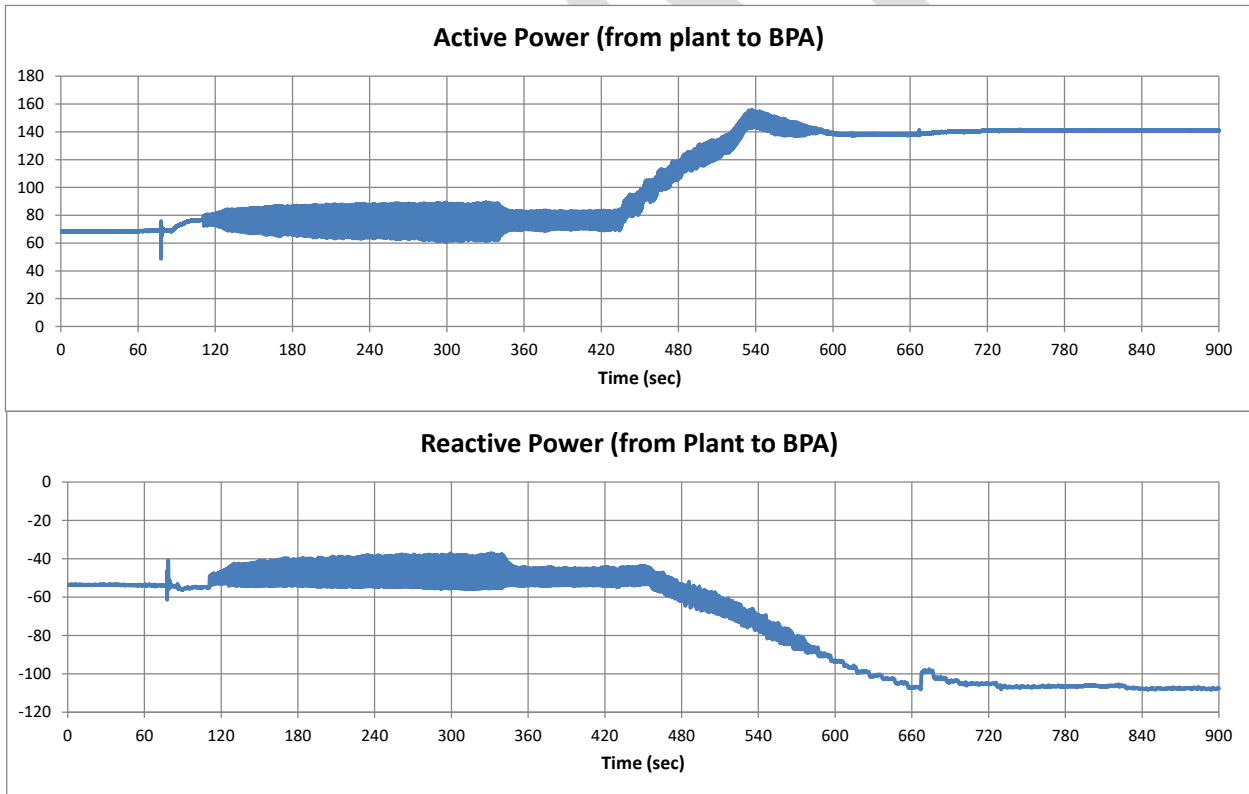
BPA technical staff notified the plant operators about the problem. The plant operator re-tuned UEL gains and re-tested for the interactions in January 2016. The problem has since been resolved.



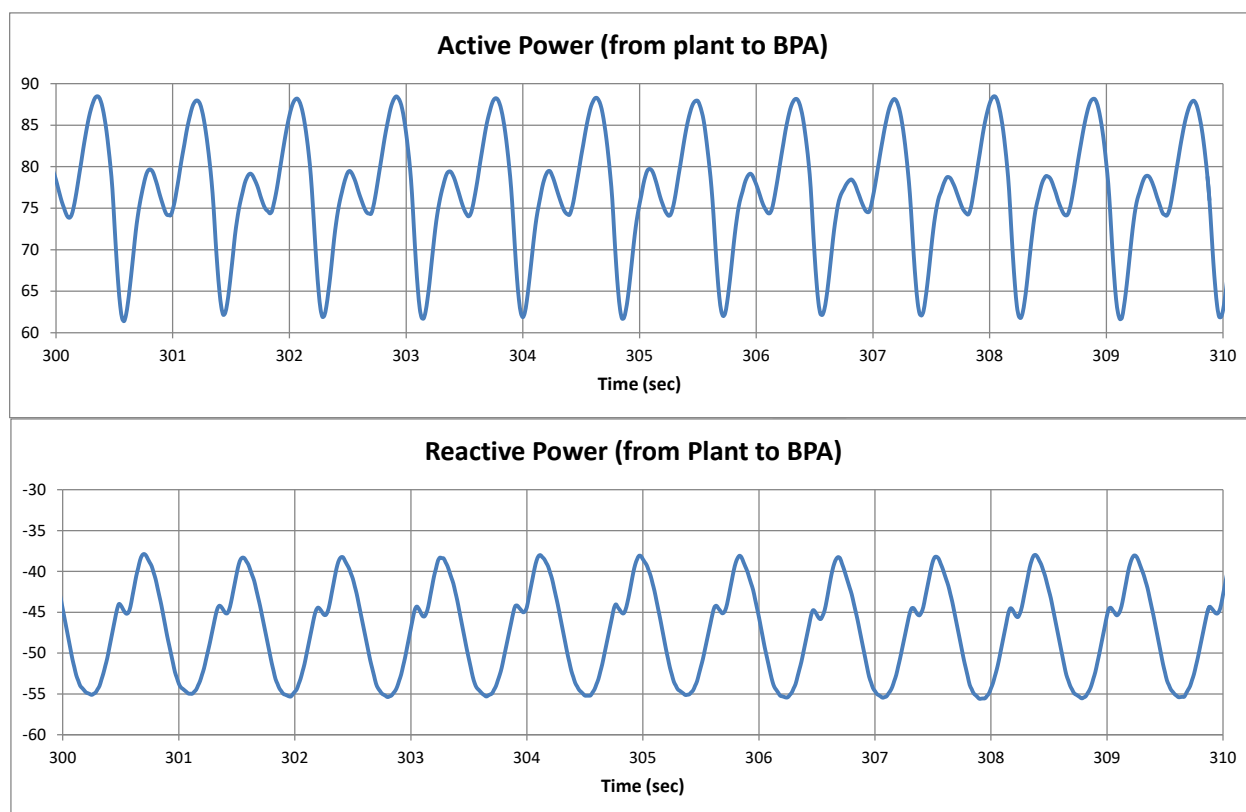
**Figure C.11: Oscillation Detection Overview Display with Control Interaction Oscillation (Source: BPA)**



**Figure C.12: Oscillation Trend Display**  
(Source: BPA)



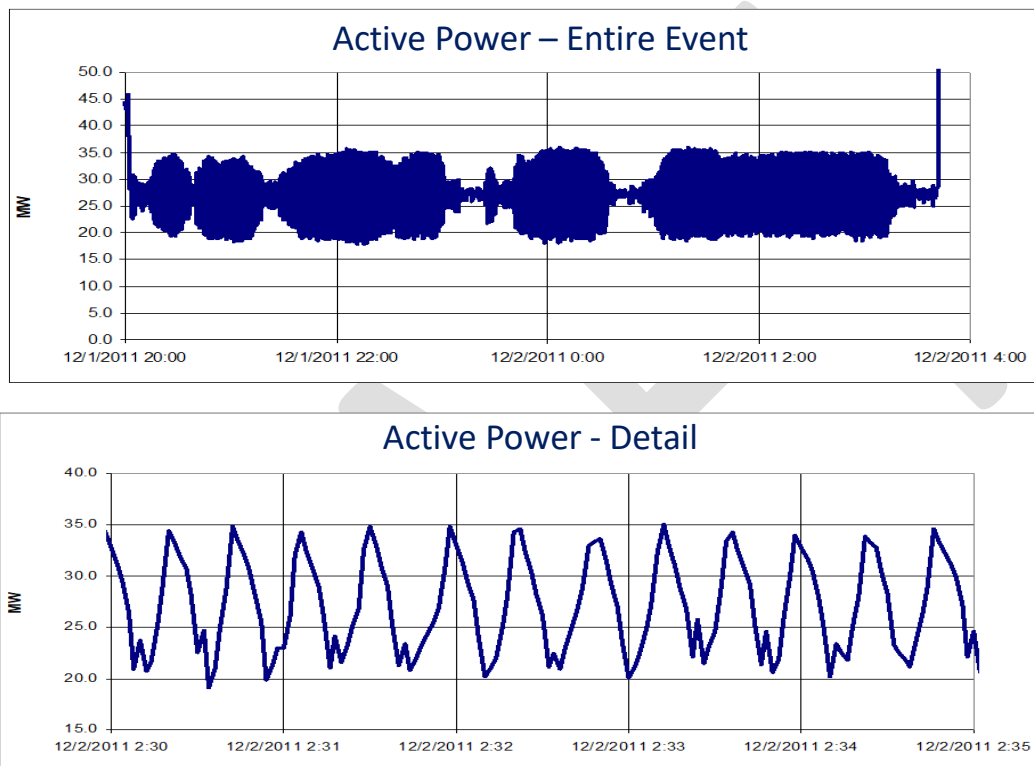
**Figure C.13: Overview of Entire Event**  
(Source: BPA)



**Figure C.14: Event Details  
(Source: BPA)**

## Event 6: Plant Controller Issues

In December 2011, BPA engineering staff observed via SCADA sustained active power oscillations at a wind power plant in Oregon (See [Figure C.15](#)). The oscillation lasted for nearly seven hours with a 15 MW peak-to-peak magnitude. The oscillation period was about 25 seconds, indicating a possible active power controller issue. Operating wind turbine blades in this condition can cause excessive wear. The plant operators and wind turbine manufacturers were notified about the issue. BPA did not have the oscillation detection application implemented at that time.

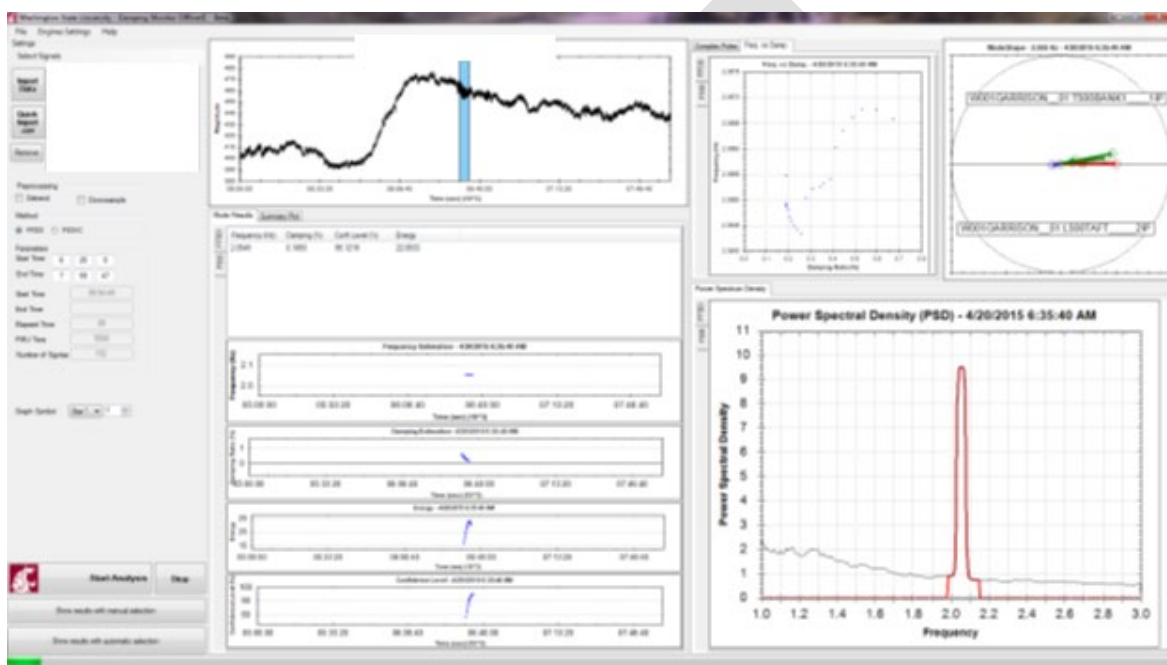


**Figure C.15: Active Power Controller Oscillation at a Wind Power Plant  
(Source: BPA)**

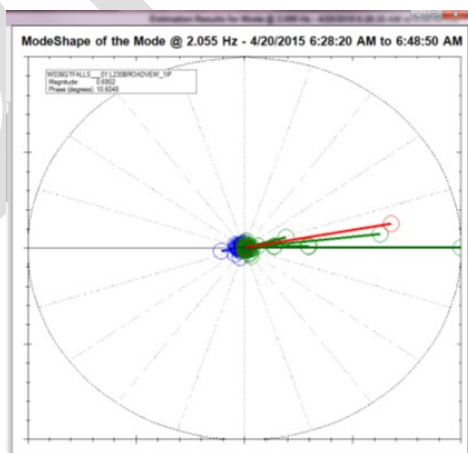
## Event 7: Hydro Generator Issue

On April 19 and 20, 2015, oscillation analysis using the Database Migration Option (DMO) tool identified a forced oscillation at around 2.05 Hz with less than a 1% damping ratio, as shown in Figure C.16. BPA's ODM alarmed on one of their substations as well. From Figure C.17, the source of the forced oscillation is not clear due to minimal PMU coverage near that substation. SCADA data of the surrounding generators supplemented the synchrophasor-based applications and showed that a hydro generator nearby may be the source of the oscillation (Figure C.18). This analysis was initially performed manually, taking about three weeks.

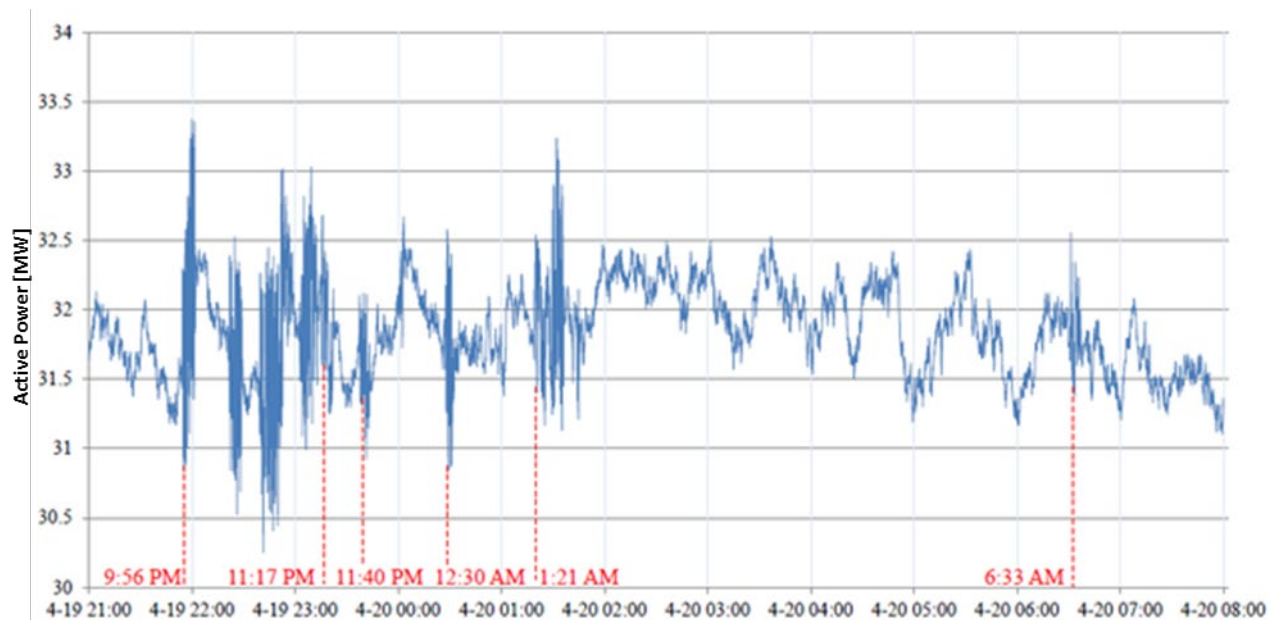
Using the PMA tool developed by Peak, the data was extracted within seconds, shown in Figure C.19, and the generator described was identified as the likely source of the oscillation. This event was not verified with the GO to identify the cause of the oscillation.



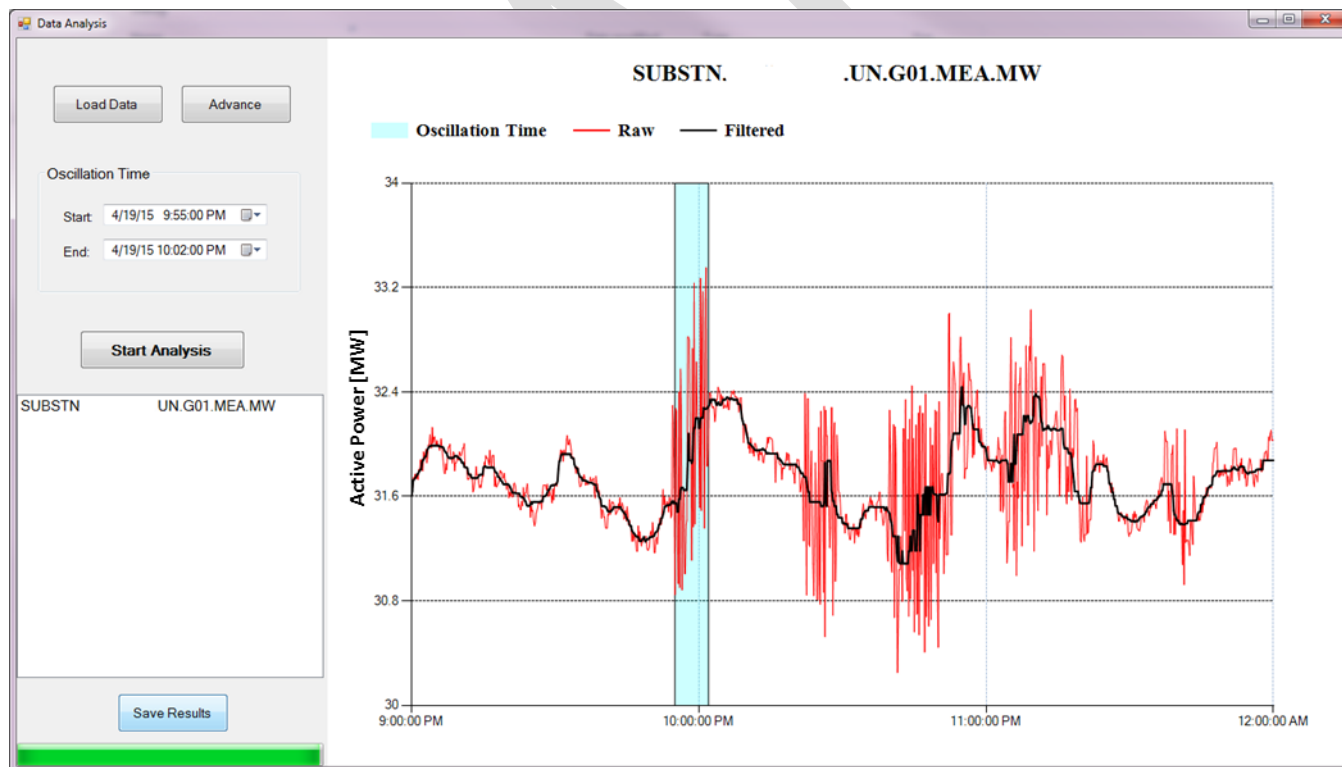
**Figure C.16: DMO Analysis for April 2015 Event  
(Source: Peak)**



**Figure C.17: Mode Shape for April 2015 Event  
(Source: Peak)**



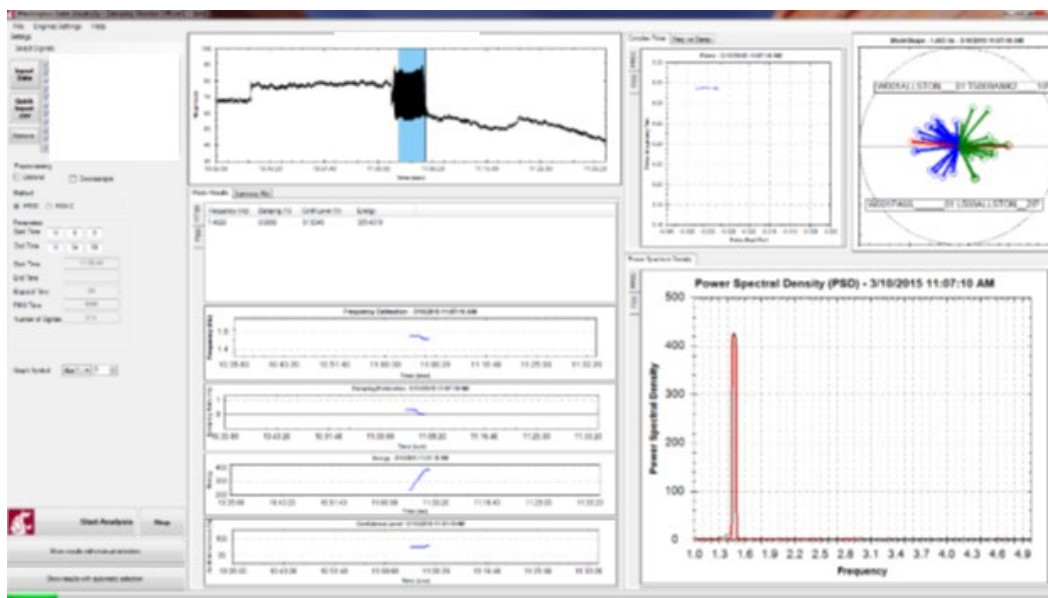
**Figure C.18: Hydro Generator Output (SCADA Data) for April 2015 Event (Source: Peak)**



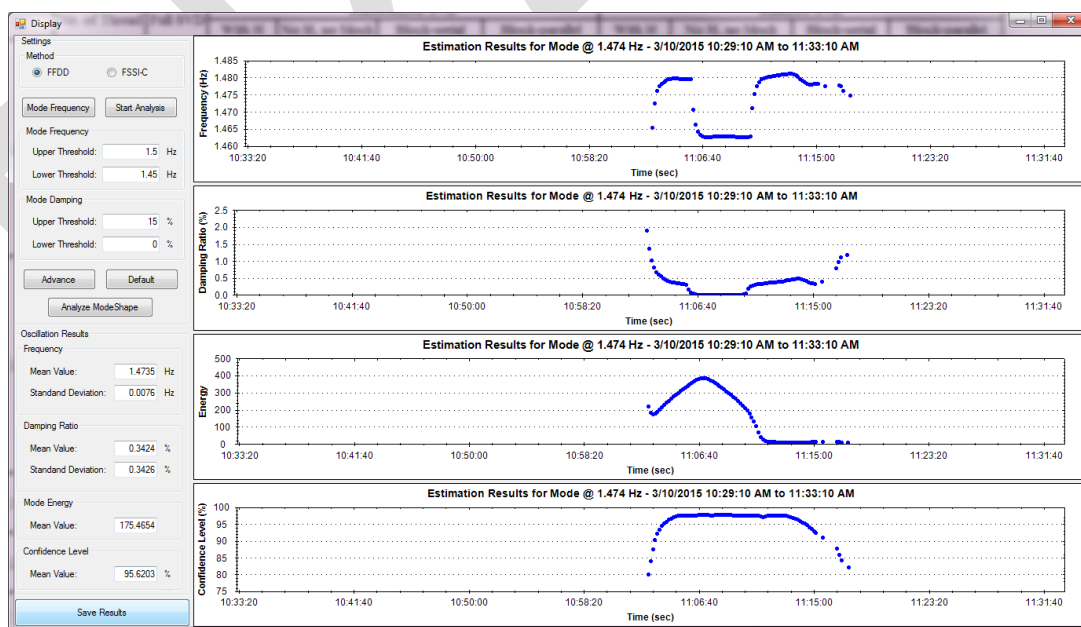
**Figure C.19: PMA Tool Results for April 2015 Event (Source: Peak)**

## Event 8: Hydro-Plant Controller Issue

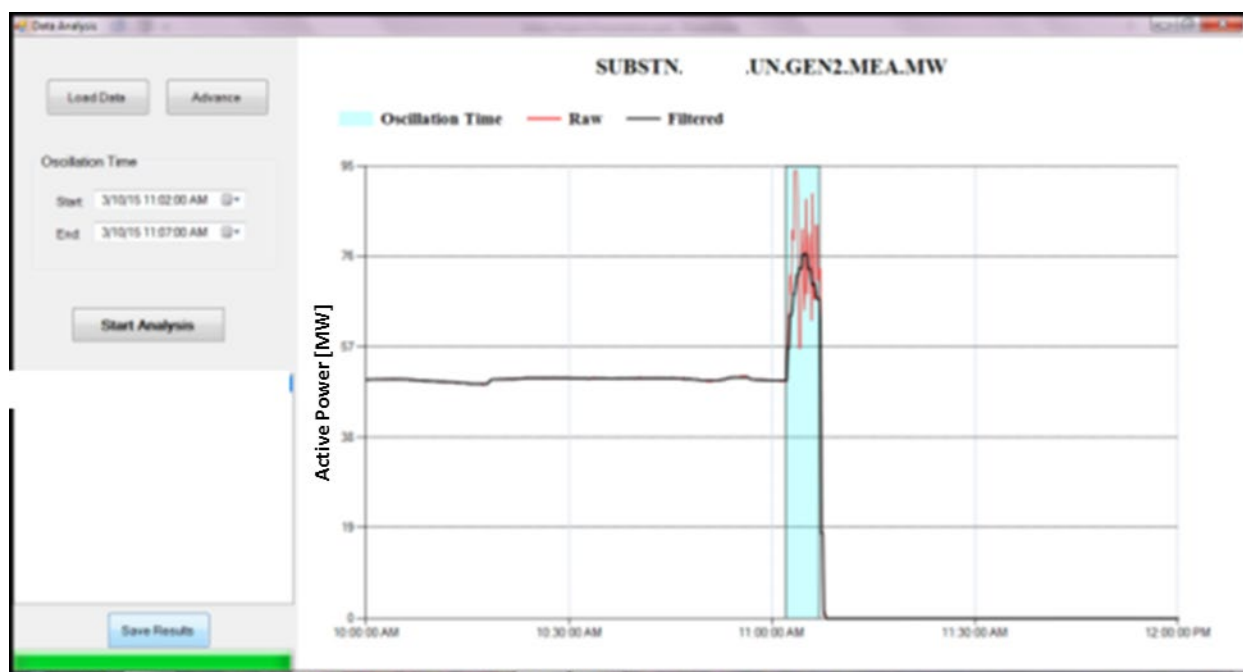
BPA ODM alarmed on several local substations; however, there was not full PMU coverage in that area to directly identify the source of the oscillation in real time. Oscillation analysis, using the DMO tool, clearly identified a forced oscillation at 1.47 Hz with less than a 1% damping ratio (Figure C.20 and Figure C.21). The PMA tool identified one hydro generator as the likely source of the oscillation (Figure C.22). Later, BPA confirmed with the GO that there was a plant controller issue during the time of the forced oscillation.



**Figure C.20: DMO Analysis  
(Source: Peak)**



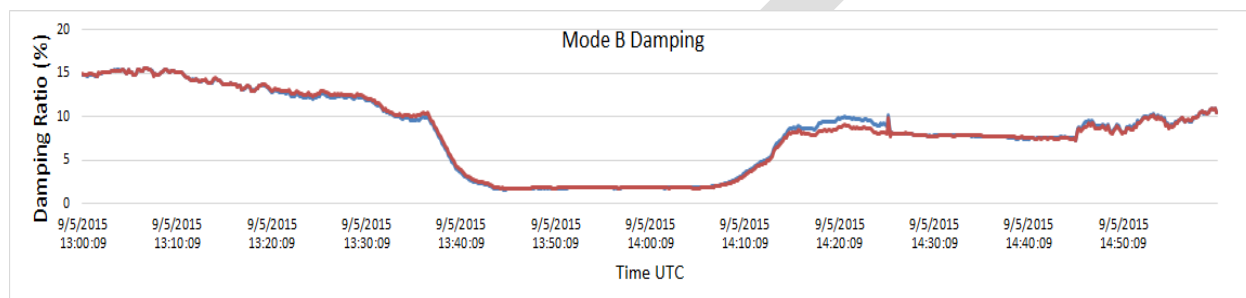
**Figure C.21: DMO Estimation Results  
(Source: Peak)**



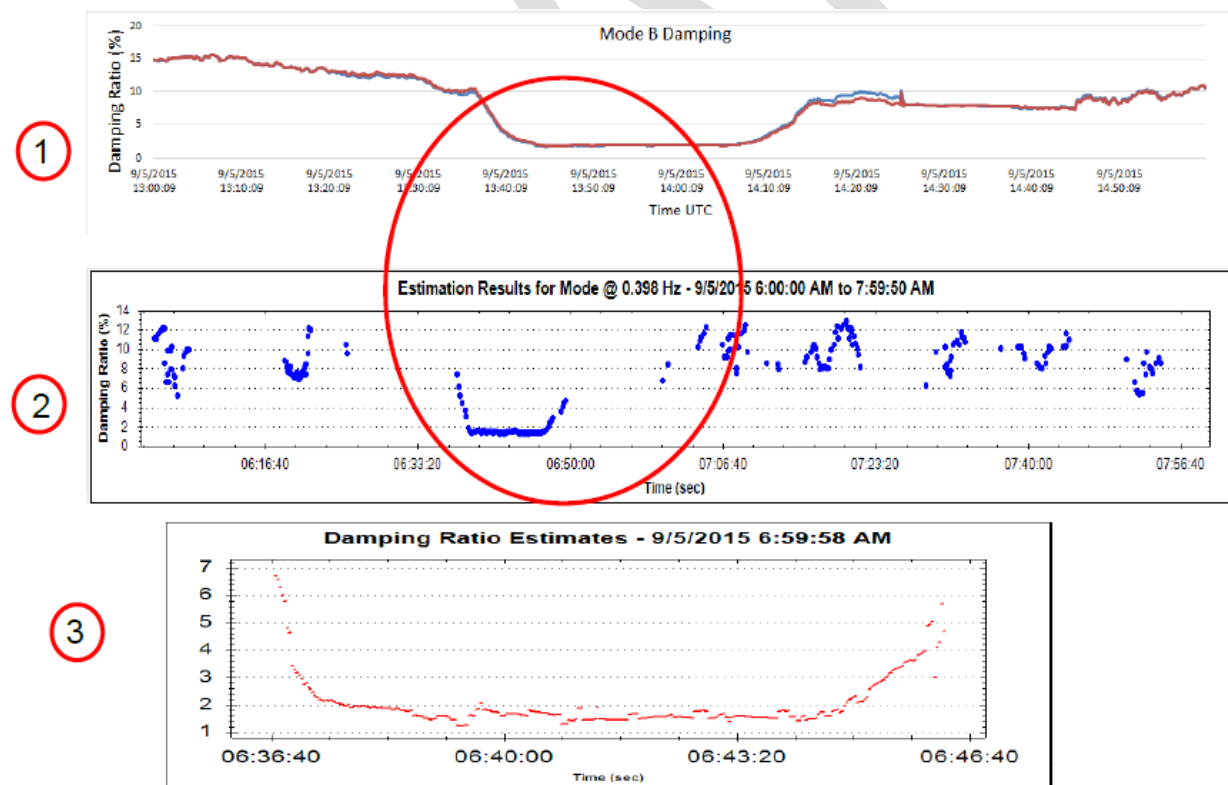
**Figure C.22: PMA Tool Results  
(Source: Peak)**

## Event 9: Interaction with Inter-Area Oscillation Tools

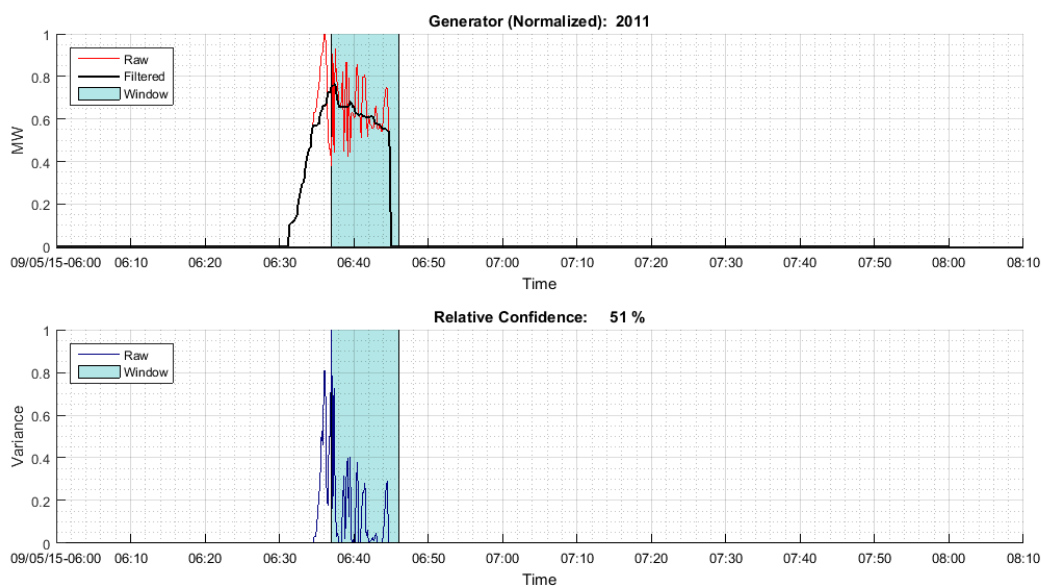
When analyzing the real-time mode meter results for N-S Mode B, a low damping ratio of about 2% for a period of ~30 minutes was identified as shown in Figure C.23. There were no significant changes to the system such as topology, generation, load, or tie-line flows during that time. Results were verified using the WSU DMO tool shown in Figure C.24. The mode shape plot (Figure C.26) clearly shows the source of the oscillation as the signal in red with highest magnitude. Using the PMA tool, it showed a generator in Nevada as the likely source of the oscillation (Figure C.25). The cause of the oscillation was verified with the GO, identified as a faulty combustion turbine MW transducer. There are two MW transducers for this unit, and the control logic uses a select logic of the two values to control the output to a MW set point. If the measured output is incorrect or oscillating, then the control system attempts to correct to the set point, resulting in active power output swings.



**Figure C.23: September 5, 2015, Event-Mode Meter Results (Source: Peak)**

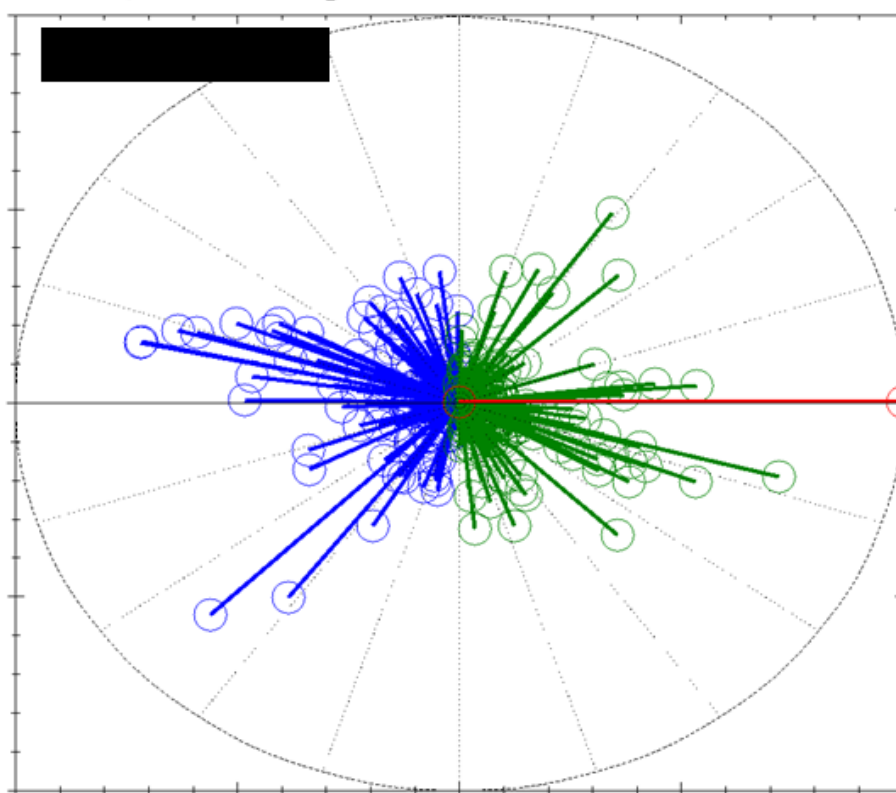


**Figure C.24: September 5, 2015 Event-DMO Results (Source: Peak)**



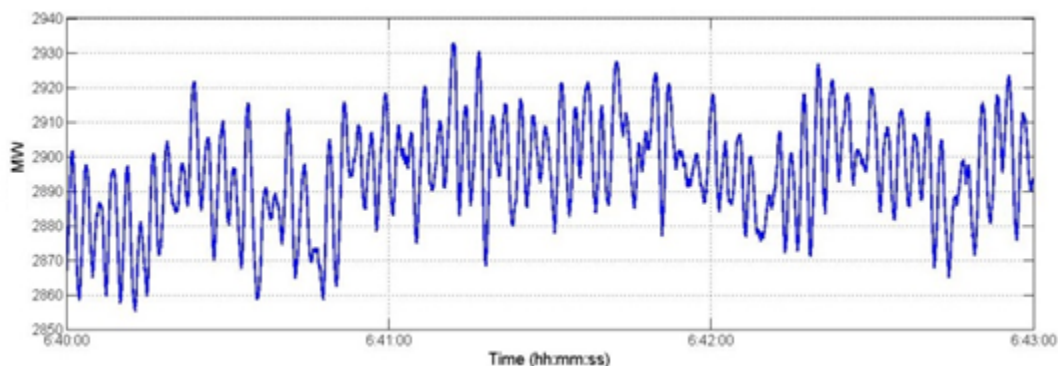
**Figure C.25: September 5, 2015 Event—PMA Results  
(Source: Peak)**

ModeShape of the Mode @ 0.383 Hz - 9/5/2015 6:00:00 AM to 6:59:58 AM



**Figure C.26: September 5, 2015 Event—Mode Shape  
(Source: Peak)**

The significance of this event is that the forced oscillation of about 6 MW had a “resonance”<sup>64</sup> effect with the system mode at 0.4 Hz (N-S mode B) that showed up on a major intertie as ~40 MW oscillations ([Figure C.27](#)).

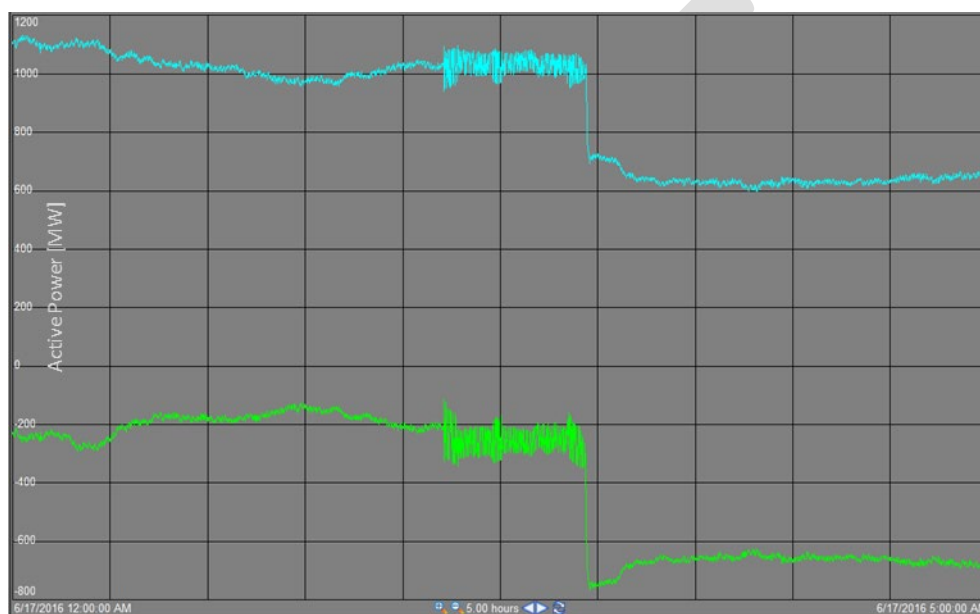


**Figure C.27: September 5, 2015, Event–Intertie MW Flow  
(Source: Peak)**

<sup>64</sup> The system has an electromechanical mode near the forced oscillation frequency. Often referred to as “resonance”; however, it is not true resonance in the academic sense because the system mode is stable. The system has high gain near the mode, so the forced oscillation shows up at many locations.

## Event 10: June 17 Forced Oscillation due to Control Failure

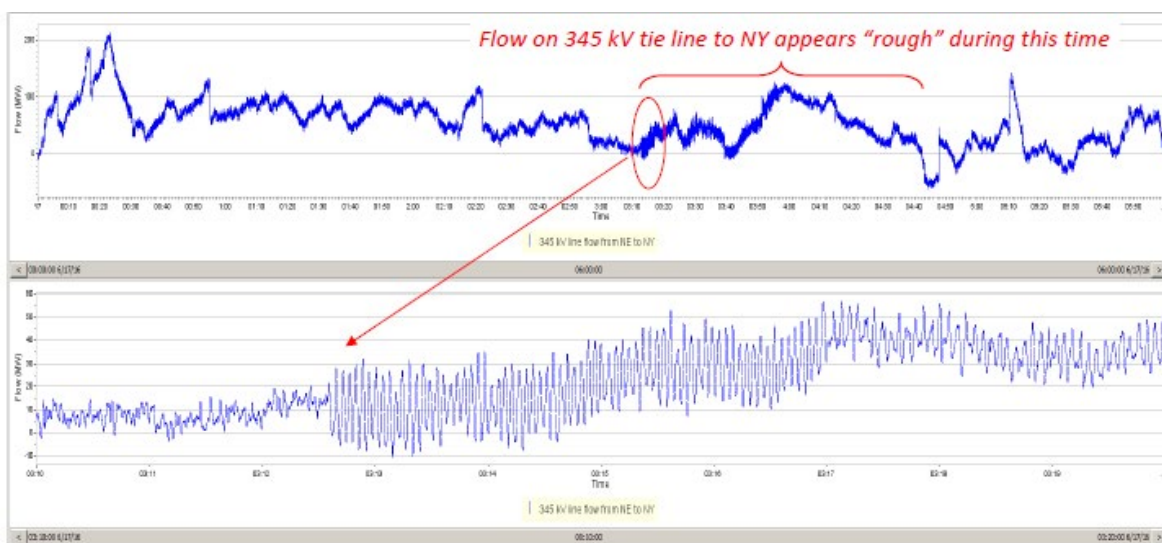
On June 17, 2016, at 02:57, the Grand Gulf Nuclear Station (GGNS) automatically scrambled<sup>65</sup> while performing the main turbine stop and control valve surveillance. The unit had reduced load to approximately 65% power for a control rod pattern adjustment. At approximately 02:15, prior to the scram and while performing the main turbine stop and control valve surveillance, one turbine stop valve would not reopen after testing due to a malfunction of a hydraulic control test valve. After the first stop valve would not reopen, a second stop valve closed. This resulted in power oscillations for about 45 minutes while the main generator was connected to the transmission system and then a subsequent reactor shutdown. The specific cause of the malfunction was not known. **Figure C.28** shows the power oscillations prior to the unit tripping off-line (the blue and green lines are line flows capturing GGNS output).



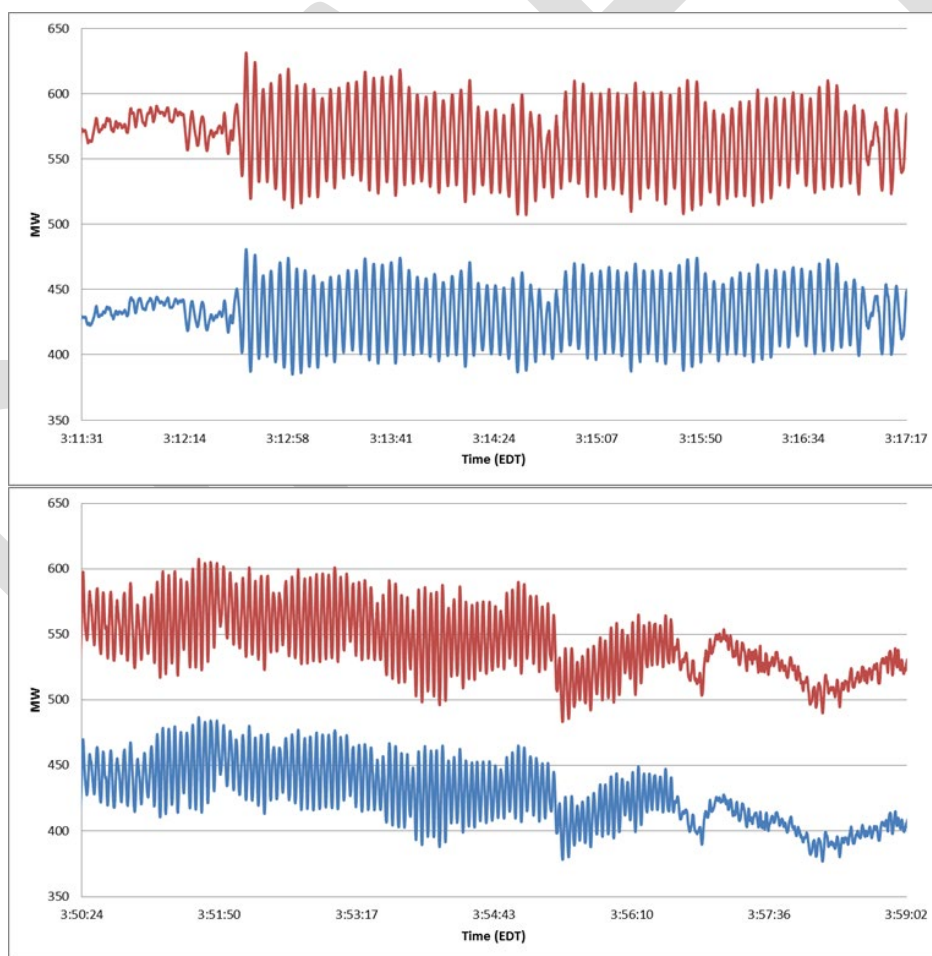
**Figure C.28: Approximately 200 MW Forced Oscillation at GGNS  
(Source: Entergy)**

The forced oscillation frequency occurred around 0.28 Hz, which is near the frequency of a system mode in the EI. This resulted in the oscillation being observed throughout the EI, as shown in **Figure C.29**, **Figure C.30**, and **Figure C.31**. Oscillations in ISO-NE area were around 40 MW, 70 MW near the Florida border, and around 50 MW in the SPP region.

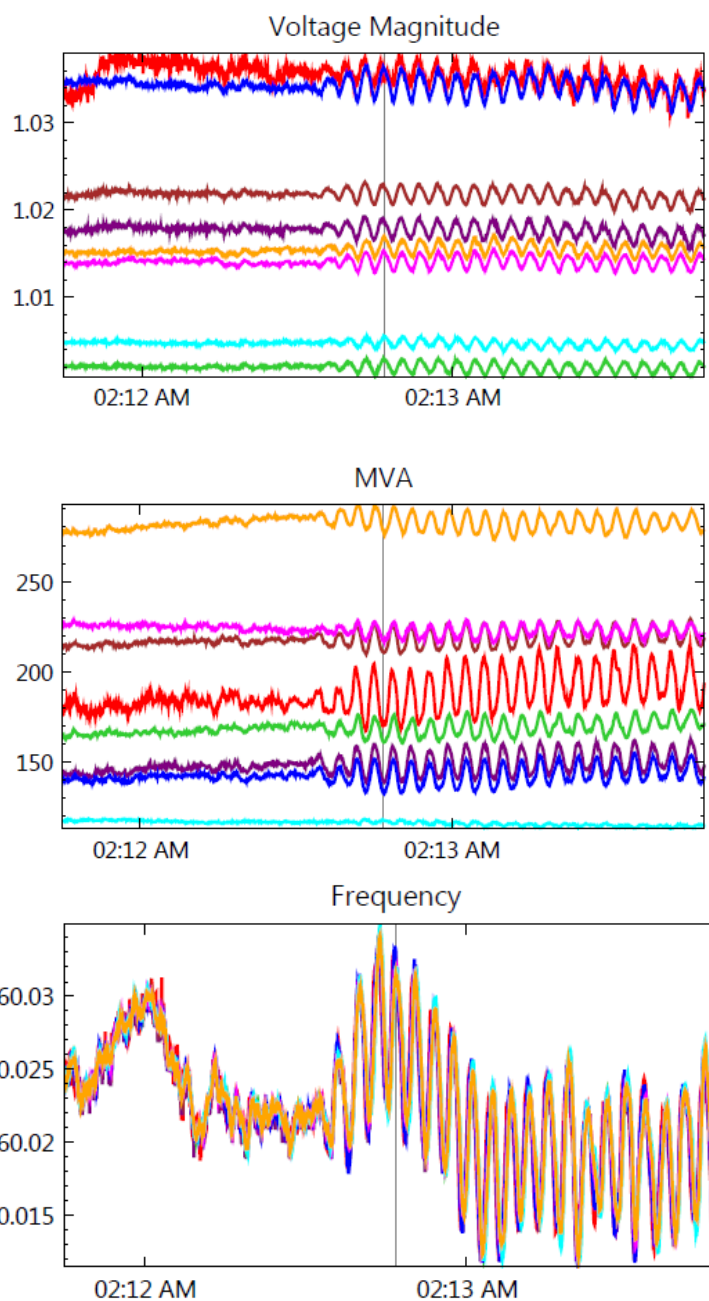
<sup>65</sup> A scram is an emergency shutdown of a nuclear reactor.



**Figure C.29: Approximately 40 MW Oscillations Seen in New England Region (Source: ISO-NE)**



**Figure C.30: Approximately 70 MW Oscillations Seen Near the Florida Border (Source: FPL)**



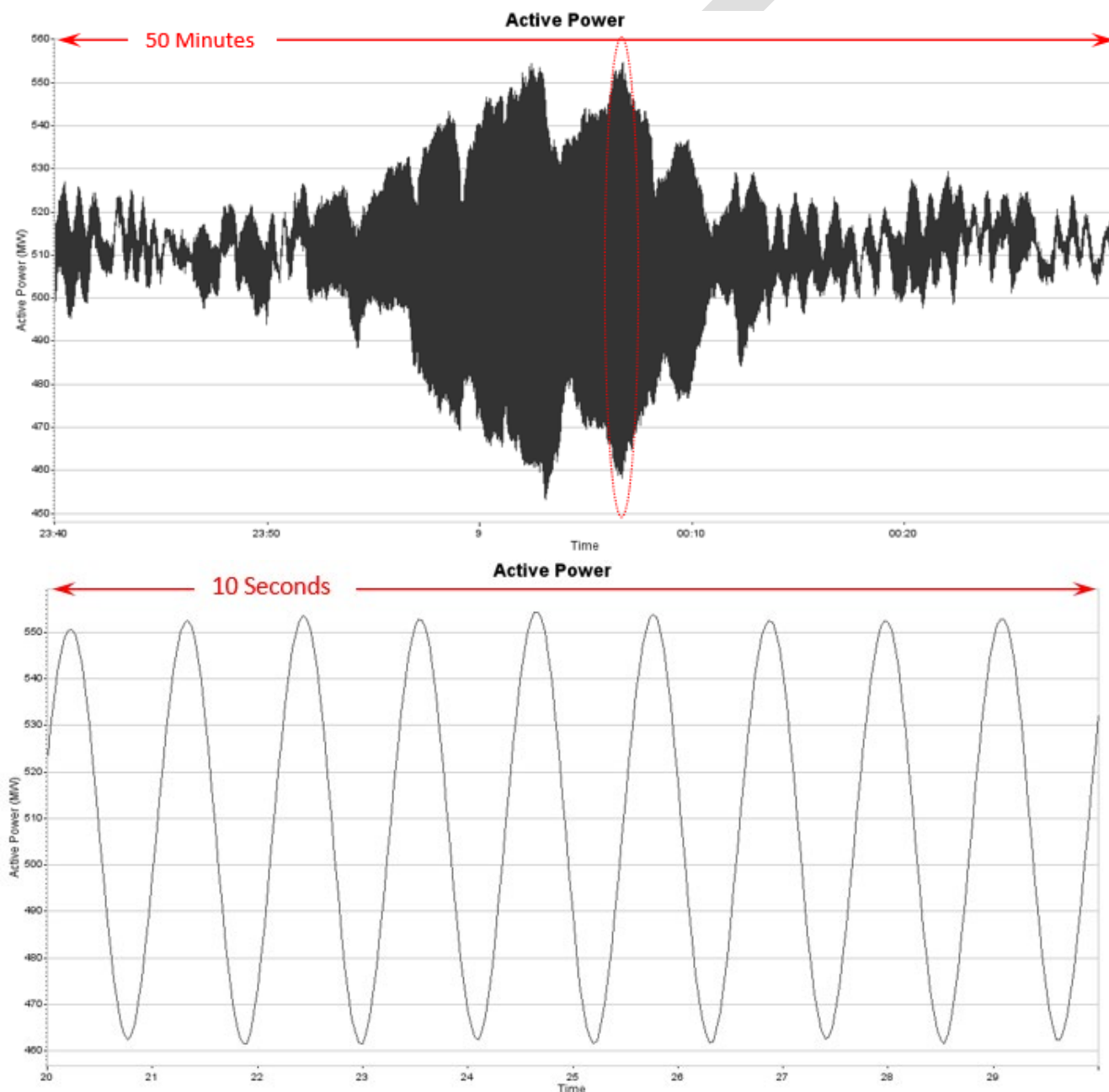
**Figure C.31: Approximately 50 MW Oscillations in the SPP Region  
(Source: OG&E)**

This oscillation persisted for about 45 minutes, causing large-magnitude oscillations across the EI for a prolonged period. Eventually the reactor was shut down and the unit removed from service. No automated oscillation detection tools identified the event to aid in detecting and mitigating the “rogue” input.

## Event 11: Combined-Cycle 1 Hz Oscillation

This forced oscillation originated at a combined cycle plant with two gas turbines and one steam turbine. ISO-NE receives voltage and current PMU signals for the combined output of the plant at the point of interconnection (POI) in the transmission substation. PMU data is streamed at 30 points per second. SCADA data is available for each machine at a rate one point every five seconds. (see [Figure C.32](#)).

PMU data showed a sustained oscillation from this plant almost continuously, usually at low amplitude of about 5 MW peak to peak and at a frequency of around 0.9~1 Hz. The oscillation amplitude would occasionally rise to much higher levels. Below is an example of the plant oscillating for 20 minutes with amplitude reaching a maximum 100 MW peak to peak. SCADA data showed oscillations from each machine. (see [Figure C.33](#)).



**Figure C.32: PMU Data at POI (full event and zoomed in)**  
(Source: ISO-NE)



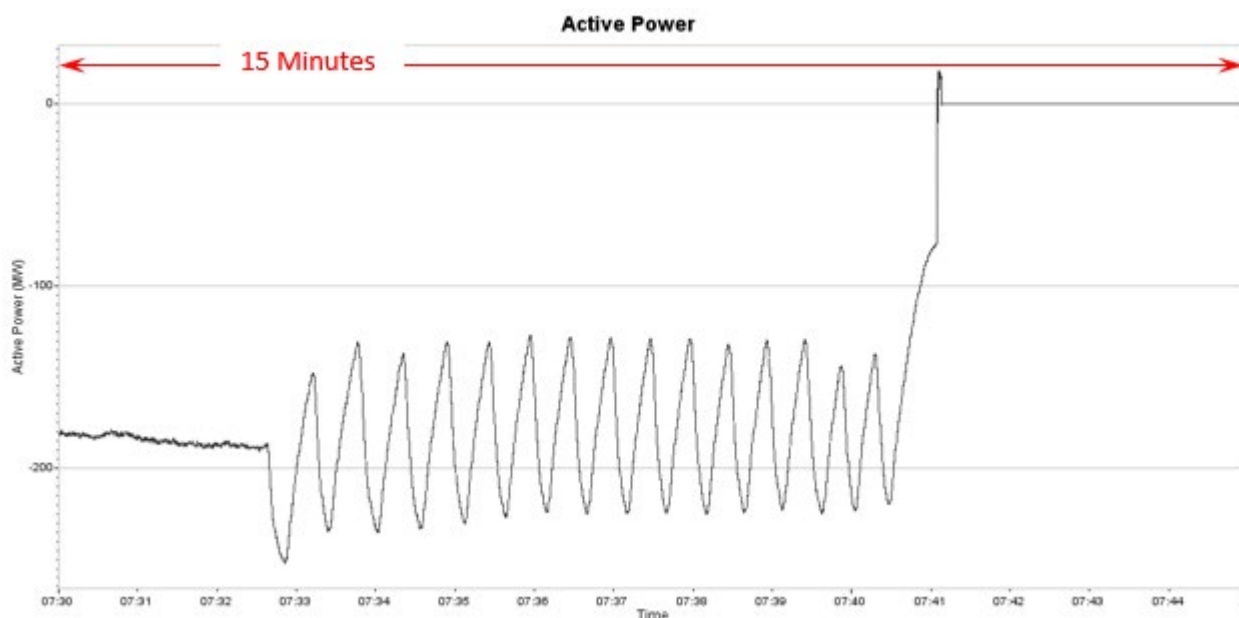
**Figure C.33: SCADA Data from Event  
(Source: ISO-NE)**

### Diagnosis

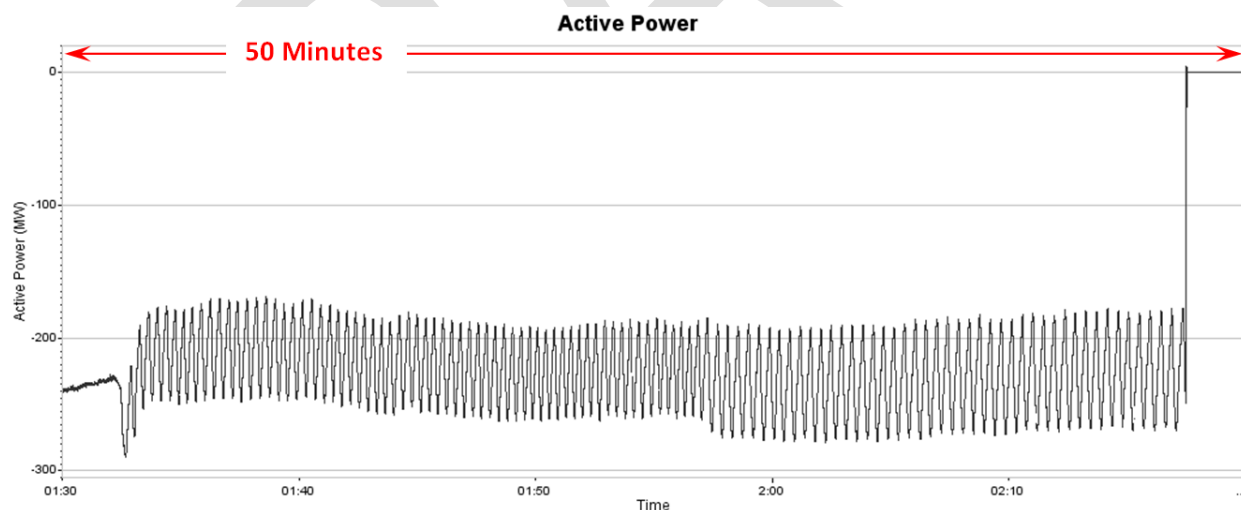
Soon after the PMU at the generator POI was installed in 2013, the real-time oscillation detection feature of the PhasorPoint software identified oscillations at this generating station. ISO engineers subsequently analyzed a number of other times when oscillations at this station grew to relatively high levels. There are several other participating generators in the local 1 Hz mode. However, the subject generator has the most consistent oscillation and especially when other generators were off-line. The oscillation was later confirmed to originate at the subject generating station using the DEF method, an analytic method for locating the source of oscillations adopted and further developed at ISO-NE.

## Event 12: Coal Unit 0.03 and 0.05 Hz Oscillation

A single coal-fired generator experienced a 100 MW peak-to-peak oscillation at 0.03 Hz for eight minutes before the unit was tripped. Three months later, a very similar 70 MW peak-to-peak oscillation at 0.05 Hz was observed for 44 minutes before unit tripping. These are the known oscillation events at this plant. See [Figure C.34](#) and [Figure C.35](#).



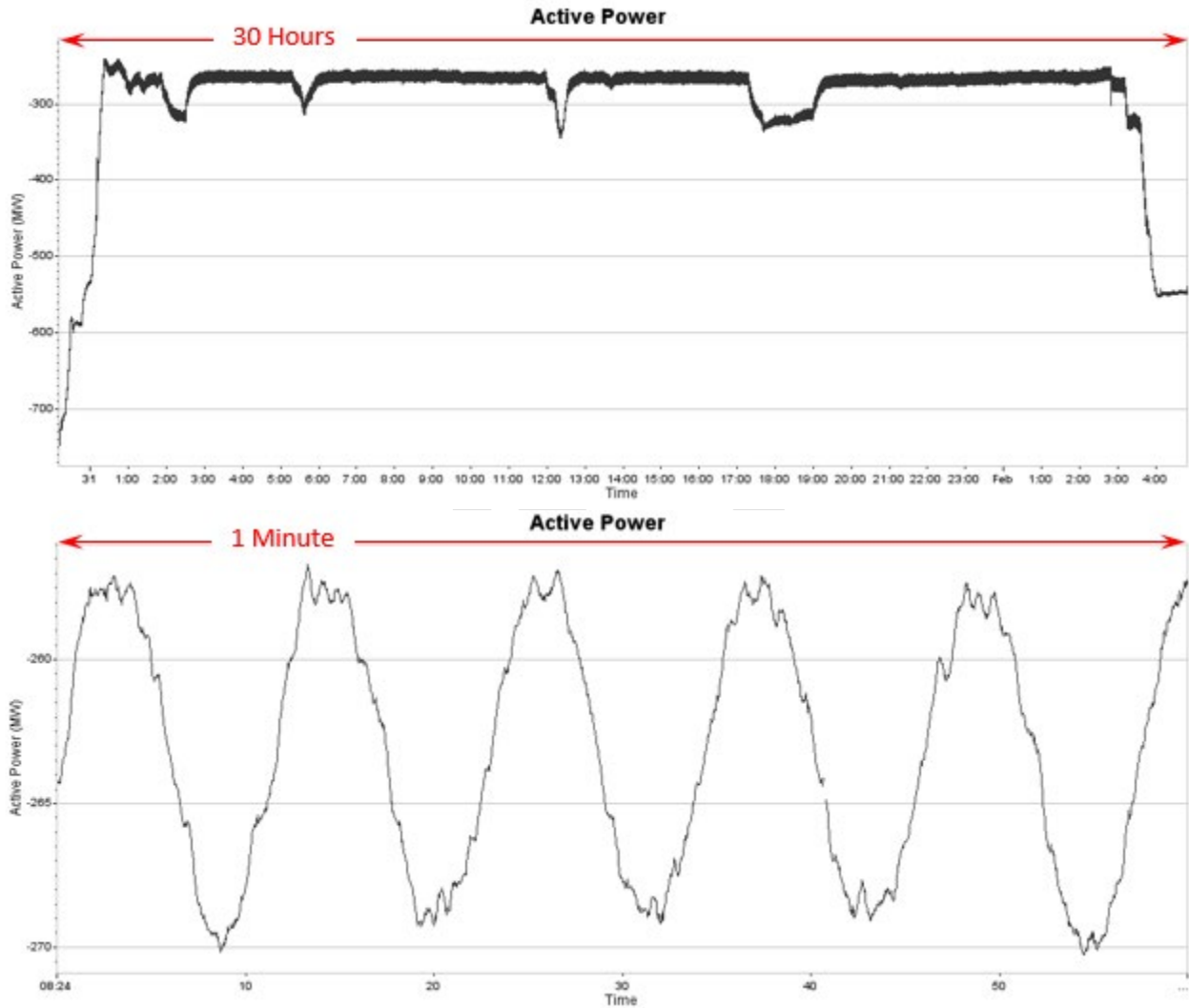
**Figure C.34: Coal Unit 0.03 Hz Oscillation**  
(Source: ISO-NE)



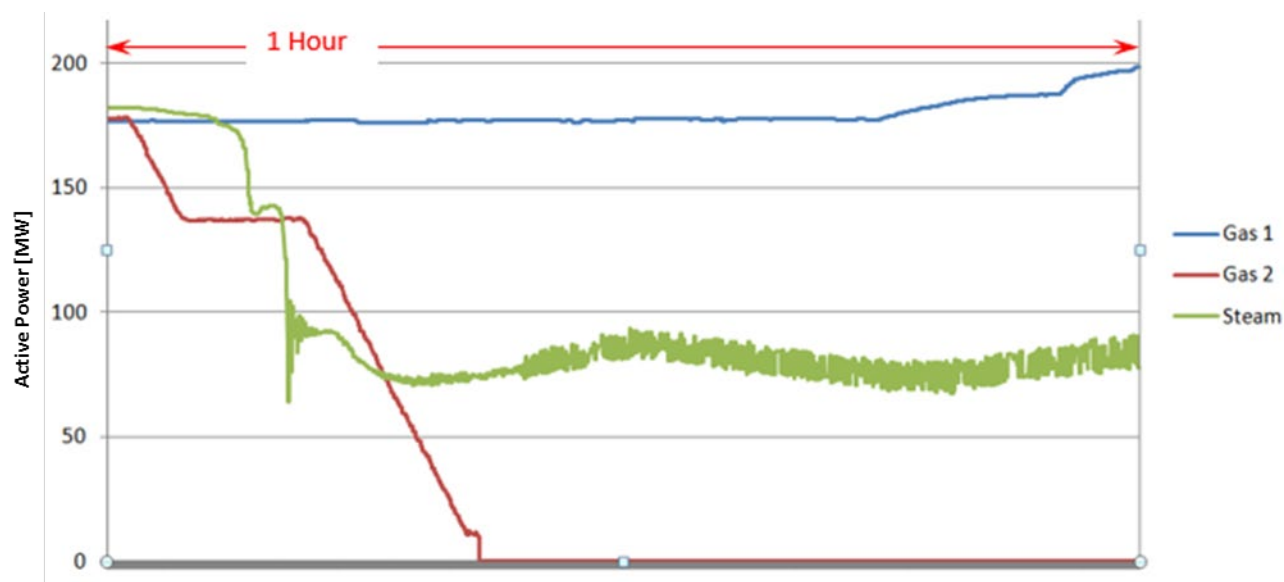
**Figure C.35: Coal Unit 0.05 Hz Oscillation**  
(Source: ISO-NE)

## Event 13: Combined-Cycle Oscillation Events

This combined-cycle generator consists of two gas turbines and one steam turbine. ISO-NE has PMU data at the POI of the plant and SCADA data from each individual generating unit. Relatively large oscillations have been observed from this unit from time to time with different oscillation frequencies. Figure C.36 shows a sustained 0.08Hz oscillation event.



**Figure C.36: PMU Data at POI (Full Event and Zoomed In)**  
(Source: ISO-NE)



**Figure C.37: SCADA Data of Event  
(Source: ISO-NE)**

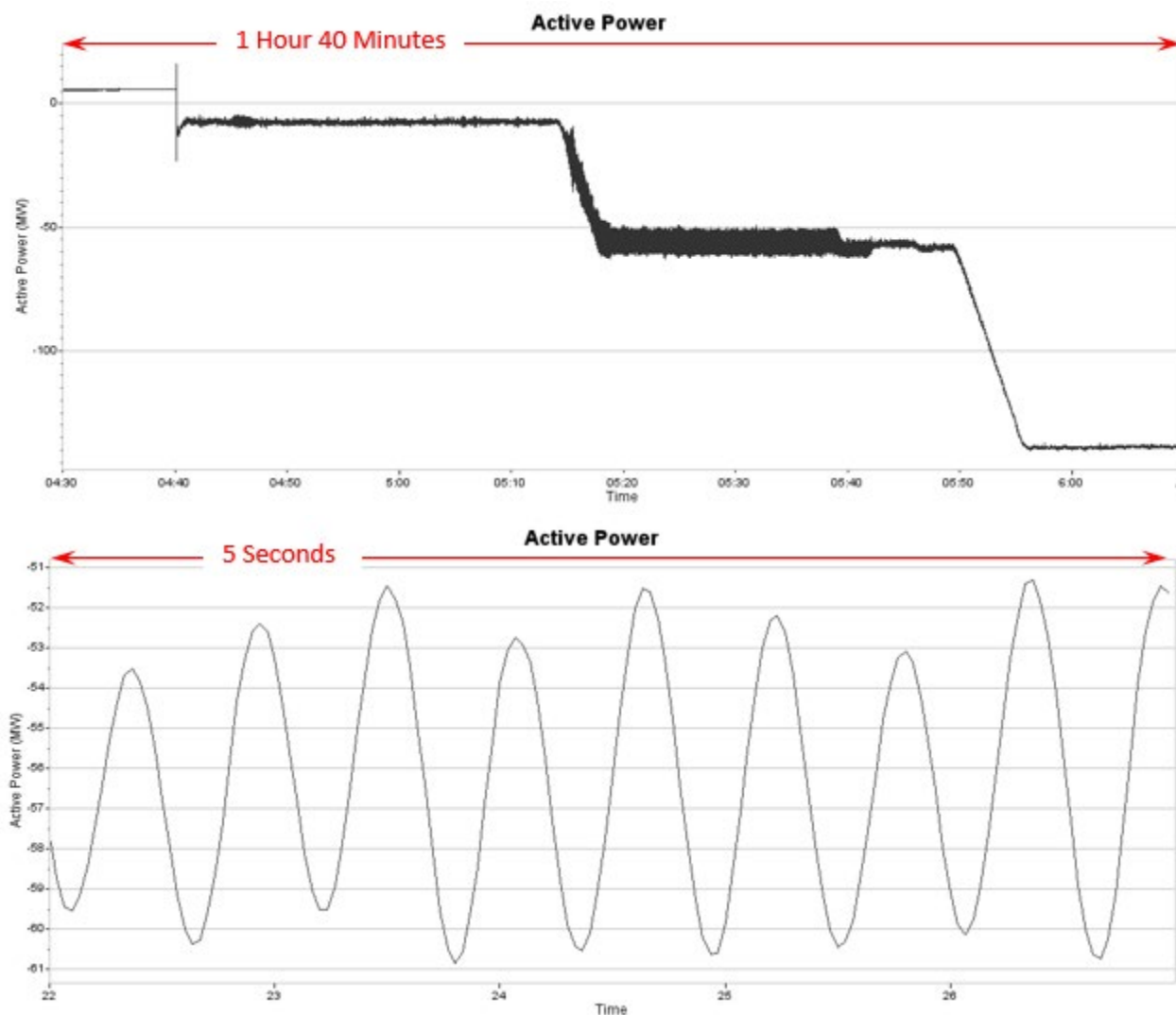
### Diagnosis

Several scenarios were analyzed, and the in-house-developed DEF tool clearly indicated that this unit was the oscillation source. Further, SCADA data (Figure C.37) showed that the steam turbine was the source inside this plant. This oscillation usually happens when one gas turbine is off-line.

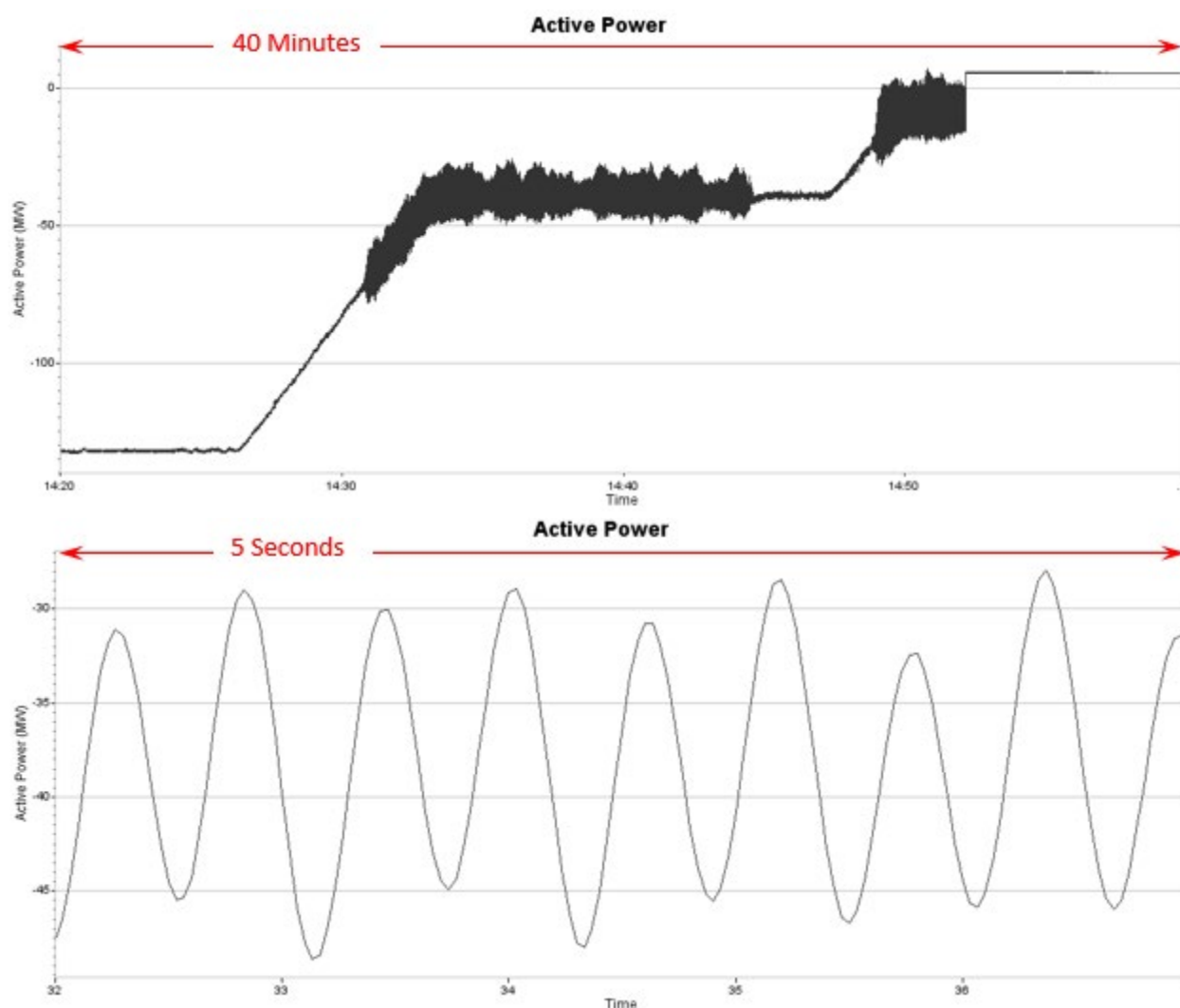
## Two Temporary 1.7 Hz Oscillation Events

The same generator also experienced a 1.7 Hz oscillation during its start-up and shut-down processes.

Figure C.38 illustrates the PMU Data at the POI.



**Figure C.38: PMU Data at POI for Startup Event (Full Event and Zoomed In)**  
(Source: ISO-NE)



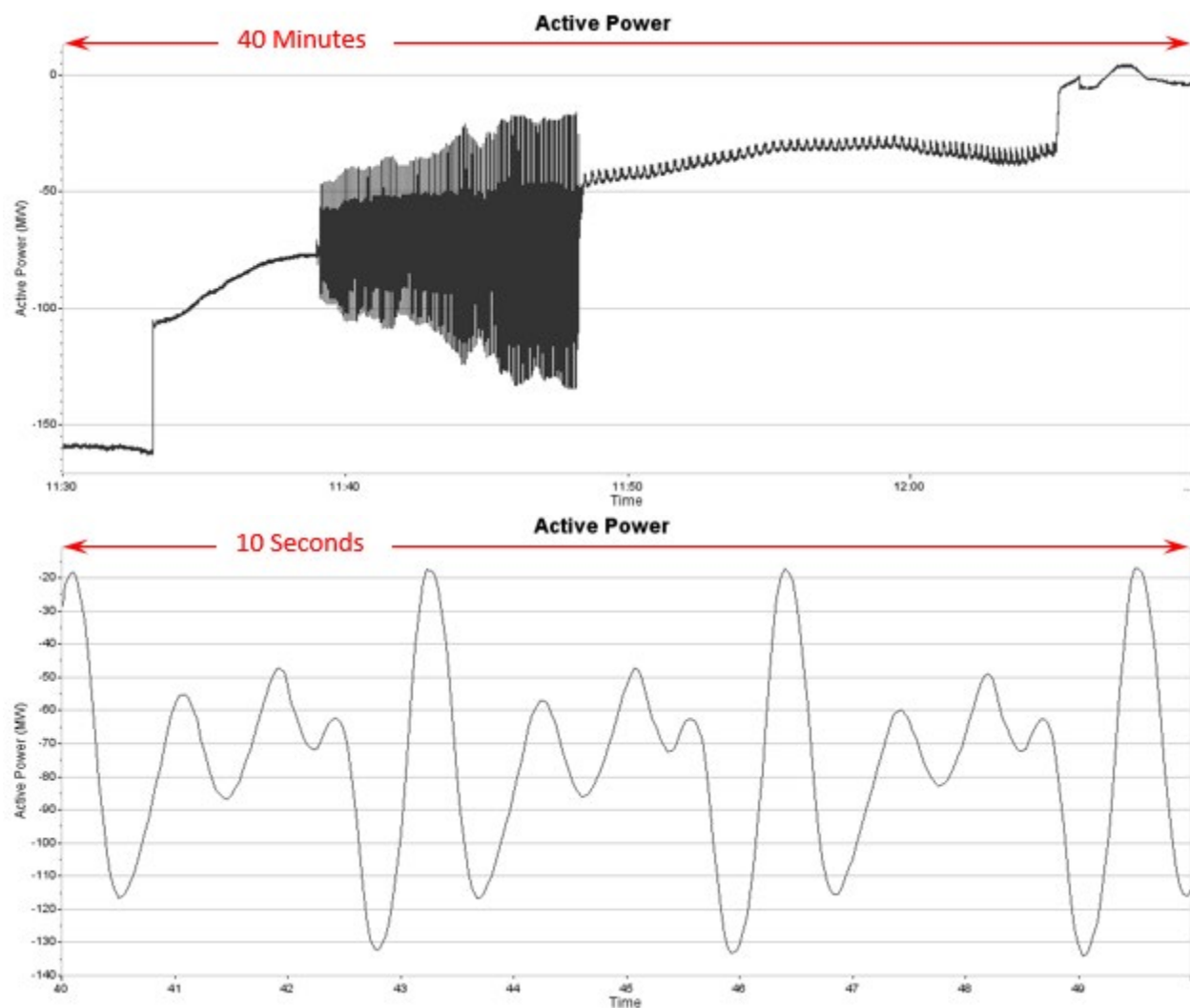
**Figure C.39: PMU Data at POI for Shutdown Event (Full Event and Zoomed In)**  
(Source: ISO-NE)

### Diagnosis

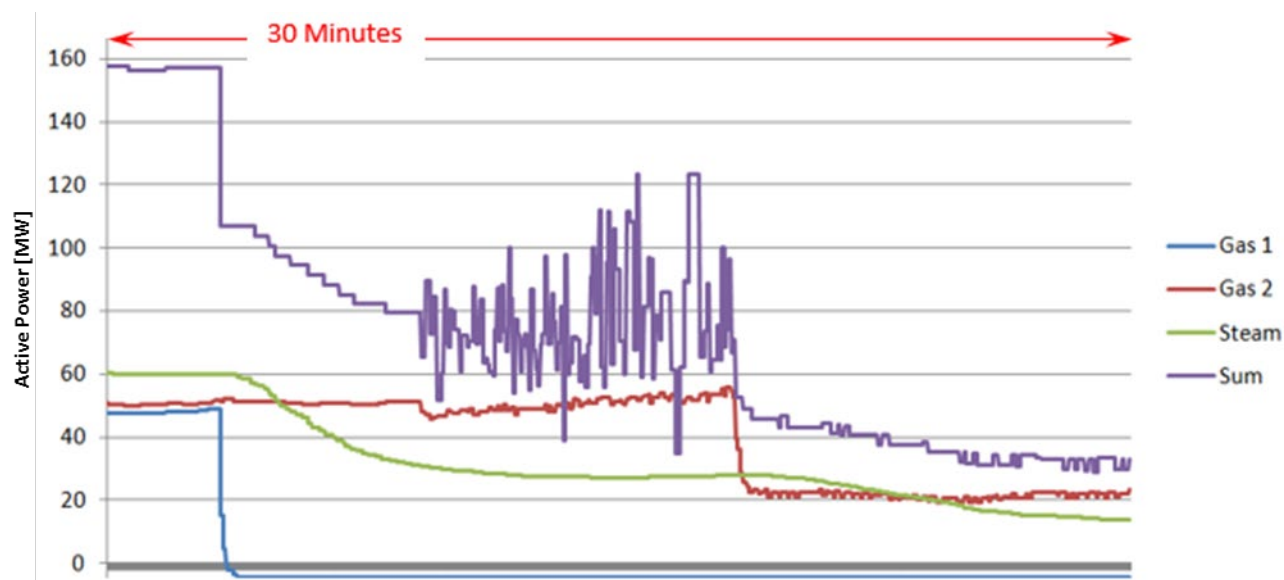
The in-house-developed DEF tool ([Figure C.39](#)) clearly indicated that the plant was the oscillation source. Since there was only one gas turbine operating at the time, it was identified as the sole source inside the plant.

## Event 14: Combined-Cycle Unit with Harmonics

This combined-cycle generator has two gas turbines and one steam turbine. ISO-NE has PMU data at POI and SCADA data for each unit. PMU data showed a growing oscillation in real-time with mixed modes for nine minutes before it disappeared as the plant was shut down (see [Figure C.40](#)).



**Figure C.40: PMU Data at POI (Full Event and Zoomed In)**  
(Source: ISO-NE)



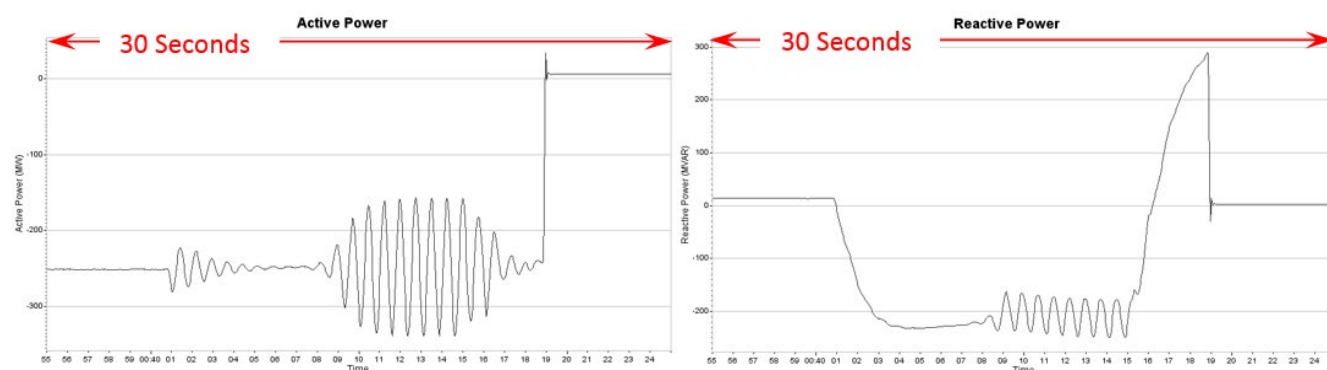
**Figure C.41: SCADA Data of Event**  
(Source: ISO-NE)

### Diagnosis

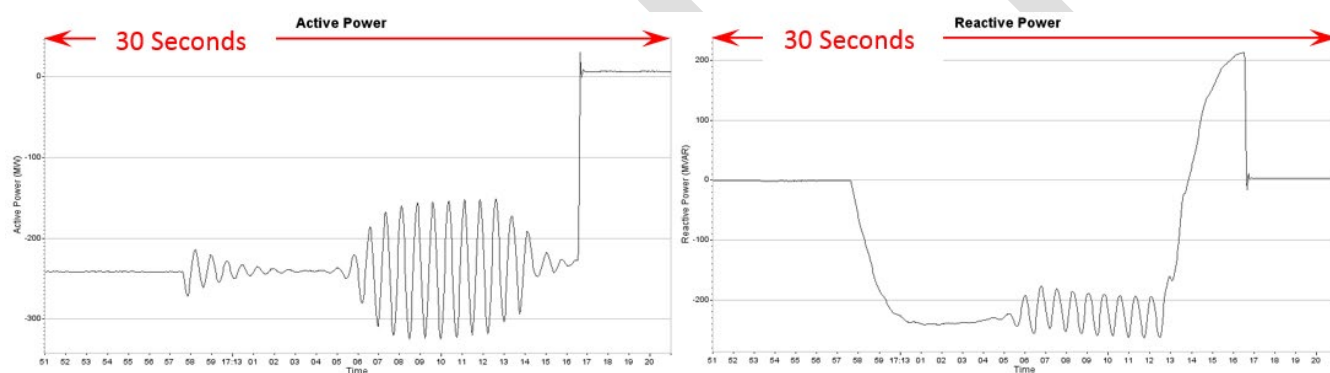
The in-house developed DEF tool clearly indicated that this power plant was the oscillation source. However, the SCADA data (Figure C.41) couldn't tell which unit was contributing inside the plant. This is a typical scenario where combined-cycle units exhibit oscillatory behaviors when one gas turbine is off-line. This also shows the limitation of the SCADA data and the need for synchrophasor measurements from each unit. This is also a typical forced oscillation when the input signal is not sinusoidal; the response is a combination of the base frequency and several harmonics.

## Event 15: Excitation Control System Failure

This single gas generator had two identical incidents of similar excitation failures, experienced two weeks apart from one another (see [Figure C.42](#)).



**Figure C.42: Active and Reactive Power Output during Oscillation #1 (Source: ISO-NE)**



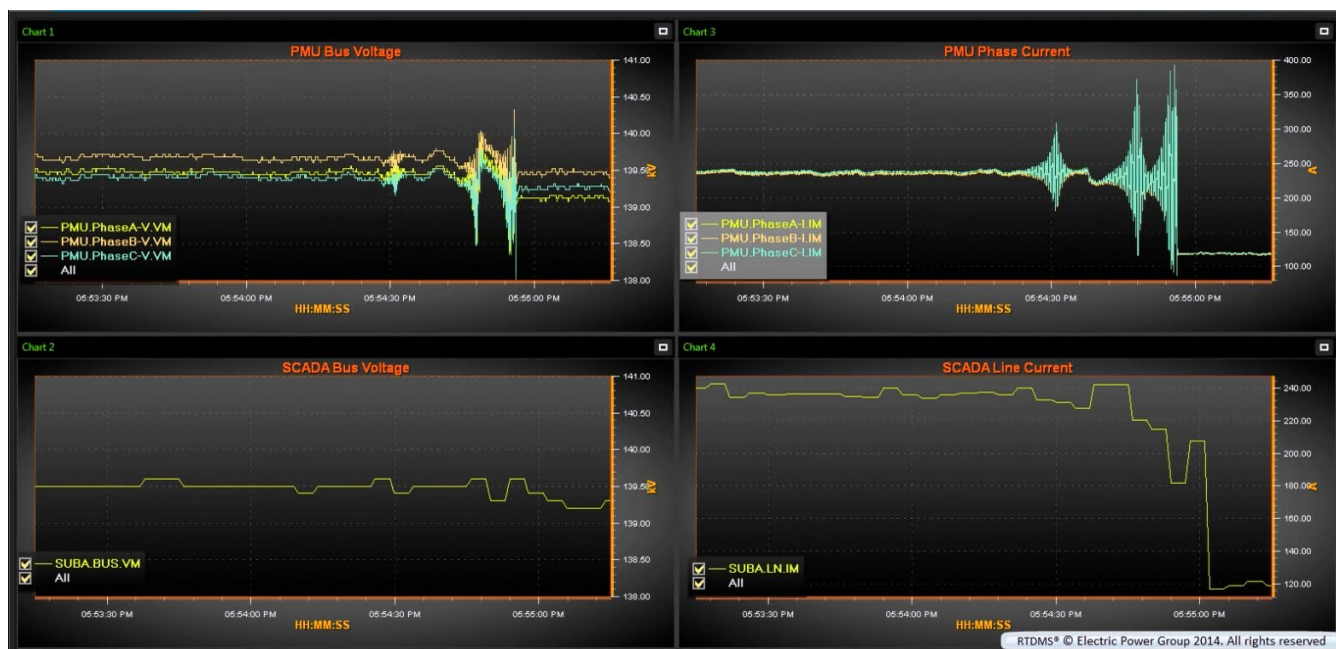
**Figure C.43: Active and Reactive Power Output during Oscillation #2 (Source: ISO-NE)**

### Diagnosis

Operations support engineers received the oscillation alarm and analyzed the event immediately. The reactive power plot (see [Figure C.43](#)) showed that it was an excitation system issue. Initial investigation by plant staff found loose connections to the excitation control system and plant staff repaired them. This repair did not prevent a second identical incident two weeks later. Further investigation identified a faulty component in the excitation control system.

## Event 16: Gas Turbine Oscillation

An oscillation event in the ATC footprint was captured with PMU data. New controls were being implemented for a gas turbine, and the unit began oscillating against the grid, eventually tripping off-line. Figure C.44 shows the event with high-resolution data. The diagnosis and solution to mitigate any controls issues are not known.



**Figure C.44: Gas Turbine Generator Oscillation  
(Source: ATC)**

## Event 17: Southern Wisconsin Oscillations

ATC also identified a 0.75 Hz oscillation using its online monitoring tools. These oscillations persist for extended periods of time on the system with period “notches” where the oscillation will disappear for short durations and then reappear. A new 0.6 Hz oscillation has also developed with the same type of response. ATC reports that both oscillations appear to have disappeared recently on their system, and they monitor for the reappearance of these oscillations for further forensic analysis (see [Figure C.45](#)).

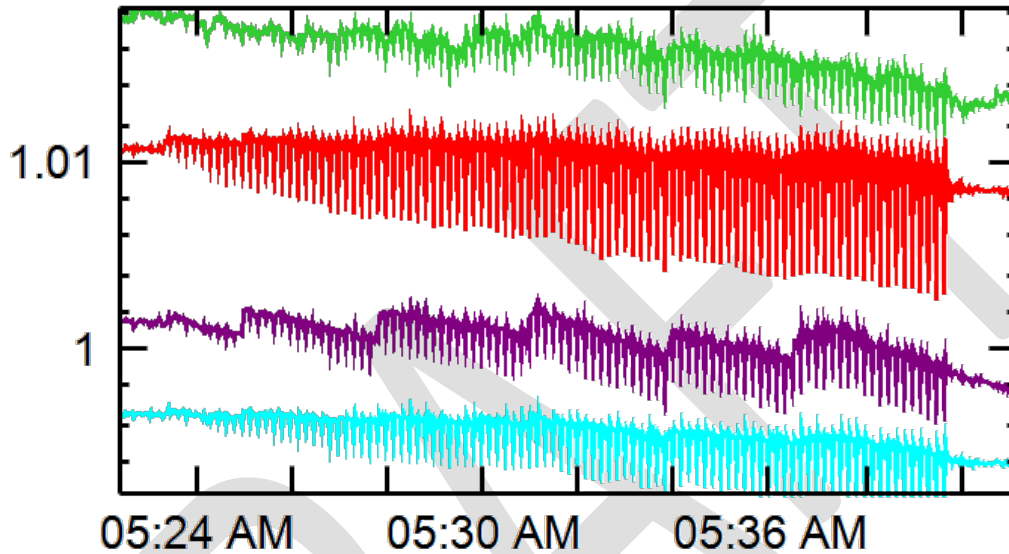


**Figure C.45: 0.6 Hz Oscillation in Southern Wisconsin  
(Source: ATC)**

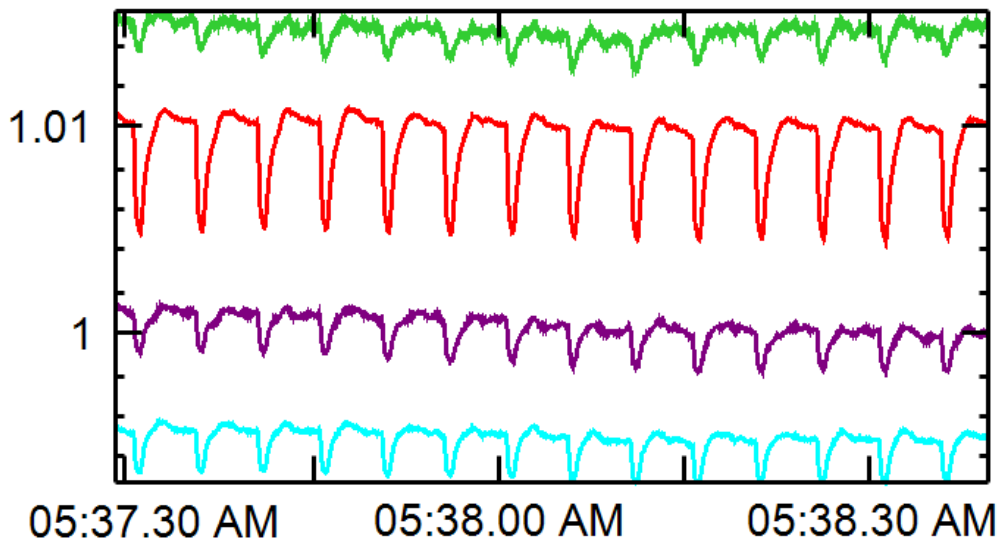
## Event 18: Redbud 0.2 Hz Oscillation

OG&E discovered a voltage oscillation on the EHV system around 0.2 Hz. PMU data quickly identified oscillations in the reactive power plots. OG&E sent snapshots of the event along with the dates and times that it occurred to nearby power plant managers, asking if the oscillations corresponded with any events, testing, or operations at the plant(s). The plant manager at the Redbud plant indicated that they found it was being caused by the Unit 4 VAR control scheme (see Figure C.46). The units start up in VAR control mode and once the unit is synchronized to the grid, it is switched by an operator to voltage control mode. The oscillations stopped when the operator switched to voltage control mode rather than VAR control mode. The plant then worked with the manufacturer to resolve the issue.

### Voltage Magnitude



### Voltage Magnitude

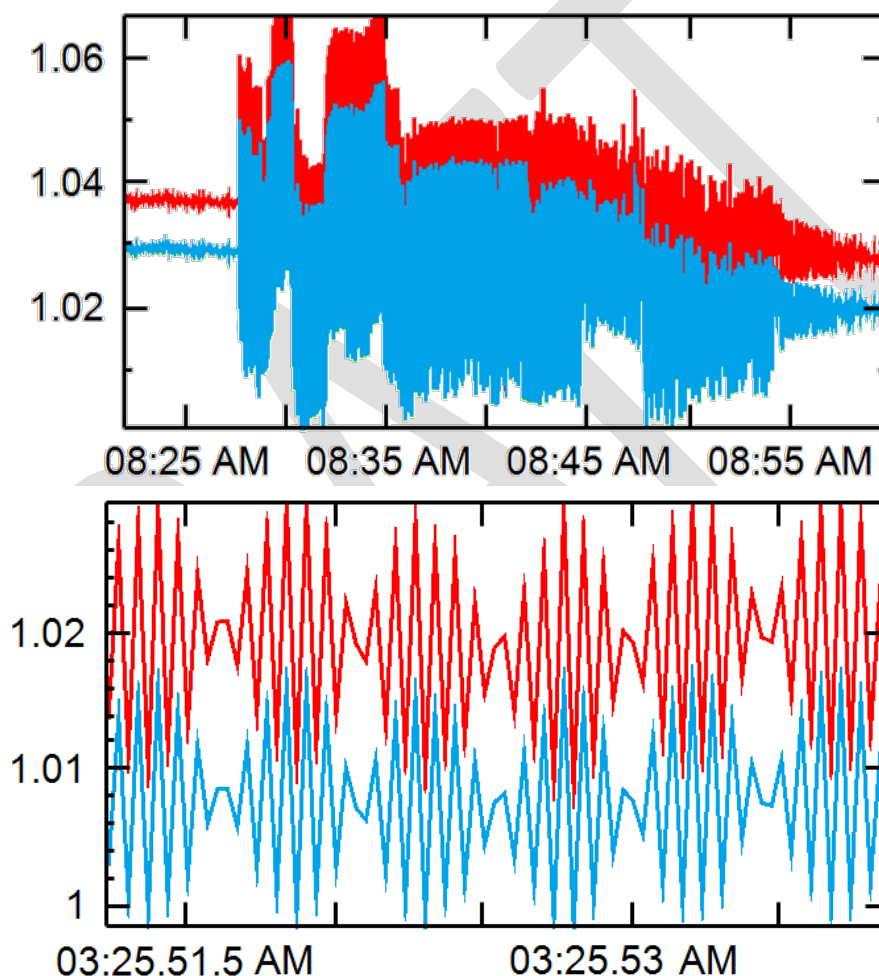


**Figure C.46: Redbud Unit 4 Oscillations in VAR Control Mode**  
(Source: OG&E)

## Event 19: Woodward Wind 13 Hz Oscillation

In December 2010, OG&E started noticing oscillations on the PMUs in northwestern Oklahoma (see [Figure C.47](#)). The oscillations were occurring only when wind farm output was greater than 80%. OG&E performed an FFT analysis to discover sub-harmonic frequencies between 13 and 14 Hz with other components of lesser magnitude. Voltage fluctuations had been as high as 5% at the 138 kV point of interconnection.

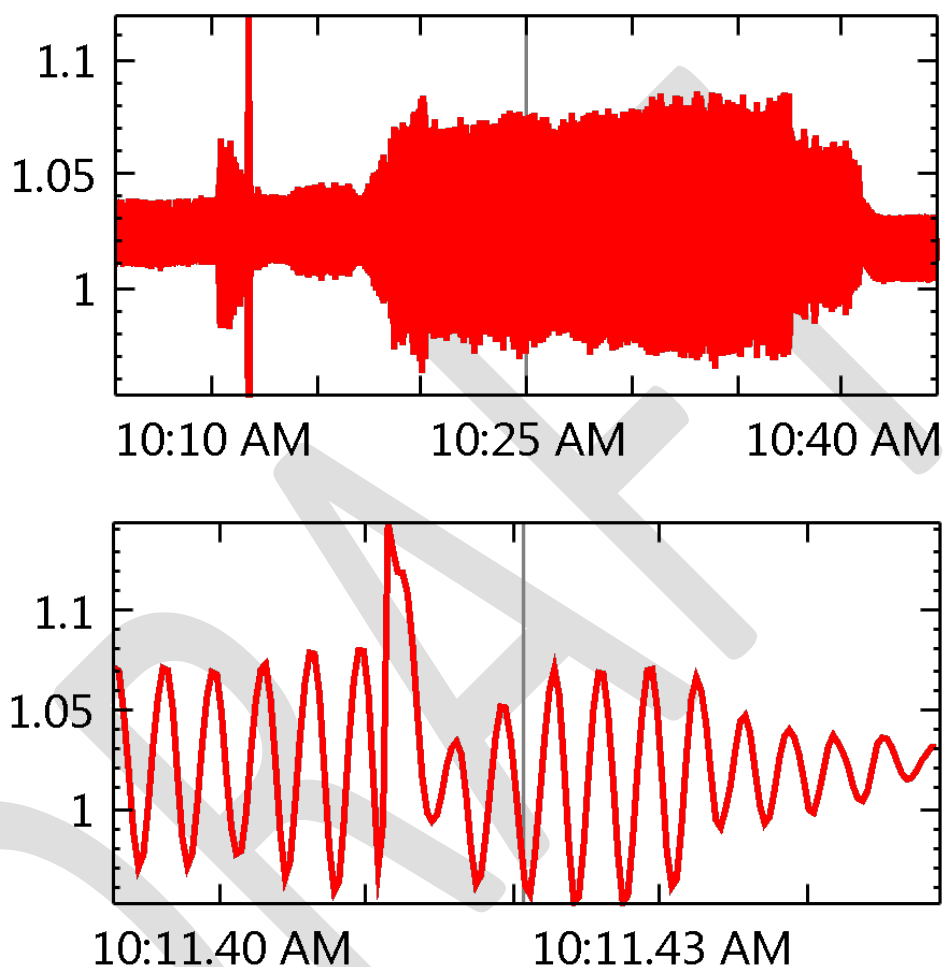
OG&E needed to find out if the oscillations were interactions between wind farms or only at one facility. It performed switching to electrically isolate the wind farms. From this exercise, it was determined to be a problem at two different wind farms that share the same turbine model. The only solution that worked consistently was to curtail the plant's output until the manufacturer made modifications to the power conversion system.



**Figure C.47: Voltages [pu] during 13 Hz Oscillation at Wind Plant  
(Source: OG&E)**

## Event 20: 3 Hz Wind Oscillation

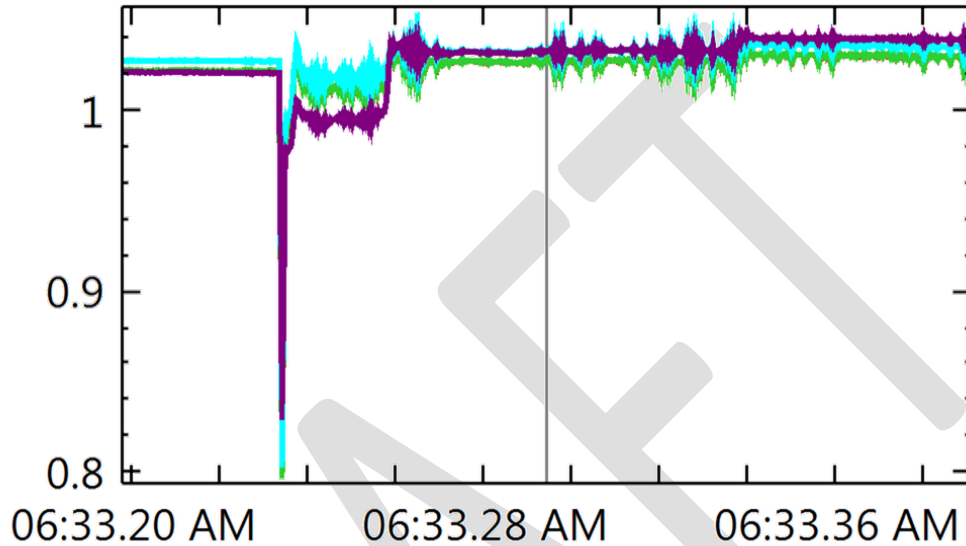
In December 2012, a line outage instigated a major forced oscillation at a new 60 MW wind farm interconnected at 69 kV. (Figure C.48) Voltage variations were as high as 18% at a frequency of 3 Hz. It took about half an hour to curtail the plant and restore the system to normal operation. The event adversely affected the operations at a nearby large industrial customer. OG&E was able to work with the manufacturer to test new power conversion parameters that successfully resolved the issue.



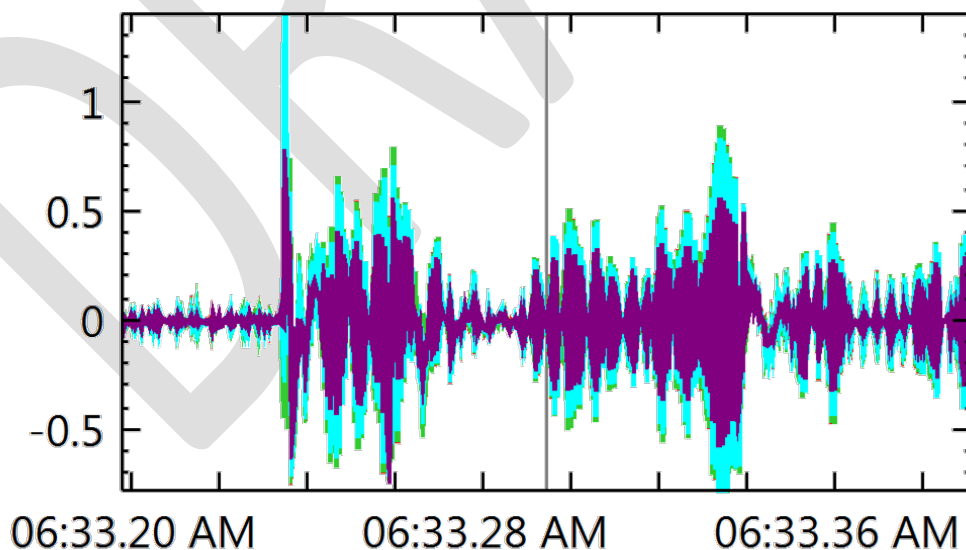
**Figure C.48: Voltage [pu] during 3 Hz Oscillation at Wind Plant  
(Source: OG&E)**

## Event 21: Wind Plant and Impedance

On December 14, 2011, a transmission line was out of service for maintenance when a permanent fault occurred on another line in the area (see [Figure C.49](#) and [Figure C.50](#)). The loss of the second line caused wind farms to begin oscillating, and this persisted for about half an hour until the plant's output was curtailed. Wind plant events are often triggered by something external, such as line faults or changes in topology. In this case, the change in electrical impedance of the system and reduction of system strength caused the oscillation, demonstrating the power conversion controls sensitivity to changes in grid topology and strength. Weaker POIs are more prone to issues with forced oscillations caused by electronically coupled resources.



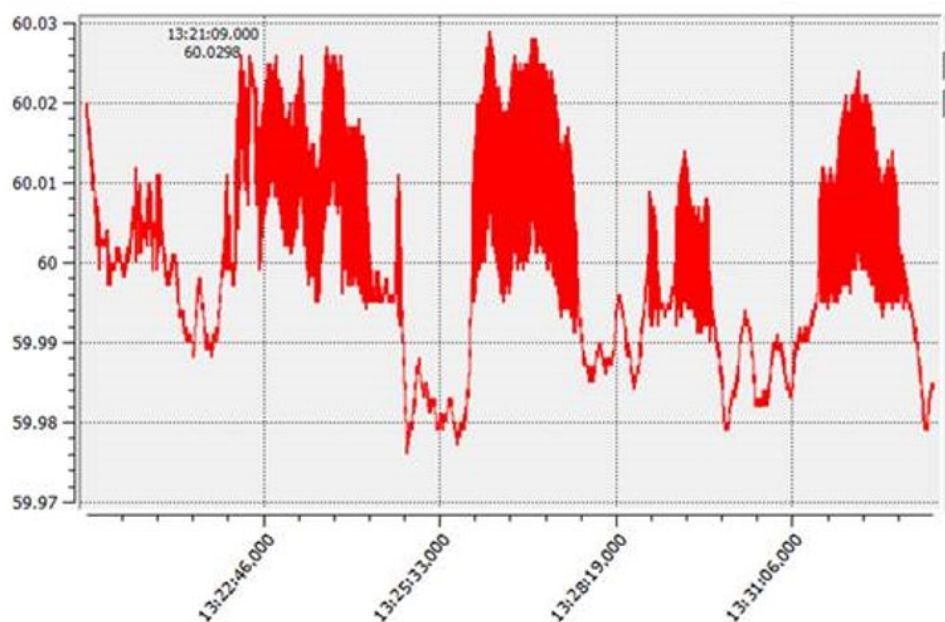
**Figure C.49: Voltage [pu] during Forced Oscillation Instigated by Topology Change**  
(Source: OG&E)



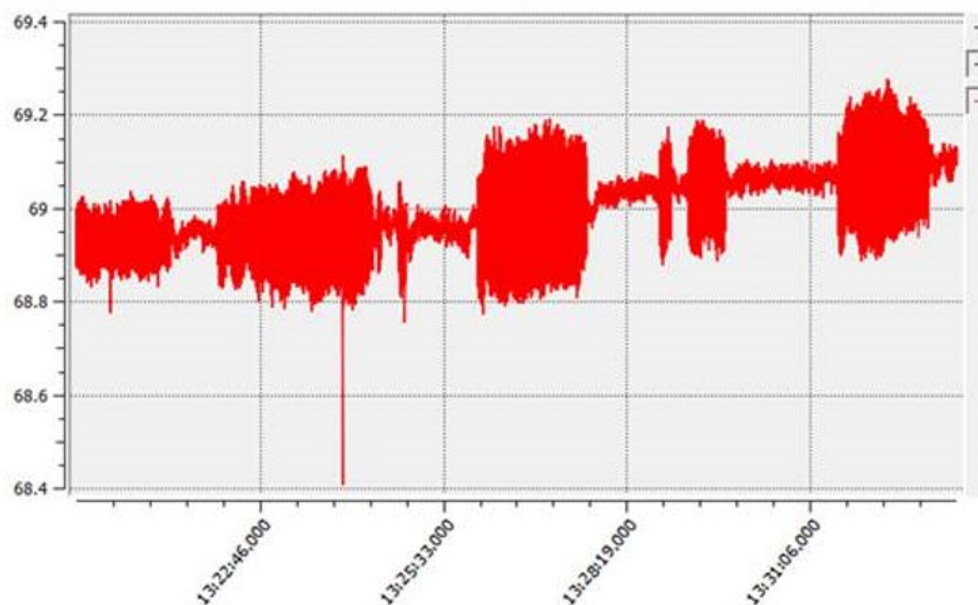
**Figure C.50: Rate-of-Change-of-Frequency [Hz] during Event**  
(Source: OG&E)

## Event 22: Solar Power Plant Oscillation

In November 2016, a forced oscillation was detected in the AEP footprint from PMU data monitoring a solar power plant. Figure C.51 and Figure C.52 shows the frequency (top) and voltage (bottom) data over multiple days. It was clear that this oscillation was occurring during the middle of the day during periods of irradiance (and likely solar output). AEP identified this oscillatory behavior and was working to address it.



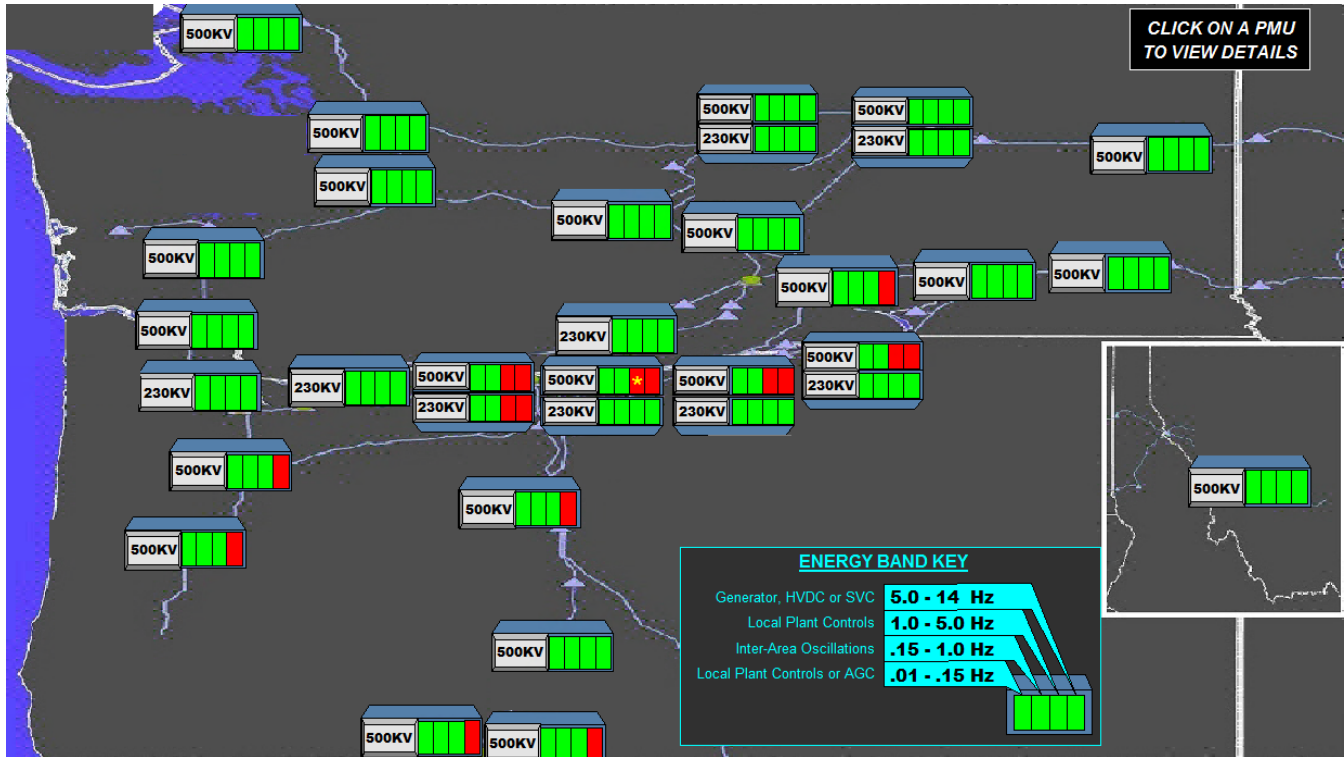
**Figure C.51: Measured System Frequency [Hz] Oscillation at Solar Power Plant (Source: AEP)**



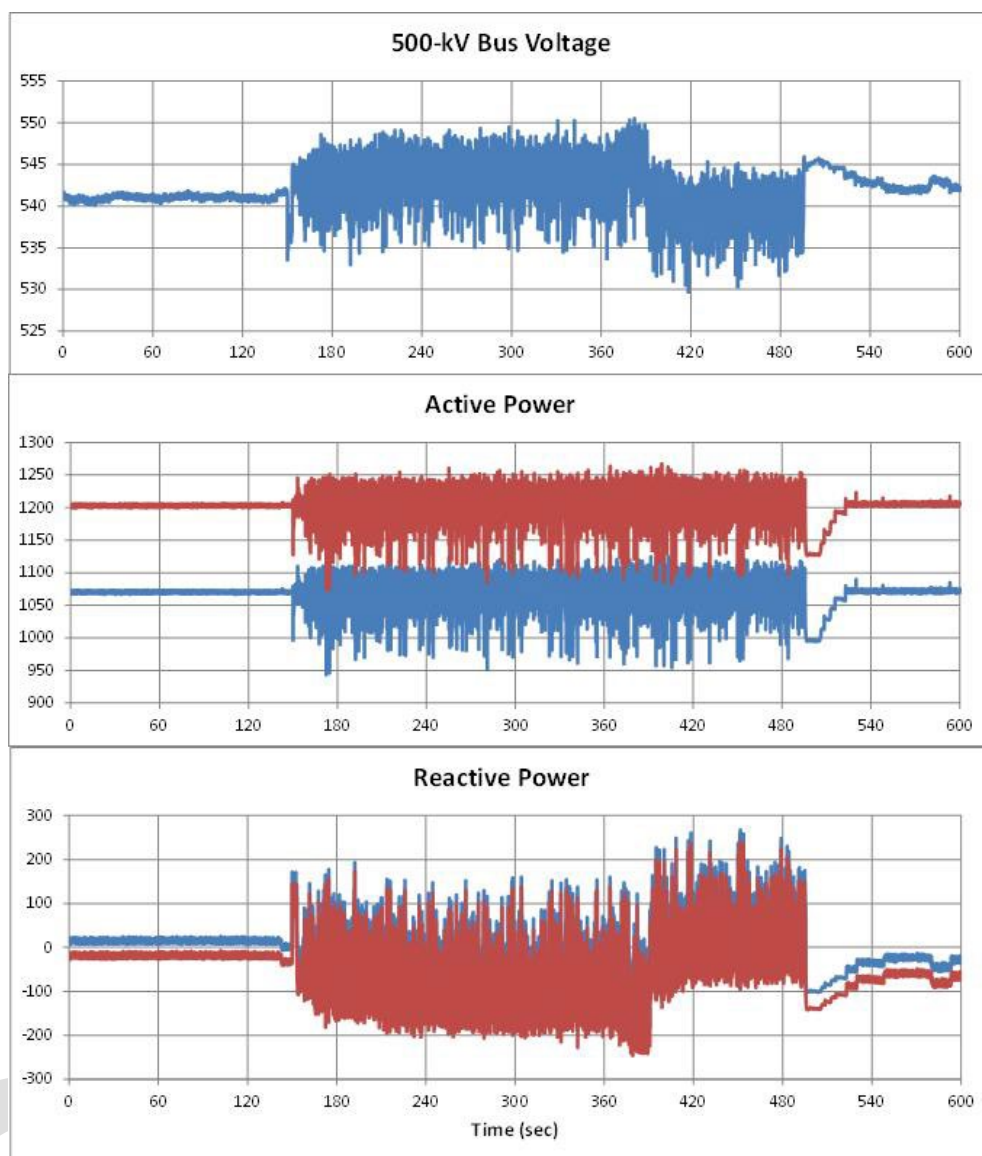
**Figure C.52: Measured System Voltage [kV] Oscillation at Solar Power Plant (Source: AEP)**

## Event 23: HVDC Oscillation

In April 2016, a high-energy forced oscillation originated at the Pacific HVDC Celilo terminal in Oregon and propagated across the Pacific Northwest system. The oscillation triggered multiple alarms, mainly in oscillation bands 3 and 4 (see [Figure C.53](#)). Further investigation showed that both DC poles were involved in the oscillation (see [Figure C.54](#)). The oscillation was due to equipment failure at Sylmar converter station at the southern end of the DC line near Los Angeles.



**Figure C.53: Overview Display for PDCI Event in April 2016**  
(Source: BPA)



**Figure C.54: Analysis of PDCI Event in April 2016  
(Source: BPA)**

## Event 24: Type 2 Wind Plant Oscillations

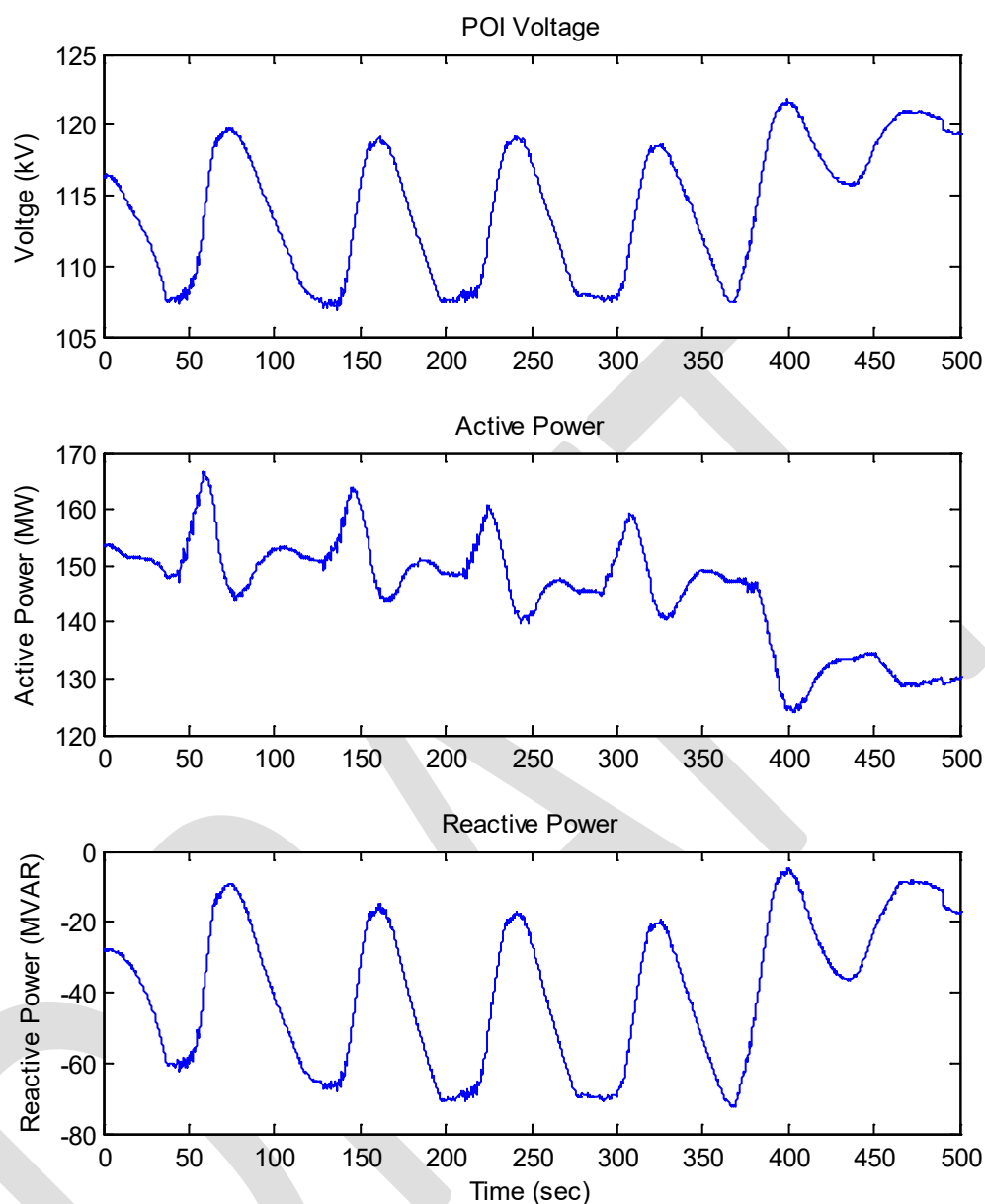
BPA has a Type 2 wind power plant connected in its Central Washington region on the 115 kV system. The project is divided into two plants of 74 and 83 MW. The project uses variable speed (Type 2) induction generators and has been in operation since November 2005.

The wind plant substation short-circuit ratio (SCR) is about 1.0 with all lines in service. Several events of voltage instability were observed over the years, mostly with all lines in service. Figure C.55 shows recordings of a voltage instability event on July 12, 2012. For no apparent reason, voltages and active and reactive power started oscillating with an 80-second period. Voltage fluctuations reached 10% peak to peak, and reactive power swings were 45 MVARs. The oscillations went away when the wind power plant output was reduced.

Following the event, the wind plant project was curtailed to operate at reduced power output.

BPA and neighboring utility engineers performed model validation studies for the event. The powerflow model included the collector system equivalent and Wind Turbine Generator (WTG) step-up transformers. Powerflow model parameters, collector system equivalent, and active and reactive power flow in each of the transformers were tuned using SCADA recordings of voltages at 115 kV and 34.5 kV substations. However, powerflow studies were unable to re-create the event, and dynamic simulations were required to gain an understanding of what occurred during the event. Dynamic simulations did not have validated generic dynamic models for Type 2 generators. While generic Type 2 WTG models reproduced the observed phenomenon in principle, the details of active power were not well matched. Nevertheless, the model was adequate to determine the causes of the voltage instability and to identify the corrective measures.

Rotor resistance controls are used in Type 2 turbines to optimize active power production. However, should turbine voltages decline, the controls slowly adjust the resistance and increase reactive power consumption by the turbine. The increased reactive power consumption would cause further voltage decline, followed by additional adjustments of rotor resistance, thereby creating positive feedback for voltage collapse. Substation transformer load tap changers (LTC) began tap adjustments when the collector bus voltages became low. LTC action restored collector bus voltages, and then the rotor resistance control started operating in the opposite direction, thereby reducing the reactive power demand and causing the overvoltages. It was concluded that the voltage instability was due to the actions of the rotor resistance controls. LTC interactions prevented a voltage collapse and limited the event to an oscillatory voltage instability.



**Figure C.55: Wind Plant Voltage Instability Event on July 12, 2009**

Joint studies concluded that a STATCOM was required to provide continuous voltage control. After the STATCOM was added, the voltage stability issues at the plant were mitigated.

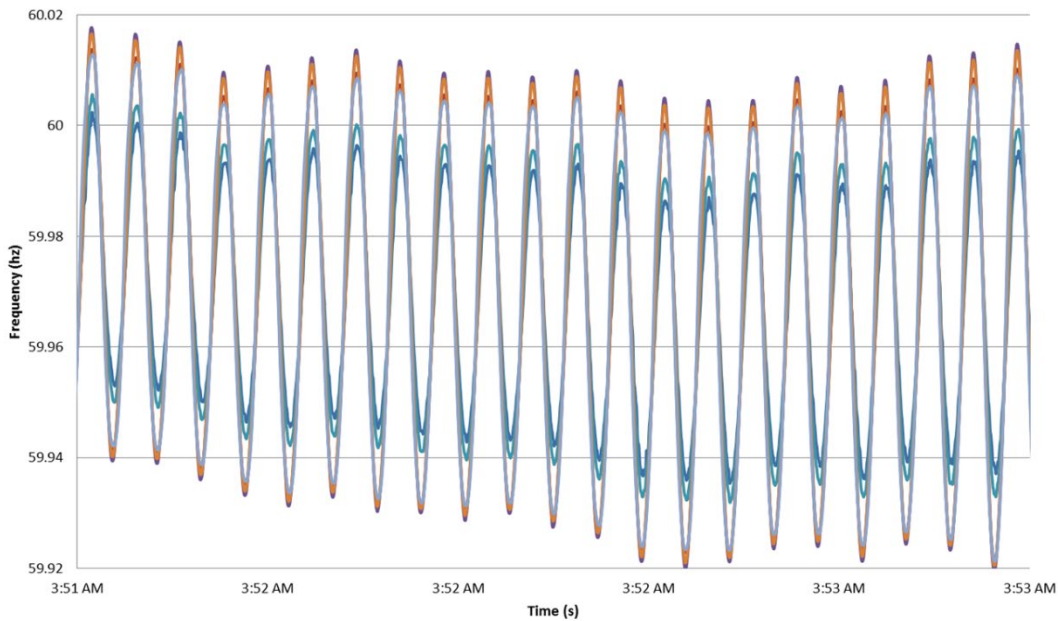
### Event 25: January 2019 Wide-Area Oscillation in the EI<sup>66</sup>

On January 11, 2019, a steam turbine at a 4x1 combined cycle power plant in Florida experienced a faulty input to its power load imbalance (PLI) control system that resulted in an approximately 200 MW peak-to-peak oscillation that persisted for around 18 minutes before the plant operator manually removed the unit from service. The 0.25 Hz oscillation (about a four-second periodic behavior) interacted with the natural system modes of the EI. For this reason,

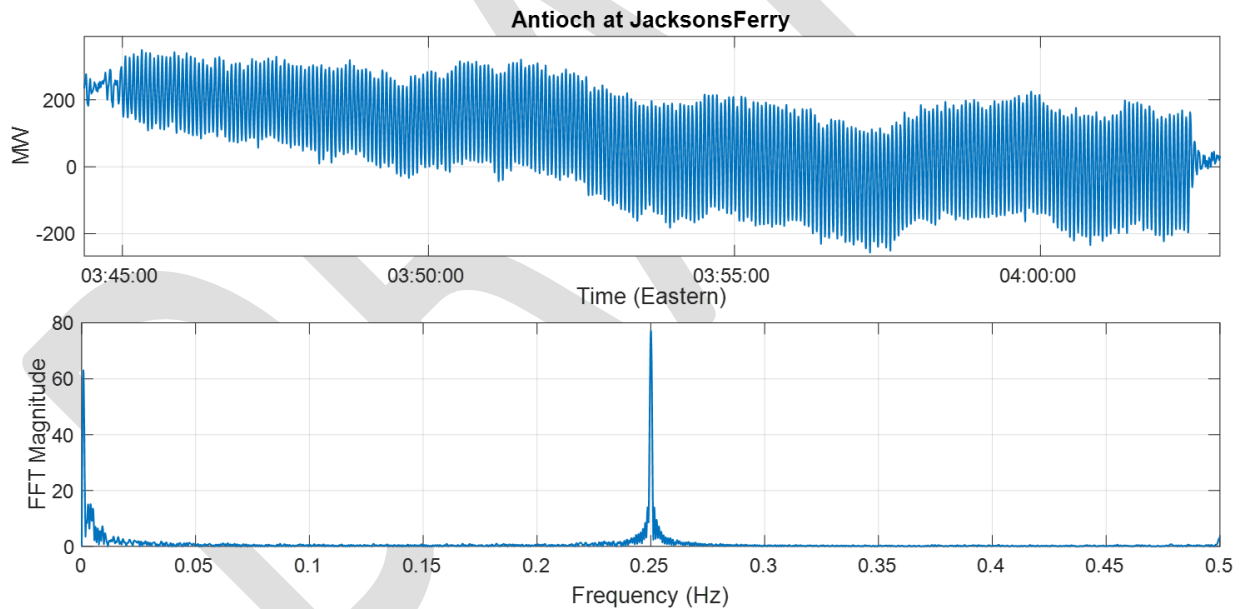
<sup>66</sup> The content in this section is a summary of the NERC report available here:

[https://www.nerc.com/pa/rrm/ea/Documents/January\\_11\\_Oscillation\\_Event\\_Report.pdf](https://www.nerc.com/pa/rrm/ea/Documents/January_11_Oscillation_Event_Report.pdf).

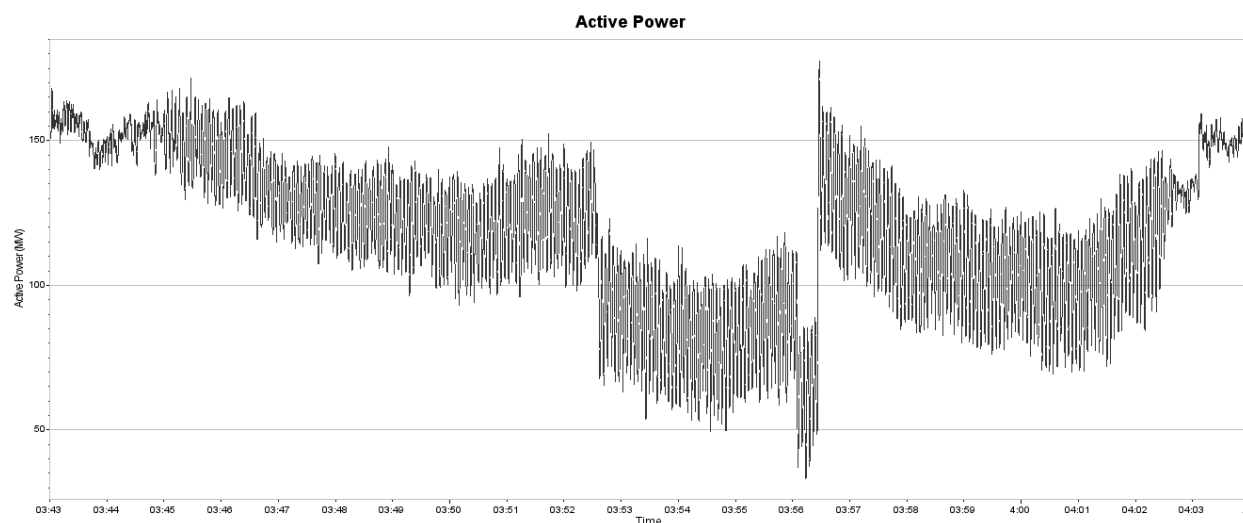
the oscillation propagated through the entire EI, impacting entities across the EI. Several PMU recording examples are shown in [Figure C.56](#), [Figure C.57](#), and [Figure C.58](#).



**Figure C.56: Frequencies in the Florida Power and Light (FPL) Footprint**



**Figure C.57: Active Power Flow in AEP Footprint (Top) and Corresponding Spectrum (Bottom)**



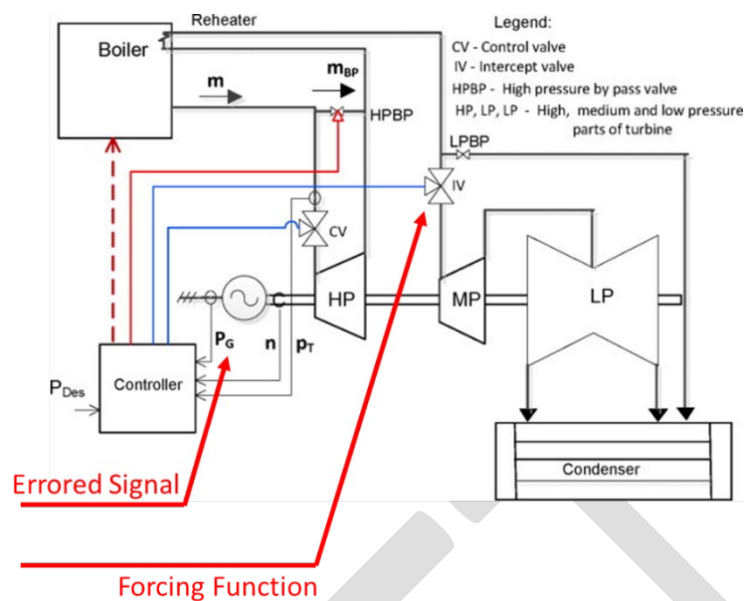
**Figure C.58: Active Power Flow in ISO-NE Footprint**

### Root Cause

The oscillation was caused by a perceived mismatch between electrical output power and steam flow (input power), a function of the PLI control. The PLI control interfaces with the fast-closing actions on the intercept valves. Refer to [Figure C.59](#) for a simplified control diagram, a marked error signal, and the location of the forced oscillation caused by the controls. The perceived mismatch was driven by the failure of a PT connection feeding the calculation of generator electrical power output.

When an unbalance is identified by the control system, the control system attempts to remedy the unbalance by controlling the valve position. The controls were programmed to only act after a 20% imbalance. When the PLI threshold is reached, the controls engage the reheat intercept valves. The valves go through an open/close cycle with a predefined period. Upon the closing of the reheat intercept valves, the electrical power output drops quickly, and the imbalance between input and output power is eliminated since both approach zero. Once the threshold is met, a timer is started and once exceeded when the valves are released. The PLI controls then reopen the valves, and the unit can again quickly ramp up power output. However, since the PT measurement still failed, the PLI controls again perceive a mismatch, and the process is repeated on a continual basis. The periodicity of this closing and reopening of the intercept values was approximately four seconds.

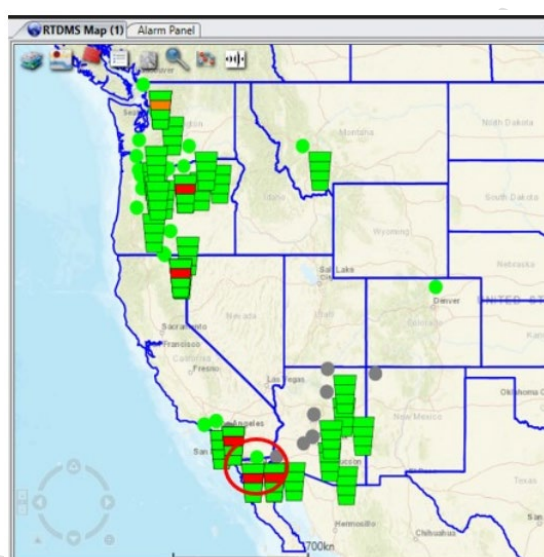
The 0.25 Hz forced oscillation interacted with the natural system mode near that frequency, causing the entire EI to experience the forced oscillation. Two out of the three conditions required for a forced oscillation to strongly resonate with a natural system mode were satisfied: The oscillation frequencies between the forced oscillation and the natural system mode matched, and the source location was in a high participation area of the natural system mode; fortunately, the natural system mode was well-damped.



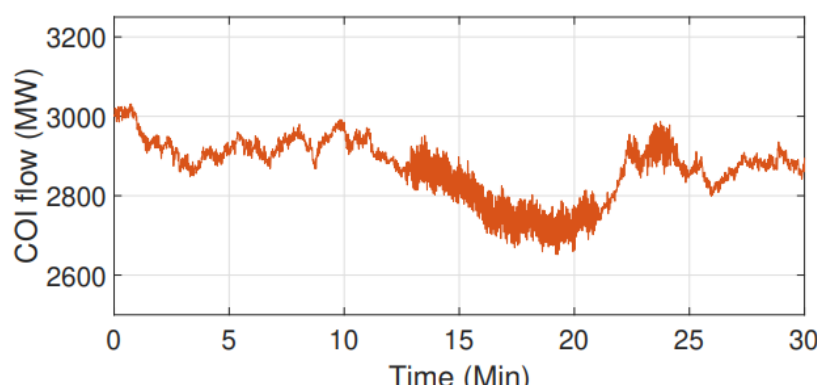
**Figure C.59: Basic Illustration of PLI Failure Mode**

## Event 26: January 2022 BESS Oscillation Event in the WI<sup>67</sup>

A system-wide forced oscillation event occurred in the RC West footprint on the 27<sup>th</sup> and 28<sup>th</sup> January 2022. The RTDMS snapshot showing areas in which this forced oscillation was observed is shown in [Figure C.60](#) with the approximate source location enclosed within the red circle. During the event, high power swings were observed in the North and the South of the WI with more than 200 MW of power swings observed in the California-Oregon Intertie (COI) flow as shown in [Figure C.61](#). The RC operators identified the source of the oscillations to be two battery units in the Southern California region in a timely manner, which then helped mitigate the oscillations. The output of the two battery units is shown in the [Figure C.62](#).



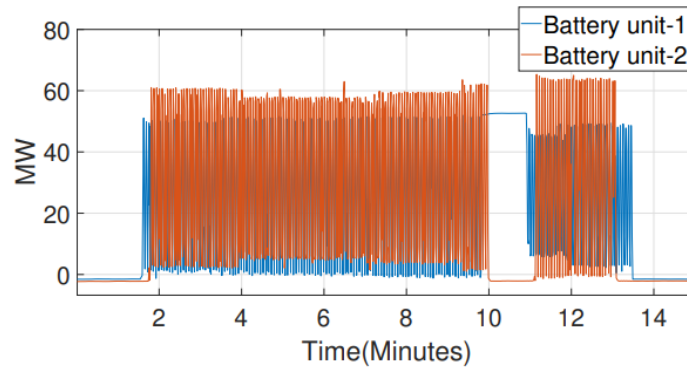
**Figure C.60: RTDMS Snapshot Showing the Extent of the Forced Oscillation Event in the WI**



**Figure C.61: 200 MW Power Swings Observed in the COI Flow during the Event**

<sup>67</sup> The content in this section is a summary of the following paper:

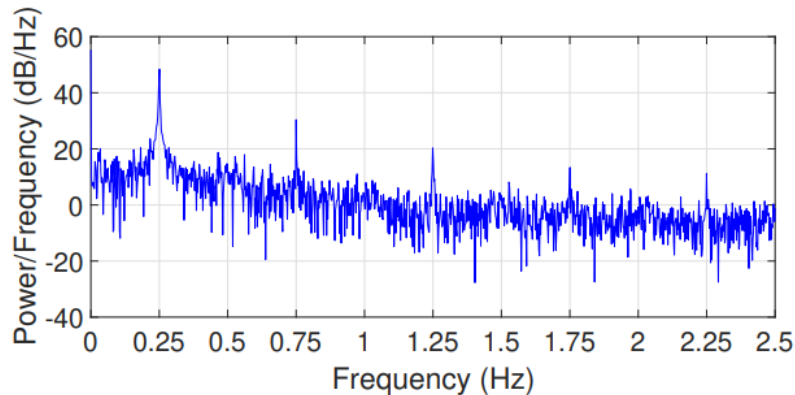
Urmila Agrawal, Jim Hiebert, Justin Bohlen, "RC West's (California ISO) Experiences with Monitoring and Mitigation of Forced Oscillations," 2023 IEEE PES GM – Submitted



**Figure C.62: Power Output Measurements of the Two Battery Units Causing the Forced Oscillations Across the WI**

### Off-line Analysis of the Event

Figure C.63 shows the periodogram calculated using the source battery power output. As seen in this figure, the frequency of the forced input was 0.25 Hz and consisted of odd harmonic components at 0.75 Hz, 1.25 Hz, and so on. During this event, the frequency of the forced input was very close to that of the dominant NS-A mode (estimated to be 0.23 Hz during the event) and the source was in the area that participates in the NS-A mode thereby meeting two of the three requirements for the resonance condition. This resulted in oscillations that were propagated across a wide area in the WI due to the system resonance at one of the system's natural frequencies. If the damping ratio of the NS-A mode were to be critical or close to 0%, then the third requirement of the resonance condition would have been met potentially resulting in significantly higher power swings and could have resulted in severe consequences. However, during the event, the damping ratio of the NS-A mode was high (>15%) and therefore the system operations were not affected much except the shutting down of the battery units causing these oscillations.



**Figure C.63: Periodogram Calculated Using Power Output Measurements of the Two Battery Units Showing the Presence of Odd Harmonic Components in the Forced Input**

## Appendix D: Modal Oscillation Event Examples

### Event 1: August 10, 1996, Breakup of the Western Interconnection<sup>68</sup>

Figure D.1 below depicts measurements leading up to the August 10, 1996, outage in western North America.<sup>69</sup>

Natural responses are dominant throughout the plot. The system's ambient response is present until the 400-second mark when a transient response occurs. At 3.46% damping, the oscillation could be considered a poorly damped oscillation. The system's ambient response resumes at approximately the 430-second mark and lasts until about 700 seconds. At that point, a second transient response dominated by a growing oscillation occurs. The growing oscillation occurred because the damping of the oscillatory mode near 0.25 Hz became negative. The oscillatory mode shape of the 0.25 Hz oscillatory mode involves generators in the northern part of the system swinging against those in the southern part of the system. As a result, power flows on north-south tie-lines oscillated with increasing amplitudes until system breakup occurred.

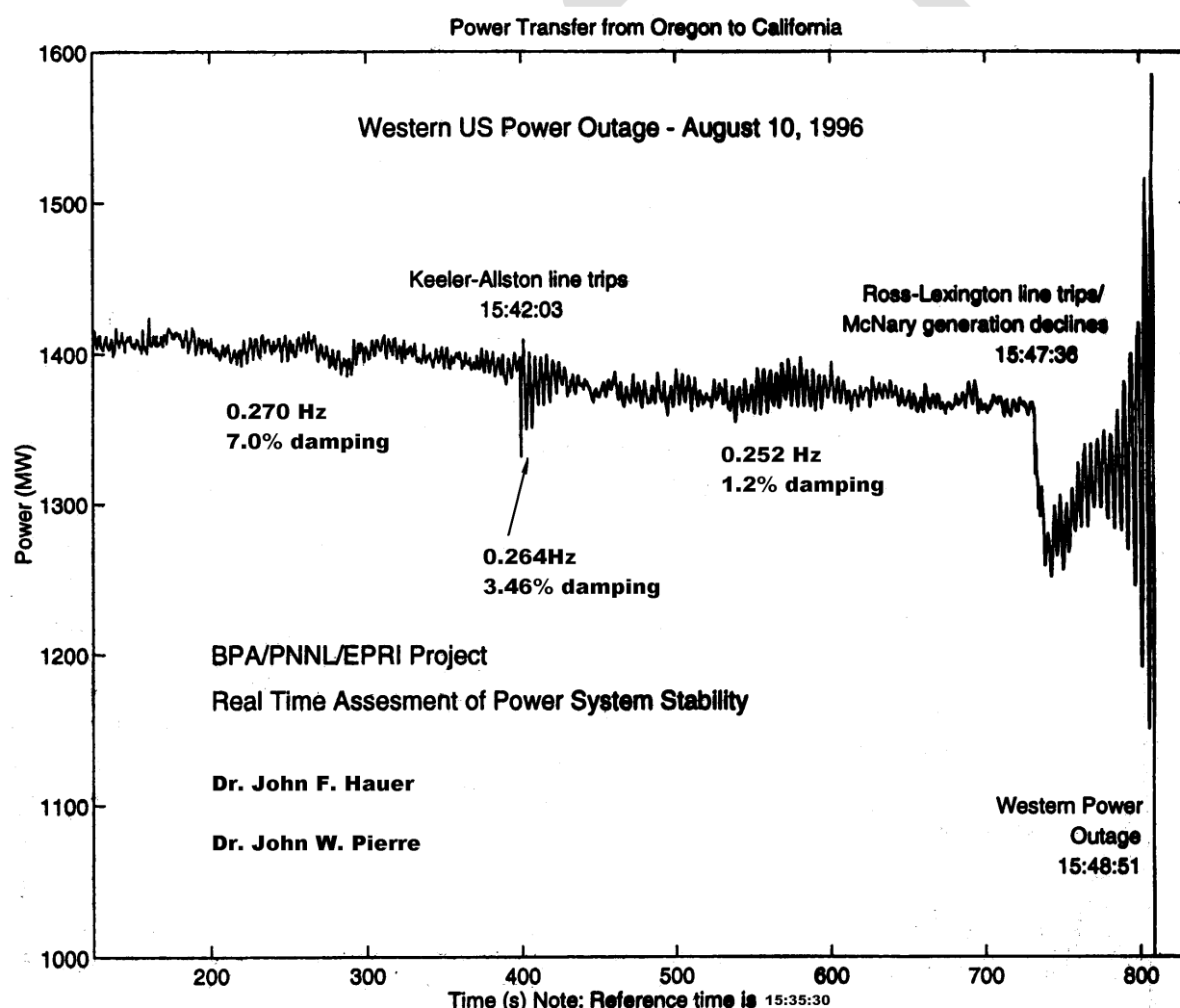


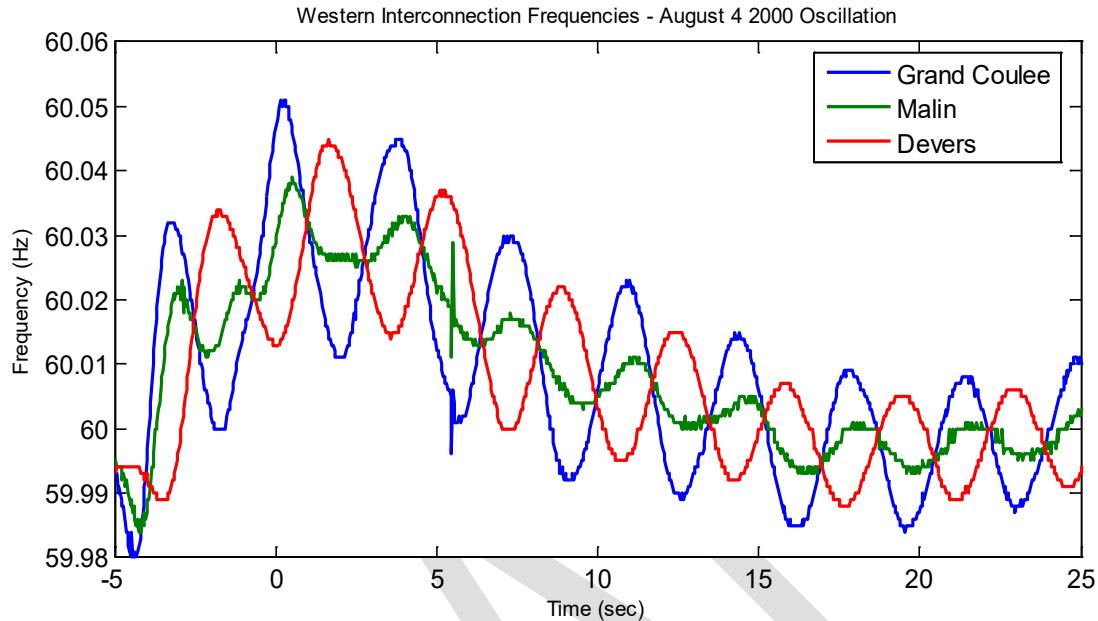
Figure D.1: Power Transfer from Oregon to California

<sup>68</sup> The event descriptions in this appendix are from the WECC report: *Bulk Power System Oscillation Terms* Western Electricity Coordinating Council (WECC) Oscillation Analysis Working Group (OAWG), August 2020.

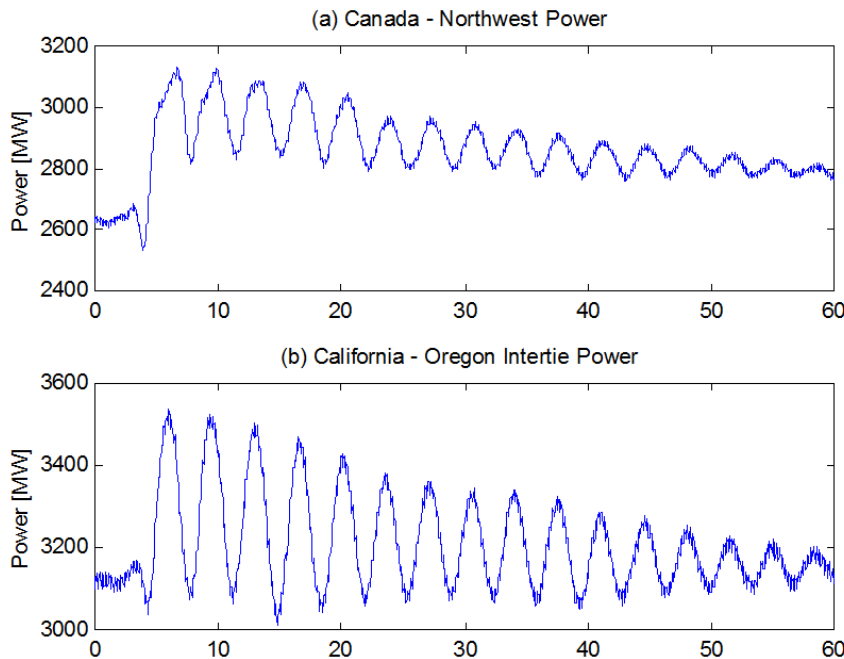
<sup>69</sup> The figure is reproduced from: *Model Validation for the August 10, 1996 WSCC System Outage* by Dmitry Kosterev, Carson Taylor, and William Mittelstadt published in IEEE Transactions on Power Systems, vol. 14, no. 3, pp. 967–979, Aug 1999.

## Event 2: August 4, 2000, WECC Inter-Area Natural Oscillations

On August 4, 2000, Alberta disconnected from the rest of the WI when the one 500 kV and two 230 kV ties between the Alberta and British Columbia systems were separated (see [Figure D.2](#) and [Figure D.3](#)). The separation resulted in a transient Response dominated by a poorly damped oscillation associated with one of the newly configured system's oscillatory modes.



**Figure D.2: Western Interconnection Frequency – August 4**



**Figure D.3: Power Graph**

## Appendix E: Contributors

NERC would like to thank all members of the NERC SMWG for their participation and guidance in developing this report. The following contributors (Table E.1) were involved in combining the previous two NERC oscillation reports into this document.

| Table E.1: Contributors       |                                        |
|-------------------------------|----------------------------------------|
| Name                          | Entity                                 |
| Jim Follum                    | Pacific Northwest National Laboratory  |
| Urmila Agrawal                | California Independent System Operator |
| Matthew Rhodes                | Salt River Project                     |
| Qiang “Frankie” Zhang         | ISO New England                        |
| Clifton R. Black (SMWG Chair) | Southern Company                       |
| Ryan Quint                    | Elevate Energy Consulting              |

For a complete list of contributors of the original materials, please refer to the original documents:  
[Forced Oscillation Monitoring and Mitigation Reliability Guideline](#)  
[Recommended Oscillation Analysis for Monitoring and Mitigation Reference Document](#)